

NEET **CHEMISTRY** MODULE - 7

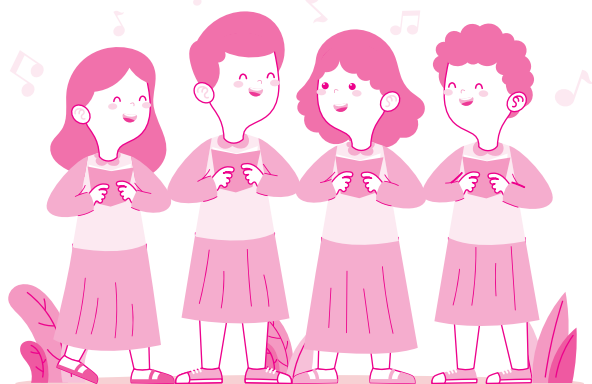
- Optical Isomerism
- Haloalkanes and Haloarenes
- Alcohols, Phenols & Ethers
- Aldehydes, Ketones, & Carboxylic Acids
- Amines
- Biomolecules

सर्वम प्रार्थना

हमको मन की शक्ति, देना मन विजय करें
दूसरों की जय से पहले खुद की जय करें
हमको मन की शक्ति, देना मन विजय करें
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भेद—भाव अपने दिल से साफ कर सकें
दूसरों से भूल हो तो माफ कर सकें
झूठ से बचे रहें, सच का दम भरें
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मुश्किलें पड़े तो हम पे, इतना कर्म करें
साथ दे तो धर्म का, चलें तो धर्म पर
खुदपे होसला रहे, बदी से ना डरें
दूसरों की जय से पहले, खुद को जय करें



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Chapter 1

OPTICAL ISOMERISM

Chapter Summary

- Introduction
- Symmetry elements
- Representation of optical isomers
- Configuration of optical isomers
- Calculation of number of optical isomers

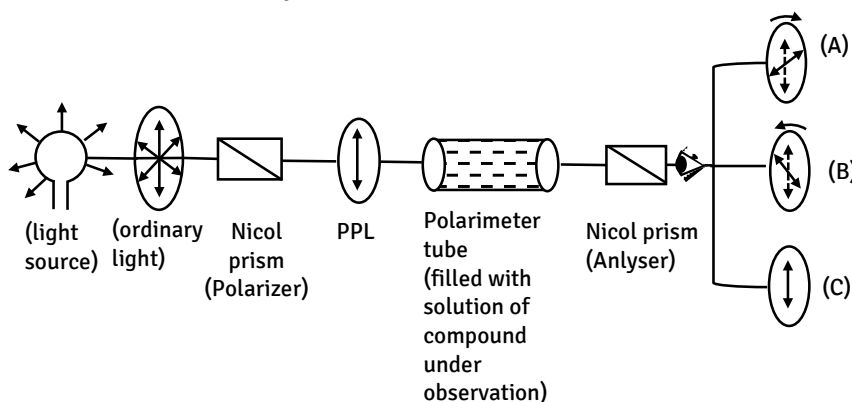
Introduction

Definition:

Compound having same molecular formula, same structural formula but different stereo chemical formula with different behaviour towards plane polarised light (PPL) are called optical isomers and the phenomenon is known as optical isomerism.

- Behaviour of optical isomers can be predicted by polarimeter experiment.

Polarimeter experiment:



Observation (A): Compound is optically active, rotating the plane of PPL in clockwise direction. It is known as dextrorotatory and is represented by $d/+$

Observation (B): Compound is optically active, rotating the plane of PPL in anti clockwise direction. It is known as laevorotatory and is represented by $l/-$.

Observation (C): This observation indicates that
(i) Compound is optically inactive and not an optical isomer.

Or

(ii) Compound is optical isomer but is optically inactive, it is known as meso isomer.

Or

(iii) Polarimeter tube was filled with equimolecular mixture of d and l mirror images molecules, it is known as racemic mixture and is represented by $dl/\pm$.

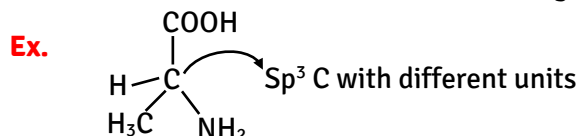
Note. The angle of rotation of plane of polarised light produced by solution depends on

- The concentration of solution (gm/ml) (C)
- The length of polarimeter tube (dm) (l)
- The temperature at the time of experiment
- The wave length of light.

Condition for optical Activity:

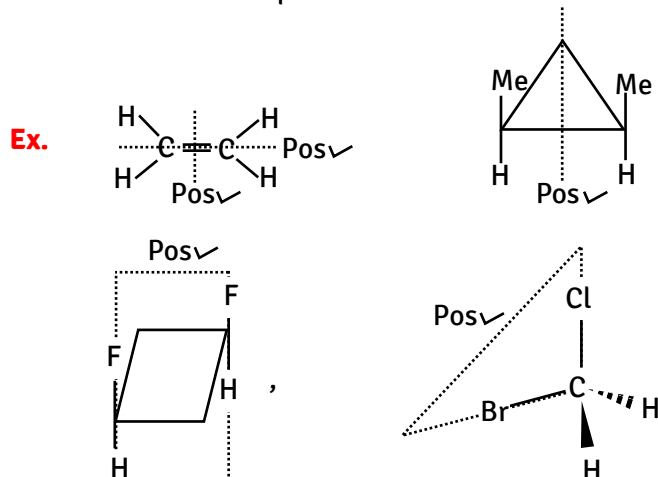
- For a molecule to be optically active, it must be chiral or asymmetrical
- A molecule is chiral or asymmetric, if it is nonsuperimposable on its mirror image
- For a molecule to be non superimposable on its mirror image, symmetry elements [plane of symmetry (pos) and centre of symmetry (Cos)] must be absent.

Note. Chirality of a molecule does not depend on presence or absence of chiral -C in it. Chiral C is Sp^3 C which is bonded to four different atoms or groups (units). It is also called asymmetric carbon.



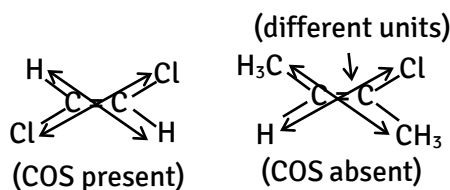
Symmetry elements

(1) **Plane of symmetry (pos):** An imaginary plane passing through a molecules which divides the molecule in two such equal halves that each half of the molecule is mirror image of the other half



(2) **Centre of symmetry (COS):** An imaginary point present in a molecule in such a way that every straight line passing through the point has similar units at its opposite ends present at the same distance from that point.

If even a single line passing through the point has different units at its opposite ends, COS is considered to be absent.

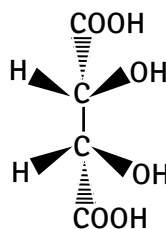
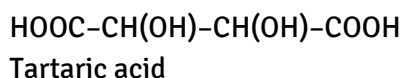
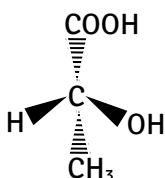
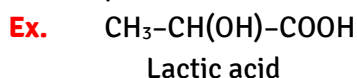


- (3) **Axis of symmetry (AOS):** If the new image obtained on rotating a molecule about an axis passing through the molecule by an angle θ is similar to the previous image, the molecule has C_n axis of symmetry where ($n = 360/\theta$)
every molecule at least has C_1 AOS, therefore this symmetry element does not decide optical activity of molecule. For a molecule to be optically active both COS & POS must be absent

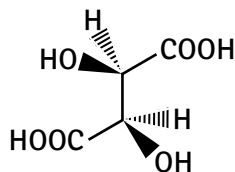
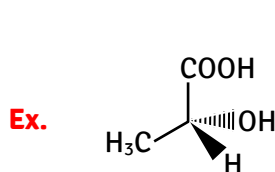
Representation of optical isomers

- (1) Perspective formula (3-D)
- (2) Wedge dash formula (3-D)
- (3) Fischer-projection formula (2-D)

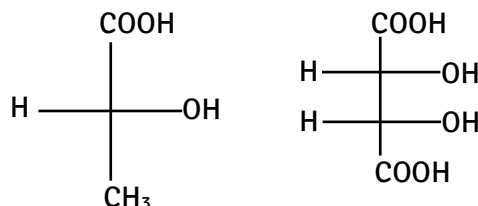
- (1) **Perspective formula (3-D):** The molecule is visualised in such a way that only chiral-c is maintained in plane and units on chiral-c are outside and inside.



- (2) **Wedge dash formula (3-D):** The molecule is visualised in such a way that out of four units on chiral carbon any two units along with chiral-c are maintained in a plane and among rest of the units, one is out side and other is inside the plane.



- (3) **Fischer-projection formula (2-D) :** It is a combination of horizontal and vertical lines. The molecule is visualised in such a way that only chiral c is maintained in a plane and units out side the plane are on horizontal line and inside the plane are on vertical line

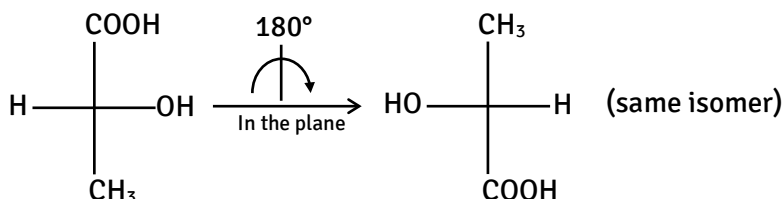


Note. Units on horizontal line are outside the plane or towards the observer in actual 3-D view and units on vertical line are inside the plane or away from the observer in actual 3-D view.

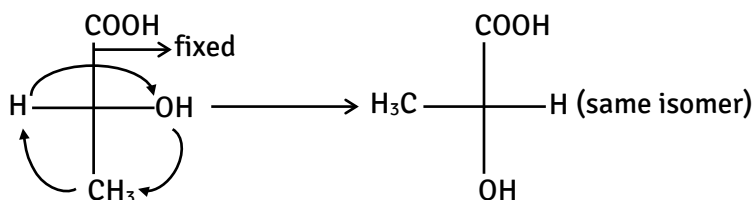
Conversion of one fisher projection formula into another:

An optical isomer with one chiral-c can be represented by twelve different fisher projection formula, so if we have one fisher projection formula then the rest of the eleven fisher projection formula can be made with the help of this fisher projection formula, It is as follows

Technique-I : If a Fischer projection formula is rotated by 180° in the plane (not by lifting from the plane) the new fisher projection formula represents same isomer



Technique-II : If in a fisher projection any one unit is kept fixed and other three are shifted clockwise or anti clock wise, the new fisher projection formula represents the same isomer



Note: A compound which show optical isomerism essentially has d & l isomers (theoretically) but may or may not have meso isomer

To have meso isomer following condition are required.

- (1) Compound should have more than one chiral-c
- (2) Compound should be symmetrical (in respect of IUPAC numbering)

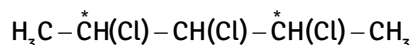
Ex. (i) $\text{HOOC} - \overset{*}{\text{C}}(\text{OH}) - (\text{H}/\text{OH}) - \text{COOH}$

- Symmetrical compound
- Number of chiral-c is two
- Possible isomers (all three types d, l & meso)

(ii) $\text{H}_3\text{C} - \overset{*}{\text{CH}}(\text{Cl}) - \overset{*}{\text{CH}}(\text{Cl}) - \overset{*}{\text{CH}}(\text{Cl}) - \text{CH}_3$

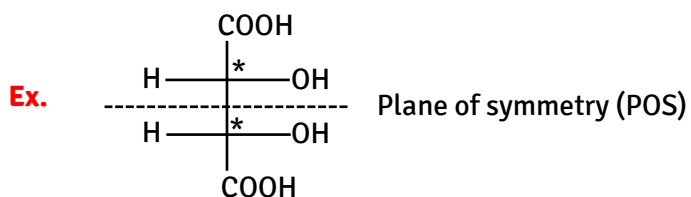
- Symmetrical compound
- Number of chiral-c is three
- Possible isomers (all three types d, l & meso)
- The centre most c is pseudo chiral -c

Pseudo chiral-c: If chirality of a carbon depends on configuration of units attached with it, is called pseudo chiral-c.



Configuration	S	achiral	S
	R	achiral	R
	R	chiral	S

Meso isomer : An optically inactive isomer having more than one chiral-c with symmetry element (Pos or cos) is called meso isomer

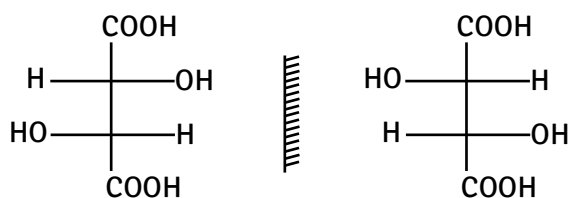


- It is always achiral molecule since it is superimposable on its mirror image
- Mirror image of a meso isomer always represents identical molecule (Homomer)
- Optical inactivity of a meso isomer is due to internal compensation / Intramolecular neutralization /molecular symmetry.

Racemic mixture : An equimolar mixture of d & l mirror image molecules is called racemic mixture.

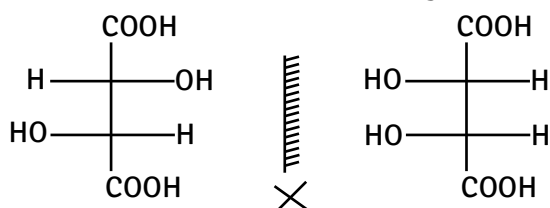
- It is a resolvable mixture and separation of d & l isomers from the mixture is called optical resolution.
- Conversion of d or l form into a racemic mix is known as racemization.
- Optical inactivity of racemic mixture is due to external compensation/Intermolecular neutralization/ optical impurity.

Enantiomers : d & l mirror image compounds are called enantiomers of each other



- Enantiomers are always chiral molecules since these are non-super imposable on mirror images.
- Enantiomers have similar physical and chemical properties but different biological properties.
- Enantiomers rotate the plane of light in opposite direction by same angle.

Diastereomers : stereoisomers which are not a mirror image of each other are called diastereomers



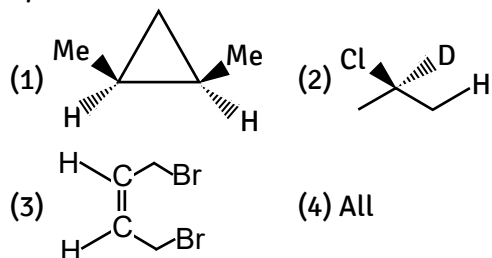
- Diastereomers maybe
 - Both optically active
 - Both optically inactive
 - One optically active & one optically inactive
- Physical properties of diastereomers like BP, MP solubility are different.
- Diastereomeric mixtures can be separated by using fractional distillation method.

Exercise 1.1

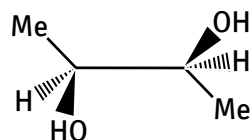
1. Which of the following can show optical isomerism :

(1) 1-Chlorobutane
(2) t-Butyl chloride
(3) sec-Butyl chloride
(4) Isobutyl chloride

2. Which of the following compound can show optical isomerism:



3. Which of the following statement is true for the given compound:

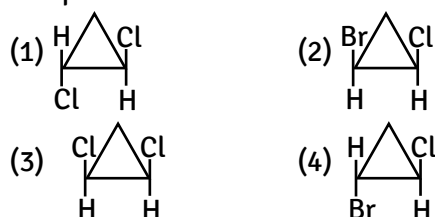


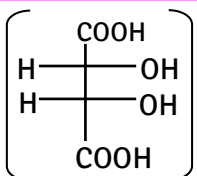
(1) Optically active
(2) Has plane of symmetry
(3) Has centre of symmetry
(4) 1 & 3 both

4. Which of the following contain chiral carbon atom:

(1) $\text{CH}_3\text{-CH-CH}_2\text{-CH}_2\text{-CH}_3$
 |
 OH
(2) $\text{CH}_3\text{-CH-CH-CH}_3$
 | |
 Br CH₃
(3) $\text{CH}_3\text{-CH}_2\text{-CH-CH}_3$
 |
 Br
(4) All the above

5. Which of the following compound is a meso compound:



6. Meso-tartaric acid  is optically

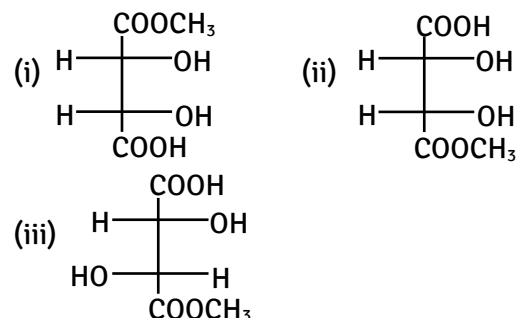
inactive due to the presence of-

(1) Molecular symmetry
(2) Molecular asymmetry
(3) External compensation
(4) Two asymmetric carbon atoms

7. Which is optically active molecule-

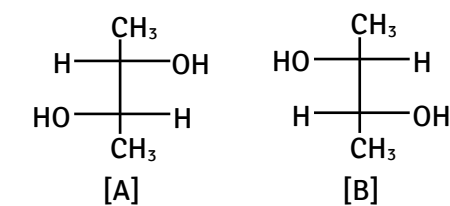
(1) $\text{C}_6\text{H}_5\text{-C(=O)-OH}$ (2) $\text{CH}_3\text{-CH(OH)-C}_2\text{H}_5$
(3) $\text{C}_6\text{H}_5\text{-CH(OH)-H}$ (4) $\text{C}_6\text{H}_5\text{-CH(OH)-CH}_3$

8. Correct statement about the compound (i), (ii) and (iii) is -



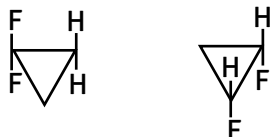
(1) (i) & (ii) are identical
(2) (i) & (ii) are diastereomers
(3) (i) & (iii) are enantiomers
(4) (i) & (ii) are enantiomers

9. If optical rotation produced by the compound [A] is + 65°, then produced by the compound [B] is -



(1) + 65° (2) -65°
(3) Zero (4) Unpredictable

10. The molecules below are:



- (1) Constitutional isomers
- (2) Enantiomers
- (3) Diastereomers
- (4) Identical

11. Which one of the following shows optical activity

- (1) $\begin{array}{c} \text{H} \\ | \\ \text{HO}-\text{C}-\text{COOH} \\ | \\ \text{H} \end{array}$
- (2) $\begin{array}{c} \text{H} \\ | \\ \text{CH}_3-\text{C}-\text{COOH} \\ | \\ \text{Cl} \end{array}$
- (3) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{COOH} \\ | \\ \text{OH} \end{array}$
- (4) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{COOH} \\ | \\ \text{Cl} \end{array}$

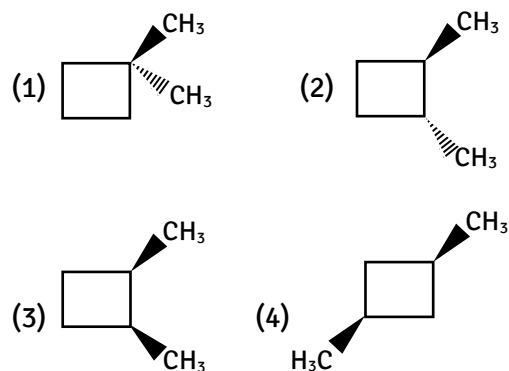
12. Which one of the following may exist in pair of enantiomers?

- (1) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{COOH} \end{array}$
- (2) $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}_3$
- (3) $\begin{array}{c} \text{NH}_2 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_3 \end{array}$
- (4) $\begin{array}{c} \text{NH}_2 \\ | \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$

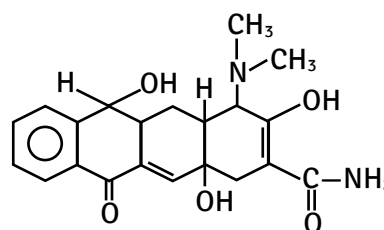
13. Which of the following compounds may not exist as enantiomers?

- (1) $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$
- (2) $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$
- (3) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$
- (4) $\text{C}_6\text{H}_5\text{CHClCH}_3$

14. Which of the following is a chiral molecule



15. How many chiral center present in tetracycline -

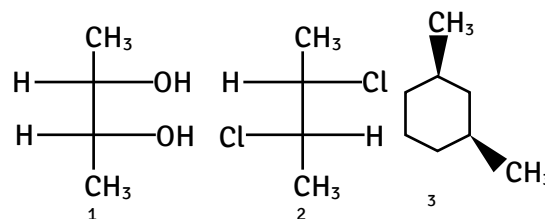


- (1) 6
- (2) 4
- (3) 8
- (4) 5

16. Among the following which one can have a meso form

- (1) $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{Cl})\text{C}_2\text{H}_5$
- (2) $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$
- (3) $\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$
- (4) $\text{HOCH}_2\text{CH}(\text{Cl})\text{CH}_3$

17. Which of the following compounds are meso forms?

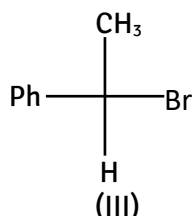
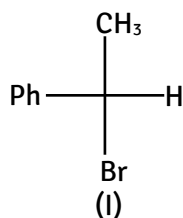


- (1) 1 Only
- (2) 1 and 3
- (3) 1 and 2
- (4) 2 and 3

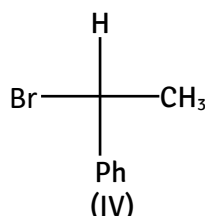
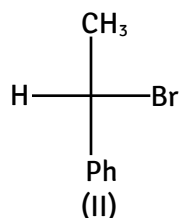
18. A compound whose molecule is superimposable on its mirror image despite containing chiral carbon atoms is called:

- (1) Threo isomer
- (2) Meso compound
- (3) Enantiomer
- (4) No special name

19. The pair of enantiomers among the following compound is:



- (1) I and IV
(3) II and III



- (2) II and IV
(4) I and I

20. Which of the following statements is correct?

- (1) The presence of chiral carbon is essential condition for enantiomerism.
- (2) Functional isomerism is a kind of stereoisomerism.
- (3) The compounds containing one chiral carbon are always chiral.
- (4) All statements are wrong.

Configuration of optical isomers

- (1) Absolute configuration (R/S configuration)
- (2) Relative configuration (D/L configuration)

(1) Absolute configuration :

R $\Rightarrow$ Rectus $\Rightarrow$ Right/clockwise

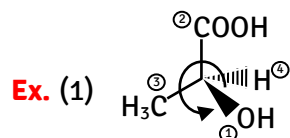
S $\Rightarrow$ Sinister $\Rightarrow$ left/Anticlockwise

R/S configuration in wedge dash formula

Rules :

- (i) Assign priority order to units attached with chiral-c according to CIP sequence rule.
- (ii) Neglect the least priority unit and check the arrangement of other three in their decreasing priority order if it is
Clock wise $\Rightarrow$ R-Configuration
Anticlockwise $\Rightarrow$ S-Configuration
- (iii) In assigning configuration, the least priority unit should be at the maximum distance (i.e. on dash)
If it is not then apply change to shift it on dash

Note. Change that can be applied is, keep any one unit fixed and swift other three in clockwise or anticlockwise direction.

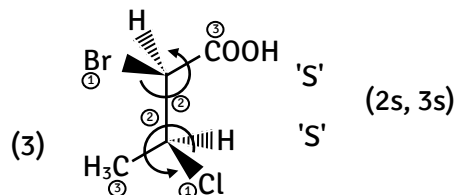
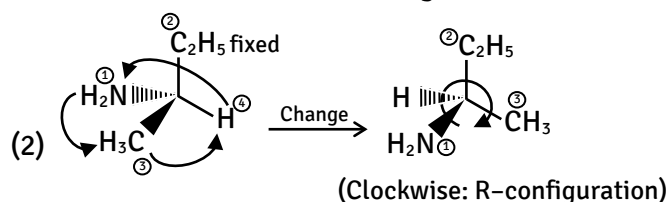


as per CIP sequence rule priority order of units is
-OH > -COOH > -CH₃ > -H

Neglect the least priority unit i.e. -H and check arrangement of other three in decreasing priority order that is



as it is anticlockwise so configuration is 'S'



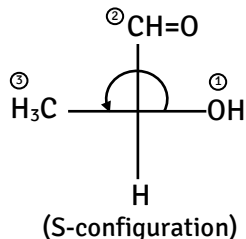
R/S configuration in fisher projection formula :

Rules are similar to wedge-dash

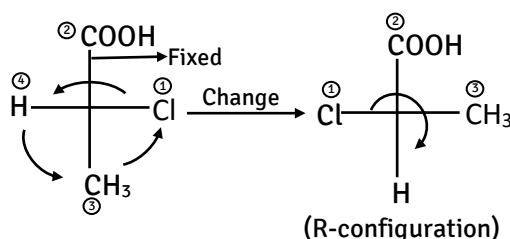
Note. As the least priority unit should be at maximum distance from observer (i.e. dash) so it should be on vertical line in fisher projection formula since units on vertical represent units away from observer (i.e. dash)

Ex.

(i)



(ii)



(2) Relative configuration (D/L configuration)

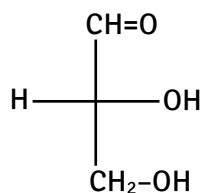
(i) Glyceraldehyde is considered as a reference molecule to assign D/L configuration to other compounds

(ii) In Fischer projection formula of glyceraldehyde in which longest c-chain should be on vertical line and most oxidised 'C' should be on top of vertical line

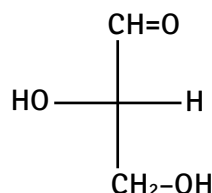
In this case 'OH' group is on

R.H.S. $\longrightarrow$ D - Configuration

L.H.S. $\longrightarrow$ L - Configuration



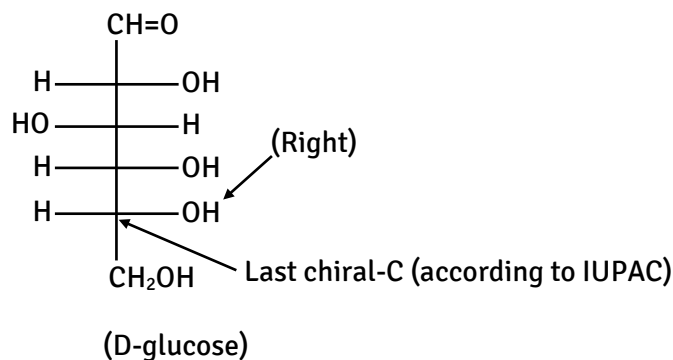
(D-Glyceraldehyde)



(L-Glyceraldehyde)

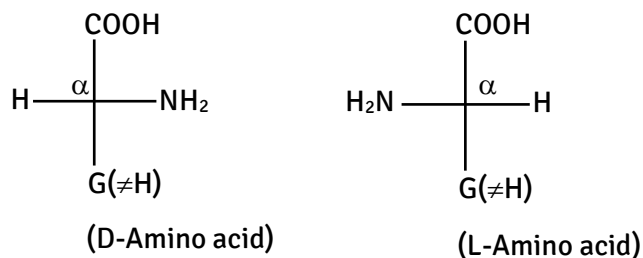
(iii) The system is used mainly for carbohydrates, Amino acids and comparable molecules.

Ex. Carbohydrates



Note. If more than one chiral c are having -OH group then D/L configuration is assigned on the basis of left/Right position of the -OH group present on last chiral -C (according to IUPAC)

Ex. Amino acid



Note. In amino acid D/L configuration is assigned on the basic of left/Right position of -NH₂ group on alpha-C

Calculation of number of optical isomers

	Type of compound	d+l	Meso isomer	Total isomers
(1)	Unsymmetrical	2^n	Zero	2^n
(2)	Symmetrical n=even no.	2^{n-1}	$2^{n/2-1}$	$2^{n-1} + 2^{n/2-1}$
	n=odd no	$2^{n-1} - 2^{(n-1)/2}$	$2^{(n-1)/2}$	2^{n-1}

n = number of chiral-c or stereo sites where optical isomerism is possible

Ex. (1) $\text{CH}_3 - \overset{*}{\text{CH}}(\text{Cl}) - \overset{*}{\text{CH}}(\text{Br}) - \text{CH}_3$

Unsymmetrical compound

n=2

(d+l) = $2^n = 2^2 = 4$

Note. Unsymmetrical compound does not has meso isomer

Ex. (2) $\text{CH}_3 - \overset{*}{\text{CH}}(\text{Cl}) - \overset{*}{\text{CH}}(\text{Cl}) - \text{CH}_3$

Symmetrical compound

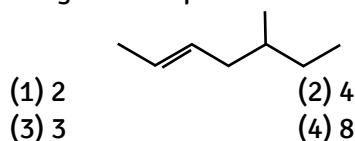
n=2 (even number)

(d+l) = $2^{n-1} = 2^{2-1} = 2$

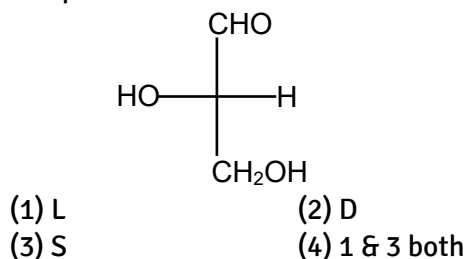
Meso = $2^{n/2-1} = 2^{2/2-1} = 2^0 = 1$

Exercise 1.2

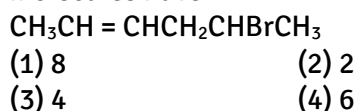
1. How many stereoisomers are possible for the given compound:



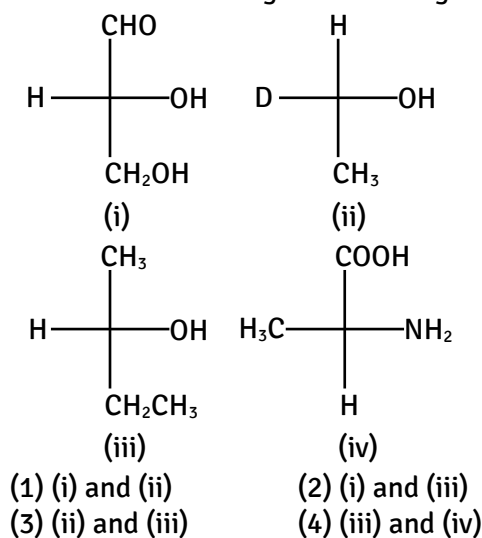
2. Absolute configuration of the given compound is:



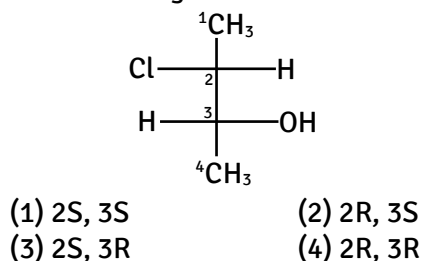
3. How many stereoisomers does this molecules have



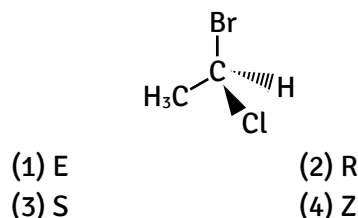
4. The R-isomers among the following are



5. The absolute configurations of the C_2 and C_3 atoms in the given molecule is



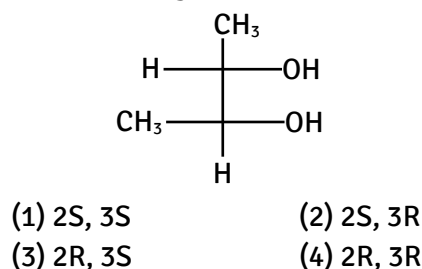
6. The configuration of the given compound is:



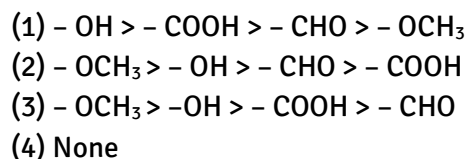
7. The total number of stereoisomers possible for 2,3-Dichlorobutane:



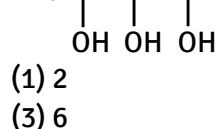
8. Correct configuration of the following is:



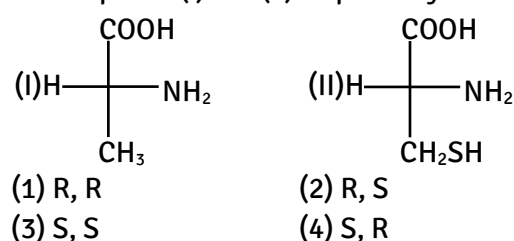
9. The priority of groups $-\text{OH}$, $-\text{COOH}$, $-\text{CHO}$, $-\text{OCH}_3$ attached to a chiral carbon is in order -



10. No. of optically active isomers possible in $\text{CH}_3-\text{CH}-\text{CH}-\text{CH}-\text{CH}_2\text{OH}$ are -



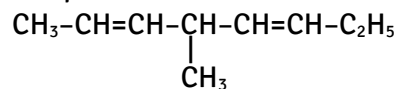
11. The correct absolute configuration assigned for compound (I) and (II) respectively is :



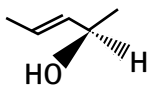
12. Which of the following has D configuration

- (1)
$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$
- (2)
$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{OH} \end{array}$$
- (3)
$$\begin{array}{c} \text{CHO} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{OH} \end{array}$$
- (4)
$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{CH}_2\text{OH} \\ | \\ \text{OH} \end{array}$$

13. Total number of stereoisomer of the given compound is :



- (1) 2 (2) 4
(3) 6 (4) 8

14.  is named as :-

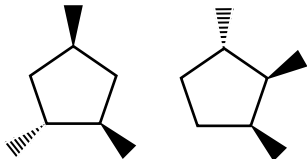
- (1) (2R, 3E)-pent-3-en-2-ol
(2) (2S, 3E)-pent-3-en-2-ol
(3) (2E, 3R)-pent-2-en-3-ol
(4) (2E, 3S)-pent-2-en-3-ol

15. How many compounds are optically active in the isomers of $\text{C}_4\text{H}_9\text{Br}$

- (1) 1 (2) 2
(3) 3 (4) 4

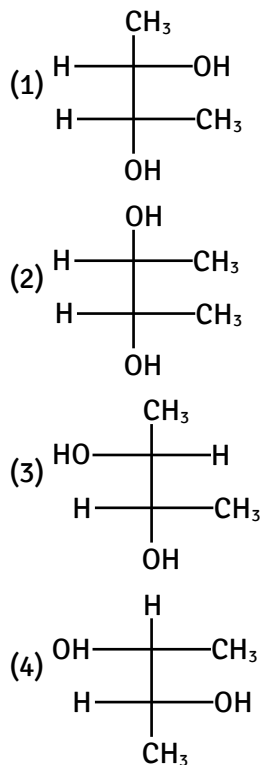
Exercise 2

1. The following pair of compounds is best described as :

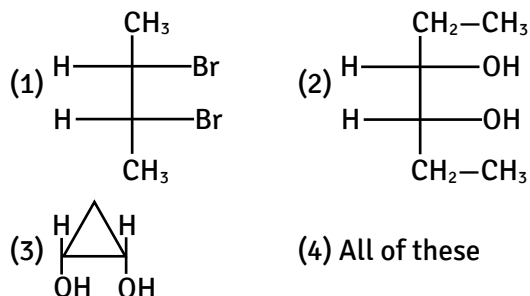


- (1) Identical
(2) Diastereomers
(3) Enantiomers
(4) Constitutional isomers

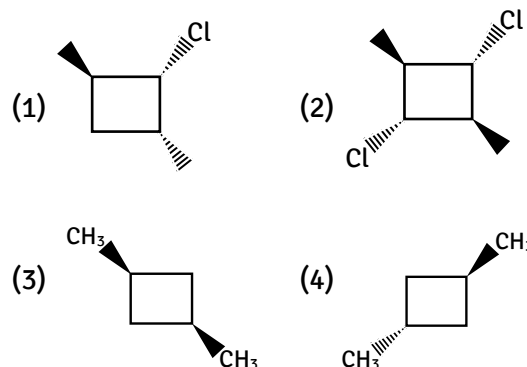
2. Which of the following structures is not meso-2,3-butanediol?



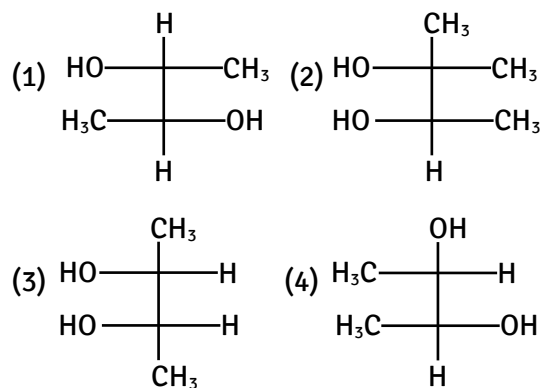
3. Which of the following is a meso compound?



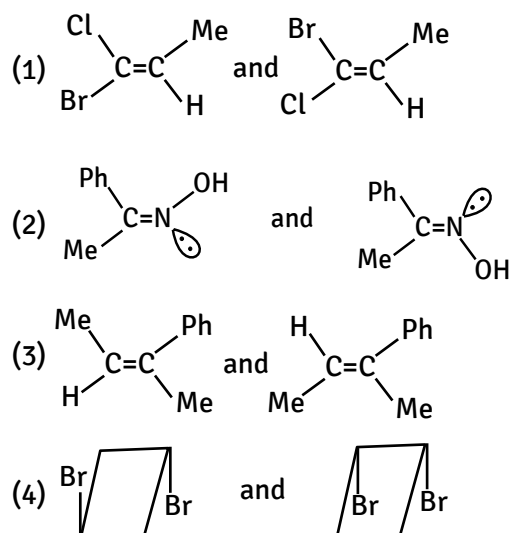
4. In which of the following compound, posses plane of symmetry as well as centre of symmetry?



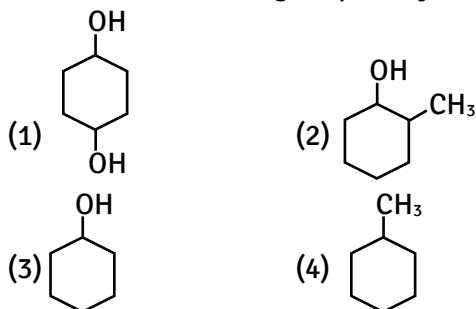
5. Which of the following is optically inactive



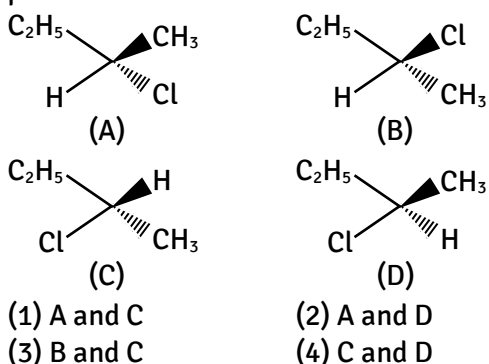
6. Which is not a pair of geometrical isomers ?



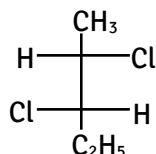
7. Which of the following is optically active:



8. Which of the following are not enantiomeric pair.

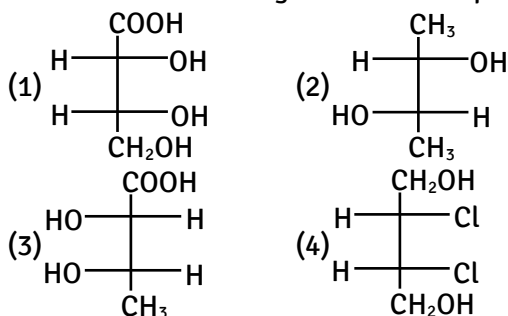


9. The absolute configuration of the given compound is-



- (1) 2S, 3R
(2) 2S, 3S
(3) 2R, 3S
(4) 2R, 3R

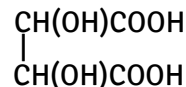
10. Which of the following is a meso-compound



11. Number of asymmetric C atoms are present in-

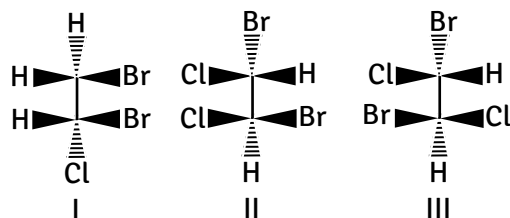
- (i) 1,2-Dimethyl cyclohexane
(ii) 3-Methyl cyclopentene
(iii) 3-Methyl cyclohexene
(1) Two, one, one
(2) One, one, one
(3) Two, none, two
(4) Two, none, one

12. Number of optically active isomers of tartaric acid is -



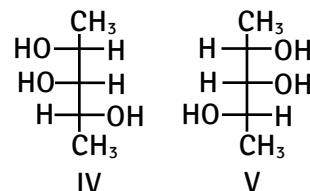
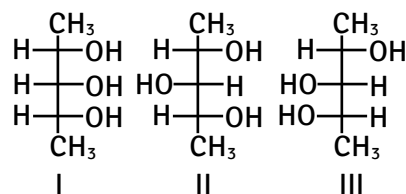
- (1) 2
(2) 3
(3) 4
(4) 5

13. Which molecule is achiral?



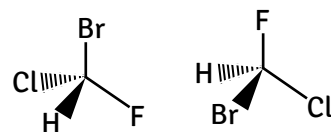
- (1) I
(2) II
(3) III
(4) All of these

14. Which pair of structures represents the same compound?



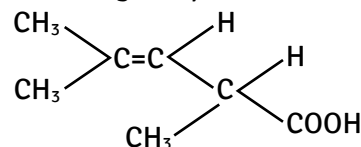
- (1) I and II
(2) II and III
(3) III and IV
(4) III and V

15. The two compounds shown below are:



- (1) Enantiomers
(2) Diastereomers
(3) Constitutional isomers
(4) Identical

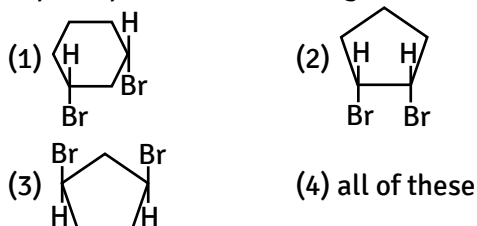
16. The following compound can exhibit



- (1) Tautomerism
(2) Optical isomerism
(3) Geometrical isomerism
(4) Geometrical and optical isomerism both

17. Out of the following the alkene that exhibits optical isomerism is
 (1) 2-Methyl-2-pentene
 (2) 3-Methyl-2-pentene
 (3) 4-Methyl-1-pentene
 (4) 3-Methyl-1-pentene

18. Which of the following compounds can have superimposable mirror image?



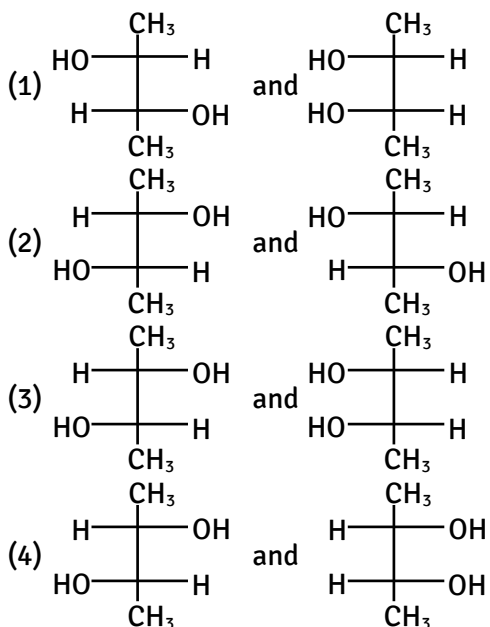
19. Predict the number of stereoisomers in $\text{CH}_2\text{OH}(\text{CHOH})_4\text{CHO}$

- (1) 16
 (2) 8
 (3) 4
 (4) 2

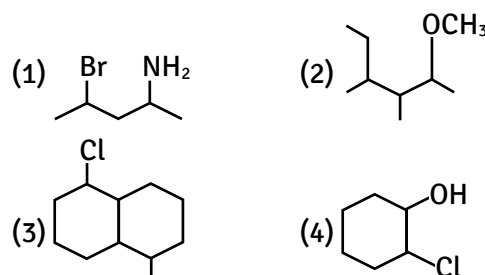
20. Meso stereoisomer is possible in which of the following compounds?

- (I) 2,4-Dibromopentane
 (II) 2,3-Dibromopentane
 (III) 3-Bromo-2-pentanol
 (IV) cis-1,3-Dimethylcyclohexane
 (V) trans-1,3-Dimethylcyclohexane
 (1) I
 (2) I and III
 (3) I and IV
 (4) None of these

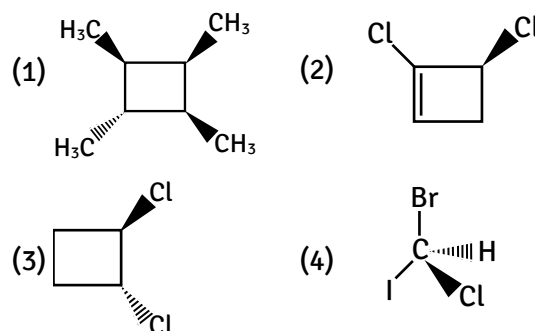
21. Which of the following pairs of compounds are enantiomers



22. The compound which has maximum number of chiral carbon is :-



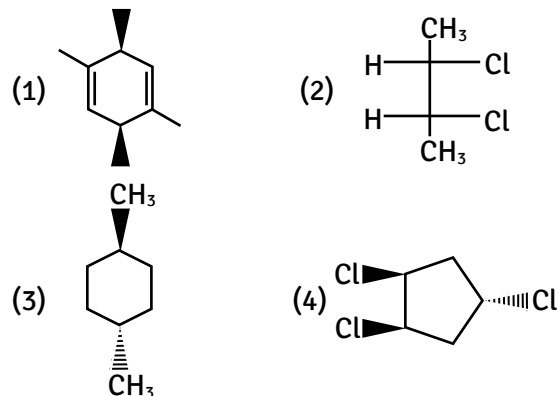
23. Which of the following compound has plane of symmetry



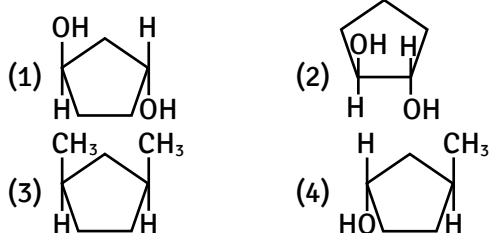
24. Which of the following statement is true :-

- (1) Compound having chiral- C essentially shows optical isomerism
 (2) Compound having one chiral- C can not have meso isomer
 (3) Chiral molecule may or may not have chiral-C
 (4) All

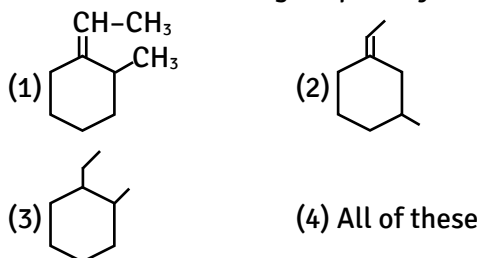
25. Which of following compound is chiral :-



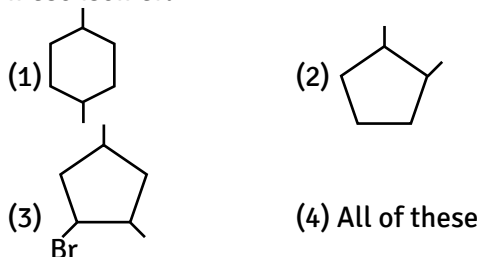
26. Which of the following cyclopentane derivative is optically inactive



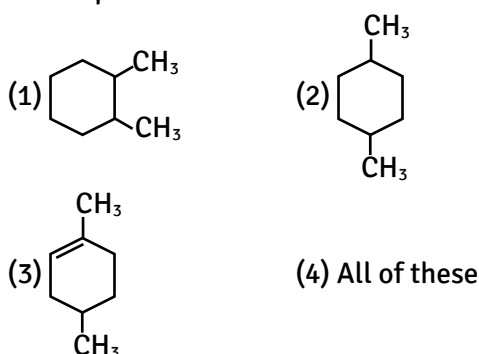
27. Which of the following is optically active?



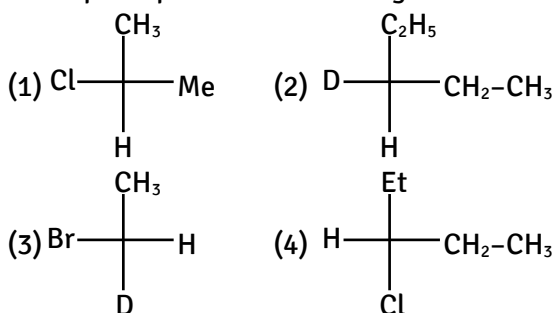
28. Which of the following compounds can have meso isomer?



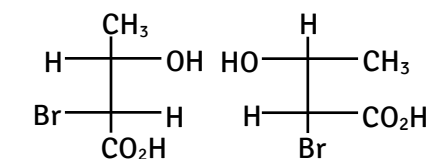
29. Which of the following compound will not show optical isomerism?



30. Which of the following molecule has nonsuperimposable mirror images?

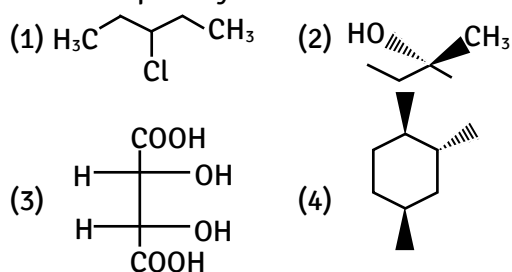


31. The molecules represented by the following two structures are



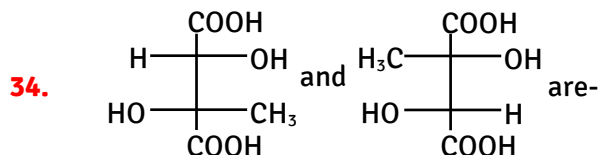
- (1) Identical (2) Enantiomers
(3) Diastereomers (4) Epimers

32. Which is optically active?



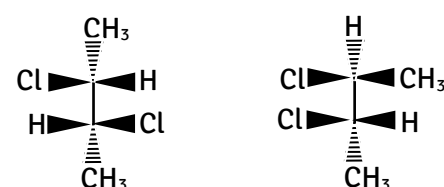
33. The necessary and sufficient condition for a molecule to be optically active is that

- (1) It must contain asymmetric carbon atoms
(2) It must be symmetric
(3) It must be identical with its mirror image
(4) It must be non-identical with its mirror image



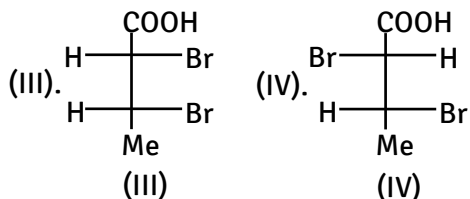
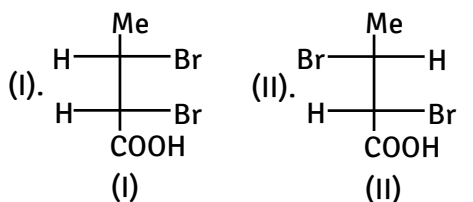
- (1) Enantiomers
(2) Position isomers
(3) Geometrical isomers
(4) Homomers

35. The molecules shown are:



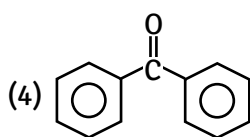
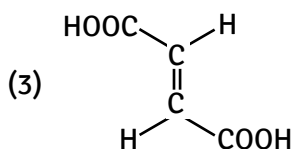
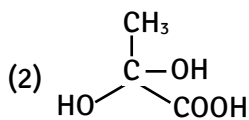
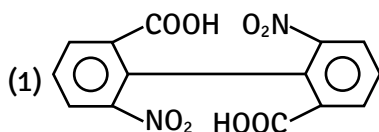
- (1) Enantiomers
(2) Diastereomers
(3) Constitutional
(4) Two different conformations of the same molecules

36. Which of the following are diastereomers?



- (1) (I) and (III) (2) (II) and (IV)
 (3) (I) and (II) (4) None

37. Which compound would exhibit optical isomers:



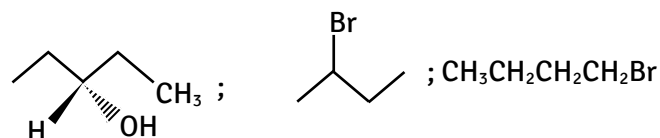
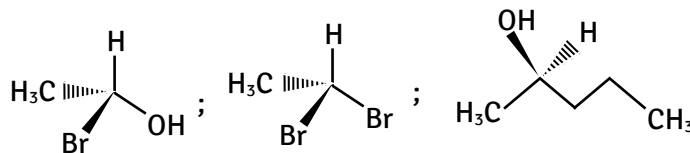
38. Which of the following is not chiral :

- (1) 2-Butanol
 (2) 2,3-Dibromopentane
 (3) 3-Bromopentane
 (4) 2-Hydroxypropanoic acid

39. A pair of diastereomers are

- (1) A pair of such stereoisomers which are not mirror image of each other
 (2) A pair of non-superimposable mirror image of each other
 (3) Always optically active
 (4) Both (1) and (3)

40. The number of chiral molecules and achiral molecules are respectively among given molecules



- (1) 3, 3
 (2) 4, 2
 (3) 2, 4
 (4) None of the above

Exercise 3

1. **Assertion:** Diastereomers are not mirror image of each other.
Reason: Diastereomers may be optically active.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) If Assertion is True but the Reason is False
(4) If both Assertion & Reason are False
2. **Assertion:** Racemic mixture is optically inactive.
Reason: Racemic mixture is mixture of meso structures.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) If Assertion is True but the Reason is False
(4) If both Assertion & Reason are False
3. **Assertion:** Chiral objects are not superimpose on their mirror image.
Reason: Chiral objects have plane of symmetry.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) If Assertion is True but the Reason is False
(4) If both Assertion & Reason are False
4. **Assertion:** An organic compound is optically active only when its mirror image is non-super imposable irrespective of the fact whether it contains a chiral carbon atom or not.
Reason: Organic compounds which do not contain chiral carbon atoms can not be optically active.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) If Assertion is True but the Reason is False
(4) If both Assertion & Reason are False
5. **Assertion:** Meso tartaric acid is optically inactive.
Reason: Its optically inactivity is due to internal compensation.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) If Assertion is True but the Reason is False
(4) If both Assertion & Reason are False
6. **Assertion :** Diastereo isomers have different physical properties
Reason: Diastereo isomers are always optically active
(1) Both Assertion and Reason are correct and Reason is correct explanation for the Assertion
(2) Both Assertion and Reason are correct but Reason is not correct explanation for Assertion
(3) Assertion is correct but Reason is incorrect
(4) Assertion is incorrect but Reason is correct

7. **Assertion:** $\text{CH}_2=\text{C}=\text{CH}_2$ exhibits optical isomerism.

Reason: Its mirror image is nonsuperimposable.

- (1) Both Assertion and Reason are correct and Reason is correct explanation for the Assertion
- (2) Both Assertion and Reason are correct but Reason is not correct explanation for Assertion
- (3) Assertion is correct but Reason is incorrect
- (4) both Assertion and Reason are incorrect

8. Given below are two statements:

Statement-I : Compound which are Non superimposable on their mirror images are called chiral & property is known as chirality.

Statement-II : All compounds which are superimposable mirror images of each other are called meso compounds.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Both statements-I and II are correct.
- (3) Both Statement-I and II are incorrect
- (4) Statement-I is incorrect but Statement-II is correct.

9. **Statement-I:** Molecules that are not superimposable on their mirror images are chiral.

Statement-II: All chiral molecules have chiral centres.

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Both statements-I and II are correct.
- (3) Both Statement-I and II are incorrect
- (4) Statement-I is incorrect but Statement-II is correct.

10. **Statement-I:** A racemic mixture shows zero optical rotation.

Statement-II: Diastereomers mixture can be separated by fractional distillation method

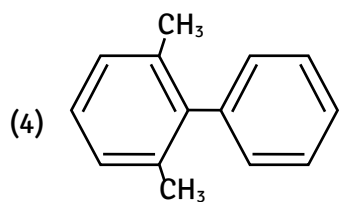
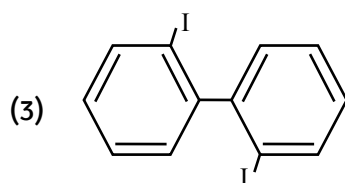
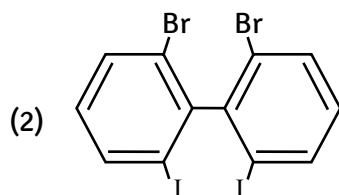
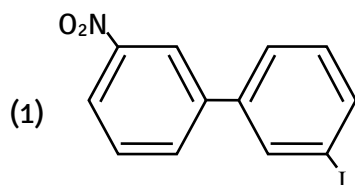
- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Both statements-I and II are correct.
- (3) Both Statement-I and II are incorrect
- (4) Statement-I is incorrect but Statement-II is correct.

Exercise 4 (Previous Year's Questions)

1. Two possible stereo-structures of $\text{CH}_3\text{CHOH.COOH}$, which are optically active, are called : **[AIPMT PRE-2015]**

(1) Enantiomers
(2) Mesomers
(3) Diastereomers
(4) Atropisomers

2. Which of the following biphenyls is optically active ? **[NEET-I 2016]**



3. Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula $\text{C}_4\text{H}_8\text{O}$ is:

[NEET- 2025]

(1) 10 (2) 11
(3) 6 (4) 8

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	1	4	3	1	2	4	2	1	2	4	3	2	4	2	2	2	3	3

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15					
Ans.	2	3	3	1	4	2	2	1	3	4	2	1	4	1	2					

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	4	4	3	4	2	1	2	4	1	1	3	4	1	2	4	4	1	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	3	1	4	1	3	4	2	2	3	1	4	4	4	1	3	1	3	1	1

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10										
Ans.	2	3	3	3	1	3	4	1	1	2										

Exercise 4 (Previous Year's Questions)

Que.	1	2	3																	
Ans.	1	2	1																	

Chapter 2

HALOALKANES AND HALOARENES

Chapter Summary

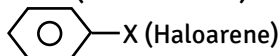
- Introduction
- Methods of preparation of haloalkanes
- Preparation of Haloarenes
- Physical properties
- Chemical Reaction of haloalkanes
- Reaction of haloarenes
- Polyhalogen compounds

Introduction

Substitution of hydrogen atom (s) of an aliphatic or aromatic hydrocarbon by halogen atom (s) results in formation of halogen derivatives.

- These are organic compound in which halogen is directly attached with carbon atom.
- These are called as haloalkanes (halogen bonded to alkyl group) and haloarene (halogen bonded to aryl group)

Ex. $R-X$ (Haloalkanes)



- In these compounds hybridisation of carbon is Sp^3 (alkyl halides) and Sp^2 (Aryl halides)

Classification:

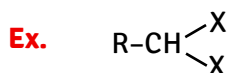
(A) On the basis of number of halogen atoms:

(i) Mono halides : consist of single halogen atom

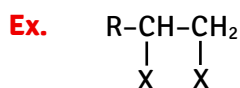
Ex. CH_3-Cl , $CH_3-CH_2-CH_2-Br$

(ii) Dihalides: consist of two halogen atoms are further categorised mainly as

- Gem dihalides (Two halogens on same carbon):



- Vicinal dihalides (Two halogens on adjacent carbon):



(iii) Polyhalides: consist of three, four etc halogen atoms

- **Trihalides:** consist of three halogen atoms

Ex. CHX_3 (Haloform)

- **Tetra halides:** Consist of four halogen atoms

Ex. CCl_4

(B) On the basis of type of C to which halogen is bonded:

(I) Compounds containing Sp^3 C-X Bond (x = F, Cl, Br, I)

(a) Alkyl halides or haloalkanes (R-X):

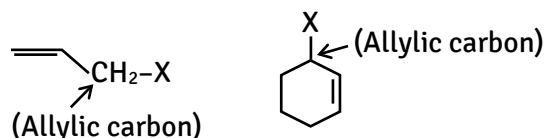
- General formula : $C_nH_{2n+1}X$
- Depending on nature of carbon to which halogen is attached, these are categorised as

Primary (1°) halide $R-\overset{1^\circ}{CH_2}-X$

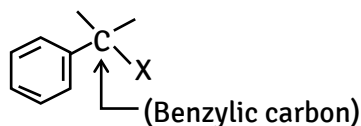
Secondary (2°) halide $R-\overset{2^\circ}{CH}-X$

Tertiary (3°) halide $R-\overset{3^\circ}{CH}-X$

(b) **Allylic halides:** Halogen atom bonded to Sp^3 C adjacent to carbon-carbon double bond (C = C)

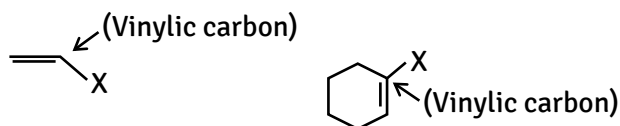


(c) **Benzylic halides:** Halogen atom bonded to Sp^3 C atom attached to benzene ring.

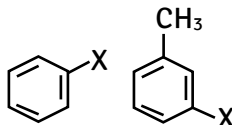


(II) Compounds containing Sp^2 C-X Bond

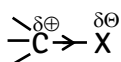
(a) **Vinyl halides:** Halogen atom bonded to Sp^2 C atom of carbon-carbon double bond.



(b) **Aryl halides:** Halogen atom bonded to Sp^2 C atom of benzene ring



Nature of C-X bond: it's a polar bond, since halogen is more electronegative than carbon so C atom bears partial positive and halogen atom bears partial negative charge



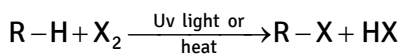
Order of C-X bond length : $C-I > C-Br > C-Cl > C-F$

Bond strength : $C-I < C-Br < C-Cl < C-F$

Dipole moment of different CH_3-X : $CH_3Cl > CH_3F > CH_3Br > CH_3I$

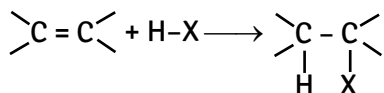
Methods of preparation of haloalkanes

(1) By halogenation of Alkanes:



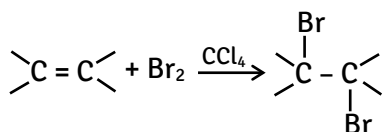
(detailed mechanism is given in hydrocarbon)

(2) By hydrohalogenation of Alkenes:



(detailed mechanism is given in hydrocarbon)

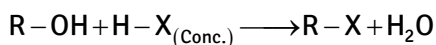
(3) By halogenation of alkenes:



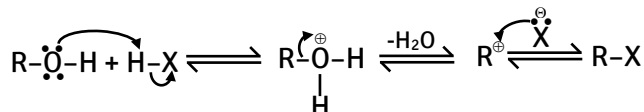
(detailed mechanism is given in hydrocarbon)

(4) From alcohols:

(a) Reaction with conc. HX



mechanism:



- Formation of carbocation is rate determining step.
- Reactivity of different HX : $\text{HI} > \text{HBr} > \text{HCl}$

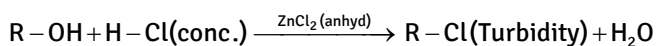
Note: In reaction of alcohol with conc. HCl, either alcohol should be highly reactive (reference 3° alcohol) or the reaction is carried out in presence of anhydrous ZnCl_2 .

Function of anhydrous ZnCl_2 :

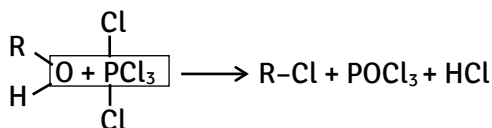
(i) Absorbs water formed in r.d.s.

(ii) Improves ionisation of HCl : $\text{HCl} + \text{ZnCl}_2 \rightleftharpoons \overset{\oplus}{\text{H}} + \overset{\ominus}{\text{ZnCl}_3}$

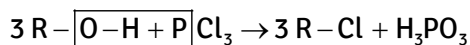
Note: Mixture of conc. HCl and anhydrous ZnCl_2 is known as **Lucas reagent**. It is used to differentiate 1°, 2° & 3° alcohols.



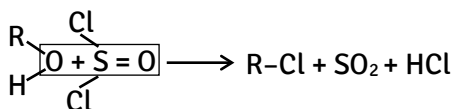
{	R-OH	time taken to appear turbidity	}
	3°	instant/immediate (2 -3 sec.)	
	2°	5-7 min.	
	1°	No turbidity at room temp. (on heating turbidity appear after 30 min.)	

(b) Reaction with PCl_5 

Note: In reaction with PCl_5 , POCl_3 is formed as by product.

(c) Reaction with PCl_3 

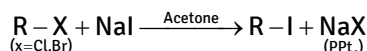
Note: In reaction with PCl_3 , H_3PO_3 is formed as by product.
For reaction with PBr_3 [$\text{P}+\text{Br}_2$] and PI_3 [$\text{P}+\text{I}_2$] is used in reaction.

(d) Reaction with SOCl_2 (Thionyl chloride)

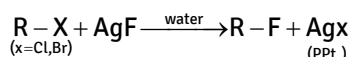
Note: In reaction with SOCl_2 , SO_2 is formed as by product.

Note: In preparation of Haloalkanes from alcohols, Thionyl chloride is preferred because in this reaction alkyl halide is formed along with gases SO_2 and HCl . The two gases products are escapable, hence the reaction gives pure alkyl halide.

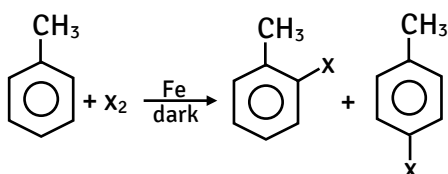
The method used to prepare alkyl halide from alcohol is not applicable for preparation of aryl halides from phenol because the carbon-oxygen bond in phenol has a partial double bond character and is difficult to break being stronger than a single bond.

(a) By halogen exchange method:**(a) Finkelstein Reaction:**

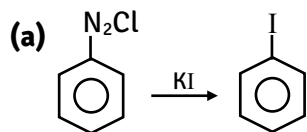
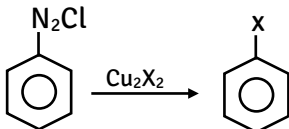
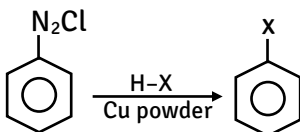
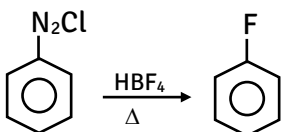
NaCl or NaBr formed is precipitated in dry acetone. It facilitates the reaction in forward direction.

(b) Swarts Reaction:

Metallic fluoride such as Hg_2F_2 , CoF_2 or SbF_3 can also be used in the reaction.

Preparation of Haloarenes**(1) From Aromatic hydrocarbon (Arene):**

- The reaction involves electrophilic substitution mechanism
- Reaction with iodine is reversible in nature and requires the presence of an oxidising agent (like $\text{HNO}_3/\text{HIO}_3$) to oxidise HI formed during iodination.
- Fluoro compounds are not prepared by this method.

(2) Form diazonium salt:**(b) Sandmeyer's reaction:****(c) Gattermann reaction:****(d) Balz Schiemann reaction****Exercise 1.1**

1. In Finkelstein reaction, which reagent is used :

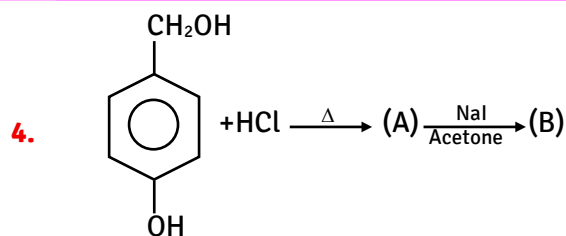
- (1) $\text{NaI} + \text{C}_2\text{H}_5\text{OH}$
- (2) $\text{NaCl} + \text{Acetone}$
- (3) $\text{NaBr} + \text{CH}_3\text{COCH}_3$
- (4) $\text{NaI} + \text{CH}_3\text{COCH}_3$

2. Which of the following leads to the formation of an alkyl halide :

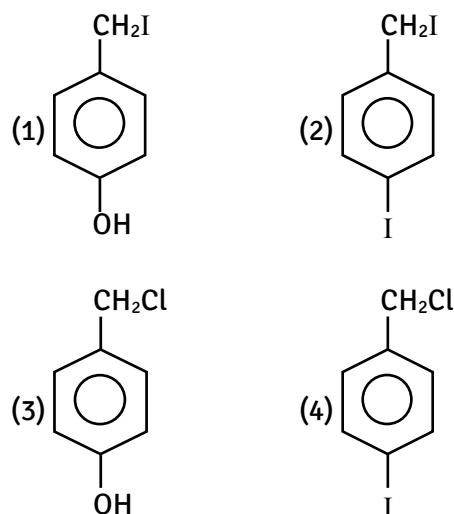
- (1) $\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{RedP} + \text{Br}_2}$
- (2) $\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{SOCl}_2}$
- (3) $\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{KBr} + \text{Conc. H}_2\text{SO}_4}$
- (4) All the above

3. The best reagent for converting ethanol to chloroethane is –

- (1) PCl_3
- (2) PCl_5
- (3) SOCl_2
- (4) $\text{HCl} + \text{ZnCl}_2$



In the above reaction product B is:

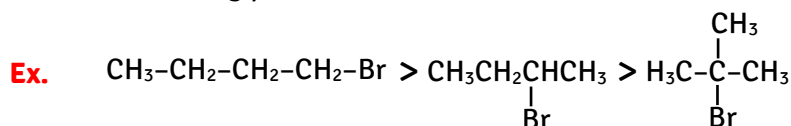


5. $\text{CH}_3\text{--CH}_2\text{--CH--CH}_3$ $\xrightarrow[\text{(iii) CH}_3\text{OH}]{\text{(i) NaBr, H}_2\text{SO}_4, \text{(ii) Mg, dry ether}}$ [X]
 OH Product
- Product [X] formed in the above reaction is
- (1) $\text{CH}_3\text{--CH}_2\text{--CH--CH}_3$
 OCH_3
 - (2) $\text{CH}_3\text{--CH=CH--CH}_3$
 - (3) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_3$
 - (4) $\text{CH}_3\text{--CH}_2\text{--CH--CH}_3$
 Br
6. The major product formed in the following reaction is :
- $\text{CH}_3\text{--CH=CH--CH(CH}_3)_2 \xrightarrow{\text{HBr}}$
- (1) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--C(CH}_3)_2\text{Br}$
 - (2) $\text{Br--CH}_2\text{--CH}_2\text{--CH}_2\text{--CH(CH}_3)_2$
 - (3) $\text{CH}_3\text{--CH(Br)--CH}_2\text{--CH(CH}_3)_2$
 - (4) $\text{CH}_3\text{--CH}_2\text{--CH(Br)--CH(CH}_3)_2$
7. Which of the following reaction occur most rapidly?
- (1) $(\text{CH}_3)_2\text{CHOH} \xrightarrow{\text{HBr}}$
 - (2) $(\text{CH}_3)_2\text{CHOH} \xrightarrow{\text{HI}}$
 - (3) $(\text{CH}_3)_3\text{C--OH} \xrightarrow{\text{HBr}}$
 - (4) $(\text{CH}_3)_3\text{C--OH} \xrightarrow{\text{HI}}$

8. In which reaction alkyl halide is not obtained?
- (1) $\text{R--OH} + \text{NaBr} \rightarrow$
 - (2) $\text{R--OH} + \text{H--Br} \rightarrow$
 - (3) $\text{R--OH} + \text{SOCl}_2 \rightarrow$
 - (4) $\text{R--OH} + \text{PCl}_5 \rightarrow$
9. In which of the following reaction alkyl halide will be obtained majorly :
- (1) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--OH} \xrightarrow{\text{HCl}}$
 - (2) $\text{CH}_3\text{--CH}_2\text{--CH(CH}_3\text{)--OH} \xrightarrow{\text{NaCl}}$
 - (3) $\text{CH}_3\text{--CH}_2\text{--CH(CH}_3\text{)--OH} \xrightarrow[\text{Conc. H}_2\text{SO}_4]{\text{KI}}$
 - (4) $\text{CH}_3\text{--CH}_2\text{--CH(CH}_3\text{)--OH} \xrightarrow[\text{Conc. H}_3\text{PO}_4]{\text{KI}}$
10. Which is correct reaction
- (1) $\text{R--X} + \text{NaI} \xrightarrow{\text{Dry Acetone}} \text{R--I} + \text{NaX}$
 $(\text{X}=\text{Cl, Br})$ (Finkelstein reaction)
 - (2) $\text{CH}_3\text{--Br} + \text{AgF} \rightarrow \text{CH}_3\text{--F} + \text{AgBr}$
 (Swarts reaction)
 - (3) $\text{R--OH} + \text{NaBr} \xrightarrow{\text{H}_2\text{SO}_4} \text{R--Br} + \text{NaHSO}_4 + \text{H}_2\text{O}$
 - (4) All

Physical properties

- Alkyl halides are colourless when pure, Bromides and Iodides develop colour when exposed to light.
- Due to stronger intermolecular forces of attraction, boiling points of halogen derivatives are higher than those of hydrocarbons of comparable molecular mass.
- For the same alkyl groups the boiling points of alkyl halides decrease in the order : $\text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-F}$, this is because with the increase in mass, the magnitude of vander waal forces increases.
- The boiling points of isomeric haloalkanes decreases with increase in branching.



- In isomeric dichlorobenzene
 Boiling point : Ortho > para > meta
 Melting point : Para > ortho > meta

- Haloalkanes are very slightly soluble in water. In order to dissolve haloalkanes in water energy is required to overcome the attraction between haloalkane molecules and break H-bonds between water molecules, but less energy is released when new attractions are set up between haloalkanes and water molecules as a result of this, solubility of haloalkanes in water is low.

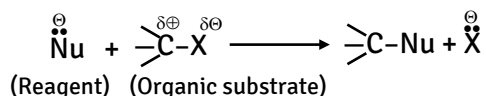
Chemical Reaction of haloalkanes

Reactions of haloalkanes can be categorized as:

- Nucleophilic Substitution reactions
- Elimination reactions
- Reaction with metals.

1. Nucleophilic substitution reaction:

In this reaction nucleophile attacks on organic substrate (haloalkane) to replace halogen called leaving group which departs as halide ion.

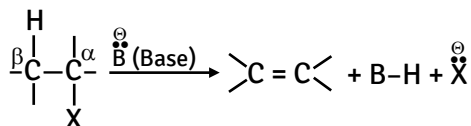


Mechanism: The reaction has been found to proceed by two different mechanisms which are

- Unimolecular nucleophilic substitution (S_N1)
- Bimolecular nucleophilic substitution (S_N2)

	Point of comparison	S_N1	S_N2
1.	Mechanism	Step-I: $\overset{\delta\oplus}{\text{R}}-\overset{\delta\ominus}{\text{L}} \xrightleftharpoons{\text{Slow}} \overset{\oplus}{\text{R}} + \overset{\ominus}{\text{L}}$ Step-II: $\overset{\oplus}{\text{R}} + \overset{\ominus}{\text{Nu}} \xrightarrow{\text{Fast}} \text{R}-\text{Nu}$	(T.S)
2.	Number of steps	Complete in two steps	Complete in one step
3.	Rate equation	$r = k [\text{R}-\text{L}]$	$r = k [\text{R}-\text{CH}_2-\text{X}] [\overset{\ominus}{\text{Nu}}]$
4.	Carbocation rearrangement	Possible since carbocation is formed as intermediate	Not possible since no intermediate is formed
5.	Solvent	Generally carried out in polar protic solvents like water, alcohol, acetic acid etc.	Can be carried out in polar protic and polar aprotic solvents
6.	Nucleophile	Rate of reaction does not depend on strength and concentration of nucleophile	Rate of reaction increases with the increase in strength or concentration of nucleophile
7.	Reactivity of halides	$\text{R}-\text{X} : 3^\circ > 2^\circ > 1^\circ > \text{CH}_3\text{X}$: $\text{R}-\text{I} > \text{R}-\text{Br} > \text{R}-\text{Cl} > \text{R}-\text{F}$	$\text{R}-\text{X} : \text{CH}_3\text{X} > 1^\circ > 2^\circ > 3^\circ$: $\text{R}-\text{I} > \text{R}-\text{Br} > \text{R}-\text{Cl} > \text{R}-\text{F}$
8.	Stereochemistry of the reaction.	Since nucleophile attack on carbocation (Planar attacking site) so both retention and inversion in configuration is observed	Since nucleophile attacks on organic substrate at 180° opposite to the leaving group so only inversion in configuration is observed.
9.	Optical activity of product	Racemic mixture that is optically inactive product is formed.	Single optically active compound with inversion of configuration is formed as product.

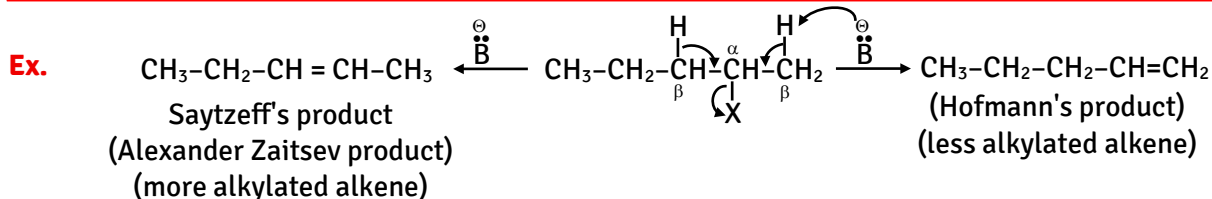
2. **Elimination reaction:** In this reaction base attacks on organic substrate that is haloalkane to remove hydrogen atom from β -carbon and halogen atom is also eliminated from α -carbon atom. as a result, an alkene is formed as product, since β -Hydrogen atom is involved in elimination. The reaction is often called β -elimination.



Mechanism: The reaction has been found to proceed with two Mechanisms which are:

- (a) Unimolecular elimination (E_1)
 (b) Bimolecular elimination (E_2)

	Point of comparison	E_1	E_2
1	Mechanism	Step-I: $\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}- \\ \quad \\ \quad \text{X} \end{array} \xrightarrow{\text{Slow}} \begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}^{\oplus}- \\ \quad \\ \quad \text{X} \end{array} + \text{X}^{\ominus}$ Step-II: $\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}^{\oplus}- \\ \quad \\ \quad \text{X} \end{array} \xrightarrow{\text{Fast}} >\text{C}=\text{C}< + \text{B-H}$	$\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}- \\ \quad \\ \quad \text{X} \end{array} \xrightarrow{\text{B}^{\ominus}} \left[\begin{array}{c} \text{H} \quad \text{B}^{\ominus} \\ \quad \vdots \\ -\text{C} \cdots \text{C}- \\ \quad \\ \quad \text{X}^{\ominus} \end{array} \right] \rightarrow >\text{C}=\text{C}< + \text{B-H} + \text{X}^{\ominus}$ (T.S.)
2.	Number of steps	Completes in two steps	Completes in one step
3.	Rate equation	$r = k \left[\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}- \\ \quad \\ \quad \text{X} \end{array} \right]$	$r = k \left[\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{C}- \\ \quad \\ \quad \text{X} \end{array} \right] [\text{B}^{\ominus}]$
4.	Carbocation rearrangement	Possible since carbocation is formed as intermediate	Not Possible since no intermediate is formed, transition state is only formed in reaction.
5.	Temperature	High temperature is a favorable condition to get product with good yield.	High temperature is favorable condition to get product with good yield.
6.	Reactivity	$\text{R-X} : 3^{\circ} > 2^{\circ} > 1^{\circ}$ $: \text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-F}$	$\text{R-X} : 3^{\circ} > 2^{\circ} > 1^{\circ}$ $: \text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-F}$
7.	Major Product	If more than one types of alkenes are expected, more alkylated alkene (Saytzeff's product) is formed as major product and less alkylated alkene (Hofmann's product) is formed as minor product.	In general saytzeff's product is major but depends on some factors as (i) If attacking base is bulky like (t-butoxide ion), Hofmann's product is formed as major product (ii) If leaving group is a bad leaving group (like-F among all halogens) major product depends on carbanion stability on two different attacking sites of β -carbon.



Elimination versus nucleophilic substitution:

It has been observed that in the same reaction mixture there is a competition between elimination reaction (E_1 and E_2) and nucleophilic substitution (S_N1 and S_N2).

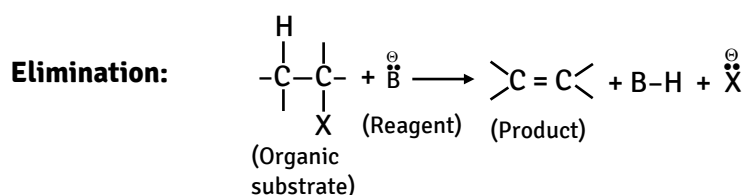
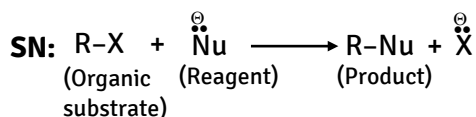
It is because elimination involves attack of base on organic substrate and nucleophilic substitution reaction involves attack of nucleophile on organic substrate and as we know all nucleophiles behave as base and vice versa, so two parallel reaction are observed in the same reaction mixture. Now there are some factors which decide that whether elimination reaction or nucleophilic substitution reaction will dominate. These factors are.

- (1) Attacking reagent:** If attacking reagent is a weak nucleophile but good base, elimination reaction dominates.
If attacking reagent is a good nucleophile as well as good base, elimination reaction dominates (on the basis of acid-base reaction)
- (2) Temperature:** High temperature is favorable condition for elimination reaction. If a reaction is elimination at a specific temperature and if temperature is decreased then reaction shifts from elimination to substitution
- (3) Solvent:** Solvent also decides the reaction whether elimination or nucleophilic substitution reaction will dominate.
- (4) Organic substrate:** degree of halides ($1^\circ/2^\circ/3^\circ$) also decides main reaction.

Note: Similar to elimination and nucleophilic substitution there is competition between S_N1 and S_N2 reaction also.

- (1)** S_N1 reaction is Preferred when attacking nucleophile is weak like H_2O , R-OH .
- (2)** S_N2 reaction is Preferred when attacking nucleophile is good like OH^- , R-S^- , X^- , NH_3 .etc.
- (3)** organic substrate also decide like
 $\text{CH}_3\text{-X}$ mainly undergoes S_N2 reaction.
 $(\text{CH}_3)_3\text{C-X}$ mainly undergoes S_N1 reaction.

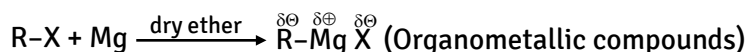
Examples of nucleophilic substitution and elimination reaction.



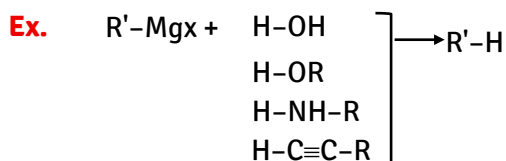
Reagent	Nature of reagent	Product	Remarks
H ₂ O	H ₂ O attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-OH	Mainly follows S _N 1 mechanism
R'-OH	R'OH attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-OR'	Mainly follows S _N 1 mechanism
KOH(aq.)	$\text{OH}^{\ominus}$ attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-OH	Mainly follows S _N 2 mechanism
KOH(alco.)	$\text{OH}^{\ominus}$ attacks as base	$\text{>C}=\text{C}<$	Mainly follows E ₂ mechanism
NaNH ₂	$\text{NH}_2^{\ominus}$ attacks as base	$\text{>C}=\text{C}<$	Mainly follows E ₂ mechanism
NH ₃	NH ₃ attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	Mixture of R-NH ₂ + R ₂ NH + R ₃ N + R ₄ N [⊕] X [⊖]	Mainly follows S _N 2 mechanism at high temperature or in 3° R-X reaction shifts to E ₂ mechanism.
R'ONa	R'O [⊖] attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-OR'	• Mainly follows S_N2 mechanism • Reaction is known as Williamson's synthesis • In 2° & 3° R-X Reaction shifts to E₂ mechanism
NaCN/KCN (ionic)	$\text{CN}^{\ominus}$ attacks as ambident $\ddot{\text{N}}^{\ominus}\text{u}$	R-CN + R-NC (Major) (minor)	Reaction follows S _N 2 mechanism
AgCN (Covalent)	Ag CN attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-NC + R-CN (Major) (minor)	Reaction follows S _N 1 mechanism
NaNO ₂ /KNO ₂ (Ionic)	$\text{O}-\text{NO}^{\ominus}$ attacks as ambident $\ddot{\text{N}}^{\ominus}\text{u}$	R-ONO + R-NO ₂ (Major) (Minor)	Reaction follows S _N 2 mechanism
AgNO ₂	Ag-O-NO attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-NO ₂ + R-ONO (Major) (Minor)	Reaction follows S _N 1 mechanism
R'-SNa	R'-S [⊖] attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-S-R'	Reaction follows S _N 2 mechanism
LiAlH ₄	H [⊖] attacks as $\ddot{\text{N}}^{\ominus}\text{u}$	R-H	• Reaction follows S_N2 mechanism • In 2° & 3° R-X E₂ dominates

3. Reaction with metals:

(a) Reaction with Mg:

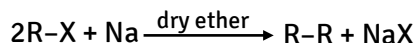


- R-Mgx is known as grignard's reagent.
- C-Mg bond is polar covalent bond, C bears negative charge.
- Mgx bond is ionic.
- R-Mgx reacts with the compound having active hydrogen (the hydrogen bonded to O/N/X/S/C_{sp}) to give hydrocarbon.



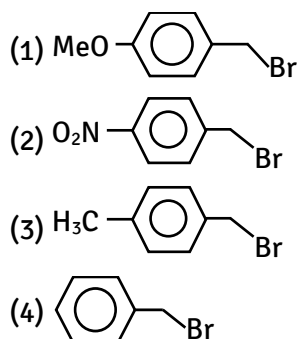
- To avoid acid-base reaction, grignard's reagent is prepared in absolute dry condition.

(b) Reaction with Na [Wurtz reaction]

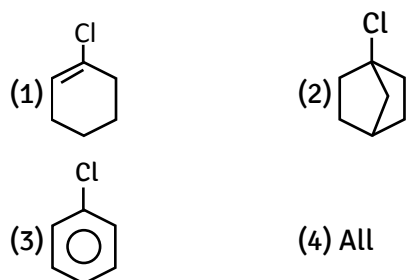


Exercise 1.2

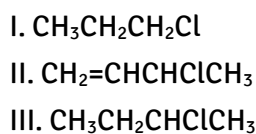
- 1.** Which of the following will give fastest S_N2 reaction ?



- 2.** Rate of S_N2 will be negligible in :

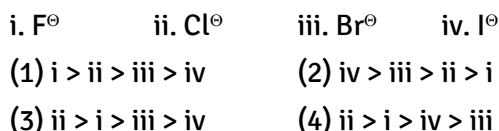


- 3.** The order of decreasing S_N1 reactivities of the halides is :

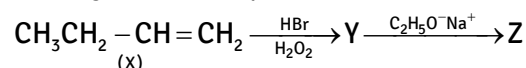


- (1) I > II > III (2) II > I > III
 (3) II > III > I (4) III > II > I

- 4.** The decreasing nucleophilic order of the following is :



- 5.** Identify Z in the sequence,



- (1) $CH_3-\underset{CH_3}{CH}-CH_2-O-CH_2-CH_3$
 (2) $CH_3-CH_2-\underset{CH_3}{CH}-O-CH_2-CH_3$
 (3) $CH_3-(CH_2)_3-O-CH_2-CH_3$
 (4) $CH_3-(CH_2)_4-O-CH_3$

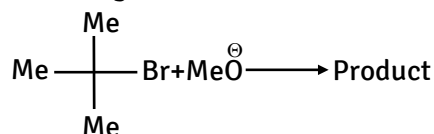
- 6.** 2,2-Dichloropropane on hydrolysis yields:

- (1) Acetone
 (2) 2,2-Propane diol
 (3) Isopropyl alcohol
 (4) Acetaldehyde

- 7.** A Halogen compound 'A' on hydrolysis with dilute alkali followed by acidification gives acetic acid. The compound A is :

- (1) $ClCH_2CH_2Cl$ (2) CH_3CHCl_2
 (3) $ClCH_2CHCl_2$ (4) CH_3CCl_3

- 8.** Which statement is correct about the following reaction?

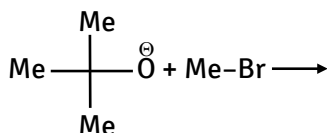


- (1) Major product $\left(\begin{array}{c} \text{Me} \\ \diagup \\ \text{C}=\text{CH}_2 \\ \diagdown \\ \text{Me} \end{array} \right)$ is formed by elimination reaction

- (2) Major product $\left(\begin{array}{c} \text{Me} \\ | \\ \text{Me}-\text{C}-\text{OMe} \\ | \\ \text{Me} \end{array} \right)$ is

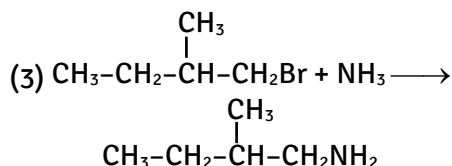
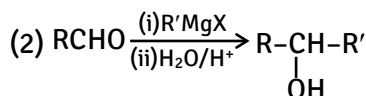
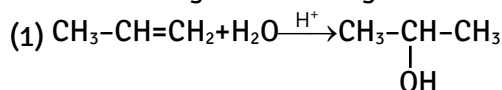
- formed by substitution reaction
 (3) Both (1) and (2) are formed in equal proportion
 (4) None

9. Which statement is correct about the given reaction?



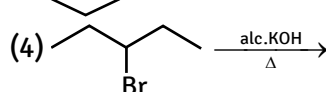
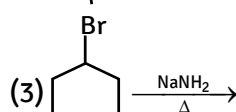
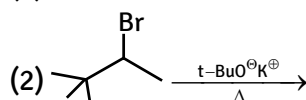
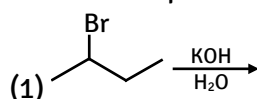
- (1) Product $\left(\text{Me}-\overset{\text{Me}}{\underset{\text{Me}}{\text{C}}}=\text{CH}_2\right)$ (I) is formed by elimination reaction
- (2) Product $\left(\text{Me}-\overset{\text{Me}}{\underset{\text{Me}}{\text{C}}}-\text{OMe}\right)$ (II) is formed by $\text{S}_{\text{N}}2$ reaction
- (3) Product (II) is formed by $\text{S}_{\text{N}}1$ reaction
- (4) Both (1) and (2) are correct

10. Which one is a nucleophilic substitution reaction among the following :-

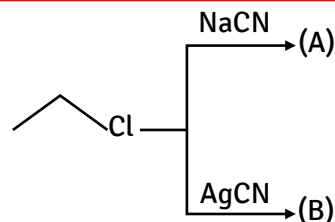


11. The given reaction is called as:
 $\text{C}_2\text{H}_5\text{ONa} + \text{C}_2\text{H}_5\text{Br} \longrightarrow \text{C}_2\text{H}_5-\text{O}-\text{C}_2\text{H}_5 + \text{NaBr}$
- (1) Frankland reaction
 (2) Wurtz reaction
 (3) Williamson's synthesis
 (4) Cannizzaro reaction

12. Which of the following reactions give substitution product?



13.



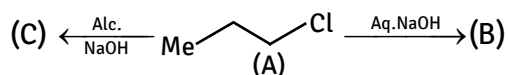
A and B will be respectively :

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{N}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{N}^+\equiv\text{C}^-$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{N}^+\equiv\text{C}^-$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{N}$
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{N}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{N}$
- (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{N}^+\equiv\text{C}^-$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{N}^+\equiv\text{C}^-$

14. Tertiary butyl halide on boiling with water gives tertiary butyl alcohol. The reaction follows -

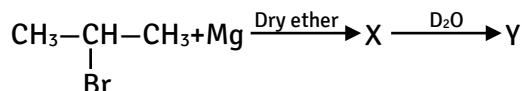
- (1) E_2 mechanism (2) $\text{S}_{\text{N}}1$ mechanism
 (3) $\text{S}_{\text{N}}2$ mechanism (4) E_1 mechanism

15. Which of the following statements is correct about the following reactions?

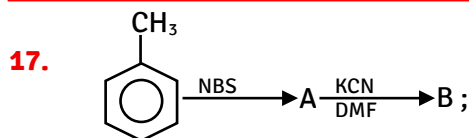


- (1) (B) is obtained by elimination reaction
 (2) (C) is obtained by substitution reaction
 (3) The molecular formula of (B) is C_3H_6 and that of (C) is $\text{C}_3\text{H}_8\text{O}$.
 (4) (B) is an isomer of ethyl methyl ether, while (C) is dehydrated compound of (B)

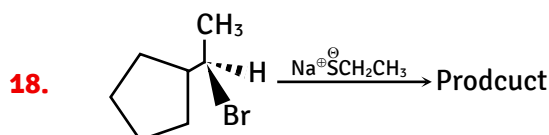
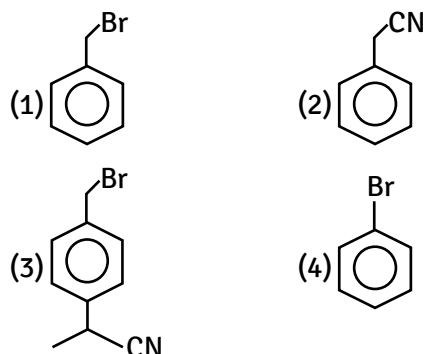
16. Identify the product X and Y in the given reaction,



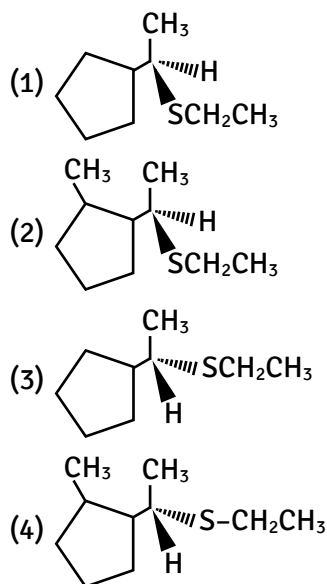
- (1) $\text{X} = \text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{CH}_2\text{Mg}$, $\text{Y} = \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- (2) $\text{X} = \text{CH}_3-\underset{\text{MgBr}}{\text{CH}}-\text{CH}_3$, $\text{Y} = \text{CH}_3-\underset{\text{D}}{\text{CH}}-\text{CH}_3$
- (3) $\text{X} = \text{CH}_3-\underset{\text{MgBr}}{\text{CH}}-\text{CH}_3$, $\text{Y} = \text{CH}_3-\underset{\text{OD}}{\text{CH}}-\text{CH}_3$
- (4) $\text{X} = \text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{CH}_2\text{Mg}$, $\text{Y} = \text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$



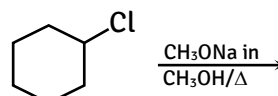
Product B is :-



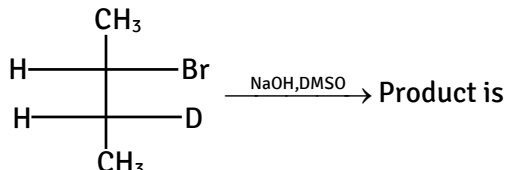
Product is :-

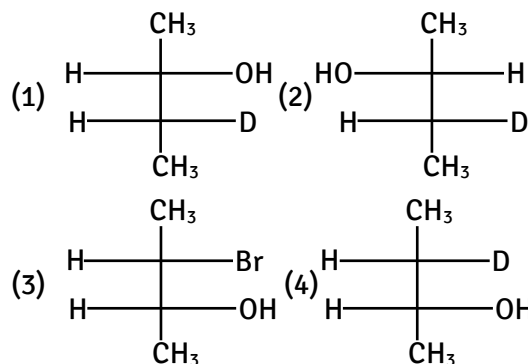


19. Given reaction mainly proceeds through -



- (1) E_1 (2) S_N1
 (3) S_N2 (4) E_2

20. 



Reactions of haloarenes

Reactions of haloarenes can be categorised as:

- (i) Nucleophilic substitution
- (ii) electrophile substitution
- (iii) Reaction with metals

(1) **Nucleophilic substitution:** Aryl halides are extremely less reactive towards nucleophilic substitution reaction due to:

- (i) **Resonance effect:** due to resonance C-X bond acquires a partial double bond character so bond cleavage in haloarenes is difficult than in haloalkanes.

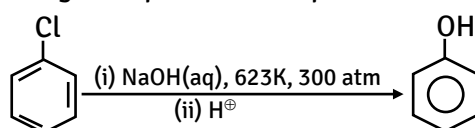
(ii) **Hybridisation of carbon:** In haloarenes sp^2 hybridised carbon with a greater s-character is more electronegative and can hold the electron pair of C-X bond more tightly than sp^3 hybridised carbon in haloalkanes with less s-character.

(iii) **Instability of phenyl carbocation:**

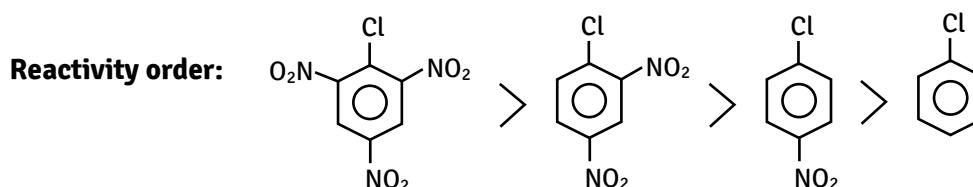
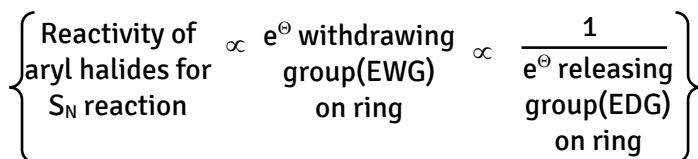
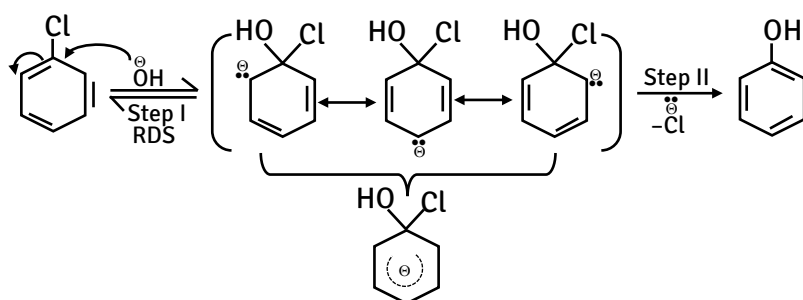
The phenyl cation formed as a result of self-ionisation will not be stabilised by resonance.

Ex. Alkaline hydrolysis of aryl halides

Since aryl halides are extremely less reactive towards nucleophilic substitution reaction, so reaction is carried out in the presence of high temperature and pressure condition.

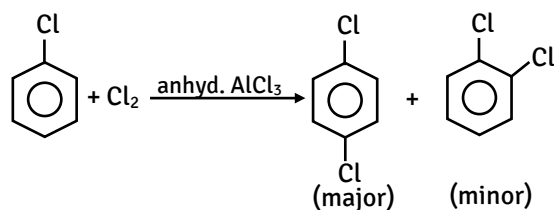


Mechanism:

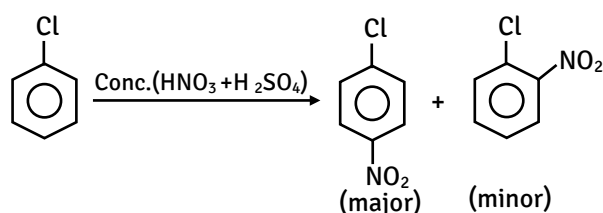


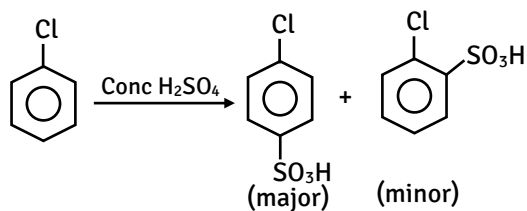
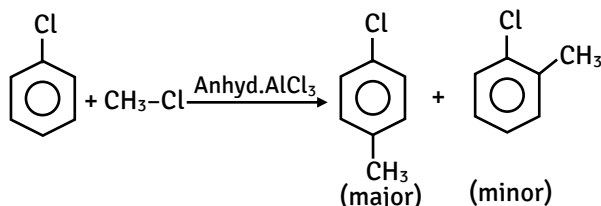
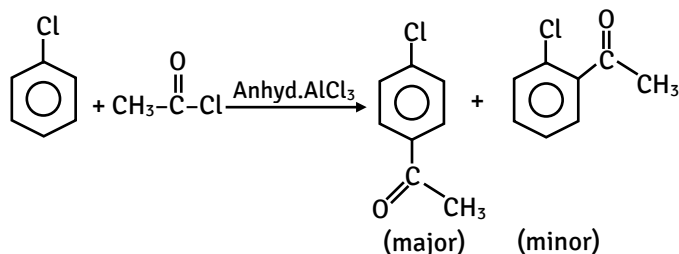
2. Electrophilic substitution reaction:

(i) **Halogenation:**

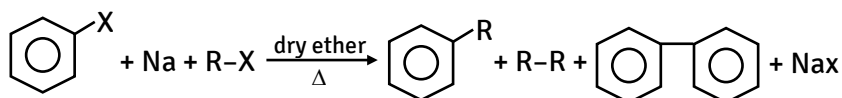
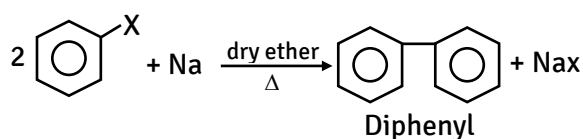


(ii) **Nitration:**



(iii) Sulphonation:**(iv) Friedel craft reaction:****a) Alkylation:****b) Acylation:**

Note: Although halogen (– X) is deactivating group but it is ortho para directing.

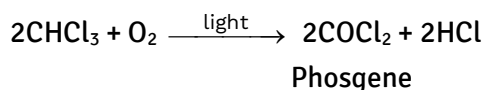
3. Reaction with metals:**A) Wurtz fittig reaction:****b) fittig reaction:****Polyhalogen compounds****(a) Dichloromethane (Methylene chloride, CH₂Cl₂)**

- (i) Dichloromethane is widely used as a solvent as a paint remover, as a propellant in aerosols, and as a process solvent in the manufacture of drugs. It is also used as a metal cleaning and finishing solvent.
- (ii) Methylene chloride harms the human central nervous system (CNS).
- (iii) In humans, direct skin contact with methylene chloride causes intense burning and mild redness of the skin. Direct contact with the eyes can burn the cornea.

(b) Trichloromethane (Chloroform, CHCl₃)

- (i) Chemically, chloroform is employed as a solvent for fats, alkaloids, iodine and other substances. The major use of chloroform today is in the production of the **freon refrigerant R-22**.
- (ii) As might be expected from its use as an anaesthetic, inhaling chloroform vapours depresses the central nervous system.

- (iii) Chronic chloroform exposure may cause damage to the liver (where chloroform is metabolised to phosgene) and to the kidneys, and some people develop sores when the skin is immersed in chloroform.
- (iv) Chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas, carbonyl chloride, also known as **phosgene**. It is therefore stored in closed dark coloured bottles completely filled so that air is kept out.



(c) **Triiodomethane (Iodoform, CHI_3)**

Iodoform was used earlier as an antiseptic but the antiseptic properties are due to the liberation of free iodine and not due to iodoform itself.

(d) **Tetrachloromethane (Carbon tetrachloride, CCl_4)**

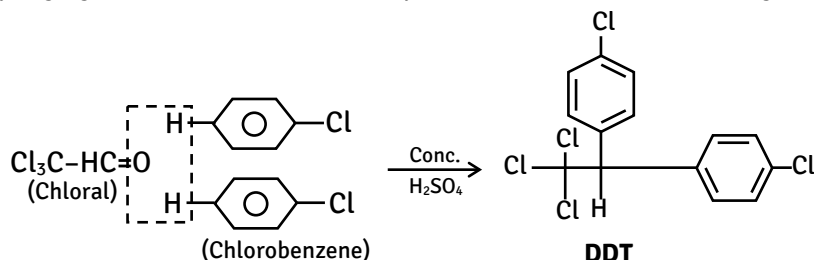
- (i) Carbontetrachloride is produced in large quantities for use in the manufacture of refrigerants and propellants for aerosol cans.
- (ii) It is also used as feedstock in the synthesis of chlorofluorocarbons and other chemicals, pharmaceutical manufacturing, and general solvent use.
- (iii) It was also widely used as a cleaning fluid, both in industry, as a degreasing agent, and in the home, as a spot remover and as fire extinguisher.
- (iv) When carbon tetrachloride is released into the air, it rises to the atmosphere and depletes the ozone layer.

(e) **Freons**

- (i) The chlorofluorocarbon compounds of methane and ethane are collectively known as freons. They are extremely stable, unreactive, non-toxic, non-corrosive and easily liquefiable gases.
- (ii) **Freon 12 (CCl_2F_2)** is one of the most common freons in industrial use. It is manufactured from tetrachloromethane by Swarts reaction.
- (iii) These are usually produced for aerosol propellants, refrigeration and air conditioning purposes.
- (iv) In stratosphere, freon is able to initiate radical chain reactions that can upset the natural ozone balance.

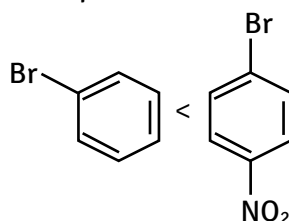
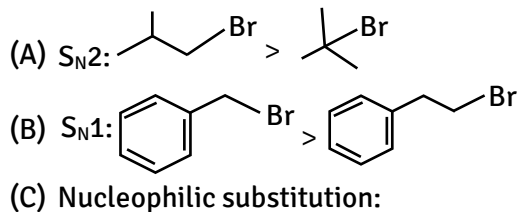
(f) **p,p'-Dichlorodiphenyltrichloroethane (DDT)**

- (i) DDT, the first chlorinated organic insecticides, was originally prepared in 1873,
- (ii) The use of DDT increased enormously on a worldwide basis after World War II, primarily because of its effectiveness against the mosquito that spreads malaria and lice that carry typhus.
- (iii) The chemical stability of DDT and its fat solubility compounded the problem. DDT is not metabolised very rapidly by animals; instead, it is deposited and stored in the fatty tissues.



Exercise 1.3

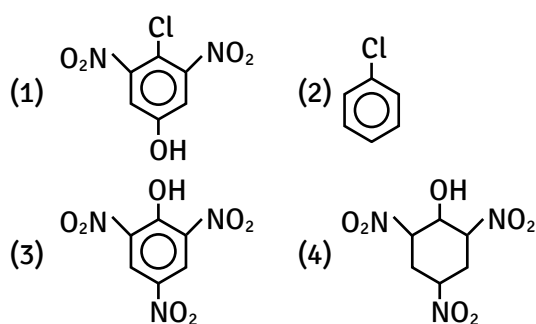
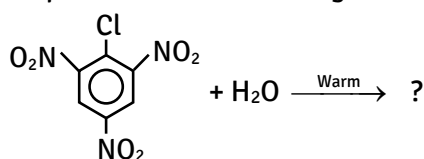
1. Identify the correct order of reactivity for the following pairs towards the respective mechanism



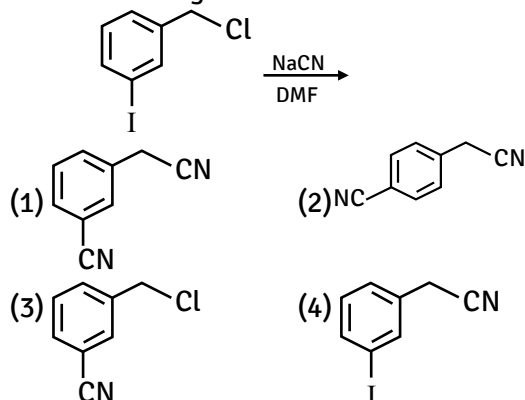
Choose the correct answer from the options given below :

- (1) (A), (B) only (2) (A), (B), (C)
(3) (A), (C) only (4) (B) Only

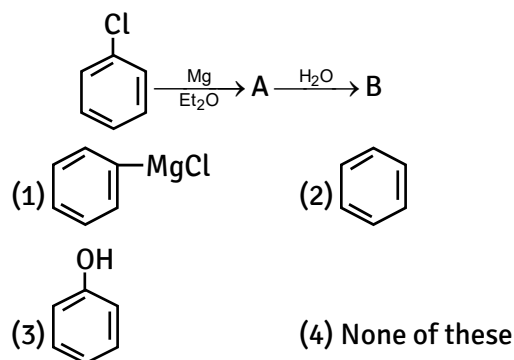
2. The product of the following reaction is :



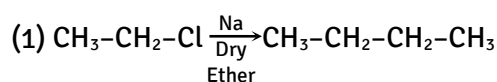
3. The structure of the major product formed in the following reaction is :



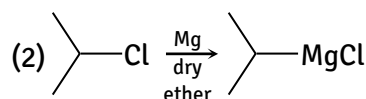
4. What is final product of the following reaction :



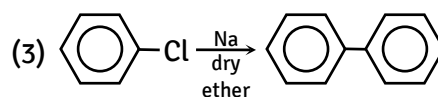
5. Which of the following is incorrect ?



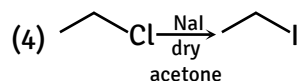
Wurtz reaction



Grignard reaction

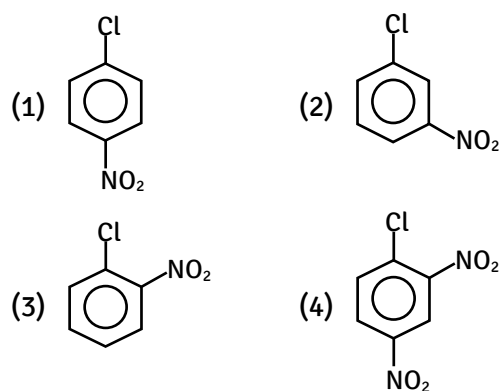


Fittig reaction

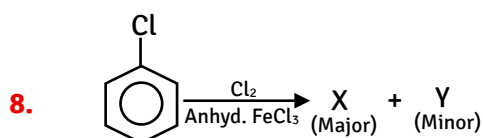
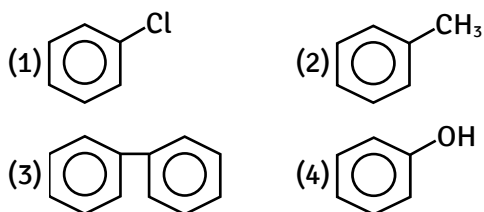
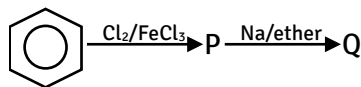


Swarts reaction

6. Which of the following compound will have the lowest rate towards nucleophilic aromatic substitution on treatment with OH^-



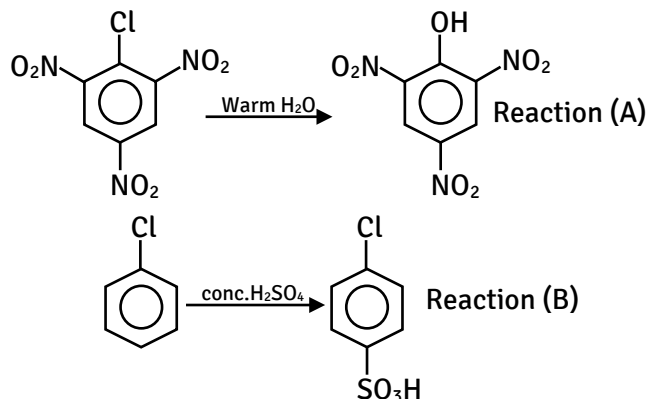
7. The end product (Q) in the following sequence of reactions is



Find the correct statement -

- (1) Boiling point of X is greater than that of Y
- (2) Melting point of X is more than that of Y
- (3) X is ortho dichlorobenzene and Y is para dichlorobenzene
- (4) All of these

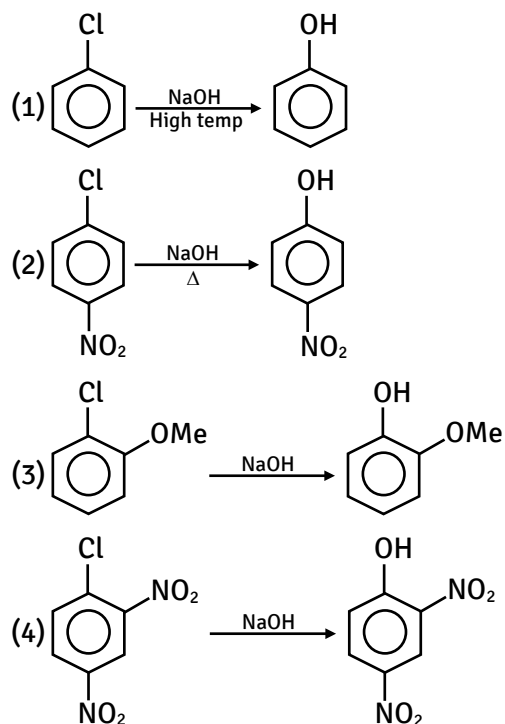
9. Consider following reaction:



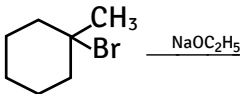
Find correct statement :-

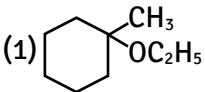
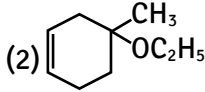
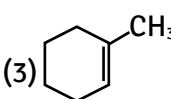
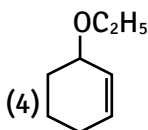
- (1) Both reaction are example of substitution reaction
- (2) Reaction A is electrophilic substitution reaction
- (3) Reaction B is nucleophilic substitution reaction
- (4) All are correct

10. Which reaction is least feasible?

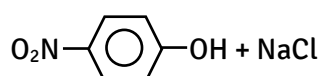
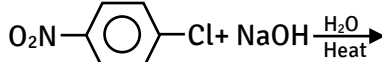


Exercise 2

1.  Major Product will be :

- (1)  (2) 
 (3)  (4) 

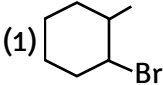
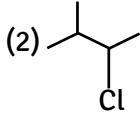
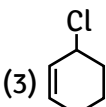
2. The reaction :



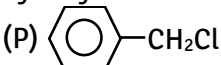
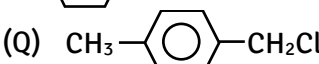
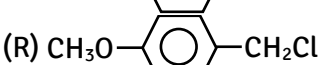
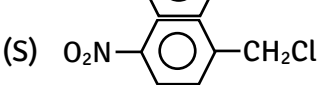
is an example of :

- (1) Nucleophilic substitution proceeding through addition elimination mechanism
 (2) Nucleophilic substitution proceeding through elimination addition mechanism
 (3) Electrophilic substitution proceeding through addition elimination mechanism
 (4) Nucleophilic substitution proceeding through $\text{S}_{\text{N}}2$ mechanism

3. Which of the following substrate can give exactly same products during $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ mechanism :

- (1)  (2) 
 (3)  (4) None of these

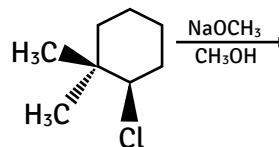
4. The following benzyl halides are subjected to hydrolysis under $\text{S}_{\text{N}}1$ reaction conditions:

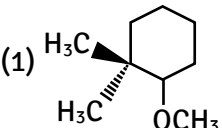
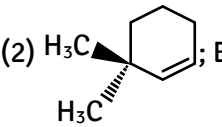
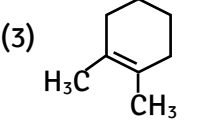
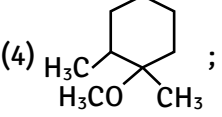
- (P) 
 (Q) 
 (R) 
 (S) 

The order of reactivity is :

- (1) $\text{S} > \text{R} > \text{Q} > \text{P}$ (2) $\text{R} > \text{Q} > \text{S} > \text{P}$
 (3) $\text{R} > \text{Q} > \text{P} > \text{S}$ (4) $\text{Q} > \text{R} > \text{P} > \text{S}$

5. The major product for the following reaction and mechanism involved is respectively :

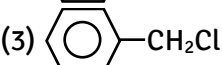


- (1)  ; $\text{S}_{\text{N}}2$ mechanism
 (2)  ; E_2 mechanism
 (3)  ; E_1 mechanism
 (4)  ; $\text{S}_{\text{N}}1$ mechanism

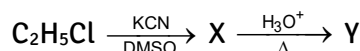
6. The order of decreasing nucleophilicities of the following species is :

- (1) $\text{CH}_3\text{S}^- > \text{CH}_3\text{O}^- > \text{CH}_3\text{COO}^- > \text{CH}_3\text{OH}$
 (2) $\text{CH}_3\text{COO}^- > \text{CH}_3\text{S}^- > \text{CH}_3\text{O}^- > \text{CH}_3\text{OH}$
 (3) $\text{CH}_3\text{OH} > \text{CH}_3\text{S}^- > \text{CH}_3\text{COO}^- > \text{CH}_3\text{O}^-$
 (4) $\text{CH}_3\text{O}^- > \text{CH}_3\text{S}^- > \text{CH}_3\text{COO}^- > \text{CH}_3\text{OH}$

7. Which will give white ppt. with AgNO_3 :

- (1)  (2) 
 (3)  (4) Both (1) & (3)

8. Consider the following sequence of reaction :



The products (X) and (Y) are, respectively:

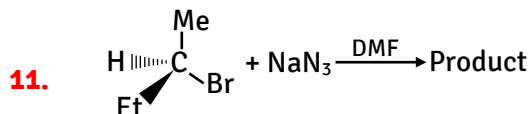
- (1) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$
 (2) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{CONH}_2$
 (3) $\text{C}_2\text{H}_5\text{NC}$ and $\text{C}_2\text{H}_5\text{NHCH}_3$
 (4) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{COOH}$

9. During reaction of alcohol with KI, which acid can be used?

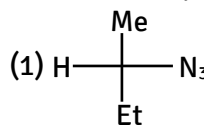
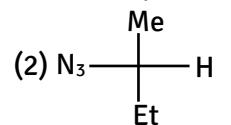
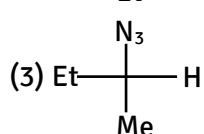
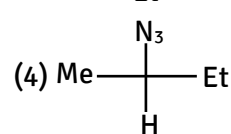
- (1) Conc. H_2SO_4 (2) Conc. H_3PO_4
 (3) Both (4) None of these

10. Mechanism involved and product obtained when 3-iodo-3-ethylpentane is treated with sodium methoxide in methanol is :

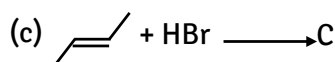
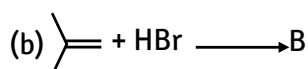
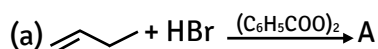
- (1) Ether, S_N1 (2) Alkene, E_1
(3) Ether, E_1 (4) Alkene, E_2



The correct representation of the product is

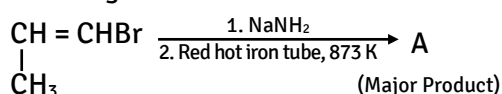
- (1)  (2) 
(3)  (4) 

12. The increasing order of the boiling points of the major products A, B and C of the following reactions will be:

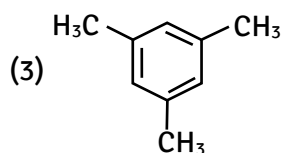
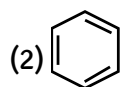
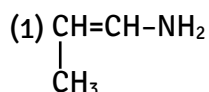


- (1) $C < A < B$ (2) $B < C < A$
(3) $A < B < C$ (4) $A < C < B$

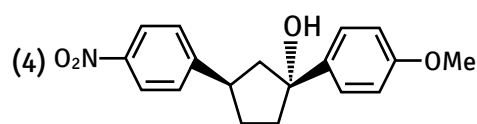
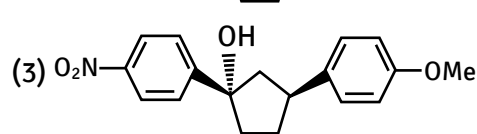
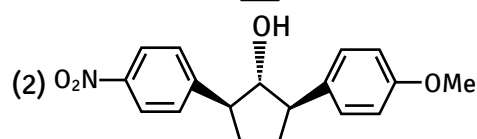
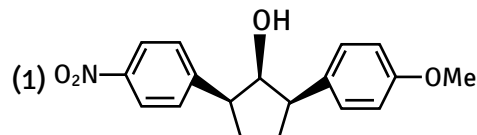
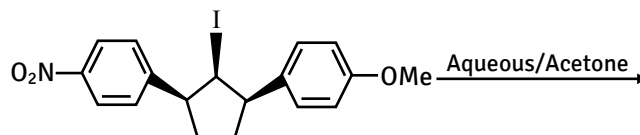
13. For the given reaction :



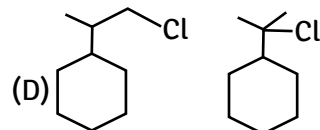
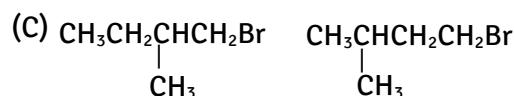
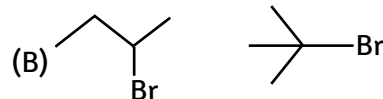
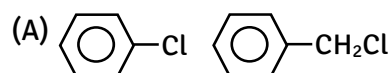
What is 'A'?



14. Identify major product for the following reactions



15. In which of the given pair, the first compound is more reactive than second towards S_N2 reaction.



- (1) B, C & D Only (2) B & D Only
(3) B & C only (4) A, B, C & D

16. (A) $\xrightarrow{\text{Cl}_2/h\nu}$ (B) $\xrightarrow{\text{aq. KOH}}$ (C) $\xrightarrow{[\text{O}]}$

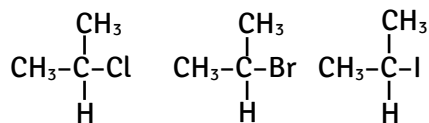
CH_3CHO , Identify A, B & C -

- (1) Ethylalcohol, Ethyl chloride & Ethane
(2) Ethane, Ethylchloride & $\text{CH}_3 - \text{CH}_2 - \text{OH}$
(3) Propane, Propylchloride & $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$
(4) All the above

17. For reaction $\text{CH}_3\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{OH} + \text{Br}^-$
The rate of reaction is given by the expression:

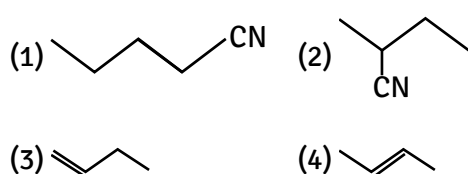
- (1) Rate = $k[\text{CH}_3\text{Br}]$
(2) Rate = $k[\text{OH}^-]$
(3) Rate = $k[\text{CH}_3\text{Br}][\text{OH}^-]$
(4) Rate = $k[\text{CH}_3\text{Br}]^\circ [\text{OH}^-]^\circ$

18. From left to right, correct statements are:

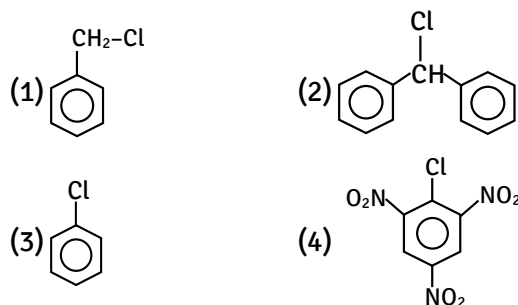


- (A) Rate of $\text{S}_{\text{N}}1$ mechanism increases in polar protic solvent
(B) Rate of $\text{S}_{\text{N}}2$ mechanism increases in DMSO
(C) Rate of E_2 mechanism increases
(D) Rate of E_1 mechanism increases
(1) Only A & D (2) Only B, C & D
(3) Only A, B & D (4) A, B, C & D

19. $\text{nBuBr} + \text{KCN} \xrightarrow{\text{EtOH-H}_2\text{O}}$ Product is

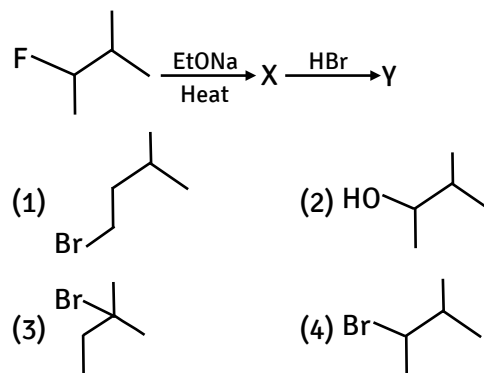


20. Which of the following is most easily hydrolyzed



21. 2-Bromopentane is heated with potassium ethoxide in ethanol. The major product is:
(1) trans-2-pentene (2) 2-Ethoxypentane
(3) 1-Pentene (4) cis-2-pentene

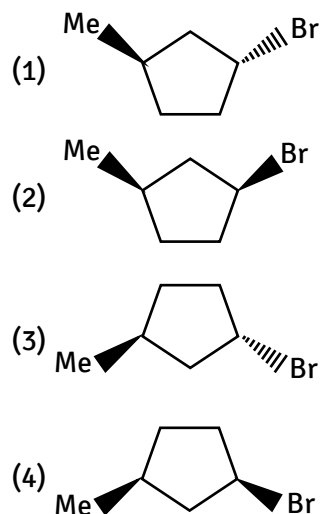
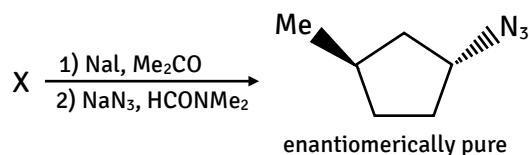
22. The major product 'Y' in the following reaction is:



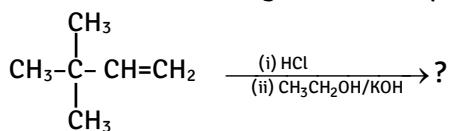
23. Which of the following reactivity order towards $\text{S}_{\text{N}}2$ reaction is incorrect.

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl} < \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} < \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{I}$
(2) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)(\text{C}_6\text{H}_5)\text{Br} < \text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br} < \text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br} < \text{C}_6\text{H}_5\text{CH}_2\text{Br}$
(3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} > (\text{CH}_3)_2\text{CHCH}_2\text{Br} > \text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3 > (\text{CH}_3)_3\text{CBr}$
(4) $\text{C}_6\text{H}_5\text{Cl} > \text{CH}_3-\text{Cl} > \text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl}$

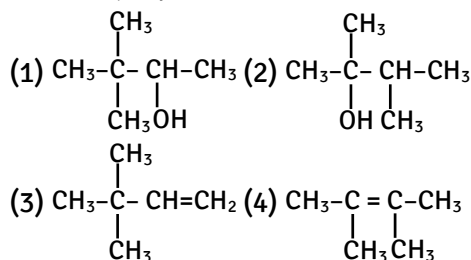
24. In the following reaction sequence, the correct structure(s) of X is



25. Consider the following reaction sequence

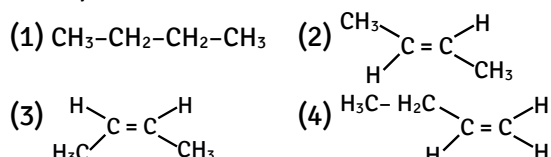


Final major product is

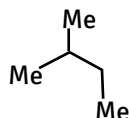


26. $\text{CH}_3-\text{CH}_2-\text{C}\equiv\text{CH} \xrightarrow[\text{(2 mole)}]{\text{HBr}} (\text{P}) \xrightarrow[\text{(ii) NaNH}_2]{\text{(i) Alc. KOH}} (\text{Q}) \xrightarrow[\text{Pd/BaSO}_4]{\text{H}_2} (\text{R})$

Final product 'R' is

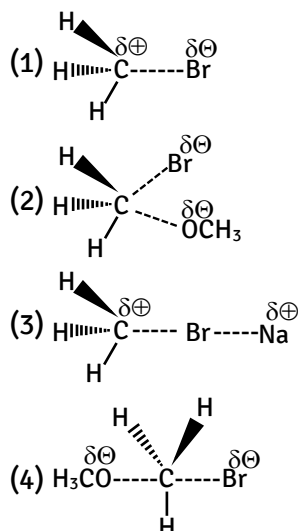


27. How many optically active compounds are formed on monochlorination of given compound



- (1) 2 (2) 4
(3) 6 (4) 8

28. Rate determining transition state can be represented for the following reaction by $\text{CH}_3-\text{Br} + \text{CH}_3\text{ONa} \rightarrow$



29. Arrange the following alkyl halides in decreasing order of the rate of β -elimination reaction with alcoholic KOH.

- (I) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2\text{Br} \end{array}$ (II) $\text{CH}_3-\text{CH}_2-\text{Br}$
- (III) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Br}$
- (1) I > II > III (2) III > II > I
(3) II > III > I (4) I > III > II

30. Arrange the following compounds in decreasing order of rate of $\text{S}_\text{N}1$ reaction

- (A) $\text{Cl}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$ (B) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{Cl}$
- (C) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}-\text{CH}_2-\text{Cl} \\ | \\ \text{CH}_3 \end{array}$ (D) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}-\text{CH}_2-\text{Cl} \\ | \\ \text{CH}_3 \end{array}$
- (1) C > B > D > A (2) C > D > A > B
(3) D > A > C > B (4) D > C > A > B

31. Which of the following alkyl halides will undergo $\text{S}_\text{N}1$ reaction most rapidly?

- (1) $(\text{CH}_3)_3\text{C}-\text{F}$ (2) $(\text{CH}_3)_3\text{C}-\text{Cl}$
(3) $(\text{CH}_3)_3\text{C}-\text{Br}$ (4) $(\text{CH}_3)_3\text{C}-\text{I}$

32. The negative part of the addendum (the molecule to be added) adds on to the carbon atom of the double bond containing the least number of hydrogen atoms. This rule is known as

- (1) Saytzeff's rule (2) Peroxide rule
(3) Markovnikov's rule (4) Hoffmann rule.

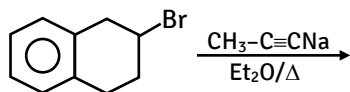
33. Which of the following molecules has highest dipole moment?

- (1) CH_3Cl (2) CH_2Cl_2
(3) CHCl_3 (4) CCl_4

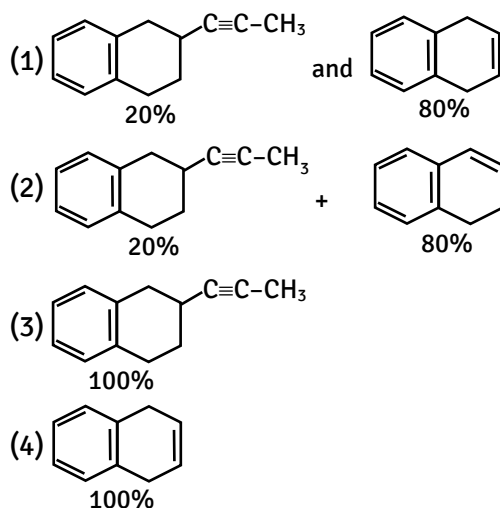
34. Arrange the following compounds in decreasing order of their boiling points.

- (i) CH_3Br
(ii) $\text{CH}_3\text{CH}_2\text{Br}$
(iii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
(iv) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
- (1) (i) > (ii) > (iii) > (iv) (2) (iv) > (iii) > (ii) > (i)
(3) (i) > (iii) > (ii) > (iv) (4) (iii) > (iv) > (i) > (ii)

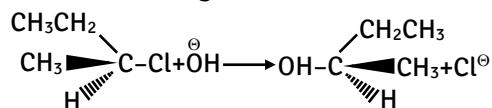
35. In the given reaction:



the products are:



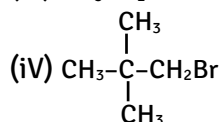
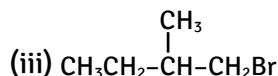
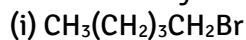
36. In the reaction given below,



which of the following statements is correct?

- (1) The reaction proceeds via $\text{S}_\text{N}2$ mechanism hence inversion of configuration takes place.
- (2) The reaction proceeds via $\text{S}_\text{N}1$ mechanism hence inversion of configuration takes place.
- (3) The reaction proceeds via $\text{S}_\text{N}2$ mechanism hence there is no change in the configuration.
- (4) The reaction proceeds via $\text{S}_\text{N}1$ mechanism hence there is no change in the configuration.

37. Arrange the following compound in order of their reactivity towards $\text{S}_\text{N}2$ reaction.

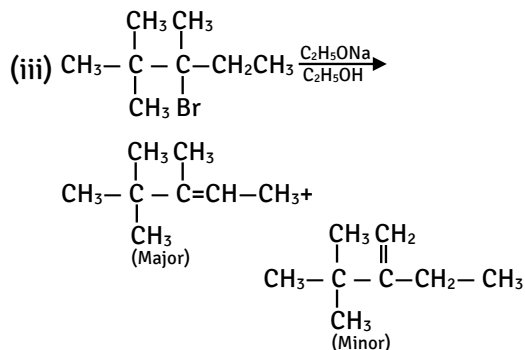
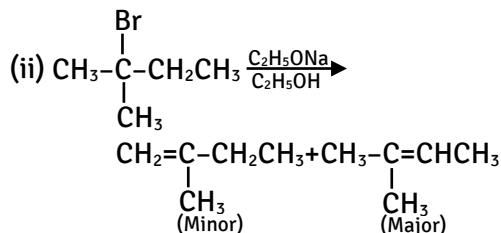
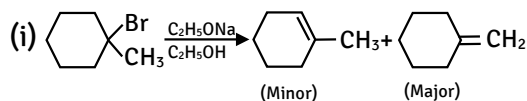


- (1) (i) > (ii) > (iii) > (iv) (2) (ii) > (iii) > (iv) > (i)
 (3) (iii) > (i) > (ii) > (iv) (4) (iv) > (ii) > (i) > (iii)

38. Which of the following statements regarding the $\text{S}_\text{N}1$ reaction shown by alkyl halide is not correct?

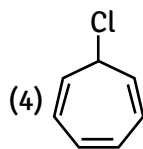
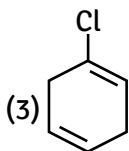
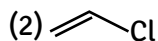
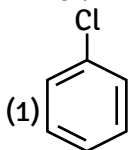
- (1) The added nucleophile plays no kinetic role in $\text{S}_\text{N}1$ reaction.
- (2) The $\text{S}_\text{N}1$ reaction involves the inversion of configuration of the optically active substrate.
- (3) The $\text{S}_\text{N}1$ reaction on the chiral carbon ends up with racemisation of the product.
- (4) The more stable the carbocation intermediate the faster the $\text{S}_\text{N}1$ reaction.

39. Which of the following products as shown by the dehydrohalogenation of alkyl halides with sodium ethoxide in ethanol is correctly marked as major product?

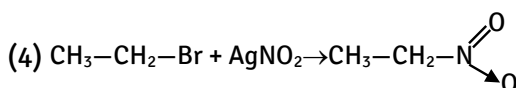
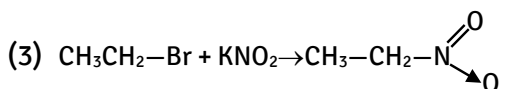


- (1) Only (i) and (ii)
- (2) Only (i) and (iii)
- (3) Only (ii) and (iii)
- (4) only (ii)

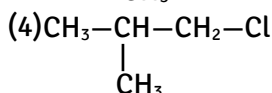
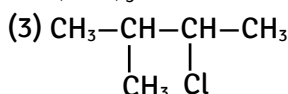
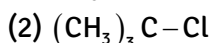
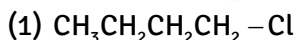
48. Which of the following compound will give curdy precipitate with AgNO_3 solution:



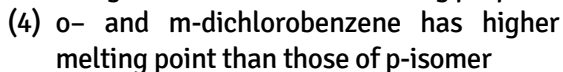
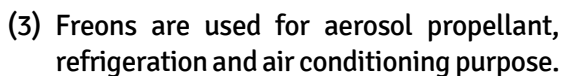
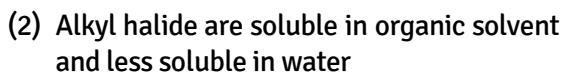
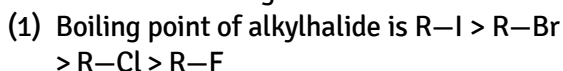
49. Which of the following reaction is correct:



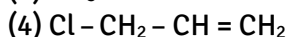
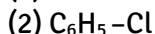
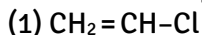
50. Which of the following compound is most reactive toward $\text{S}_{\text{N}}2$ displacement?



51. Which of the following statement is incorrect:-



52. Which of the following is most reactive towards nucleophilic substitution reaction?



53. A $\text{S}_{\text{N}}2$ reaction at an asymmetric carbon of a compound always give:

- (1) An enantiomer of the substrate
- (2) A product with opposite rotation
- (3) A mixture of diastereomer
- (4) A single stereoisomers

54. DDT is manufactured by the acid catalysed condensation of:-

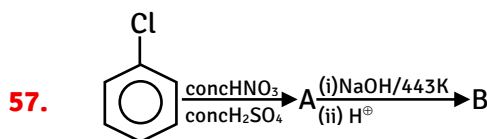
- (1) Chloroform and acetone
- (2) Chloroform and nitric acid
- (3) Chlorobenzene and chloral
- (4) Chlorobenzene and chloroform

55. Which of the following compound is used as refrigerant?

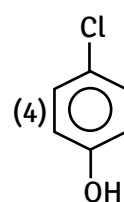
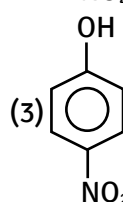
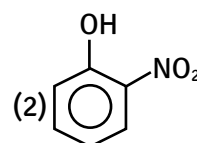
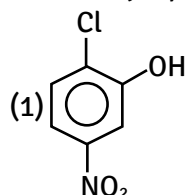
- (1) Urea
- (2) Acrolein
- (3) CCl_2F_2
- (4) $\text{CH}_2(\text{OH})-\text{CH}_2(\text{OH})$

56. Which of the following is added to chloroform in a small quantity before is bottled for sale?

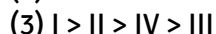
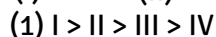
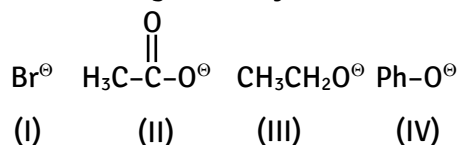
- (1) CH_3OH
- (2) $\text{C}_2\text{H}_5\text{Cl}$
- (3) $\text{C}_2\text{H}_5\text{OH}$
- (4) AgNO_3



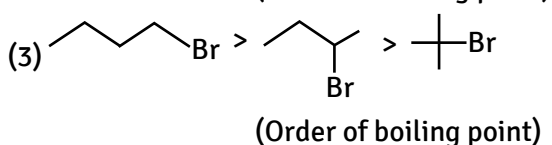
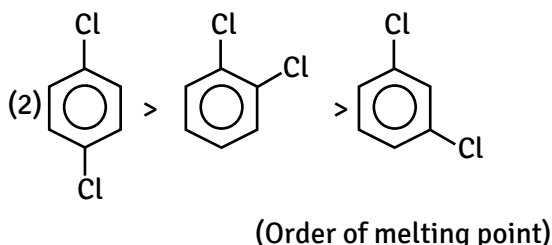
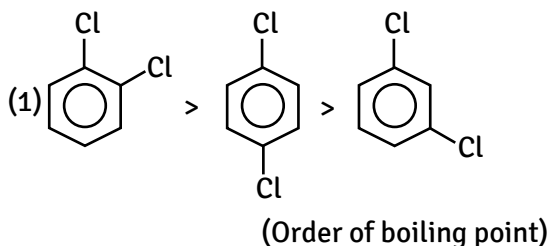
B as a major product is-



58. Arrange the following in decreasing order of their leaving tendency



59. Which is correct order

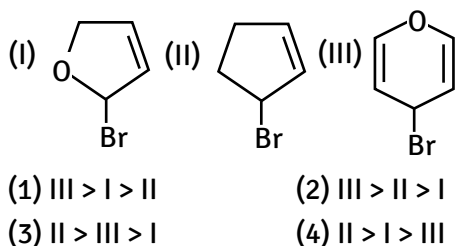


(4) All

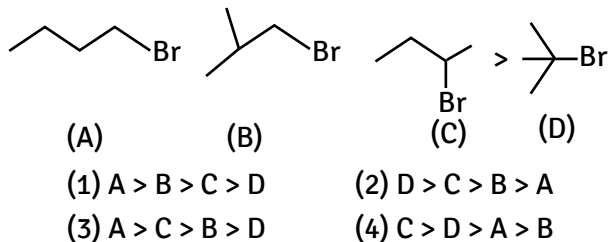
60. The order of reactivity of the following alkyl halides for S_N1 and S_N2 reaction is -

- (1) $RF > RCl > R-Br > R-I$
- (2) $R-F > R-Br > R-Cl > R-I$
- (3) $R-Cl > R-Br > R-F > R-I$
- (4) $R-I > R-Br > R-Cl > R-F$

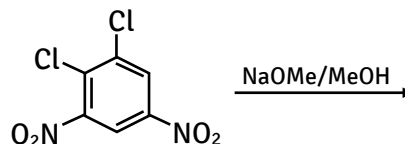
61. Among the bromides given below, the order of reactivity of S_N1 reaction is -



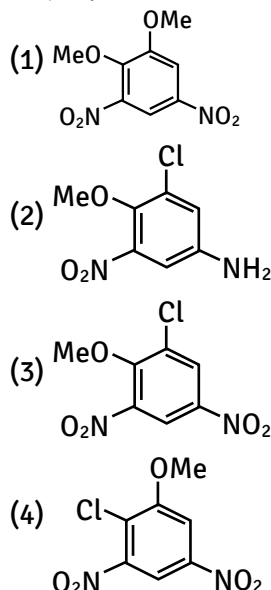
62. The correct order of rate of following compounds towards S_N2 reaction



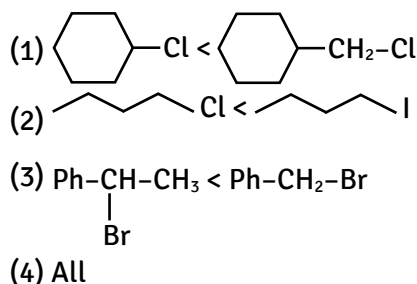
63. Identify major product for the following reaction.



Major product is?



64. Which is correct order of reactivity in S_N2 reaction of following compound?



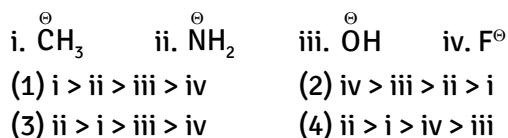
65. Which of the following gives cyanide when reacts with KCN

- (A) Chloro benzene (B) Allyl bromide
(C) Vinyl chloride (D) Benzyl bromide
(1) A, B and C only
(2) A and B only
(3) B and D only
(4) A and C only

66. The decreasing basic strength order of the following is :

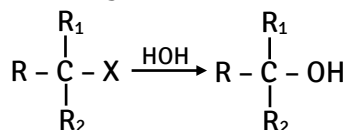
- i. $F^\ominus$ ii. $Cl^\ominus$ iii. $Br^\ominus$ iv. $I^\ominus$
(1) $i > ii > iii > iv$ (2) $iv > iii > ii > i$
(3) $ii > i > iii > iv$ (4) $ii > i > iv > iii$

67. The decreasing nucleophilic order of the following compounds is:

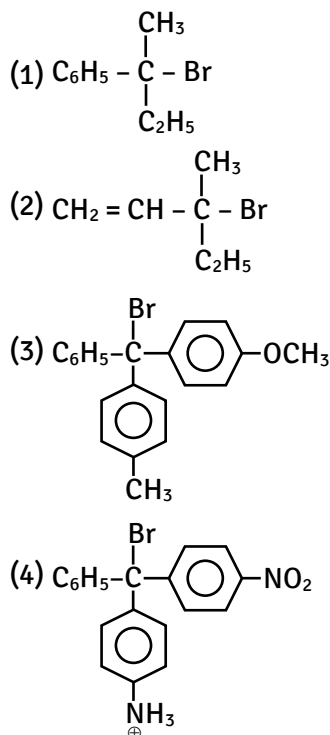


68. $\text{S}_{\text{N}}1$ & $\text{S}_{\text{N}}2$ are not favourable in
 (A) $\text{H}_2\text{C}=\text{CH}-\text{Cl}$ (B) $\text{Ph}-\text{CH}_2-\text{Cl}$
 (C) $\text{Ph}-\text{Cl}$ (D) $\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{Cl}$
 (1) A, B & C Only (2) A & C Only
 (3) B & D Only (4) A, B, C & D

69. For the given reaction



Which substrate will give maximum racemisation?



70. $\text{C}_4\text{H}_9\text{Br} \xrightarrow[\Delta]{\text{alc. KOH}} \text{(B)} \xrightarrow[\text{(ii) Zn/H}_2\text{O}]{\text{(i) O}_3} \text{CH}_3-\text{CHO}$
 (A) and (B) in the above reaction sequence are:
 (1) sec-Butyl bromide, α -Butylene
 (2) tert-Butyl bromide, Isobutylene
 (3) sec-Butyl bromide, β -Butylene
 (4) n-Butyl bromide, α -Butylene

71. In dehydrohalogenations the base (alcoholic KOH) abstracts:

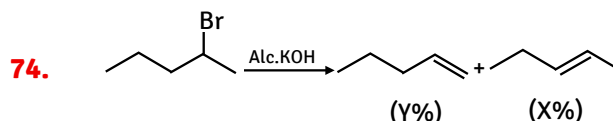
- (1) The halide ion.
 (2) The proton present on the carbon next to the carbon to which the halogen is attached.
 (3) The proton present on the carbon to which the halogen is attached.
 (4) The proton as well as halide ion

72. In the elimination reactions, the reactivity order of halogens in alkyl halides is:

- (1) $\text{R-I} > \text{R-Br} > \text{R-Cl}$ (2) $\text{R-Cl} > \text{R-Br} > \text{R-I}$
 (3) $\text{R-Br} > \text{R-Cl} > \text{R-I}$ (4) $\text{R-I} > \text{R-Cl} > \text{R-Br}$

73. The elimination of HX from an alkyl halide forms an alkene. The order of the elimination reaction is:

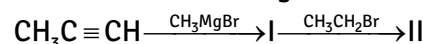
- (1) $3^\circ \text{ halide} > 2^\circ \text{ halides} > 1^\circ \text{ halides}$
 (2) $1^\circ \text{ halide} > 2^\circ \text{ halides} > 3^\circ \text{ halides}$
 (3) $1^\circ \text{ halide} = 2^\circ \text{ halides} > 3^\circ \text{ halides}$
 (4) $2^\circ \text{ halide} > 1^\circ \text{ halides} > 2^\circ \text{ halides}$



X and Y respectively are:

- (1) 81, 19 (2) 50, 50
 (3) 19, 81 (4) 100, 0

75. Consider the following reaction sequence.



The final product (II) formed is:

- (1) $\text{CH}_3\text{C}\equiv\text{CCH}_3$
 (2) $\text{CH}_3\text{C}\equiv\text{CCH}_2\text{CH}_3$
 (3) $\text{CH}_3\text{C}\equiv\text{CMgBr}$
 (4) $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CHCH}_2\text{CH}_3$

Exercise 3

1. **Assertion:** The boiling points of alkyl halides decrease in the order :
 $RI > RBr > RCl > RF$

Reason: The boiling points of alkyl chlorides, bromides and iodides are considerably higher than that of the hydrocarbon of comparable molecular mass.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

2. **Assertion:** Aryl halides cannot be prepared by replacement of hydroxyl group of phenol by halogen atom.

Reason: Phenols react with halogen acids violently.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

3. **Assertion:** The boiling point of the compounds increases in the order:
 $\text{Isopropylchloride} < \text{1-Chloropropane} < \text{1-Chlorobutane}$.

Reason: Boiling point depends upon the molecular mass and surface area.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

4. **Assertion:** Aryl halides are highly reactive towards nucleophilic substitution reactions.

Reason: In case of haloarenes, halogen atom is attached to sp hybridised carbon atom.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

5. **Assertion:** Electrophilic substitution reactions in haloarenes occur slowly and require more drastic conditions as compared to those in benzene.

Reason: Halogens are ortho and para-directors.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

6. **Assertion:** Presence of a nitro group at ortho or para position increases the reactivity of haloarenes towards nucleophilic substitution.

Reason: Nitro group, being an electron withdrawing group decreases the electron density over the benzene ring.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

7. **Assertion:** It is difficult to replace chlorine by $-OH$ in chlorobenzene in comparison to that in chloroethane.

Reason: Chlorine-carbon ($C-Cl$) bond in chlorobenzene has a partial double bond character due to resonance.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

8. **Assertion:** Rate of alkaline hydrolysis of methyl chloride to methanol is higher in DMF than in water

Reason: alkaline hydrolysis of methyl chloride follows second order kinetics.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

9. **Assertion:** Alkyl iodide can be prepared by treating alkyl chloride/bromide with NaI in acetone.

Reason: formed $NaCl/NaBr$ is precipitated in dry acetone.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

10. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion: Alkyl chlorides are best prepared from alcohols by reaction of $SOCl_2$.

Reason: The other two products are easily escapable. So reaction gives pure Alkyl chlorides. In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

11. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion: Boiling points of chlorides, bromides and iodides are considerably higher than those of the parent hydrocarbons.

Reason: Due to greater polarity as well as higher molecular mass as compared to parent hydrocarbon, the intermolecular forces of attraction are stronger in halogen derivatives.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

12. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion: Chlorobenzene does not react with NaOH whereas ethyl chloride reacts.

Reason: The partial double bond between carbon and chlorine in chlorobenzene causes less reactivity towards Nucleophilic substitution reactions.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

13. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion : Alkyl iodides are often prepared by the reaction of alkyl chlorides/bromides with NaI in dry acetone.

Reason : The synthesis of alkyl fluorides is best accomplished by heating an alkyl chloride/ bromide in presence of metallic fluoride such as AgF, Hg₂F₂, CoF₂ or SbF₃.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

14. Given below are two statements:

Statement-I: 2-Bromo pentane on reaction with alcoholic KOH gives 1-pentene as a major product.

Statement-II: In dehydrohalogenation reactions the preferred product is always that alkene which has lesser number of alkyl groups attached to the doubly bonded c-atoms.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statements-I and II are correct.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Both Statement-I and II are incorrect
- (4) Statement-I is incorrect but Statement-II is correct.

15. Given below are two statements:

Statement-I: NaCl or NaBr formed is precipitated in dry acetone in Finkelstein reaction.

Statement-II: It facilitates the reverse reaction according to Le-chatliers principle.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Both statements-I and II are correct.
- (3) Both Statement-I and II are incorrect
- (4) Statement-I is incorrect but Statement-II is correct.

16. Match the reactions given in column I with the type of reaction mentioned in column II and mark the appropriate choice.

	Column I		Column II
A	$\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}-\text{CH}_3 + \text{C}_2\text{H}_5\text{OH} \\ \quad \\ \text{Br} \quad \text{CH}_3 \\ \longrightarrow \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{OC}_2\text{H}_5 \\ \\ \text{CH}_3 \end{array}$	i	β -Elimination
B	$\text{CH}_3\text{CH}_2\text{Br} \xrightarrow[\text{Acetone}]{\text{NaI}}$ $\text{CH}_3\text{CH}_2\text{OH}$	ii	$\text{S}_{\text{N}}1$ Reaction
C	$\text{CH}_3\text{CH}=\text{CH}_2 + \text{HBr} \xrightarrow{\text{Peroxide}}$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	iii	$\text{S}_{\text{N}}2$ Reaction
D	$\text{CH}_3\text{CH}_2\text{Br} + \text{alc.KOH} \rightarrow$ $\text{CH}_2=\text{CH}_2$	iv	Kharasch effect

- (1) A $\rightarrow$ (iv), B $\rightarrow$ (i), C $\rightarrow$ (ii), D $\rightarrow$ (iii)
 (2) A $\rightarrow$ (ii), B $\rightarrow$ (iii), C $\rightarrow$ (iv), D $\rightarrow$ (i)
 (3) A $\rightarrow$ (i), B $\rightarrow$ (ii), C $\rightarrow$ (iv), D $\rightarrow$ (iii)
 (4) A $\rightarrow$ (iii), B $\rightarrow$ (i), C $\rightarrow$ (ii), D $\rightarrow$ (iv)

17. Match the column I with column II and mark the appropriate choice.

	Column I		Column II
A	Carbon tetrachloride	(i)	Paint remover
B	Methylene chloride	(ii)	Refrigerators and air conditioners
C	DDT	(iii)	Fire-extinguisher
D	Freons	(iv)	Non-biodegradable insecticide

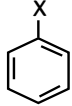
- (1) A $\rightarrow$ (ii), B $\rightarrow$ (iii), C $\rightarrow$ (i), D $\rightarrow$ (iv)
 (2) A $\rightarrow$ (iv), B $\rightarrow$ (iii), C $\rightarrow$ (ii), D $\rightarrow$ (i)
 (3) A $\rightarrow$ (i), B $\rightarrow$ (ii), C $\rightarrow$ (iii), D $\rightarrow$ (iv)
 (4) A $\rightarrow$ (iii), B $\rightarrow$ (i), C $\rightarrow$ (iv), D $\rightarrow$ (ii)

18. Match the following

Column-I	Column-II
(a) $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{Cl} + \text{KOH}_{(\text{Aq.})} \rightarrow$	(p) n-Propyl bromide
(b) $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \rightarrow$	(q) Propene
(c) $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \xrightarrow{\text{Peroxide}}$	(r) Isopropyl bromide
(d) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl} + \text{KOH}_{(\text{Alco.})} \rightarrow$	(s) n-Propyl alcohol

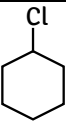
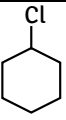
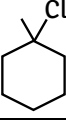
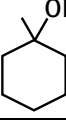
- (1) (a-q), (b-r), (c-p), (d-s)
 (2) (a-s), (b-r), (c-p), (d-q)
 (3) (a-s), (b-p), (c-r), (d-q)
 (4) (a-q), (b-p), (c-r), (d-s)

19. Match the structures of compounds given in column I with the classes of compounds given in column II.

Column-I	Column-II
(i) $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_3 \\ \\ \text{X} \end{array}$	(a) Aryl halide
(ii) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{X}$	(b) Alkyl halide
(iii) 	(c) Vinyl halide
(iv) $\text{CH}_2=\text{CH}-\text{X}$	(d) Allyl halide

- (1) (i-b), (ii-c), (iii-a), (iv-d)
 (2) (i-d), (ii-b), (iii-a), (iv-c)
 (3) (i-b), (ii-d), (iii-a), (iv-c)
 (4) (i-c), (ii-b), (iii-a), (iv-d)

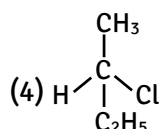
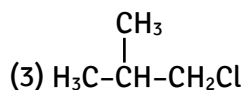
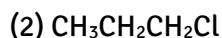
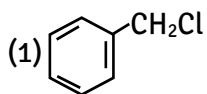
20. Match the column-I and column-II and select the correct answer :

Column-I	Column-II
(a)  $\xrightarrow{\text{aq.KOH}}$	(p) E_1
(b)  $\xrightarrow{\text{alc.KOH}}$	(q) $\text{S}_{\text{N}}1$
(c)  $\xrightarrow{\text{H}_2\text{O}}$	(r) $\text{S}_{\text{N}}2$
(d)  $\xrightarrow{\text{H}^+/\Delta}$	(s) E_2

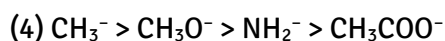
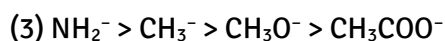
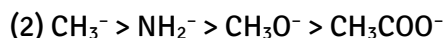
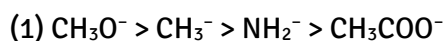
- (1) (a-s); (b-r); (c-p); (d-q)
 (2) (a-r); (b-s); (c-q); (d-p)
 (3) (a-r); (b-s); (c-p); (d-q)
 (4) (a-q); (b-p); (c-r); (d-s)

Exercise 4 (Previous Year's Questions)

1. Which of the following compounds will undergo racemisation when hydrolysed by solution of KOH [AIPMT-2014]



2. Correct order of nucleophilicity :- [AIIMS-2014]



3. In an $\text{S}_{\text{N}}1$ reaction on chiral centres, there is: [Re-AIPMT-2015]

(1) 100 % retention

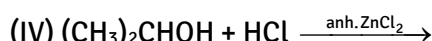
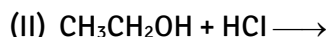
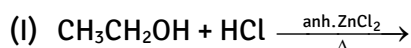
(2) 100 % inversion

(3) 100 % racemization

(4) inversion more than retention leading to partial racemization

4. Which of the following reaction(s) can be used for the preparation of alkyl halides?

[Re-AIPMT-2015]



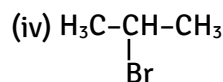
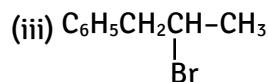
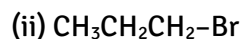
(1) (IV) only

(2) (III) and (IV) only

(3) (I), (III) and (IV) only

(4) (I) and (II) only

5. Correct order of reactivity towards elimination reaction [AIIMS-2015]



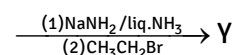
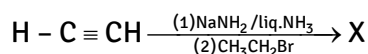
(1) i > ii > iii > iv

(2) ii > i > iii > iv

(3) iii > iv > i > ii

(4) iii > i > iv > ii

6. In the reaction



X and Y are :

[NEET-I-2016]

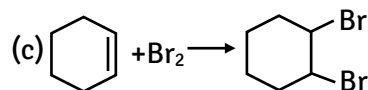
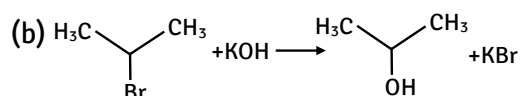
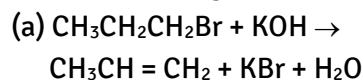
(1) X = 1-Butyne ; Y = 3-Hexyne

(2) X = 2-Butyne ; Y = 3-hexyne

(3) X = 2-Butyne ; Y = 2-Hexyne

(4) X = 1-Butyne ; Y = 2-Hexyne

7. For the following reactions :-



Which of the following statement is correct? [NEET-I-2016]

(1) (a) and (b) are elimination reaction and (c) is addition reaction

(2) (a) is elimination, (b) is substitution and (c) is addition reaction

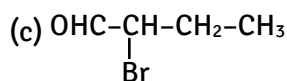
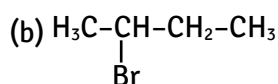
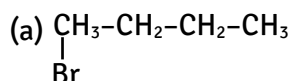
(3) (a) is elimination, (b) and (c) are substitution reactions

(4) (a) is substitution, (b) and (c) are addition reaction

8. Consider the reaction
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{NaCN} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{NaBr}$
[NEET- II-2016]

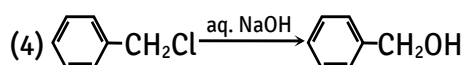
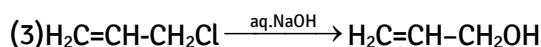
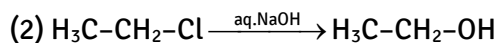
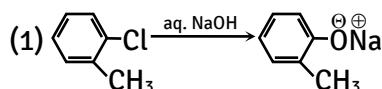
This reaction will be the fastest in
 (1) N, N'-dimethylformamide (DMF)
 (2) Water
 (3) Ethanol
 (4) Methanol

9. Arrange the following compounds in order of reactivity towards $\text{S}_{\text{N}}1$ reaction.
[AIIMS-2016]



- (1) $a > b > c$ (2) $b > a > c$
 (3) $c > b > a$ (4) $a > c > b$

10. The hydrolysis reaction that takes place at the slowest rate, among the following is
[NEET-2019(ODISHA)]

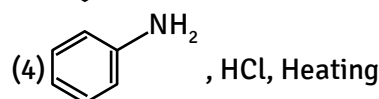
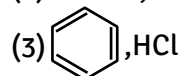


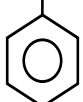
11. The correct sequence of bond enthalpy of 'C-X' bond is: **[NEET(UG)-2021]**
 (1) $\text{CH}_3-\text{F} < \text{CH}_3-\text{Cl} < \text{CH}_3-\text{Br} < \text{CH}_3-\text{I}$
 (2) $\text{CH}_3-\text{F} > \text{CH}_3-\text{Cl} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
 (3) $\text{CH}_3-\text{F} < \text{CH}_3-\text{Cl} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
 (4) $\text{CH}_3-\text{Cl} > \text{CH}_3-\text{F} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$

12. The incorrect statement regarding chirality is : **[NEET(UG)-2022]**
 (1) $\text{S}_{\text{N}}1$ reaction yields 1 : 1 mixture of both enantiomers.
 (2) The product obtained by $\text{S}_{\text{N}}2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration.
 (3) Enantiomers are superimposable mirror images on each other.
 (4) A racemic mixture shows zero optical rotation.

13. Which of the following is suitable to synthesize chlorobenzene?
[NEET(UG)-2022]

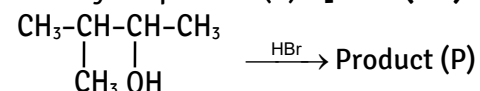
- (1) Benzene, Cl_2 , anhydrous FeCl_3
 (2) Phenol, NaNO_2 , HCl , CuCl



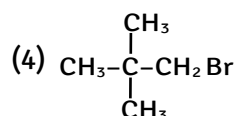
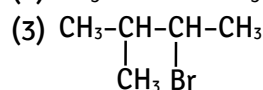
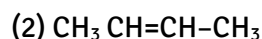
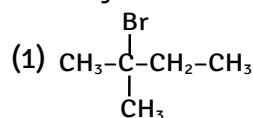
14. The given compound
 $\text{CH}=\text{CH}-\text{CH}-\text{CH}_2\text{CH}_3$


Is an example of _____ **[NEET(UG)-2023]**
 (1) Benzylic halide (2) Aryl halide
 (3) Allylic halide (4) Vinylic halide

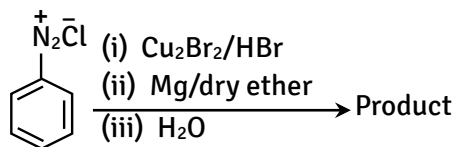
15. Consider the following reaction and identify the product (P). **[NEET(UG)-2023]**



3-Methylbutan-2-ol



16. Identify the product in the following reaction [NEET(UG)-2023]



- (1)
 (2)
- (3)
 (4)

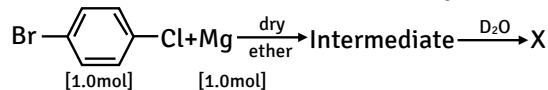
17. Which amongst the following reactions of alkyl halides produces isonitrile as a major product? [NEET(Manipur)-2023]

- (A) $R-X + HCN \rightarrow$
 (B) $R-X + AgCN \rightarrow$
 (C) $R-X + KCN \rightarrow$
 (D) $R-X + NaCN \xrightarrow[C_2H_5OH]{H_2O}$

Choose the most appropriate answer from the options given below:

- (1) (D) only
 (2) (C) and (D) only
 (3) (B) only
 (4) (A) and (B) only

18. Identify 'X' in the following reaction. [NEET(Manipur)-2023]



- (1)
 (2)
- (3)
 (4)

19. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). [NEET-2025]

Assertion (A): undergoes S_N2 reaction faster than Cl

Reason (R): Iodine is a better leaving group because of its large size.

In the light of the above statements, choose the correct answer from the options given below :

- (1) A is true but R is false
 (2) A is false but R is true
 (3) Both A and R are true and R is the correct explanation of A
 (4) Both A and R are true but R is not the correct explanation of A

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	4	4	3	1	3	3	4	1	4	4	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	3	2	3	1	4	1	2	3	3	1	1	2	4	2	2	3	4	2

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	2	3	4	2	4	2	3	2	1	3	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	3	3	2	1	4	4	2	4	1	2	3	4	2	2	3	4	1	2
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	3	4	1	4	3	2	4	4	2	4	3	1	2	2	1	1	2	3	3
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	1	4	1	4	2	4	4	1	4	4	4	3	3	3	3	3	4	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75					
Ans.	1	1	3	4	3	1	1	2	3	3	2	1	1	1	2					

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	1	4	2	1	1	1	1	1	1	1	2	3	1	2	4	2	3	2

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
Ans.	4	2	4	3	4	1	2	1	2	1	2	3	1	3	1	2	3	1	3	

Chapter 3

ALCOHOLS, PHENOLS & ETHERS

Chapter Summary

- Introduction
- Preparation of alcohols
- Preparation of phenol
- Physical properties
- Chemical Reaction of Alcohols
- Chemical reaction of phenol
- Preparation of ethers
- Chemical reactions of ether

Introduction

- Alcohols and phenols are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively is replaced by -OH group
- An alcohol contains one or more hydroxyl (OH) groups directly attached to carbon atom(s) of an aliphatic system while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system (C_6H_5OH)

Classification of alcohol:-

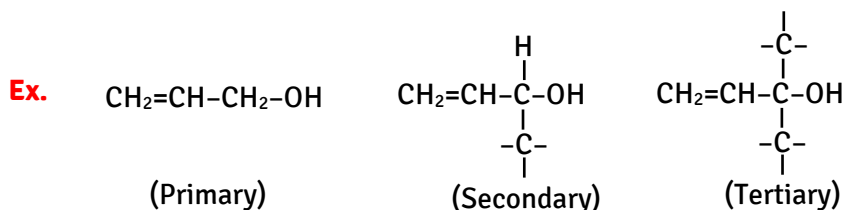
(a) Classification According to Number of -OH Groups:

- (i) Monohydric [one -OH] $\longrightarrow$ CH_3CH_2-OH
- (ii) Dihydric [two -OH] $\longrightarrow$ $\begin{array}{c} CH_2-CH_2 \\ | \quad | \\ OH \quad OH \end{array}$
- (iii) Trihydric [three -OH] $\longrightarrow$ $\begin{array}{c} CH_2-CH-CH_2 \\ | \quad | \quad | \\ OH \quad OH \quad OH \end{array}$
- (iv) Polyhydric [n -OH] $\longrightarrow$ $\begin{array}{c} CH_2-CH \cdots \cdots CH_2 \\ | \quad | \quad | \\ OH \quad OH \quad OH \end{array}$

(b) Classification According to Nature of Carbon:

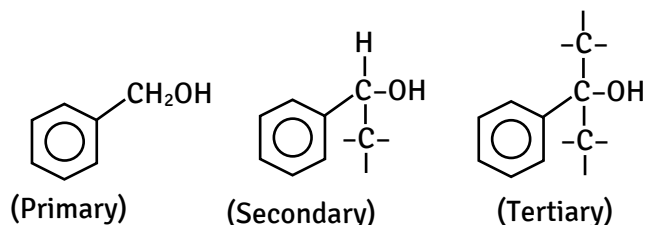
- (i) p or 1° - alcohol $\longrightarrow$ CH_3CH_2-OH
- (ii) s or 2° - alcohol $\longrightarrow$ $(CH_3)_2CH-OH$
- (iii) t or 3° - alcohol $\longrightarrow$ $(CH_3)_3C-OH$

- **Allylic alcohols:-** OH group is attached to a Sp^3 hybridized carbon atom adjacent to the carbon-carbon double bond.



- **Benzylic alcohols:-** OH group is attached to a Sp^3 hybridized carbon atom next to an aromatic ring.

Ex.



Allylic & benzylic alcohols may be primary, secondary or tertiary

- **Vinylic alcohols:-** OH group bonded to a carbon-carbon double bond

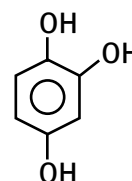
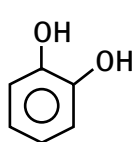
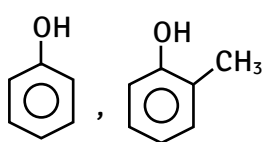
Ex. $CH_2=CH-OH$

Classification of phenol on the basis of number of -OH group

(i) monohydric (One-OH)

(ii) Dihydric (two-OH)

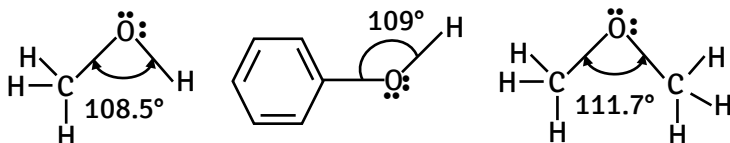
(iii) Trihydric(three-OH)



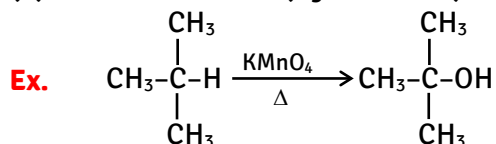
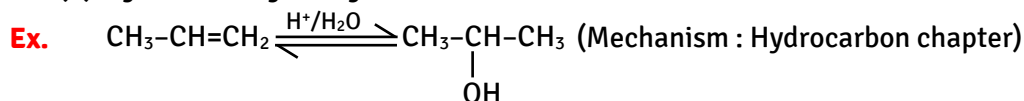
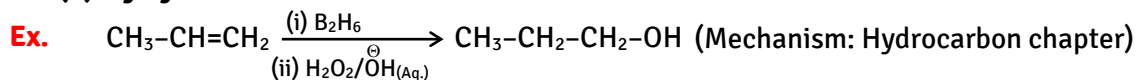
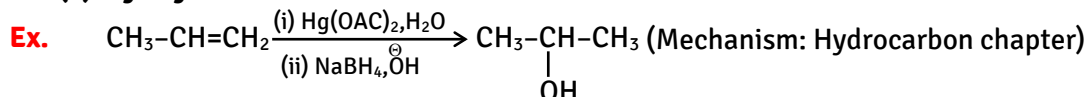
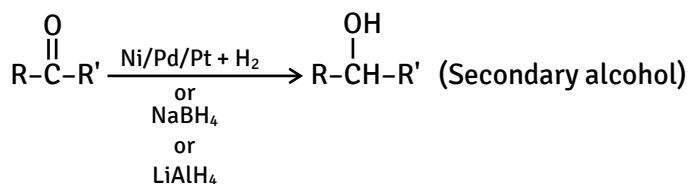
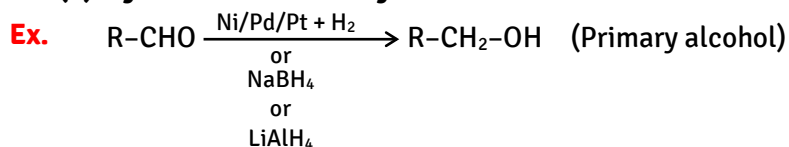
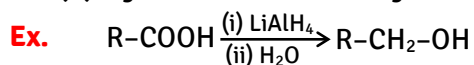
Some important common name

	Phenol		O-cresol
	m-cresol		p-cresol
	Catechol		Resorcinol
	Hydroquinone or quinol	CH_3OH	Methyl alcohol
CH_3CH_2OH	Ethyl alcohol	$CH_3-CH_2-CH_2-OH$	n-propyl alcohol
$CH_3-\underset{\substack{ \\ OH}}{CH}-CH_3$	Iso propyl alcohol	$CH_3-CH_2-CH_2-CH_2-OH$	n-butyl alcohol
$CH_3-\underset{\substack{ \\ OH}}{CH}-CH_2-CH_3$	Sec. Butyl alcohol	$CH_3-\underset{\substack{ \\ CH_3}}{CH}-CH_2-OH$	Iso butyl alcohol

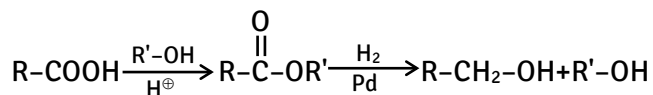
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$	tert-butyl alcohol	$\begin{array}{cc} \text{CH}_2 & - & \text{CH}_2 \\ & & \\ \text{OH} & & \text{OH} \end{array}$	ethylene glycol
$\begin{array}{ccc} \text{CH}_2 & - & \text{CH} & - & \text{CH}_2 \\ & & & & \\ \text{OH} & & \text{OH} & & \text{OH} \end{array}$	Glycerol		

Comparison of Bond Angles in Phenols, Alcohols and Ethers:

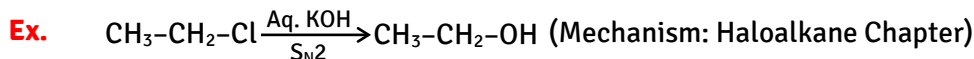
Bond angle increases with the increase in hindrance.

Preparation of alcohols**(1) From alkanes (By oxidation):****(2) From alkenes****(a) By acid catalysed hydration:****(b) By hydroboration oxidation:****(c) By oxymercuration Demercuration:****(3) From carbonyl compounds:****(a) By reduction of aldehyde and ketones:****(b) By reduction of carboxylic acids and esters:**

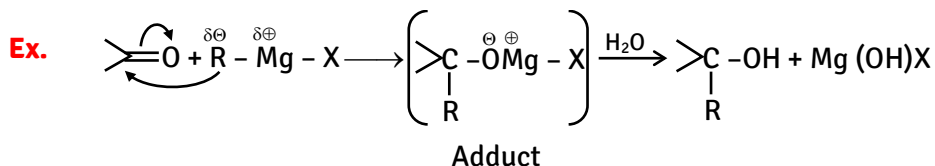
However, LiAlH_4 is an expensive reagent and therefore used for preparing special chemical only. commercially, acids are reduced to alcohol by converting them to the esters followed by their reduction using Hydrogen in the presence of catalyst (catalytic Hydrogenation)



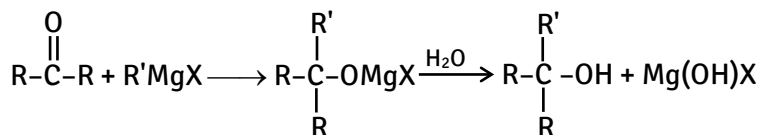
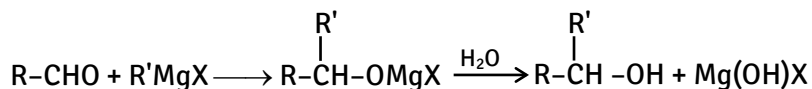
(4) From Alkyl halides:



(5) From Grignard reagents:



The overall reaction using different aldehyde and ketones are as follows:

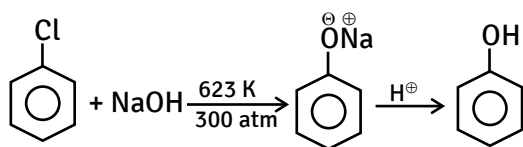


Note. The reaction of grignard reagents with methanal produces a primary alcohol, with other aldehyde secondary alcohols and with ketones tertiary alcohols.

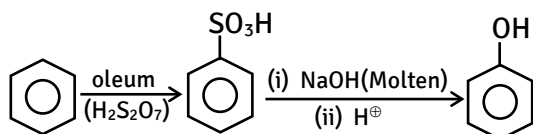
Preparation of phenol

Phenol also known as carbolic acid, was first isolated in the early nineteenth century from cool tar. Nowadays, phenol is commercially produced synthetically. In the laboratory, phenols are prepared from benzene derivatives by following methods.

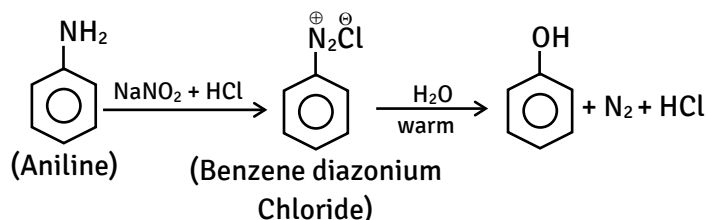
(1) From Haloarenes



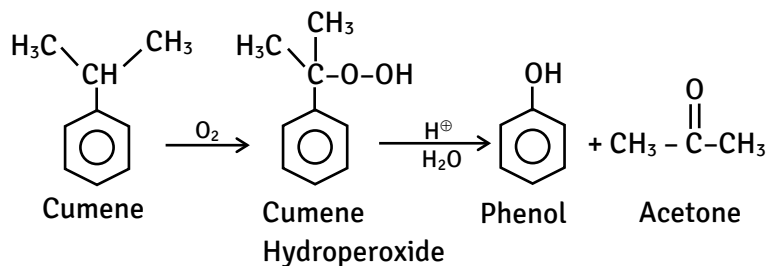
(2) From benzene sulphonic acid



(3) From diazonium salts



(4) From cumene

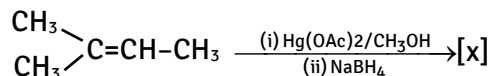


Note. Acetone is by-product of this reaction which is obtained in large quantities by this method

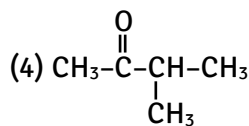
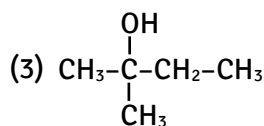
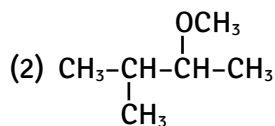
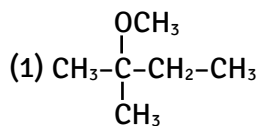
Exercise 1.1

- A tertiary alcohol is obtained when CH_3MgI reacts with:
 - CH_3CHO
 - $\text{CH}_3\text{CH}_2\text{CHO}$
 - $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CHO}$
 - CH_3COCH_3
- $$\begin{array}{ccc}
 \text{Cyclohexene} & \xrightarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{BD}_3} & \text{Product A; A is:} \\
 \text{(1) } \text{Cyclohexyl-CH}_3\text{OH} & & \text{(2) } \text{Cyclohexyl-CH}_2\text{OH} \\
 \text{(3) } \text{Cyclohexyl-CH}_2\text{OH} & & \text{(4) } \text{Cyclohexyl-CH}_2\text{OH}
 \end{array}$$
- Alkene $\xrightarrow{\text{B}_2\text{H}_6} \xrightarrow[\text{OH}^-]{\text{H}_2\text{O}_2} 2^\circ \text{ alcohol}$
The alkene would be:
 - $\text{CH}_3-\text{CH}=\text{CH}_2$
 - $\text{CH}_3\text{CH}_2-\text{CH}=\text{CH}_2$
 - $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CH}_2$
 - $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
- Iso-butylene when subjected to hydroboration oxidation reaction yields
 - Sec-butyl alcohol
 - Tert-butyl alcohol
 - Iso-butyl alcohol
 - n-butyl alcohol
- 3-methyl-1-butene on oxymercuration-demercuration yields _____ as the major product.
 - 3-methyl-2-butanol
 - 2-methyl-2-butanol
 - 3-methyl-1-butanol
 - 2-methyl-1-butanol
- Which reagent converts propene to 1-propanol?
 - $\text{H}_2\text{O}, \text{H}_2\text{SO}_4$
 - $\text{B}_2\text{H}_6, \text{H}_2\text{O}_2, \text{OH}^-$
 - $\text{Hg}(\text{OAc})_2, \text{NaBH}_4/\text{H}_2\text{O}$
 - Aq. KOH
- When $\text{CH}_2=\text{CH}-\text{COOH}$ is reduced with LiAlH_4 , the compound obtained will be:
 - $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{OH}$
 - $\text{CH}_3-\text{CH}_2-\text{CHO}$
 - $\text{CH}_3-\text{CH}_2-\text{COOH}$
 - $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$

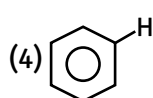
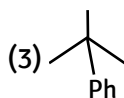
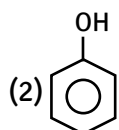
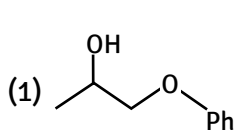
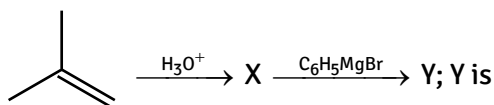
8. In the given reaction:



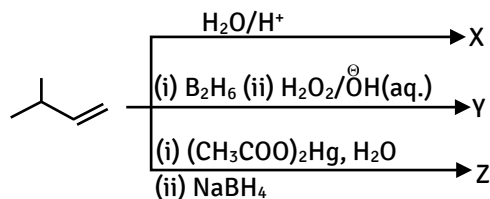
[X] will be:



9.



10.



Most acidic compound is

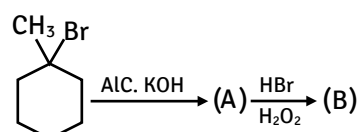
(1) X

(2) Y

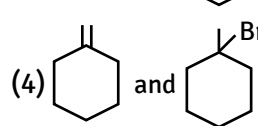
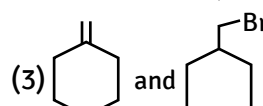
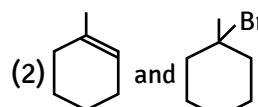
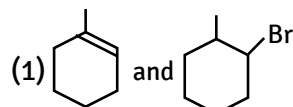
(3) Z

(4) All are equally acidic

11.

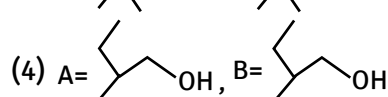
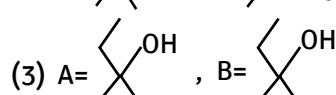
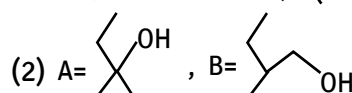
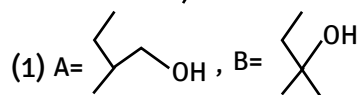
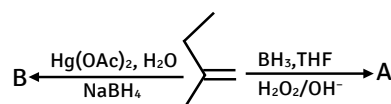


Identify 'A' and 'B' respectively



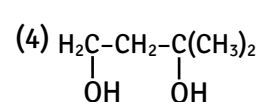
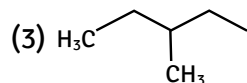
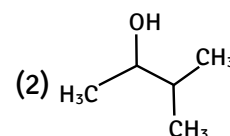
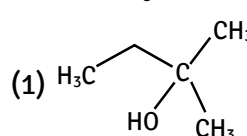
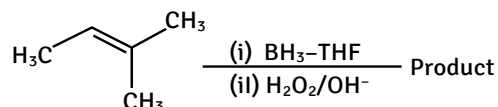
12.

Find out the major products from the following reactions.

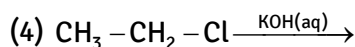
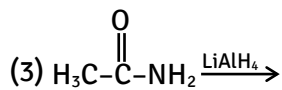
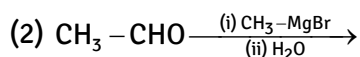
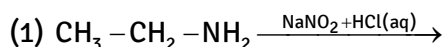


13.

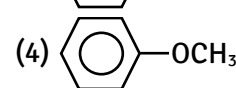
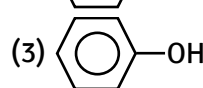
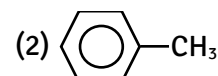
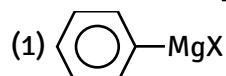
The product formed in the following reaction is



14. Alcohol is not formed as product in



15. $\text{HCHO} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) A}} \rightarrow \text{C}_6\text{H}_5 - \text{CH}_2 - \text{OH}$; A is :



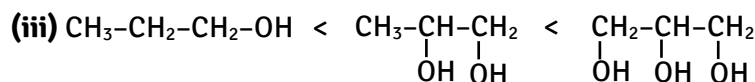
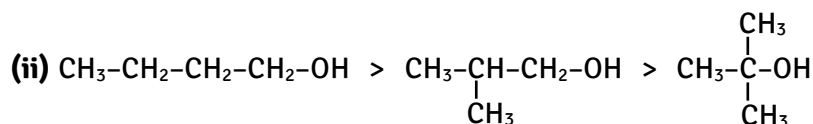
Physical properties

(a) **Boiling point:-**

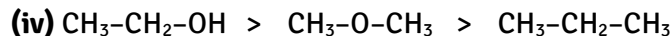
B.P $\propto$ molecular weight

B.P $\propto \frac{1}{\text{Branching (Sidechain)}}$ (when M_w is same)

Ex. **Order of B.P**



(Number of OH increase, H-bonding increases)



Note. Boiling point of alcohols and phenols are higher in comparison to other classes of compound, namely Hydrocarbons, ether, haloalkanes and Haloarenes of comparable molecular masses due to inter molecular H-bonding

(b) **Solubility:-** Solubility of alcohols and phenols in water is due to their ability to form Hydrogen bonds with water molecules.

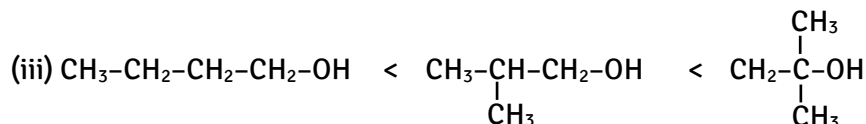
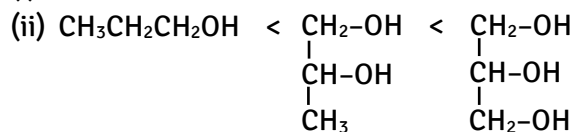
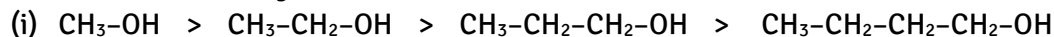
Solubility $\propto$ tendency to form H-bonds with water

Solubility $\propto \frac{1}{\text{Molecular weight of Alkyl / Aryl Part}}$

When M_w is same then

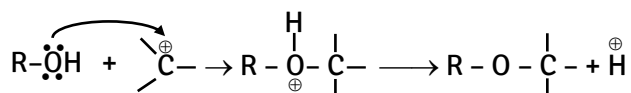
Solubility $\propto$ Branching (side chain)

Ex. **Order of solubility:**

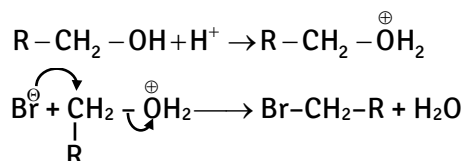


Alcohols are versatile compounds. They react both as nucleophile and electrophile.

The bond between O-H is broken when alcohol react as nucleophiles



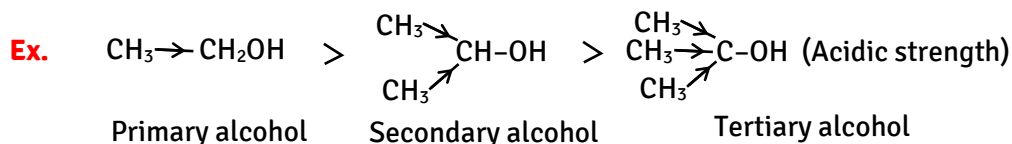
The bond between C-O is broken when they react as electrophiles. Protonated alcohols react in this manner



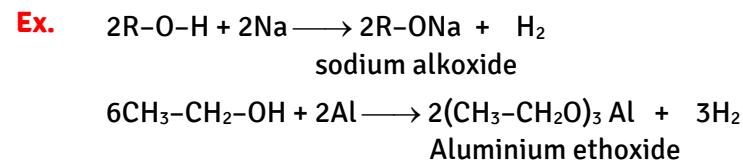
- Based on the cleavage of O-H and C-O bonds, the reaction of alcohols and phenols may be divided into two groups

(1) Acidity of alcohols:

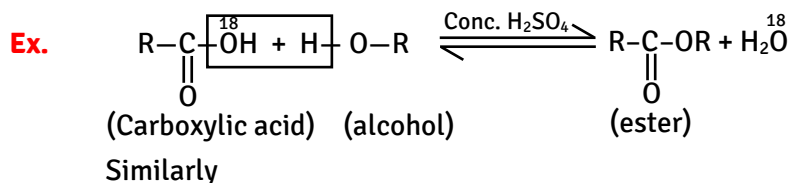
The acidic character of alcohol is due to the polar nature of O-H bond. An electron-releasing group ($-\text{CH}_3$, $-\text{C}_2\text{H}_5$) increases electron density on oxygen tending to decrease the polarity of O-H bond. This decreases the acid strength.

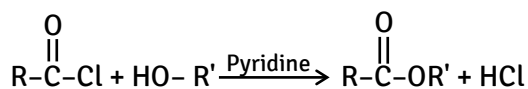
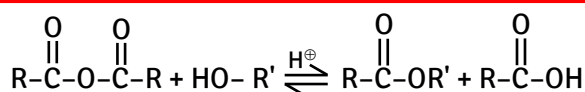


Alcohol reacts with active metals such as sodium, potassium and Aluminium to yield corresponding alkoxide and Hydrogen gas



- alcohol react with carboxylic acid to give ester
- conc. H_2SO_4 is used as catalyst and dehydrating agent

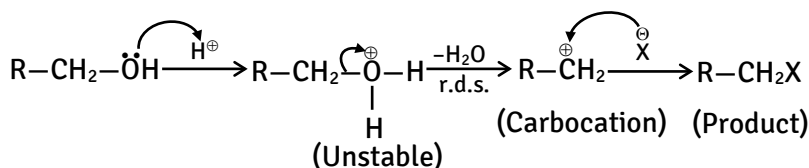
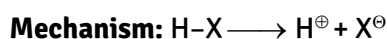
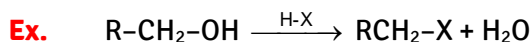




reaction of alcohol with acid chloride is carried out in the presence of a base (pyridine) so as to neutralise HCl which is formed during the reaction. It shifts the equilibrium to the right hand side.

(B) Reaction involving cleavage of C-O bond

(1) Reaction with hydrogen halides (H-X):



- Formation of carbocation is r.d.s step.
- Since carbocation is formed as intermediate hence, rearrangement can take place
- Reactivity order of alcohol towards the reaction

Reactivity $\propto$ stability of carbocation

so reactivity order : 3° alcohol > 2° alcohol > 1° alcohol

- Reactivity order for H-X: HI > HBr > HCl
- Mechanism used:

3° alcohol	2° alcohol	1° alcohol
$\text{S}_{\text{N}}1$	$\text{S}_{\text{N}}1$	$\text{S}_{\text{N}}2$

- Catalyst used in reactions:

conc. H-I (usually no. Catalyst is used)

conc. H-Br + H_2SO_4

conc. H-Cl + Anhyd. ZnCl_2 → It is called **Lucas Reagent (L.R.)**

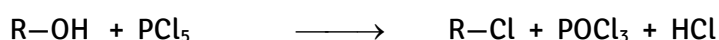
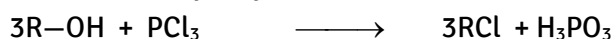
- Alcohol are soluble in Lucas reagent but their halides are immiscible and produce turbidity in solution therefore this reaction is used to identify 1°, 2° and 3° alcohol by noticing time taken to appear turbidity

3° alcohol or allylic alcohol or benzylic alcohol $\xrightarrow{\text{L.R.}}$ Immediate turbidity

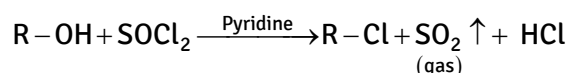
2° alcohol $\xrightarrow{\text{L.R.}}$ turbidity in 5-7 min.

1° alcohol $\xrightarrow{\text{L.R.}}$ No turbidity at room temperature (turbidity appear only on heating after 30 Min.)

(2) Reaction with phosphorus trihalides

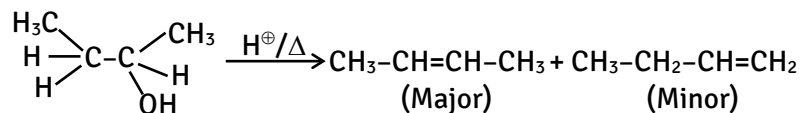
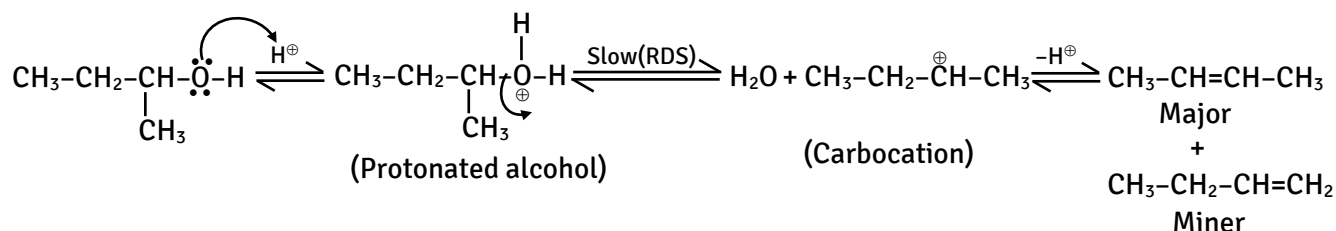


(3) Reaction with thionyl chloride (SOCl_2)



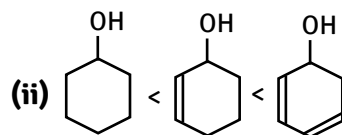
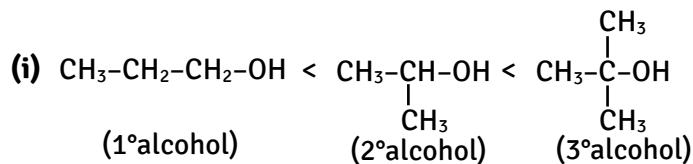
(4) Acidic dehydration of Alcohol.

Alcohols undergo dehydration (removal of a water molecule) to form alkenes on treating with a protic acid i.e. Concentrated H_2SO_4 or H_3PO_4 , or catalysts such as anhydrous zinc chloride or alumina

**Mechanism:**

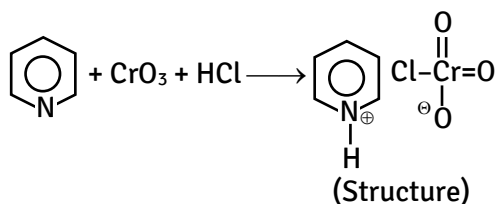
- Formation of carbocation is r.d.s step
- Rearrangement is possible
- Reaction follow E_1 mechanism
- Ease of dehydration of alcohol $\propto$ Stability of carbocation formed in r.d.s. step
- Order of rate of dehydration of alcohol $\rightarrow 3^\circ \text{OH} > 2^\circ \text{OH} > 1^\circ \text{OH}$
- Saytzeff alkene is formed as major product

Ex. correct order of ease of dehydration of alcohol is-

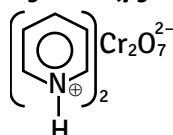
**(5) Oxidation of Alcohol.**

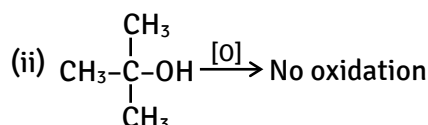
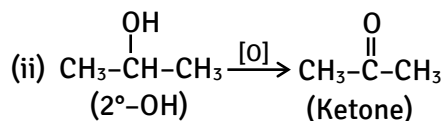
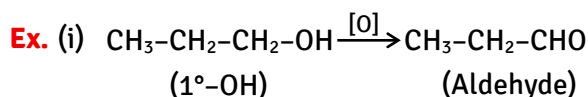
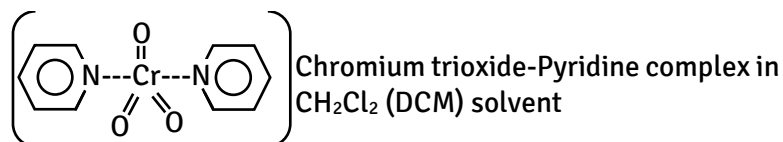
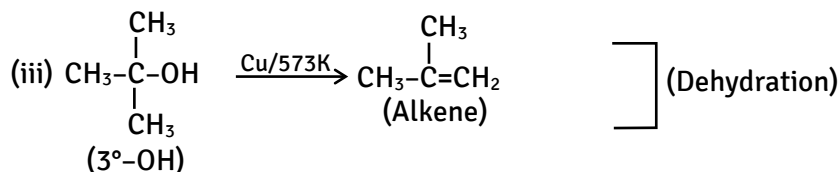
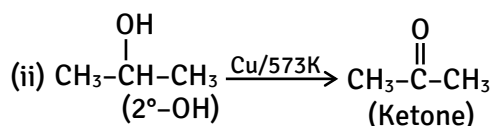
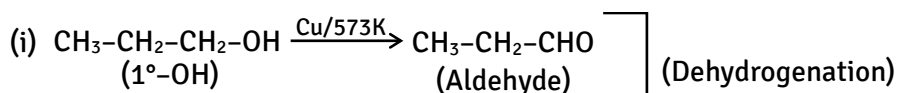
(a) By mild oxidising agent

(I) By PCC (Pyridinium chloro chromate)

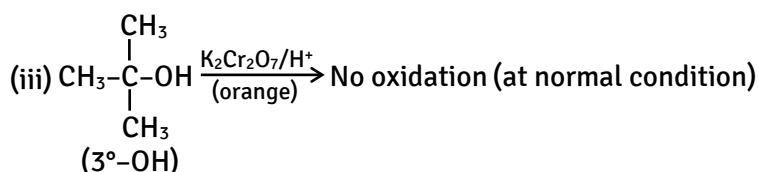
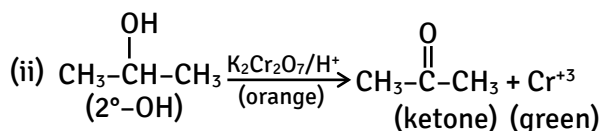
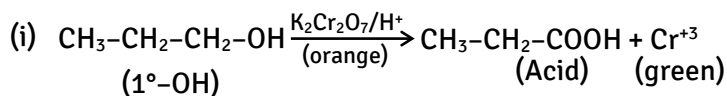


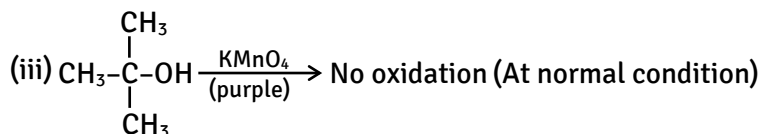
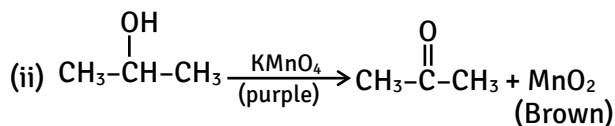
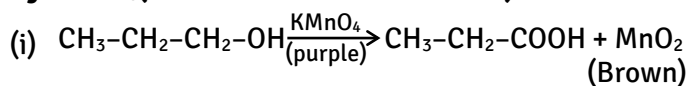
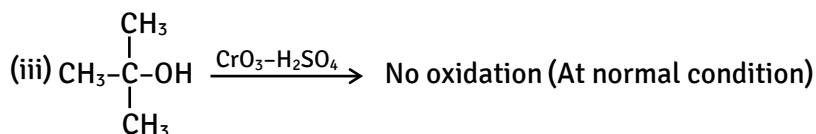
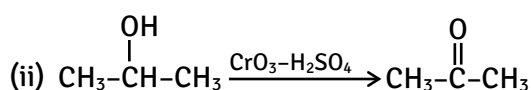
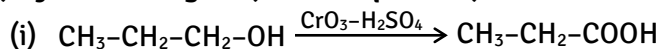
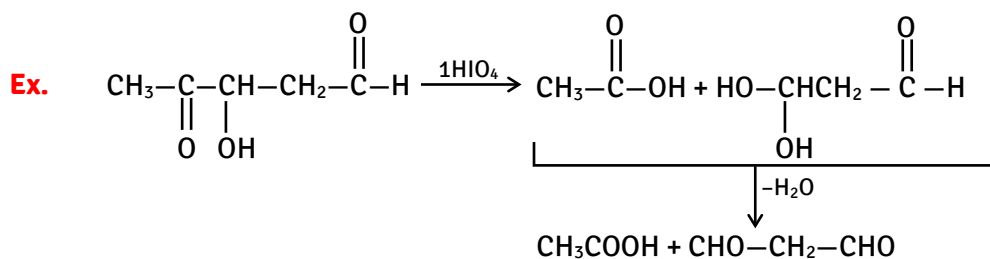
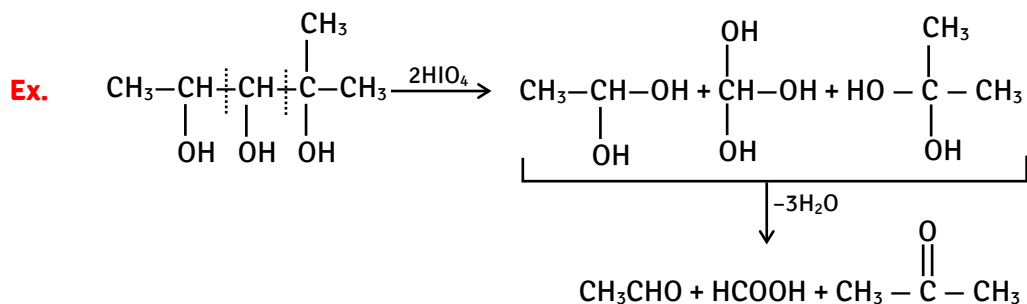
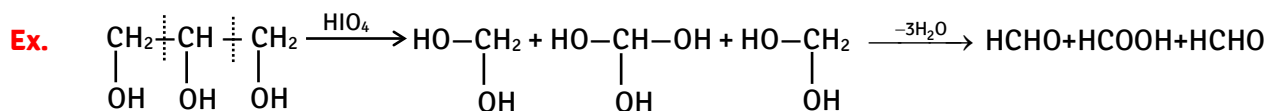
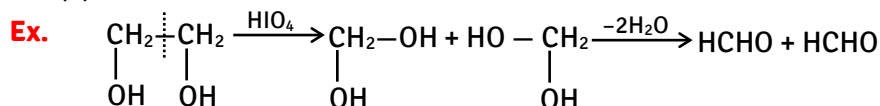
(II) By PDC (pyridinium dichromate)

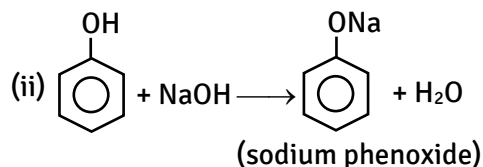
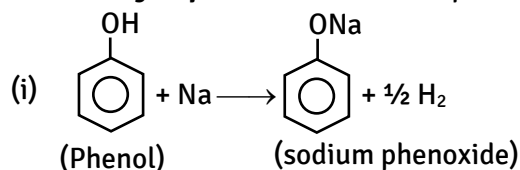


(III) By collin's reagent**(IV) By Cu/ Δ** 

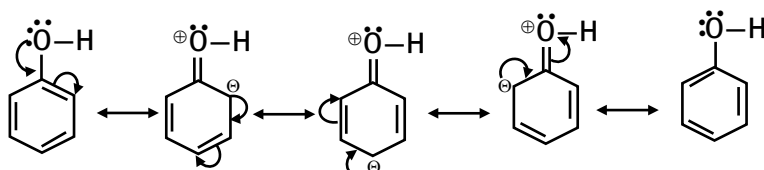
Note. When the vapour of a primary or a secondary alcohol are passed over heated copper at 573K, dehydrogenation takes place and an aldehyde or ketone is formed while tertiary alcohols undergo dehydration.

(b) By strong oxidising agent**(I) $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$** 

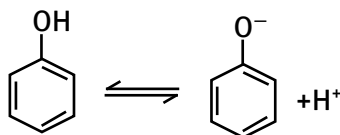
(II) By KMnO_4 (Acidic or alkaline or neutral)**(III) By Jones Reagent ($\text{CrO}_3 + \text{Aq. H}_2\text{SO}_4$) $\rightarrow$ in acetone solvent****(c) oxidation by HIO_4 :**Condition for Oxidation By HIO_4 (i) At least 2 --OH or 2 >C=O or 1 --OH and 1 >C=O should be at vicinal carbons.(ii) One HIO_4 breaks one bond and adds one --OH to each carbon.

Chemical reaction of phenol**(1) Acidity of phenols:** Reaction of phenol with metals and sodium hydroxide indicates its acidic nature

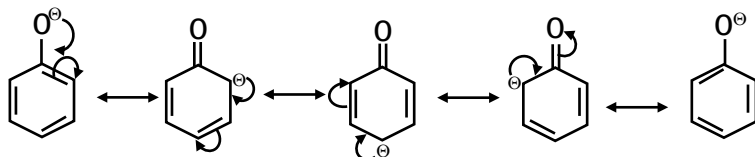
Note. The hydroxyl group of phenol is directly attached to sp² hybridised carbon of benzene ring which acts as an electron withdrawing group.



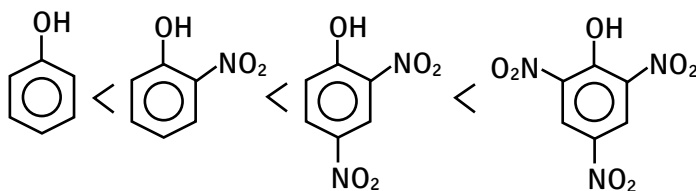
- Due to resonance oxygen acquires positive charge thus facilitate the release of H⁺.



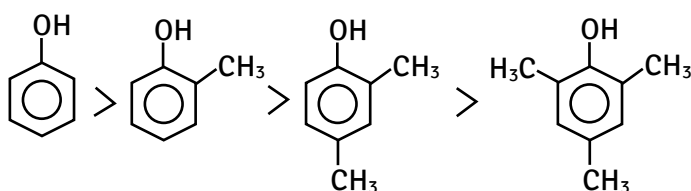
- Phenoxide ion is also resonance stabilised



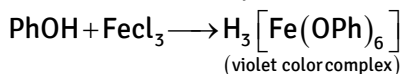
- Both phenol and phenoxide ion are resonance stabilised but due to charge separation phenol is less stable than phenoxide ion so it readily releases H⁺ and forms phenoxide ion.
- In substitute phenols, the presence of electron withdrawing groups such as -NO₂, -CN, -CHO enhances the acidic strength of phenol. This effect is more pronounced when such a group is present at ortho and para position. it is due to the effective delocalization of negative charge.



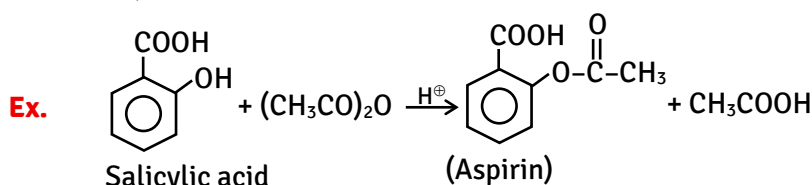
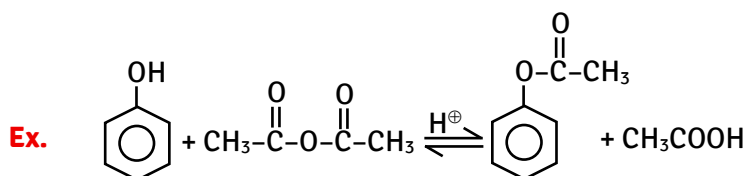
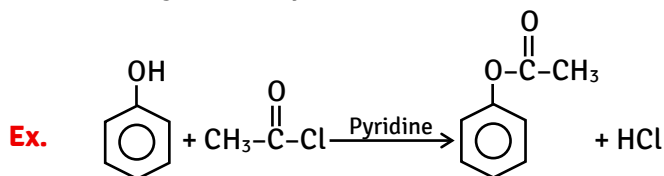
- Electron releasing group such as alkyl groups does not favour the formation of phenoxide ion resulting in decrease in acidic strength.



- (2) **Reaction with neutral FeCl_3 :** Phenol gives violet colouration with FeCl_3 solution (neutral) due to formation of a complex.



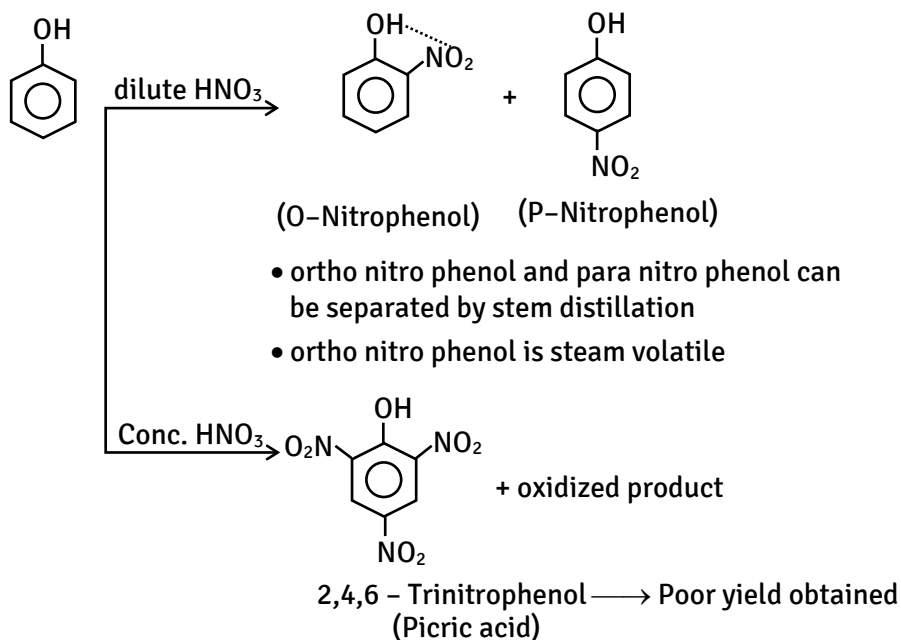
- (3) **Acetylation of phenol**



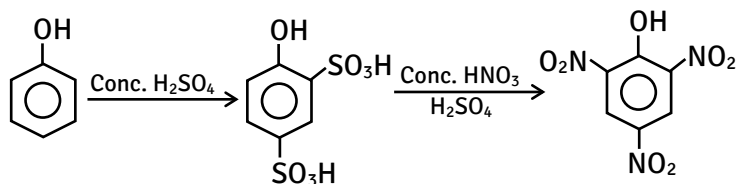
- Note.** (i) Acetylation of salicylic acid produces aspirin
(ii) Aspirin has analgesic, anti inflammatory and antipyretic properties

- (4) **Electrophilic aromatic substitution reaction**

- (i) **Nitration:**

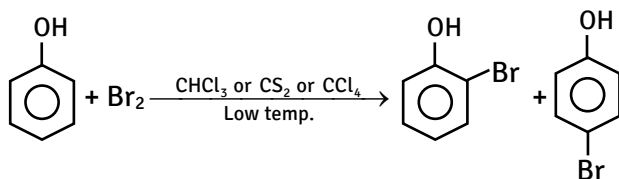


- Note.** To get picric acid in good yield, first do sulphonation so that phenol becomes less activated then do nitration

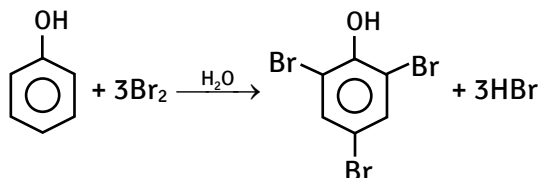
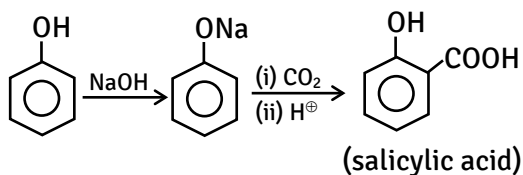
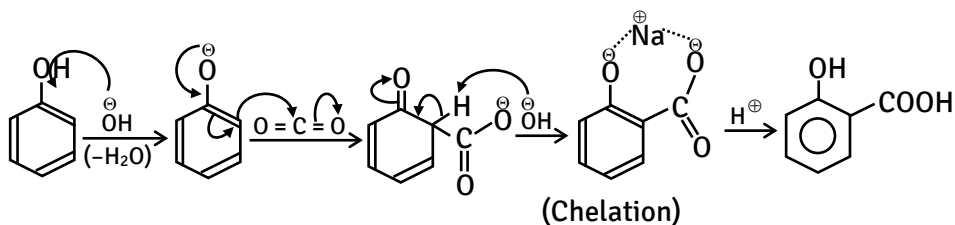
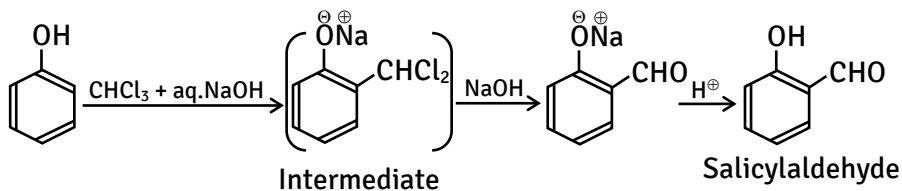
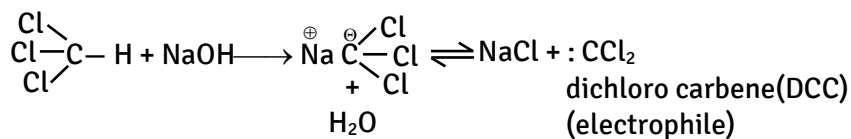


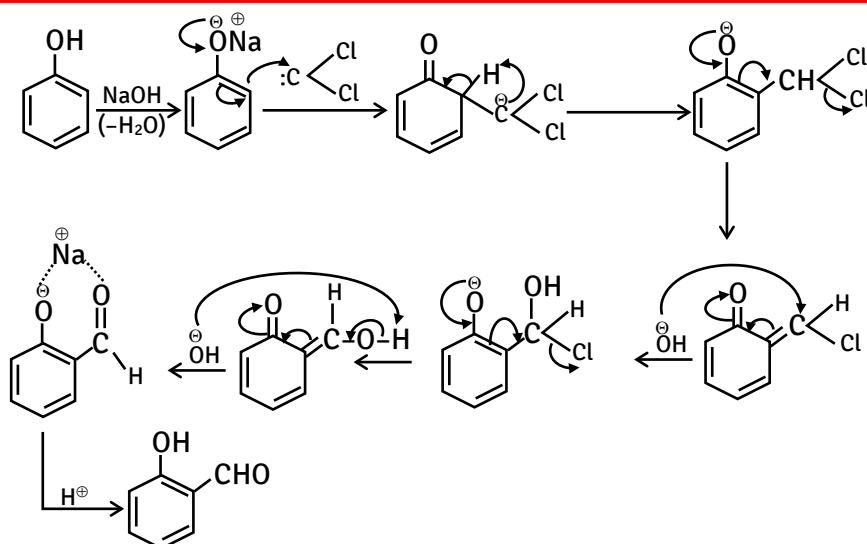
(ii) Halogenation:

- Phenol reacts with bromine in CCl_4 to form mixture of o- and p-bromo phenol.

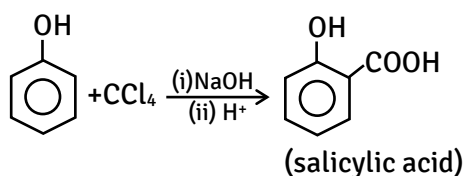


- Phenol reacts with bromine water to form a white ppt. of 2,4,6 tribromophenol.

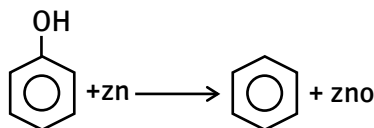
**(5) Kolbe's reaction:****Mechanism:****(6) Reimer - Tiemann reaction:****Mechanism:**



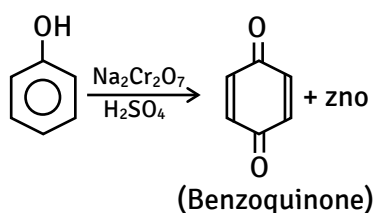
Note. Abnormal Reimer- Tiemann rxn→



(7) Reaction with zn dust



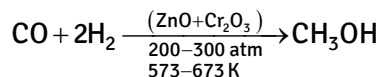
(8) oxidation



- oxidation of phenol with chromic acid produces a conjugated diketone known as benzoquinone. In the presence of air, phenols are slowly oxidised to dark colored mixtures containing quinones.

Some commercially important alcohols

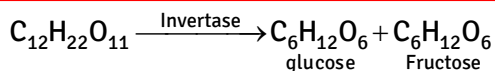
1. Methanol : (Wood spirit):



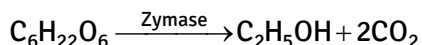
It is highly poisonous in nature, causes blindness and even death. It is used in paints and varnishes.

2. Ethanol:

It is obtained commercially by fermentation of sugars. The sugar in molasses, sugarcane or fruits such as grapes is converted to glucose and fructose



Glucose and fructose undergoes fermentation in the presence of zymase enzyme, which is found in yeast.



The action of zymase is inhibited once the percentage of alcohol reaches 14% under anaerobic (absence of O_2) condition. If air gets into fermentation mixture, the O_2 oxidises ethanol to ethanoic acid, which destroys the taste of alcoholic drinks.

Exercise 1.2

1. The -OH group of methyl alcohol cannot be replaced by chlorine by the action of:

(1) NaCl
(2) Hydrogen chloride
(3) Phosphorus trichloride
(4) Phosphorus pentachloride

2. Which of the following Reaction of alcohol does not show cleavage of R-O linkage:

(1) $\text{ROH} + \text{PCl}_5$ (2) $\text{ROH} + \text{SOCl}_2$
(3) $\text{ROH} + \text{HCl}$ (4) $\text{ROH} + \text{Na}$

3. Phenol reacts with bromine in CS_2 at a low temperature, the product is:

(1) m-Bromophenol
(2) p-Bromophenol only
(3) o-and p-Bromophenol
(4) 2, 4, 6-Tribromophenol

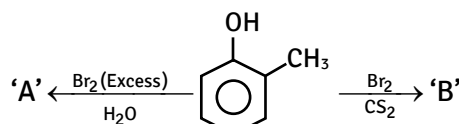
4. Phenol reacts with conc. HNO_3 in the presence of conc. H_2SO_4 to give:

(1) Meta nitrophenol
(2) Ortho nitrophenol
(3) Ortho and para nitrophenol
(4) Picric acid

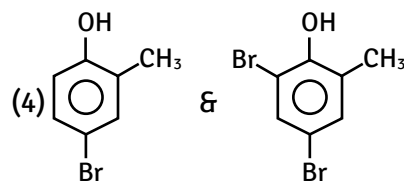
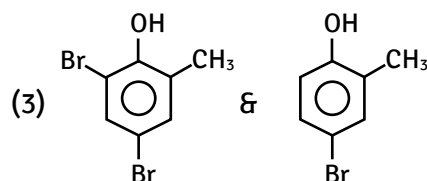
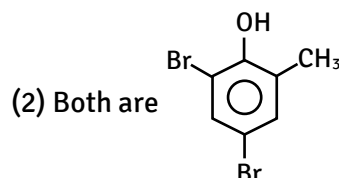
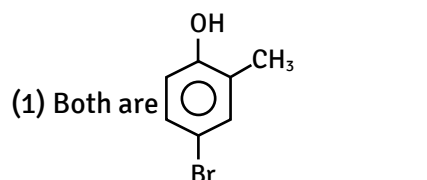
5. Phenol can be converted into 1,3,5-tribromo benzene by using the sequence

(1) $\text{Br}_2/\text{H}_2\text{O}$
(2) (i) $\text{Br}_2/\text{H}_2\text{O}$ (ii) Zn-Dust
(3) (i) Br_2/CS_2 (ii) Zn-Dust
(4) (i) Zn-Dust (ii) $\text{Br}_2/\text{H}_2\text{O}$

6.

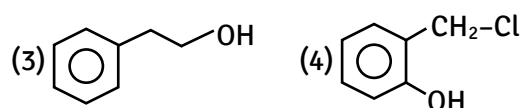


'A' & 'B' respectively are:



7.

An alcohol gives turbidity with HCl/ZnCl_2 (Lucas reagent) only on heating. The alcohol is:

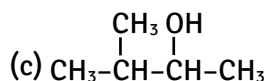
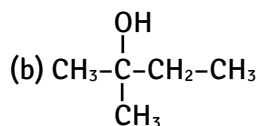
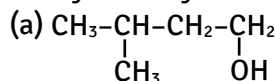


8. The following reaction is known as
 Phenol $\xrightarrow[\text{(ii) } \text{H}^+]{\text{(i) } \text{CHCl}_3/\text{NaOH}}$ Salicylaldehyde

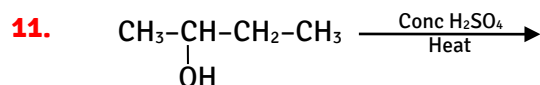
- (1) Gattermann aldehyde synthesis
- (2) Sandmeyer's reaction
- (3) Kolbe's reaction
- (4) Reimer-Tiemann reaction

9. HBr reacts fastest with
 (1) 2-methyl propan-2-ol
 (2) Propan-2-ol
 (3) Propan-1-ol
 (4) 2-methylpropan-1-ol

10. Arrange the following alkanols a, b and c in order of their reactivity towards acid catalysed dehydration:



- (1) $a > b > c$
- (2) $b > a > c$
- (3) $b > c > a$
- (4) $c > b > a$

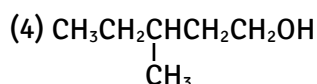
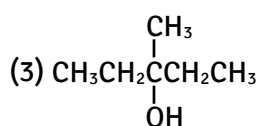
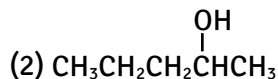


Product (I) + Product (II)

Which is not true regarding the products:

- (1) Product (I) and (II) are position isomers
- (2) Product (I) and (II) contains the same number of sp^3 and sp^2 carbon atoms
- (3) The yield of the product I and II is same
- (4) Reaction obeys Saytzeff rule

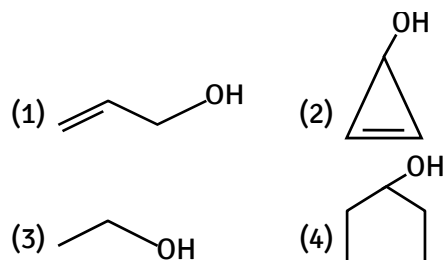
12. Among the following compounds which can be dehydrated very easily?



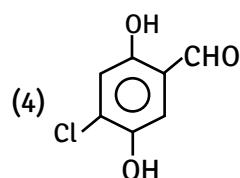
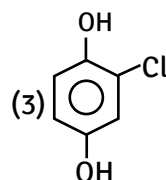
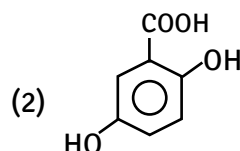
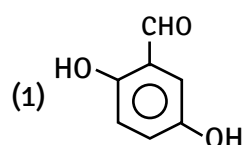
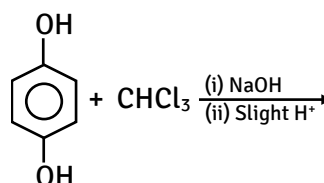
13. n-propyl alcohol and isopropyl alcohol can be chemically distinguished by which reagent:

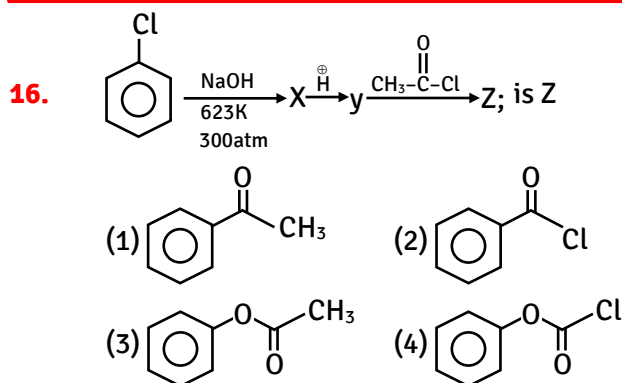
- (1) PCl_5
- (2) Reduction
- (3) Oxidation with Pot. Dichromate, followed by I_2/NaOH
- (4) Ozonolysis

14. An alcohol gives turbidity with HCl/ZnCl_2 (Lucas reagent) only on heating. The alcohol is:

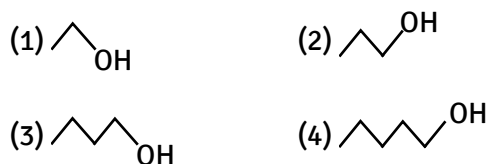


15. The major product obtained in the following reaction is



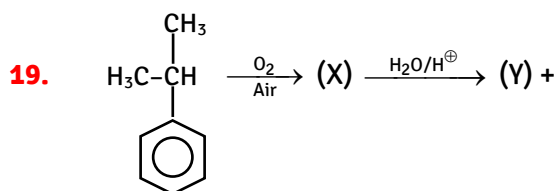


17. Least soluble alcohol in water is:



18. Which of the following statement is incorrect.

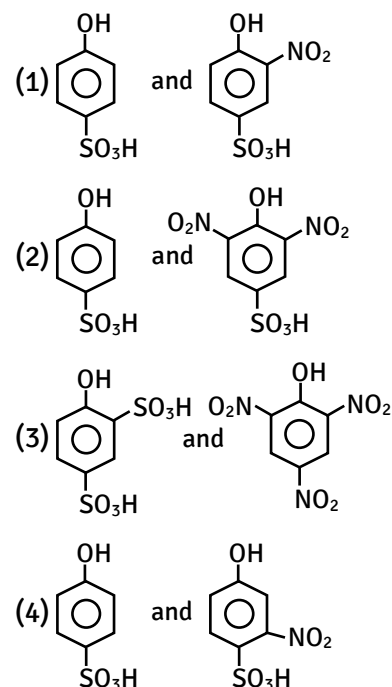
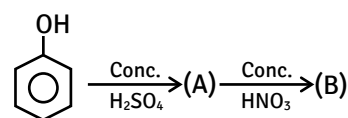
- (1) Ethanol and phenol can be differentiated by their reaction with bromine water.
- (2) O-Nitrophenol and p-nitrophenol can be separated by steam distillation method
- (3) Vapours of t-butyl alcohol when passed over copper at 573K forms isobutylene by dehydrogenation
- (4) Primary alcohols do not produce turbidity with Lucas reagent at room temperature



carboxylic acid; y is

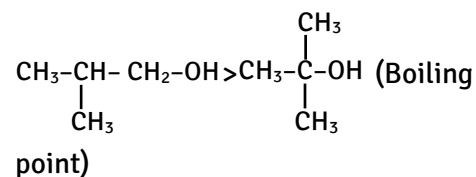
- (1) Phenol (2) Benzyl alcohol
- (3) Propanone (4) Acetophenone

20. Identify the major product A and B respectively in the following reaction of phenol

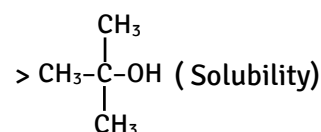


21. Which of the following options is not correctly showing the trend of the properties mentioned?

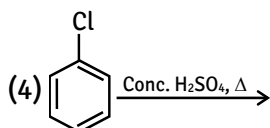
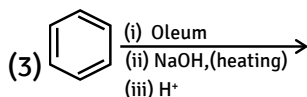
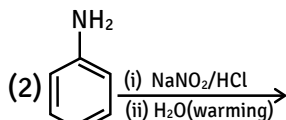
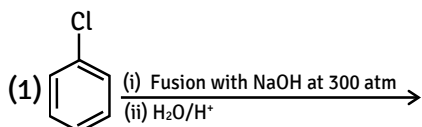
- (1) $\text{CH}_3\text{-CH}_2\text{-OH} > \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH} > \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$ (Solubility)
- (2) $\text{CH}_3\text{-CH}_2\text{-OH} < \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH} < \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$ (Boiling point)
- (3) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH} >$



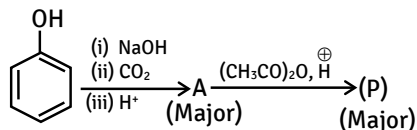
- (4) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH} > \text{CH}_3\text{-CH(CH}_3\text{)-CH}_2\text{-OH}$



22. Which of the following reaction will not yield phenol?

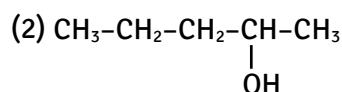
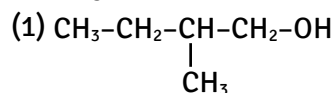


23. In the following reactions, product 'P' is

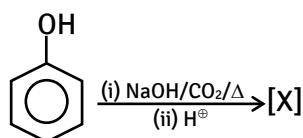


- (1) Salicylaldehyde (2) Benzoquinone
(3) Salicylic acid (4) Aspirin

24. Among the following compounds which can be dehydrated most difficulty.



25. In the given reaction :

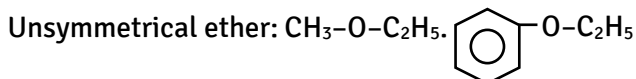
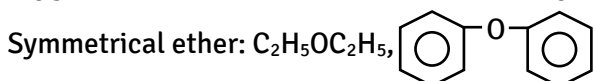


Major product [X] is :

- (1) Salicylic acid
(2) m-hydroxybenzoic acid
(3) Mixture of 1 and 2
(4) Salicylaldehyde

Ether:

Classification: Ether are classified as simple or symmetrical, if the alkyl or aryl group attached to the oxygen atom are the same, and mixed or unsymmetrical, if the two groups are different.



Naming of Ether: Common names of ethers are derived from the names of alkyl/aryl groups written as separate words in alphabetical order and adding the word ether at the end.

Ex. $\text{CH}_3\text{-O-C}_2\text{H}_5$ ethyl methyl ether
 $\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$ diethyl ether

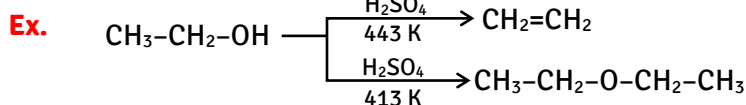
Structure of Functional Groups:



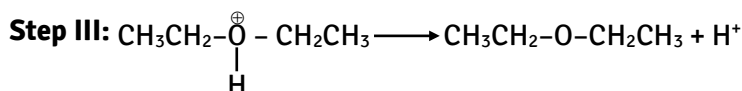
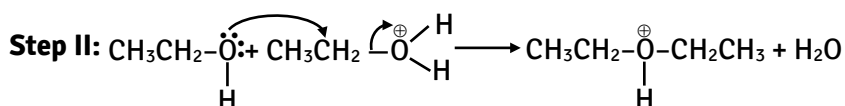
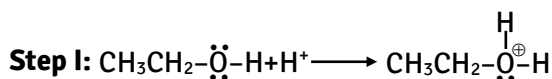
- In ethers, the two bond pairs and two lone pairs of electrons on oxygen are arranged approximately in a tetrahedral arrangement. The bond angle is slightly greater than the tetrahedral angle due to the repulsive interaction between the two bulky (-R) groups.
- The (C-O) bond length is almost the same as in alcohols.

Preparation of ethers**(1) By intermolecular dehydration of alcohol:**

Alcohols undergo dehydration in the presence of protic acids (conc. H_2SO_4 , or cone. H_3PO_4). The formation the product, alkene or ether depends on the reaction condition.



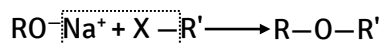
The formation of ether is a nucleophilic bimolecular reaction, as indicated below



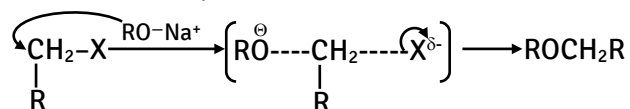
This method is suitable for the preparation of ethers having 1° alkyl groups only and the alkyl group should be unhindered and temp. be kept low other wise reaction favour the formation of alkene. The dehydration of 2° and 3° alcohol give alkenes as major product.

(2) Williamson's synthesis:

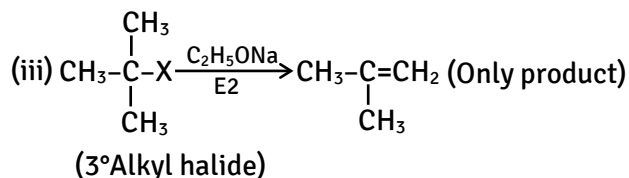
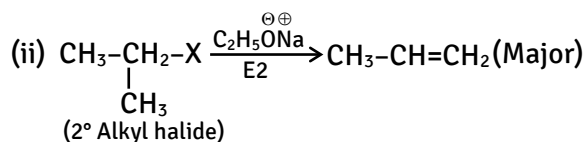
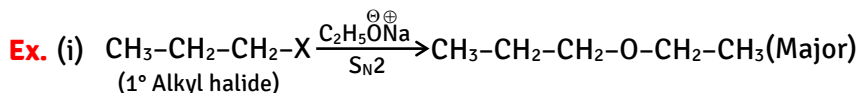
It is an important method to prepare all type of ether. Alkyl halide is allowed to react with sodium alkoxide.



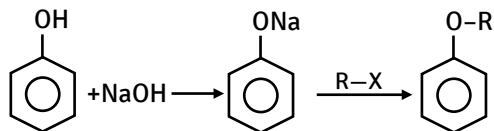
This is a nucleophilic substitution reaction which undergoes via $\text{S}_{\text{N}}2$



Note. Better results are obtained if the alkyl halide is 1° . In case of 2° alkyl halide, elimination dominate over substitution. If 3° alkyl halide is used, an alkene is the only product and no ether is formed.



- Phenols are also converted to ethers by this method



Physical properties:

Due to large size of alkyl groups (than H) R-O-R bond angle in ethers is large ($> 110^\circ$) therefore net dipole-moment becomes less and therefore ethers show:

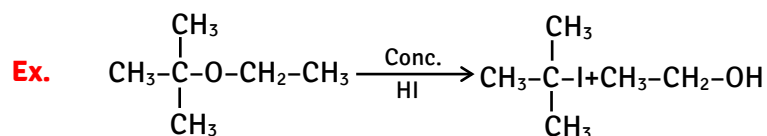
- (i) Low boiling points
- (i) Low solubility in water

Solubility in water further decreases with increasing molar mass of ether as oxygen of ether become more hindered and become less available for inter-molecular H-bonding with water.

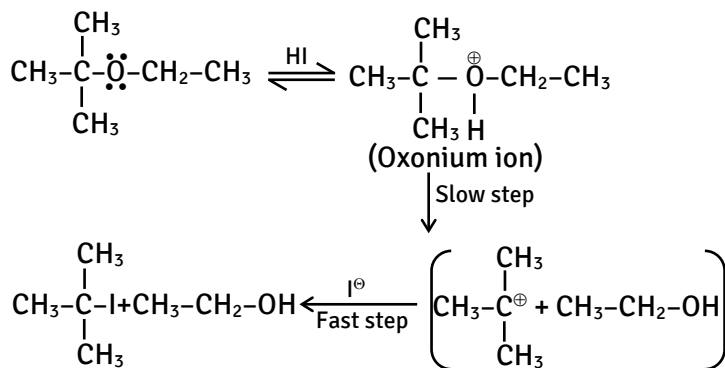
Chemical reactions of ether

(1) Reaction with conc. HX:

- Ether due to low polarity are less reactive functional group
- With cold & Conc. HX, they form alkyl halide and alcohols by cleavage of C-O bond
- reactivity of hydrogen halides is $\rightarrow \text{HI} > \text{HBr} > \text{HCl}$

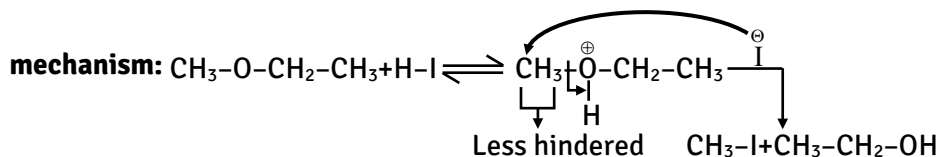
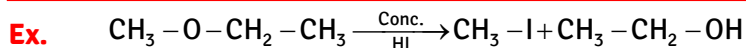


Mechanism:

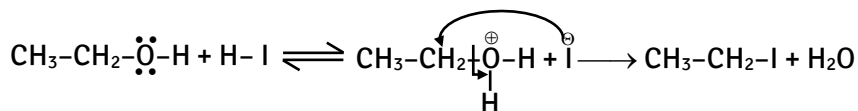


- Reaction with cold & conc. HI is known as Zeisel method.

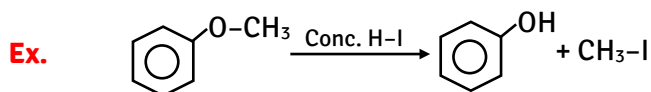
- Note.** (i) If oxonium ion gives resonance stabilised carbocation $\left(\text{Ph}-\text{CH}_2^\oplus, \text{CH}_2=\text{CH}-\text{CH}_2^\oplus \right)$ or 3° carbocation then reaction follow $\text{S}_{\text{N}}1$ mechanism.
- (ii) If oxonium ion does not give resonance stabilised or 3° carbocation then reaction follow $\text{S}_{\text{N}}2$ mechanism and $\text{I}^\ominus$ attack at less hindered carbon



If conc. HI is taken in excess then alcohol is also converted to Alkyl Iodide



Note. (iii) Alkyl aryl ethers are cleaved at the alkyl oxygen bond due to the more stable aryl-oxygen bond.

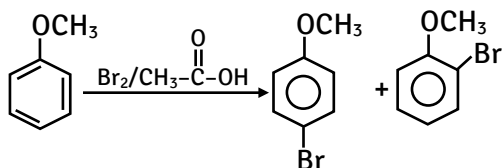


(2) Electrophilic substitution :

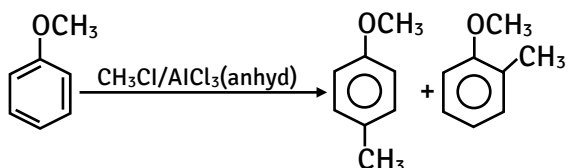
Aryl ethers show the reactions like phenols as $-\text{OR}$ is also an activating group.

The electrophilic substitution shown by aryl ethers are

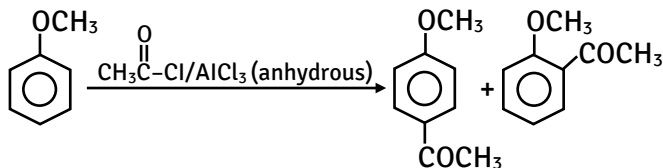
(i) Halogenation



(ii) Friedel-Crafts Reaction

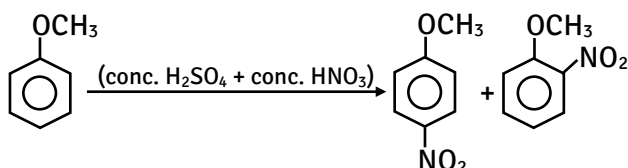


This is known as Friedel-Craft alkylation



This is known as Friedel-Craft acylation

(iii) Nitration

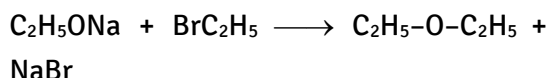


Exercise 1.3

1. Which of the following is not expected to give ether on reaction with sodium methoxide:

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
- (2) $\text{CH}_2=\text{CHCH}_2\text{Cl}$
- (3) PhCH_2Cl
- (4) $\text{CH}_2=\text{CHCl}$

2. The given reaction is called as:



- (1) Frankland reaction
- (2) Wurtz reaction
- (3) Williamson's Ether synthesis
- (4) Cannizzaro reaction

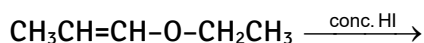
3. Among the following sets of reactants which one produces anisole?

- (1) CH_3CHO ; RMgX
- (2) $\text{C}_6\text{H}_5\text{OH}$; NaOH ; CH_3I
- (3) $\text{C}_6\text{H}_5\text{OH}$; neutral FeCl_3
- (4) $\text{C}_6\text{H}_5\text{--CH}_3$; CH_3COCl ; AlCl_3

4. The order of reactivity of halogen acids with ether is:

- (1) $\text{HCl} > \text{HBr} > \text{HI}$
- (2) $\text{HI} > \text{HBr} > \text{HCl}$
- (3) $\text{HCl} > \text{HI} > \text{HBr}$
- (4) $\text{HI} > \text{HCl} > \text{HBr}$

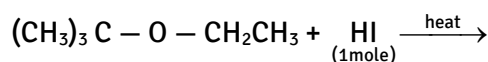
5. Consider the following transformation:



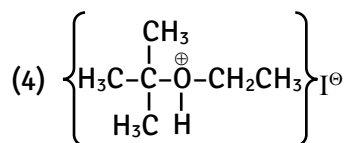
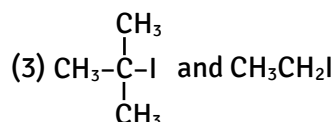
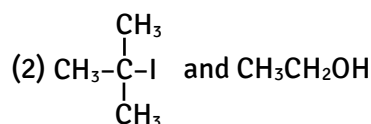
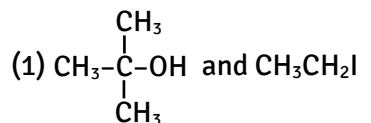
The major product(s) formed is (are):

- (1) $\text{CH}_3\text{CH}=\text{CHI}$ and $\text{CH}_3\text{CH}_2\text{I}$
- (2) $\text{CH}_3\text{CH}=\text{CHI}$ and $\text{CH}_3\text{CH}_2\text{OH}$
- (3) $\text{CH}_3\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{I}$
- (4) $\text{CH}_3\text{CH}_2\text{CH--O--CH}_2\text{CH}_3$
I

6. In the reaction



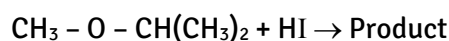
the product(s) formed is (are):



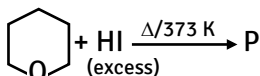
7. What will happen when diethyl ether treated with hot and conc. HI:

- (1) Ethyl alcohol
- (2) Ethyl iodide
- (3) Methyl iodide
- (4) Methanol

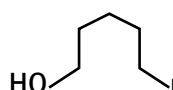
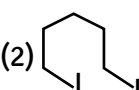
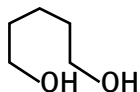
8. The major organic product in the given reaction is.



- (1) $\text{CH}_3\text{OH} + (\text{CH}_3)_2\text{CHI}$
- (2) $\text{ICH}_2\text{OCH}(\text{CH}_3)_2$
- (3) $\text{CH}_3\text{OC}(\text{CH}_3)_2$
I
- (4) $\text{CH}_3\text{I} + (\text{CH}_3)_2\text{CHOH}$

9. 

Product P of the reaction

- (1) 
- (2) 
- (3) 
- (4) No reaction

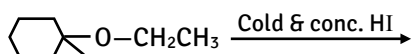
10. $\text{HO}-\text{C}_6\text{H}_4-\text{O}-\text{CH}_3$ (with ^{18}O on the phenolic oxygen) $\xrightarrow[\text{Heat}]{\text{HBr}}$ The major product obtained in the following reaction is:

- (1) $\text{HO}-\text{C}_6\text{H}_4-\text{Br}$, CH_3-OH (with ^{18}O on the methanol oxygen)
 (2) $\text{HO}-\text{C}_6\text{H}_4-\text{OH}$, CH_3-Br
 (3) $\text{HO}-\text{C}_6\text{H}_4-\text{OH}$ (with ^{18}O on the phenolic oxygen), CH_3-Br
 (4) $\text{Br}-\text{C}_6\text{H}_4-\text{OH}$ (with ^{18}O on the phenolic oxygen), CH_3-OH (with ^{18}O on the methanol oxygen)

11. Mark the correct statement:

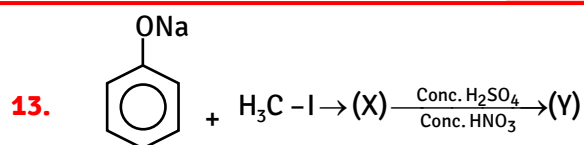
- (1) Ethers behave as Lewis base
 (2) Ethers form coordinated complexes with Lewis acids
 (3) With cold HI, diethyl ether gives ethyl alcohol & ethyl iodide
 (4) All are correct

12. In the reaction



the product(s) formed is (are):

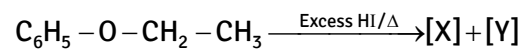
- (1) $\text{Cyclohexyl-OH} + \text{CH}_3\text{CH}_2\text{-I}$
 (2) $\text{Cyclohexyl-I} + \text{CH}_3\text{CH}_2\text{-OH}$
 (3) $\text{Cyclohexyl-I} + \text{CH}_3\text{CH}_2\text{-I}$
 (4) $\left[\text{Cyclohexyl-O}^+\text{H-CH}_2\text{CH}_3 \right] \text{I}^-$



major product Y is:

- (1) $\text{C}_6\text{H}_4(\text{OCH}_3)(\text{NO}_2)$ (ortho)
 (2) $\text{C}_6\text{H}_4(\text{OCH}_3)(\text{SO}_3\text{H})$ (ortho)
 (3) $\text{C}_6\text{H}_4(\text{OCH}_3)(\text{SO}_3\text{H})$ (para)
 (4) $\text{C}_6\text{H}_4(\text{OCH}_3)(\text{NO}_2)$ (para)

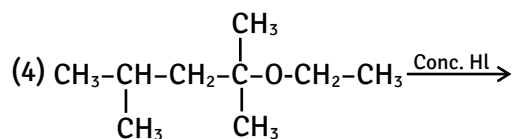
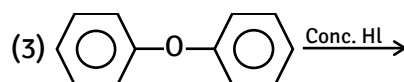
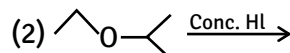
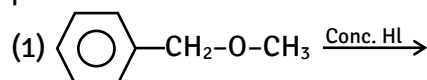
14. In the given reaction



[X] and [Y] will be :

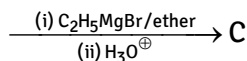
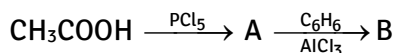
- (1) $\text{C}_6\text{H}_5\text{I}$ and $\text{CH}_3\text{CH}_2\text{I}$
 (2) $\text{C}_6\text{H}_5\text{OH}$ and $\text{CH}_3\text{-CH}_2\text{-I}$
 (3) $\text{C}_6\text{H}_5\text{I}$ and $\text{CH}_3\text{CH}_2\text{OH}$
 (4) $\text{C}_6\text{H}_5\text{OH}$ and $\text{CH}_2=\text{CH}_2$

15. Which of the following reaction follows $\text{S}_\text{N}2$ path.



Exercise 2

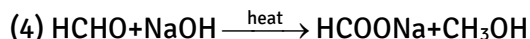
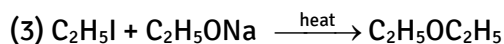
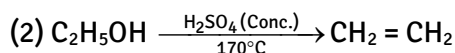
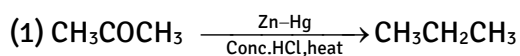
1. In a set of the given reactions, acetic acid yielded a product C.



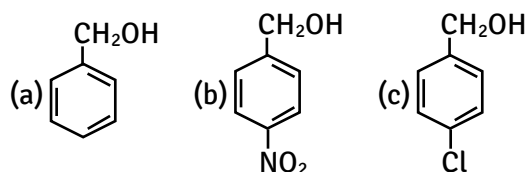
product C would be:

- (1) $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$ (2) $\text{CH}_3\text{COC}_6\text{H}_5$
 (3) $\text{CH}_3\text{CH}(\text{OH})\text{C}_6\text{H}_5$ (4) $\text{CH}_3-\overset{\text{C}_2\text{H}_5}{\underset{\text{C}_6\text{H}_5}{\text{C}}}(\text{OH})\text{C}_6\text{H}_5$

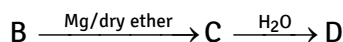
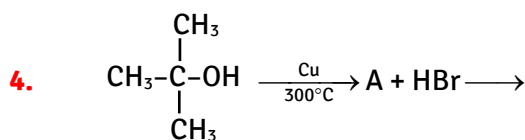
2. Which one among the following reactions is an example of Williamson synthesis -



3. Mark the correct increasing order of reactivity of the following compounds with HBr:



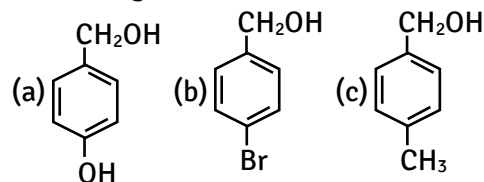
- (1) $a < b < c$ (2) $b < a < c$
 (3) $b < c < a$ (4) $c < b < a$



What is D?

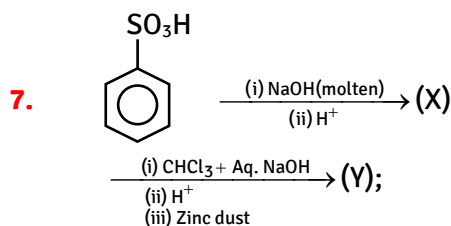
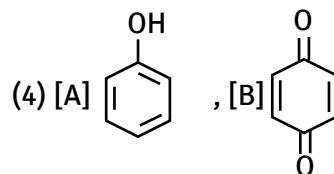
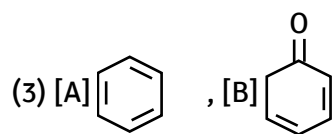
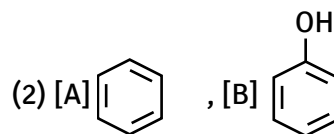
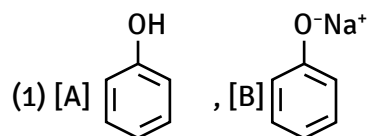
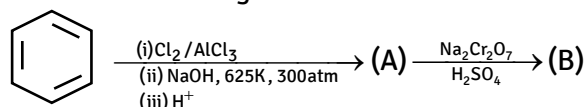
- (1) $\text{CH}_3-\overset{\text{CH}_2}{\underset{\text{CH}_3}{\text{C}}}=\text{CH}_2$ (2) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{Br}$
 (3) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{MgBr}$ (4) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_3$

5. Mark the correct decreasing order of reactivity of the following compounds with Lucas reagent

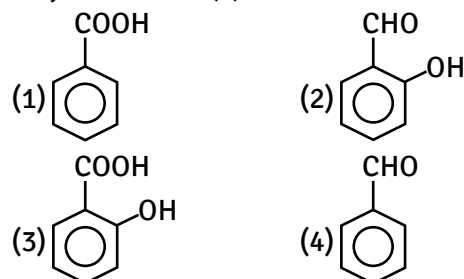


- (1) $c > b > a$ (2) $c > a > b$
 (3) $a > c > b$ (4) $a > b > c$

6. Identify the products (A) and (B) respectively in the following reaction.



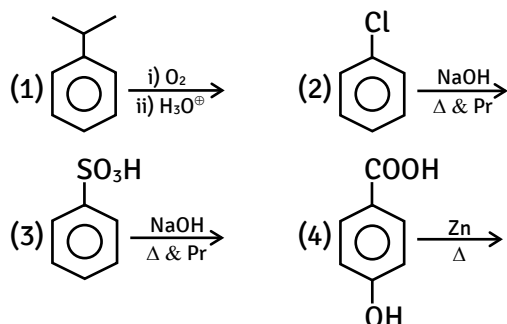
Major Products (Y) is



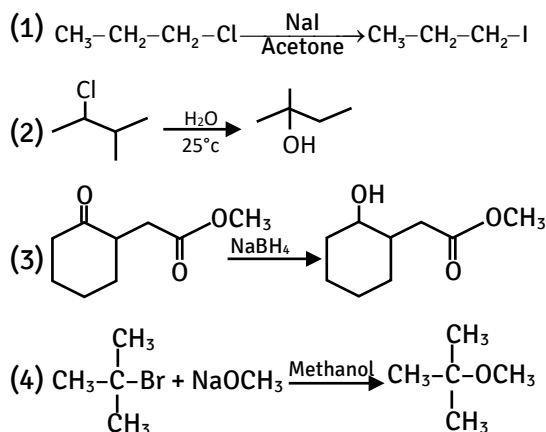
8. Grignard's reagent can be used to prepare tertiary alcohol by reaction with

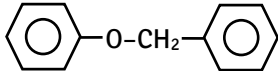
- (1) Acetone (2) Acetaldehyde
(3) Benzaldehyde (4) Formaldehyde

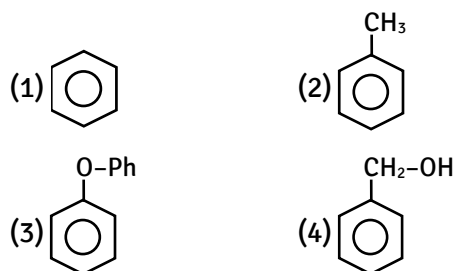
9. Phenol can be prepared by



10. Which of the following conversion is incorrect?



11.  $\xrightarrow{\text{HI}} \text{X} + \text{Y}$
 $\text{X} \xrightarrow[\text{dry ether}]{\text{Mg}} \text{Z (organometallic compound)}$
 $\text{Z} \xrightarrow{\text{Y}} \text{P; P is}$

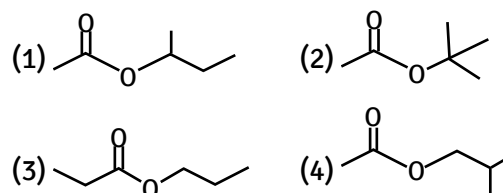


12. Arrange the following compounds in increasing order of boiling point.

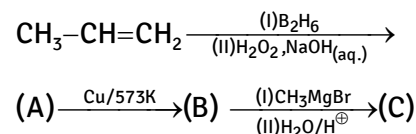
Propan-1-ol, butan-1-ol, butan-2-ol, pentan-1-ol

- (1) Propan-1-ol, butan-2-ol, butan-1-ol, pentan-1-ol
(2) Propan-1-ol, butan-1-ol, butan-2-ol, pentan-1-ol
(3) Pentan-1-ol, butan-2-ol, butan-1-ol, propan-1-ol
(4) Pentan-1-ol, butan-1-ol, butan-2-ol, propan-1-ol

13. An organic compound (A) with molecular formula $\text{C}_6\text{H}_{12}\text{O}_2$ is hydrolysed with dil. H_2SO_4 to give a carboxylic acid (B) and an alcohol (C). 'C' gives white turbidity immediately when treated with anhydrous ZnCl_2 and conc. HCl . The organic compound (A) is

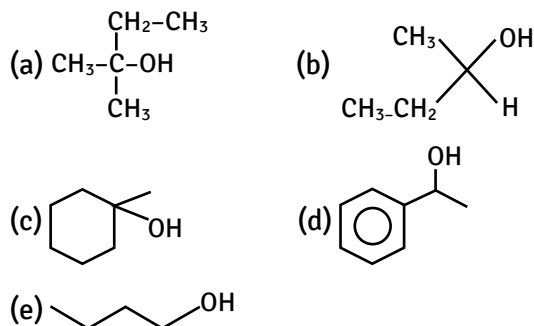


14. The final product formed in the following multistep reaction is :



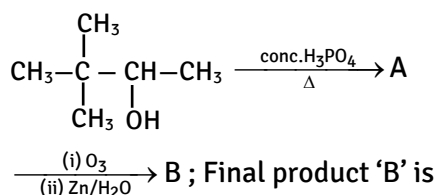
- (1) $\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---OH}$
(2) $\text{CH}_3\text{---CH}_2\text{---CH(OH)---CH}_3$
(3) $\text{CH}_3\text{---C(OH)(CH}_3\text{)---CH}_3$
(4) $\text{CH}_3\text{---CH(OH)---CH}_3$

15. Which of the following compounds give instant turbidity with anhyd. $\text{ZnCl}_2/\text{Conc. HCl}$ (Lucas reagent)



- (1) Only a and c (2) Only a, b and c
(3) Only a, c and d (4) Only e

16.



- (1) $2\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$
- (2) $\text{HCHO} + \text{CH}_3-\text{C}(=\text{O})-\text{CH}(\text{CH}_3)-\text{CH}_3$
- (3) $\text{CH}_3-\text{CHO} + \text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$
- (4) $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CHO} + \text{CHO}$

17. A primary alcohol, $\text{C}_3\text{H}_8\text{O}$ (A) on heating with sulphuric acid undergo dehydration to give an alkene (B). 'B' when reacts with HCl gives C, which on treatment with aqueous KOH gives compound D ($\text{C}_3\text{H}_8\text{O}$).

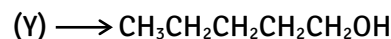
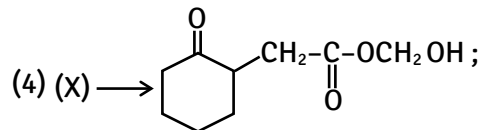
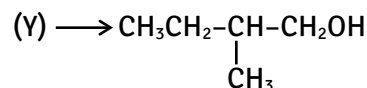
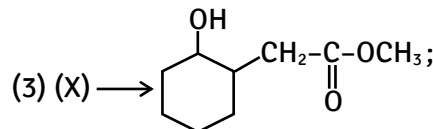
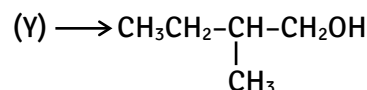
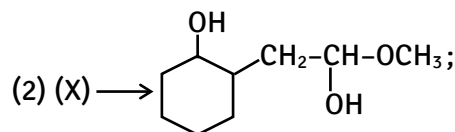
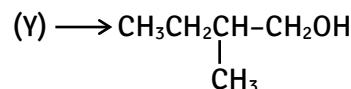
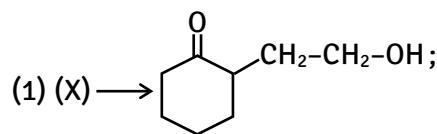
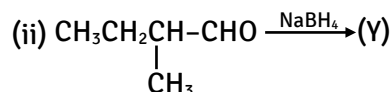
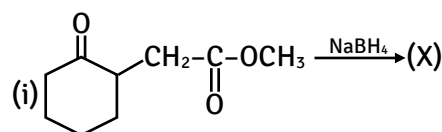
A and D are

- (1) Functional group isomers
(2) Position isomers
(3) Chain isomers
(4) stereoisomers.

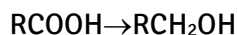
18. Which of the following statement is correct?

- (1) The reaction of methyl magnesium iodide with acetone followed by hydrolysis gives secondary butanol.
(2) Primary alcohols are dehydrated easily than secondary and tertiary alcohols.
(3) Tertiary alcohol is more acidic than primary alcohol.
(4) Tertiary butyl alcohol gives turbidity fastest with Lucas reagent.

19. Choose the correct X and Y in the given reactions.

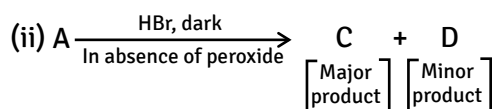
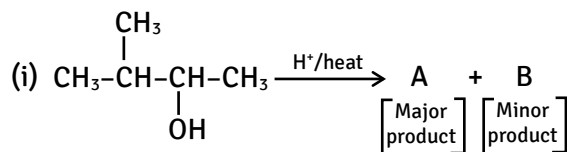


20. Which reducing agent is used for the following conversion?

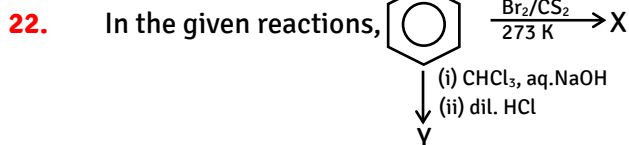
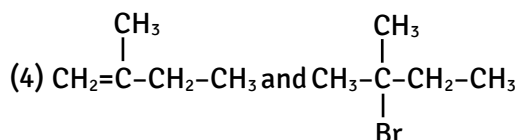
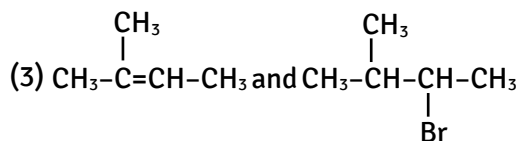
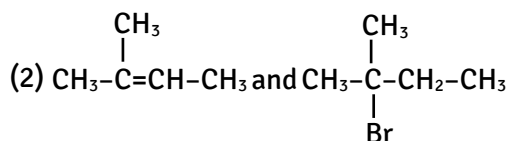
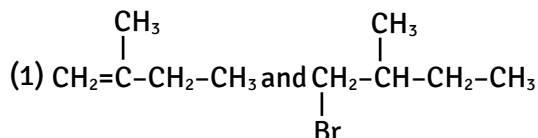


- (1) LiAlH_4 (2) NaBH_4
(3) $\text{K}_2\text{Cr}_2\text{O}_7$ (4) KMnO_4

21. In the following reactions,



The major product A and C are respectively



X and Y are respectively

- (1) bromobenzene and acetophenone
(2) o- and p-bromophenol and salicylaldehyde
(3) p-bromophenol and salicylic acid
(4) o-bromophenol and benzoic acid.

23. Which of following is incorrect for reactivity of 1°, 2° and 3° alcohols with given reagents.

- (1) Esterification $3^\circ > 2^\circ > 1^\circ$
(2) Lucas reagent $3^\circ > 2^\circ > 1^\circ$
(3) With Na $1^\circ > 2^\circ > 3^\circ$
(4) All of these

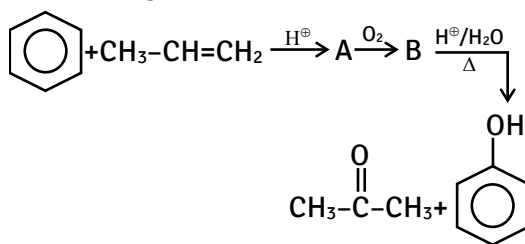
24. Benzene is converted into para-nitrophenol by using reagents:

- (I) dil. NaOH (II) $\text{HNO}_3/\text{H}_2\text{SO}_4$
(III) Br_2/Fe

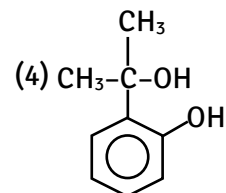
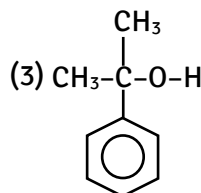
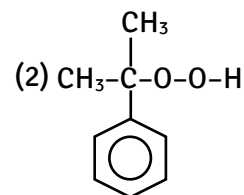
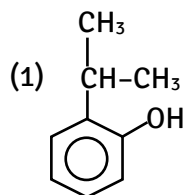
The most suitable sequence of these reagents are respectively

- (1) I, II, III (2) I, III, II
(3) II, III, I (4) III, II, I

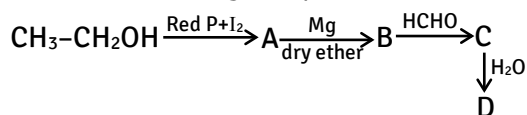
25. In following reaction



The structure of 'B' is :-



26. In the following sequence of reaction



the compound D is-

- (1) Butanal (2) n-butyl alcohol
(3) n-propyl alcohol (4) propanal

27. An ether is more volatile than alcohol having the same molecular formula this is due to :-
 (1) Inter molecular H -bonding in ether
 (2) Inter molecular H -bonding in alcohols
 (3) Dipolar character of ether
 (4) Alcohol has resonating structure

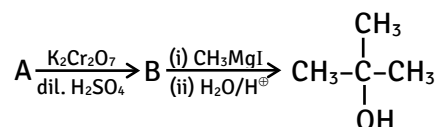
28. n-propyl bromide reacts with aq. KOH to form:-
 (1) Propane (2) Propene
 (3) Propyne (4) 1-propanol

29. When phenol is treated with bromine/H₂O, it gives:-
 (1) o-and p-dibromophenol
 (2) 2,3,4-tribromophenol
 (3) 2,4,6-tribromophenol
 (4) None

30. Among the following the one that gives positive iodoform test upon reaction with I₂ and NaOH is:-
 (1) $\text{CH}_3\text{-CH}_2\text{-CH(OH)-CH}_2\text{-CH}_3$
 (2) $\text{C}_6\text{H}_5\text{-CH}_2\text{-CH}_2\text{-OH}$
 (3) $\text{CH}_3\text{-CH(CH}_3\text{)-CH}_2\text{OH}$
 (4) PhCHOHCH_3

31. In the given reaction :
 $\text{CH}_3\text{-CH(CH}_3\text{)-CH}_2\text{-O} \xrightarrow[\text{(ii) HOH/H}^+]{\text{(i) CH}_3\text{MgBr}} [\text{X}]$
 [X] will be :
 (1) $\text{CH}_3\text{-CH(CH}_3\text{)-CH}_2\text{-OH}$
 (2) $\text{CH}_3\text{-CH(OH)-CH}_2\text{-CH}_3$
 (3) $\text{CH}_3\text{-C(OH)(CH}_3\text{)-CH}_3$
 (4) $\text{CH}_2\text{=CH-CH}_3$

32. The reactant A in the following reactions is:

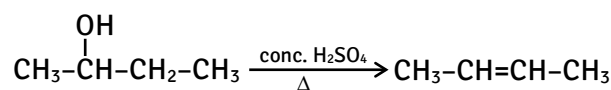


- (1) $\text{CH}_3\text{CHOHCH}_3$ (2) $\text{CH}_3\text{CH}_2\text{CHO}$
 (3) $\text{C}_2\text{H}_5\text{OH}$ (4) CH_3COOH

33. In the given reaction
 $\text{CH}_3\text{-CH=CH-CH}_2\text{-OH} \xrightarrow{\text{PCC}} \text{'P'}$
 Product 'P' is :
 (1) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$
 (2) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CHO}$
 (3) $\text{CH}_3\text{-CH=CH-CHO}$
 (4) $\text{CH}_3\text{-CH=CH-COOH}$

34. In the given reaction
 $\text{CO} + \text{H}_2 \xrightarrow[573-673\text{K}]{\text{ZnO} + \text{Cr}_2\text{O}_3} \text{Product}$
 (1) CH_3OH (2) $\text{CH}_2\text{-OH}$
 (3) CHO (4) COOH
 CHO COOH

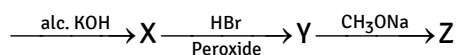
35. The following reaction :



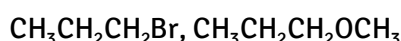
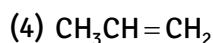
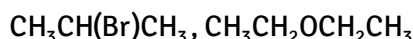
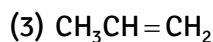
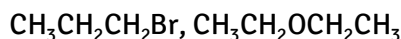
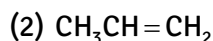
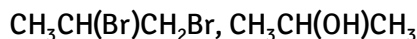
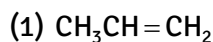
is known as :

- (1) Friedel-Craft reaction
 (2) Acidic dehydration of alcohol
 (3) Kolbe reaction
 (4) Gatterman-Koch reaction
36. Ethylene glycol on oxidation with periodic acid gives :
 (1) Oxalic acid (2) Glyoxal
 (3) Formaldehyde (4) Glycolic acid
37. Amongst the following alcohols which would react fastest with conc. HCl and ZnCl₂ ?
 (1) 2-Methylbutan-1-ol
 (2) 2-Pentanol
 (3) 1-Pentanol
 (4) 2-Methyl-2-butanol

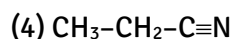
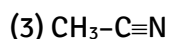
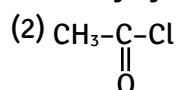
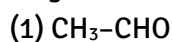
38. Complete the missing links.



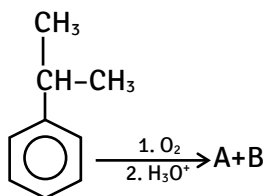
X, Y, Z are respectively ?



39. Butan-2-one is obtained, if following compound is reacted with methyl magnesium bromide followed by hydrolysis?



40.



A and B can't be distinguished by :-

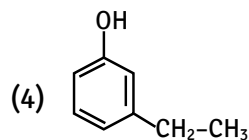
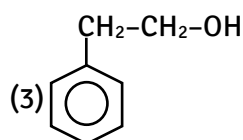
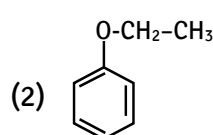
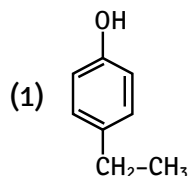
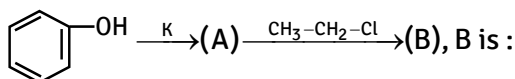
(1) Neutral FeCl_3 test

(2) 2, 4-DNP test

(3) Bromine water test

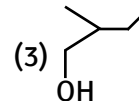
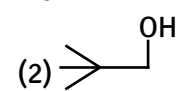
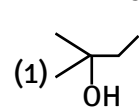
(4) Tollen's test

41.



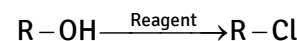
42.

Which compound on reaction with Cu at 300°C may not give aldehyde or ketone

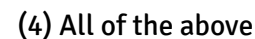
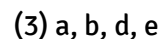
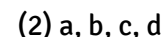
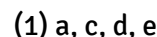
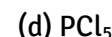
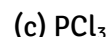
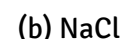
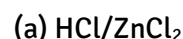


(2) All of these

43.

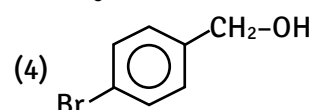
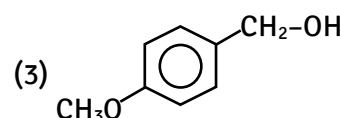
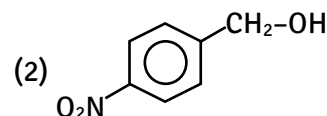
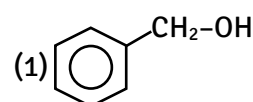


Which of the following reagents can be used for this reaction.

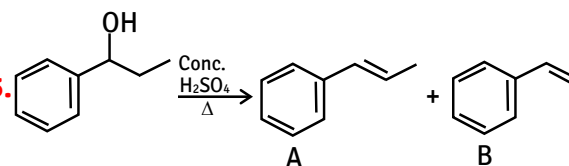


44.

Which of the following alcohols is most reactive towards substitution with HBr ?



45.



consider the above reaction, and choose the correct statement :

(1) The reaction is not possible in acidic medium

(2) Both compounds A and B are formed equally

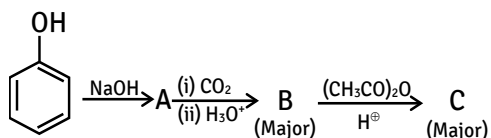
(3) Compound A will be the major product

(4) Compound B will be the major product

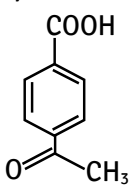
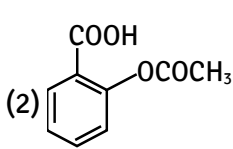
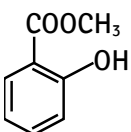
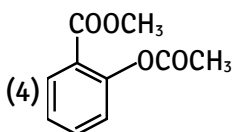
46. Which compound give alcohol on reaction with one mole of phenyl magnesium bromide (PhMgBr) followed by acidic hydrolysis?

- (1) $\text{CH}_3\text{-C}\equiv\text{N}$ (2) $\text{Ph-C}\equiv\text{N}$
 (3) $\text{CH}_3\text{-}\overset{\text{O}}{\parallel}\text{C-O-Ph}$ (4) $\text{Ph-C}\overset{\text{O}}{\parallel}\text{-CH}_3$

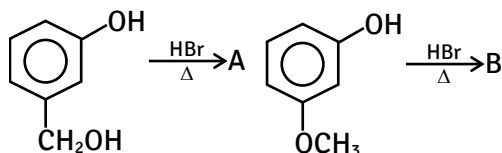
47. Consider the following reaction sequence.

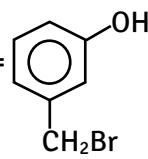
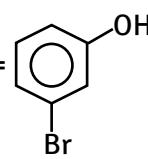
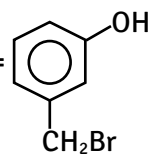
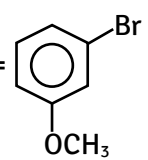
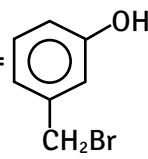
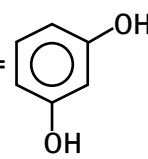
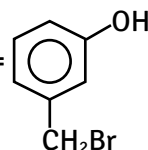
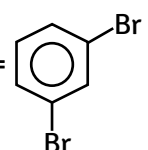


The product C is

- (1)  (2) 
 (3)  (4) 

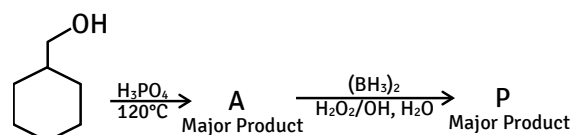
48. 'A' and 'B' formed in the following set of reactions are:



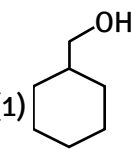
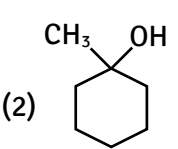
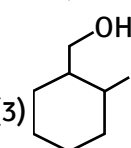
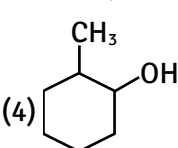
- (1) A = , B = 
 (2) A = , B = 
 (3) A = , B = 
 (4) A = , B = 

49. $\text{H}_3\text{C-CH}_2\text{-CH-CH}_3 \xrightarrow[\text{(iii) D}_2\text{O}]{\text{(i) NaI, H}_3\text{PO}_4, \text{(ii) Mg, Dry ether}} [\text{X}]$
 Product [X] formed in the above reaction is :

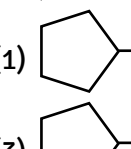
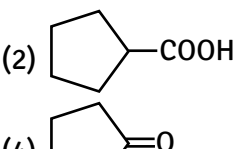
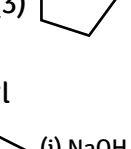
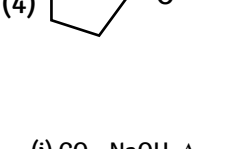
- (1) $\text{H}_3\text{C-CH}_2\text{-CH-CH}_3$ (2) $\text{H}_3\text{C-CH}_2\text{-}\overset{\text{H}}{\underset{\text{D}}{\text{C}}}\text{-CH}_3$
 (3) $\text{H}_3\text{C-CH}_2\text{-CH=CH}_2$ (4) $\text{H}_3\text{C-CH=CH-CH}_3$

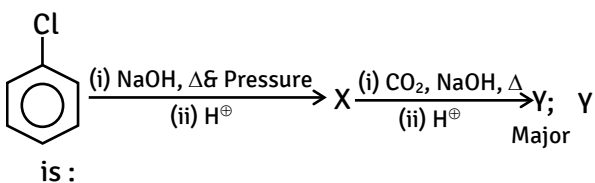
50. 

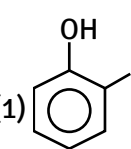
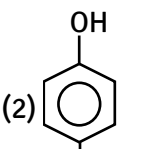
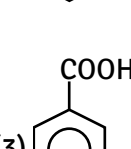
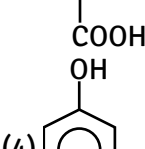
Consider the above reaction and identify the Product P:

- (1)  (2) 
 (3)  (4) 

51. $\text{Cyclopentyl-CH}_2\text{-OH} \xrightarrow[\text{anhydrous}]{\text{CrO}_3} \text{X} ; \text{X}$

- (1)  (2) 
 (3)  (4) 

52. 

- (1)  (2) 
 (3)  (4) 

53. $\text{CH}_3 - \text{CH}_2 - \text{OH} \xrightarrow{\text{conc. H}_2\text{SO}_4}$
 $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$
 Suitable temperature in the above reaction to get ether with good yield is :

- (1) 443 K (2) 413 K
 (3) 100 K (4) 470 K

54. $\text{CH}_3 - \text{CH}_2 - \text{OH} \xrightarrow{\text{Na}} \text{X} \xrightarrow{\text{CH}_3 - \text{CH}_2 - \text{Cl}} \text{Y}$; Y is :

- (1) $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$
 (2) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{Cl}$
 (3) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{OH}$
 (4) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{ONa}$

55. $\text{CH}_3 - \text{O} - \text{CH}_2\text{CH}_3 \xrightarrow[\text{Excess}]{\text{HI}/\Delta} (\text{A}) + (\text{B})$

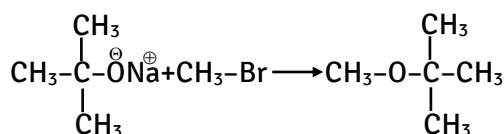
Product A and B are

- (1) $\text{CH}_3\text{OH} + \text{CH}_3\text{CH}_2\text{I}$
 (2) $\text{CH}_3\text{I} + \text{CH}_3\text{CH}_2\text{OH}$
 (3) $\text{CH}_3\text{I} + \text{CH}_3\text{CH}_2\text{I}$
 (4) $\text{CH}_3\text{OH} + \text{CH}_3\text{CH}_2\text{OH}$

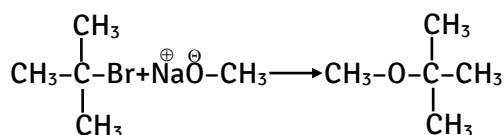
56. Ethyl chloride is converted into diethyl ether by

- (1) Hell-Volhard Zelinsky reaction
 (2) Kucherov's reaction
 (3) Wurtz reaction
 (4) Williamson's synthesis

57. R-1



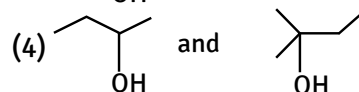
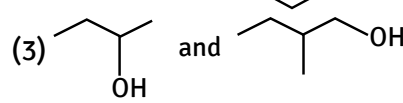
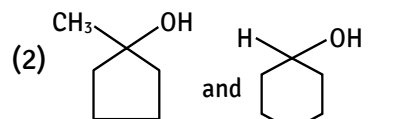
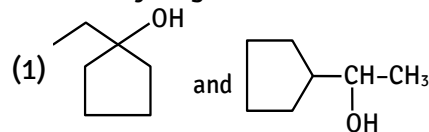
R-2



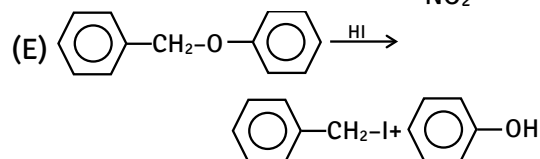
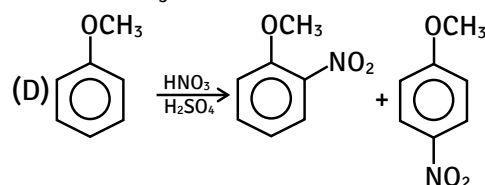
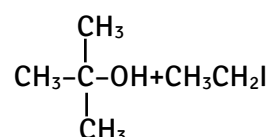
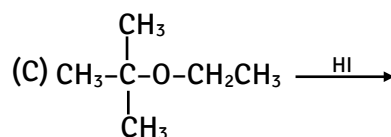
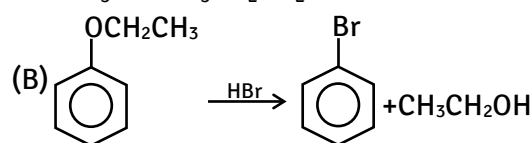
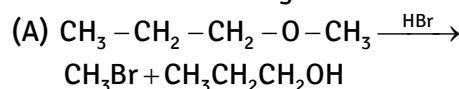
Which reaction is appropriate for the preparation of an ether

- (1) Both R_1 & R_2
 (2) Only R_1
 (3) Only R_2
 (4) Both R_1 & R_2 wrong

58. In the given pair of alcohols, in which pair second alcohol is more reactive than first towards hydrogen bromide?



59. Which of the following is correct



- (1) Only A & B correct
 (2) Only B & C incorrect
 (3) Only A & D incorrect
 (4) Only D & E incorrect

60. Which of the following reagent can't be used to distinguish between phenol and ethyl alcohol ?

- (1) Na (2) Neutral FeCl_3
 (3) $\text{Br}_2/\text{H}_2\text{O}$ (4) NaOH

Exercise 3

1. **Assertion:** Phenol is stronger acid than alcohols.
Reason: Conjugate base of phenol is stabilized by resonance whereas alcohol are not.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If Assertion & Reason are false.
2. **Assertion:** Anisole on reaction with HI gives phenol and CH_3I .
Reason: Phenol-oxygen bond is stronger than methyl-oxygen bond in anisole and hence it is not cleaved by HI.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If Assertion & Reason are false.
3. **Assertion:** In Lucas test, 3° alcohols react immediately.
Reason: A mixture of anhydrous ZnCl_2 and conc. HCl is called Lucas reagent.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If Assertion & Reason are false.
4. **Assertion:** Phenols give picric acid on nitration with conc. HNO_3 .
Reason: $-\text{OH}$ group in phenol shows $-M$ effect.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If Assertion & Reason are false.
5. **Assertion:** o-nitrophenol is more volatile than p-nitrophenol
Reason: Intramolecular hydrogen bonding is present in o-nitrophenol while intermolecular H-bonding is present in p-nitrophenol.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If Assertion & Reason are false.
6. **Assertion:** CH_3OCH_3 and $\text{C}_2\text{H}_5\text{OH}$ has comparable molecular weight but boiling point of $\text{C}_2\text{H}_5\text{OH}$ is more than dimethyl ether.
Reason: $\text{C}_2\text{H}_5\text{OH}$ forms intermolecular H-bonding while CH_3OCH_3 forms intramolecular H-bonding.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If Assertion & Reason are false.

7. **Assertion:** Ethers behave as bases in the presence of mineral acids

Reason: It is due to the presence of lone pair of electrons on the oxygen.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If Assertion & Reason are false.

8. **Assertion:** The major products formed by heating $C_6H_5CH_2OCH_3$ with HI are $C_6H_5CH_2I$ and CH_3OH

Reason: Benzyl cation is more stable than methyl cation.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If Assertion & Reason are false.

9. **Assertion:** 2-Butanol on heating with H_2SO_4 gives 2-butene as major product.

Reason: Dehydration of 2-butanol follows Saytzeff's rule.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If Assertion & Reason are false.

10. **Assertion:** Depending on condition alcohol can react both as nucleophile and electrophile

Reason: Alcohol react with active metals such as sodium, potassium to yield corresponding alkoxide and liberate hydrogen

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

11. **Assertion:** Treatment of bromine water with propene yields 1-bromo Propan-2-ol.

Reason: Attack of water on bromonium ion follows Markovnikov rule and results in 1-bromo Propan-2-ol.

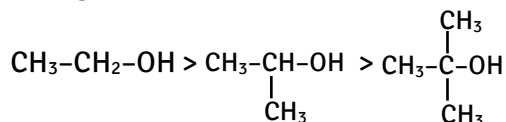
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

12. **Assertion:** synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

Reason: Reaction of bromobenzene with sodium ethoxide yields ethyl phenyl ether.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

- 13. Assertion:** The correct order of acidic strength for the alcohols is



Reason : Electron releasing group ($-\text{CH}_3$) increases electron density on oxygen tending to decrease the polarity of O-H bond.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct.
- 14. Assertion:** Anisole undergoes electrophilic substitution at ortho and para-positions.
Reason: Anisole is less reactive than phenol towards electrophilic substitution reaction.
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct.
- 15. Assertion :** When alkyl aryl ethers react with excess of hydrogen halides, phenol and alkyl halide are produced.
Reason: Alkylarylethers are cleaved at the alkyl-oxygen bond due to more stable aryl-oxygen bond.
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct

- 16. Assertion:** Phenol forms 2, 4, 6-tribromophenol on treatment with Br_2 -water at 273 K.

Reason: Phenol is o, p-directing group

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

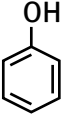
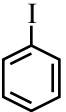
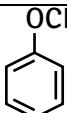
- 17.** Given below are two statement.

Statement I:- Bromination of phenol in solvent with low polarity such as CHCl_3 or CS_2 requires lewis acid catalyst

Statement II:- The lewis acid catalyst polarizes the bromine to generate Br^+

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false.

18. Match the starting materials given in Column-I with the products formed by these (Column II) in the reaction with HI.

Column I		Column II	
a	$\text{CH}_3\text{-O-CH}_3$	i	 + CH_3I
b	$\text{CH}_3\text{-CH(CH}_3\text{)-O-CH}_3$	ii	$\text{CH}_3\text{-C(CH}_3\text{)}_2\text{-I} + \text{CH}_3\text{OH}$
c	$\text{H}_3\text{C-C(CH}_3\text{)}_2\text{-O-CH}_3$	iii	 + CH_3OH
d		iv	$\text{CH}_3\text{-OH} + \text{CH}_3\text{-I}$
		v	$\text{CH}_3\text{-CH(CH}_3\text{)-OH} + \text{CH}_3\text{I}$
		vi	$\text{CH}_3\text{-CH(CH}_3\text{)-I} + \text{CH}_3\text{OH}$
		vii	$\text{CH}_3\text{-C(CH}_3\text{)}_2\text{-OH} + \text{CH}_3\text{I}$

- (1) a $\rightarrow$ v; b $\rightarrow$ iii; c $\rightarrow$ vi; d $\rightarrow$ i
 (2) a $\rightarrow$ vii; b $\rightarrow$ vi; c $\rightarrow$ v; d $\rightarrow$ ii
 (3) a $\rightarrow$ iv; b $\rightarrow$ v; c $\rightarrow$ ii; d $\rightarrow$ i
 (4) a $\rightarrow$ iii; b $\rightarrow$ i; c $\rightarrow$ iv; d $\rightarrow$ ii

19. Match list I with list II:

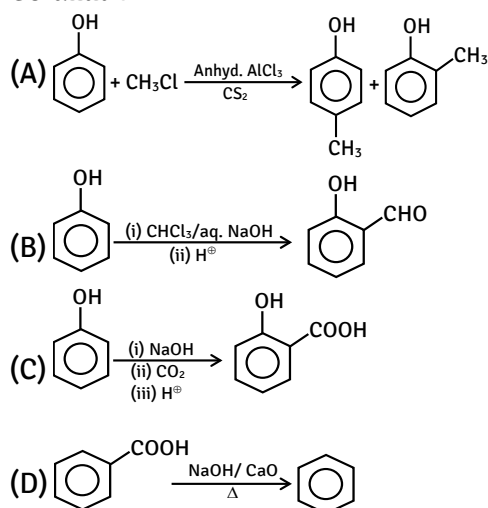
List I (Compound)		List II (pK _a value)	
A.	Ethanol	I.	10.0
B.	Phenol	II.	15.9
C.	m-Nitrophenol	III.	7.1
D.	p-Nitrophenol	IV.	8.3

Choose the correct answer from the options gives below:

- (1) A-I, B-II, C-III, D-IV
 (2) A-II, B-I, C-IV, D-III
 (3) A-III, B-IV, C-I, D-II
 (4) A-IV, B-I, C-II, D-III

20. Match the column I with column II and mark the appropriate choice.

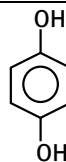
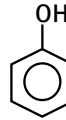
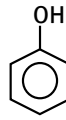
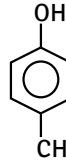
Column-I



Column-II

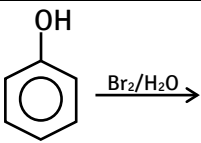
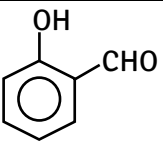
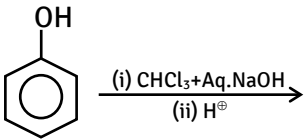
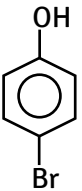
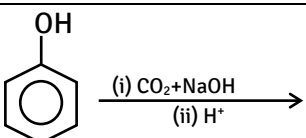
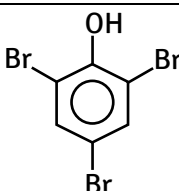
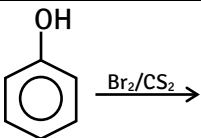
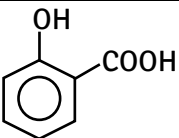
- (i) Decarboxylation
 (ii) Friedel-Crafts reaction
 (iii) Reimer-Tiemann reaction
 (iv) Kolbe's reaction
 (1) (A) $\rightarrow$ (i), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (iv)
 (2) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (i)
 (3) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (i), (D) $\rightarrow$ (ii)
 (4) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (i)

21. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	Catechol	(i)	
(B)	Resorcinol	(ii)	
(C)	p-Cresol	(iii)	
(D)	Quinol	(iv)	

- (1) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (i)
 (2) (A) $\rightarrow$ (i), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (iv)
 (3) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (i)
 (4) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (i), (D) $\rightarrow$ (iii)

22. Match the column

Column-I (Reactions)	Column-II (Product)
(a) 	(P) 
(b) 	(Q) 
(c) 	(R) 
(d) 	(S) 

(1) a-R, b-P, c-S, d-Q

(2) a-P, b-S, c-R, d-Q

(3) a-P, b-Q, c-R, d-S

(4) a-S, b-Q, c-P, d-R

23. Match the following :

(a)	Phenol + Neutral FeCl_3	(q)	No reaction
(b)	Phenol + $\text{Br}_2(\text{aq.})$	(p)	Violet colour
(c)	Phenol + NaHCO_3	(r)	White ppt.
(d)	Picric acid + NaHCO_3	(s)	CO_2 gas is evolved

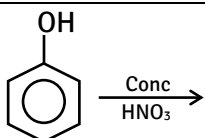
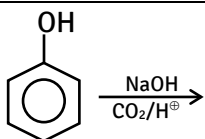
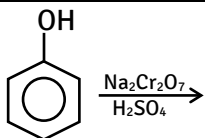
(1) (a) -p, (b) -r, (c) -q, (d) -s

(2) (a) -r, (b) -p, (c) -s, (d) -q

(3) (a) -p, (b) -s, (c) -q, (d) -r

(4) (a) -s, (b) -q, (c) -r, (d) -p

24. Match the column

Column-I (Reactions)	Column-II (Product)
(a) 	(q) Aspirin
(b) 	(p) Picric acid
(c) 	(r) Salicylic acid
(d) Salicylic acid $\xrightarrow[\text{H}^+]{(\text{CH}_3\text{CO})_2\text{O}}$	(s) Benzoquinone

(1) a-p, b-r, c-s, d-q

(2) a-q, b-s, c-r, d-p

(3) a-s, b-r, c-p, d-q

(4) a-r, b-s, c-q, d-p

25.

Column-I (Reactions)	Column-II (Product)
(a) $\text{CO} + 2\text{H}_2 \xrightarrow[573-673\text{K}]{\text{ZnO}-\text{Cr}_2\text{O}_3}$	(q) $\text{C}_2\text{H}_5-\text{O}-\text{C}_2\text{H}_5$
(b) $\text{C}_6\text{H}_{12}\text{O}_6 \xrightarrow{\text{Zymase}}$	(p) CH_3-OH
(c) $\text{CH}_3\text{CH}_2-\text{OH} \xrightarrow[443\text{K}]{\text{Conc. H}_2\text{SO}_4}$	(r) $\text{C}_2\text{H}_5\text{OH} + \text{CO}_2$
(d) $\text{CH}_3\text{CH}_2-\text{OH} \xrightarrow[413\text{K}]{\text{Conc. H}_2\text{SO}_4}$	(s) $\text{CH}_2=\text{CH}_2$

(1) a-q, b-r, c-p, d-s

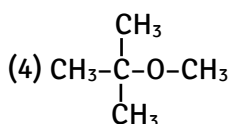
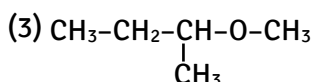
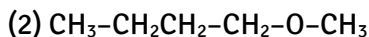
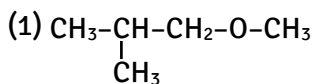
(2) a-p, b-r, c-s, d-q

(3) a-r, b-s, c-p, d-q

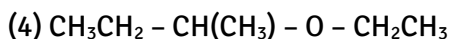
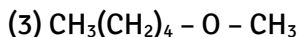
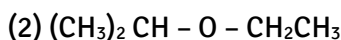
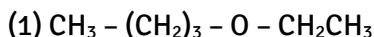
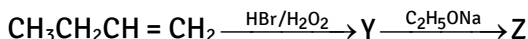
(4) a-r, b-p, c-q, d-s

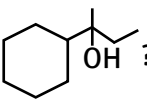
Exercise 4 (Previous Year's Questions)

1. Among the following ethers, which one will produce methyl alcohol on treatment with hot concentrated HI? [NEET-UG – 2013]

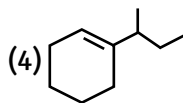
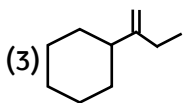
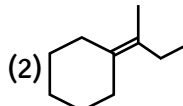
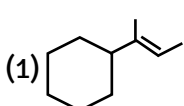


2. Identify Z in the sequence of reactions: [AIPMT – 2014]



3. Which of the following is not the product of dehydration of  ?

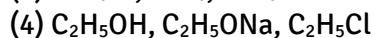
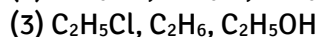
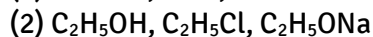
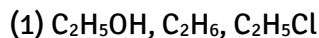
[Re-AIPMT – 2015]



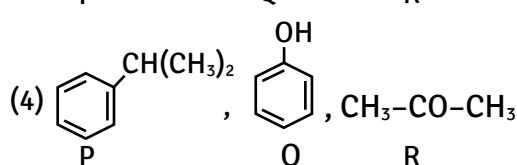
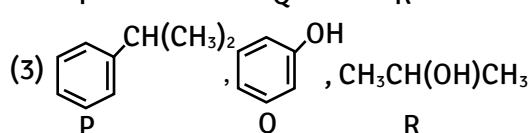
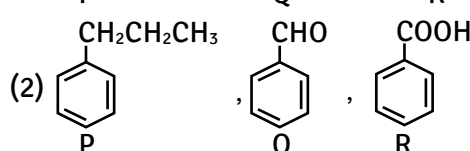
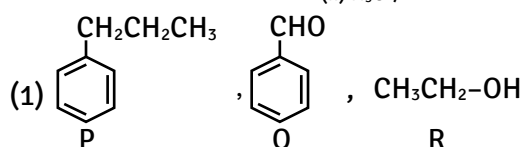
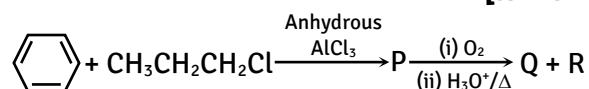
4. The heating of methyl phenyl ether with HI produces: [NEET – 2017]

- (1) ethyl chlorides (2) Iodobenzene
(3) phenol (4) benzene

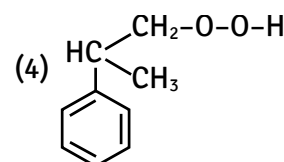
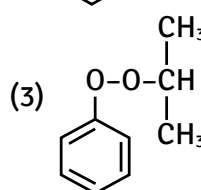
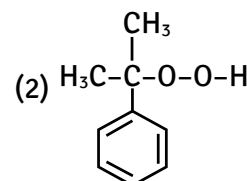
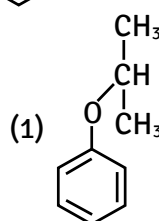
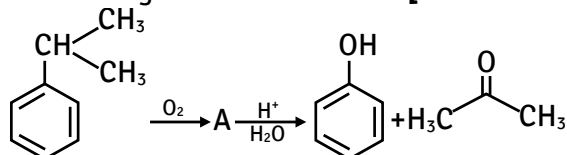
5. The compound A on treatment with Na gives B and with PCl_5 gives C, B and C react together to give diethyl ether. A, B and C are in the order [NEET-2018]



6. Identify the major products P, Q and R in the following sequence of reaction: [NEET - 2018]

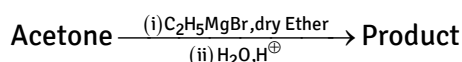


7. The structure of intermediate A in the following reaction is: [NEET - 2019]



8. When vapours of a secondary alcohol is passed over heated copper at 573 K. the product formed is: [NEET - 2019(ODISHA)]
 (1) a carboxylic acid (2) an aldehyde
 (3) a ketone (4) an alkene

9. What is the IUPAC name of the organic compound formed in the following chemical reaction? [NEET-2021]



- (1) 2-methyl propan-2-ol
 (2) pentan-2-ol
 (3) pentan-3-ol
 (4) 2-methyl butan-2-ol
10. Given below are two statements:

[NEET-2022]

Statement I: In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. HCl + ZnCl₂, known as Lucas Reagent.

Statement II: Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

11. Given below are two statements:

[NEET-2022]

Statement I: The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

Statement II: o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

12. Match the reagent (List-I) with the product (List - II) obtained from phenol.

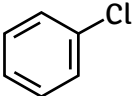
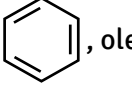
[Re-NEET-2022]

List-I		List-II	
a	(i) NaOH (ii) CO ₂ (iii) H ⁺	I	Benzoquinone
b	(i) Aqueous NaOH + CHCl ₃ (ii) H ⁺	II	Benzene
c	Zn dust, Δ	III	Salicylaldehyde
d	Na ₂ Cr ₂ O ₇ , H ₂ SO ₄	IV	Salicylic acid

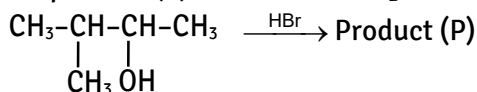
Choose the correct answer from the options given below :

- (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
 (2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

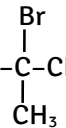
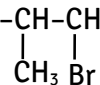
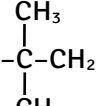
13. Which one of the following reaction sequence is incorrect method to prepare phenol? [Re-NEET-2022]

- (1) Cumene, O_2 , H_3O^+
 (2) , NaOH, STP condition
 (3) , oleum, NaOH, H_3O^+
 (4) Aniline, $NaNO_2 + HCl$, heating

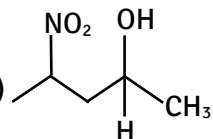
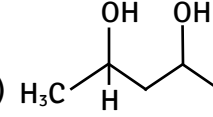
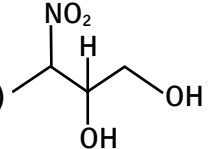
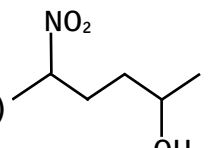
14. Consider the following reaction and identify the product (P). [NEET-2023]



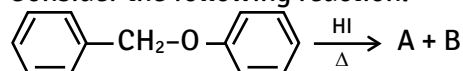
3-Methylbutan-2-ol

- (1) 
 (2) $CH_3CH=CH-CH_3$
 (3) 
 (4) 

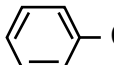
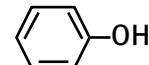
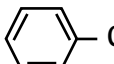
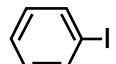
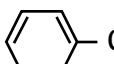
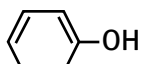
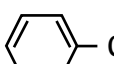
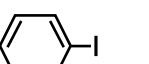
15. Which amongst the following will be most readily dehydrated under acidic conditions? [NEET-2023]

- (1) 
 (2) 
 (3) 
 (4) 

16. Consider the following reaction:



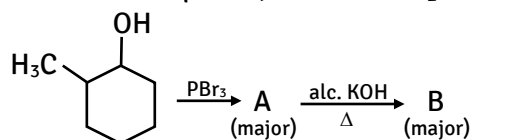
Identify products A and B [NEET-2023]

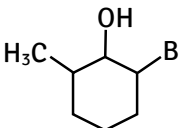
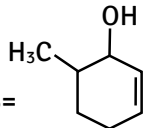
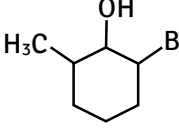
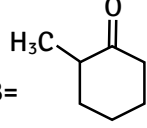
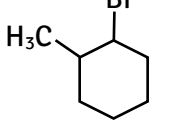
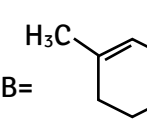
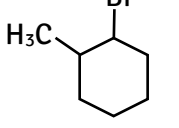
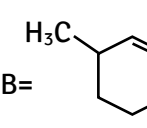
- (1) A =  and B = 
 (2) A =  and B = 
 (3) A =  and B = 
 (4) A =  and B = 

17. Which one of the following alcohols reacts instantaneously with Lucas reagent? [NEET-2024]

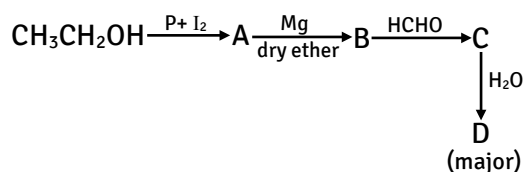
- (1) $CH_3-\underset{\substack{| \\ CH_3}}{CH}-CH_2OH$
 (2) $CH_3-\underset{\substack{| \\ CH_3}}{C}-OH$
 (3) $CH_3-CH_2-CH_2-CH_2OH$
 (4) $CH_3-CH_2-\underset{\substack{| \\ CH_3}}{C}H-OH$

18. Major products A and B formed in the following reaction sequence, are: [NEET-2024]



- (1) A =  ; B = 
 (2) A =  ; B = 
 (3) A =  ; B = 
 (4) A =  ; B = 

19. Identify D in the following sequence of reactions : **[RE-NEET-2024]**



- (1) n-propyl alcohol
- (2) isopropyl alcohol
- (3) propanal
- (4) propionic acid

20. given below are two statements:

[RE-NEET-2024]

Statement-I: propene on treatment with diborane gives an addition product with the formula $\left[(\text{CH}_3)_2\text{CH}\right]_3\text{B}$

Statement II: oxidation of $\left[(\text{CH}_3)_2\text{CH}\right]_3\text{B}$

with hydrogen peroxide in presence of NaOH gives propan-2-ol.

in the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	4	2	4	3	1	2	4	1	4	2	1	1	2	3	1	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	3	4	2	3	3	4	1	3	3	3	3	3	1	3	4	3	3	3
Que.	21	22	23	24	25															
Ans.	4	4	4	3	1															

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	4	3	2	2	3	2	2	4	2	3	4	2	4	2	2	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	4	3	4	4	1	1	4	2	1	2	2	3	1	2	4	3	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	2	1	4	2	3	2	4	3	4	2	1	3	1	2	3	4	4	4	4
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	1	1	3	3	4	2	3	1	4	1	1	2	1	3	4	2	4	2	1

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	2	3	1	3	1	1	1	2	1	3	1	2	1	2	2	3	2	2
Que.	21	22	23	24	25															
Ans.	1	1	1	1	2															

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	4	3	4	4	2	3	4	3	3	2	2	1	2	3	2	3	1	2

Chapter 4

ALDEHYDES, KETONES, AND CARBOXYLIC ACIDS

Chapter Summary

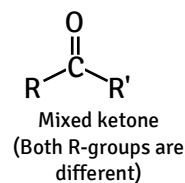
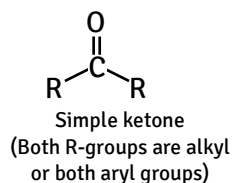
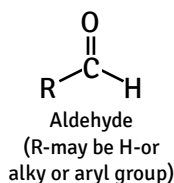
- Introduction
- Preparation of Aldehydes and ketones
- Physical properties of aldehyde and ketones
- Chemical reaction of aldehydes and ketones
- Uses of Aldehyde and ketones
- Methods of Preparation of carboxylic acids
- Physical properties of carboxylic acids
- Chemical reactions of carboxylic acids

Introduction

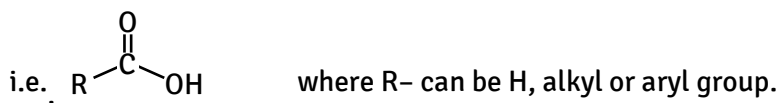
This chapter includes knowledge of carbonyl compounds i.e.

The compound containing $\text{>C}=\text{O}$ as functional group. In this we study aldehydes, ketones, carboxylic acids and their derivatives. In aldehydes carbonyl group is attached to either two H-atoms or one R-or Ar-group and one H-atom.

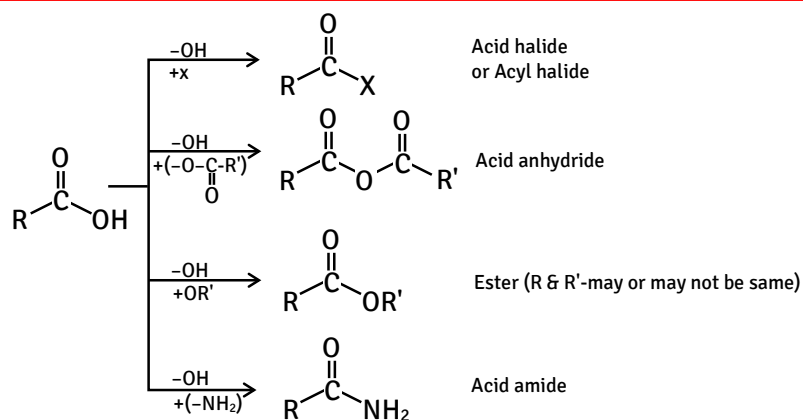
In ketones carbonyl group is attached to two alkyl or aryl groups. If both the groups are similar then this is a simple ketone but if both the groups are different it is a mixed ketone.



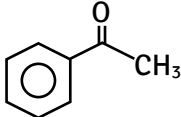
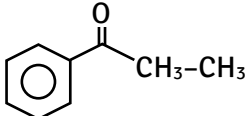
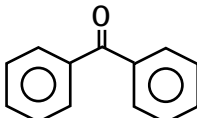
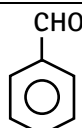
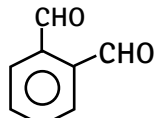
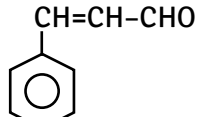
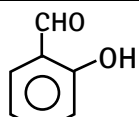
The carbonyl compound in which carbonyl group is attached to an -OH group, are called carboxylic acids



The compounds in which -OH group of carboxylic acid is replaced by other groups such as -X, -OR, -OCOR, -NH₂ etc. are called acid derivatives. All these derivatives form carboxylic acid again on hydrolysis.

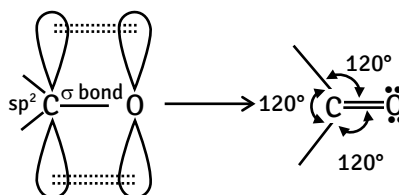


Some important common name

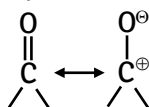
Structure	Common name	Structure	Common name
HCHO	Formaldehyde	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$	Acetone
CH ₃ CHO	Acetaldehyde		Acetophenone
CH ₃ -CH ₂ -CHO	Propionaldehyde		Propiophenone
$\text{CH}_3-\underset{\text{CH}_3}{\underset{ }{\text{CH}}}-\text{CHO}$	Isobutyraldehyde		Benzo phenone
CH ₂ =CH-CHO	Acrolein		
	Benzaldehyde		
	Phthalaldehyde		
	Cinnamaldehyde		
	Salicylaldehyde		

Structure of the carbonyl group:

Carbon atom of carbonyl group is sp^2 hybridised having triangular planar geometry. Carbon atom forms three sigma bonds and one π (pi) bond, out of which 3σ bonds are located in the same plane whereas fourth π bond. Which is formed by lateral or sideways overlapping, is situated above and below the plane. Bond angles are approximately 120°

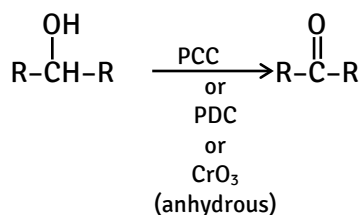
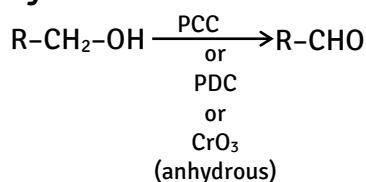


Carbonyl group is polarized due to difference in electronegativity between carbon and oxygen. Oxygen being more electronegative pulls the shared pair of electrons more towards itself making oxygen as a nucleophilic centre and carbon as an electrophilic centre. Hence carbonyl compound have substantial dipole moment and their polarity can be expressed on the basis of resonance.

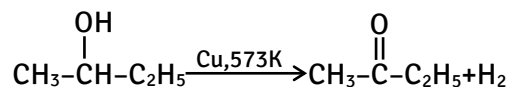
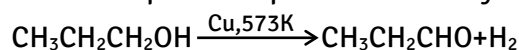


Preparation of Aldehydes and ketones

(1) By oxidation of alcohols



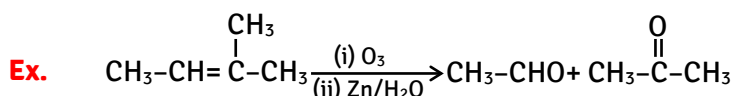
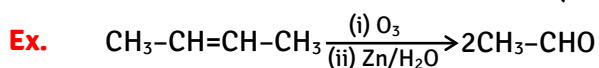
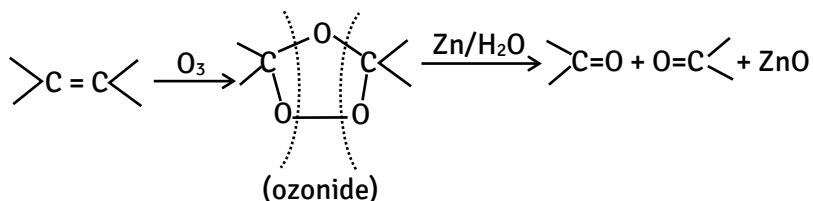
(2) **By catalytic dehydrogenation:** This method is suitable for volatile alcohols and is of industrial application. In this method alcohol vapours are passed over heavy metal catalyst such as Ag or copper.



(3) From Hydrocarbons

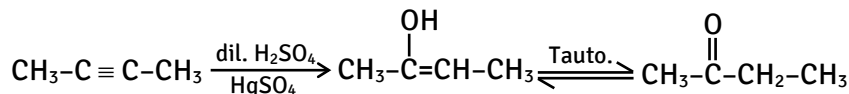
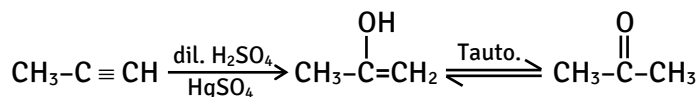
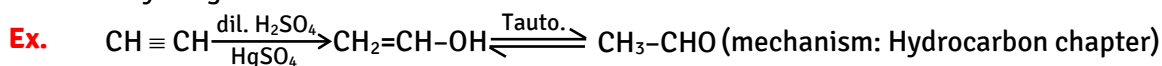
(i) By ozonolysis of alkenes:

ozonolysis of alkenes followed by reaction with zinc dust and water gives aldehyde, ketones or a mixture of both depending on the substitution pattern of the alkenes

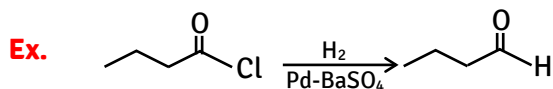
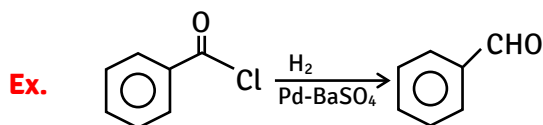


(ii) By hydration of Alkynes

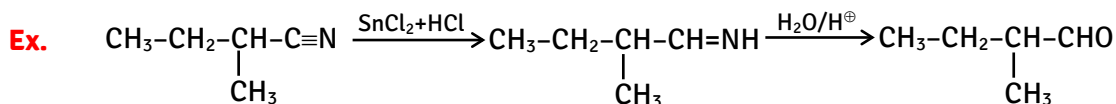
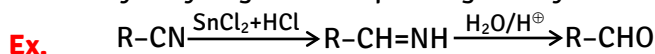
Addition of water to ethyne in the presence of dil. H_2SO_4 and HgSO_4 gives acetaldehyde. All other alkynes give ketones in this reaction

**Specific methods of preparation of aldehyde****(1) From acyl chloride (acid chloride)**

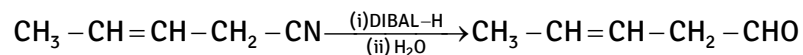
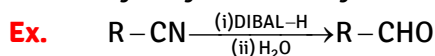
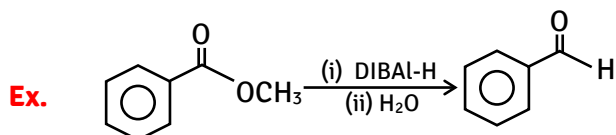
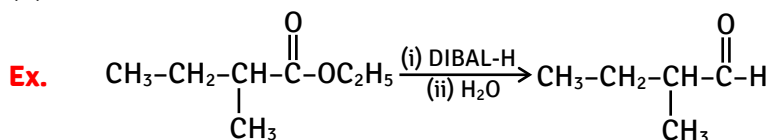
Acid chloride (Acyl chloride) is hydrogenated over catalyst, Palladium on barium sulphate. This reaction is called Rosenmund reduction.

**(2) From nitriles**

Nitriles are reduced to corresponding imine with stannous chloride in the presence of HCl. Which on hydrolysis give corresponding aldehyde. This reaction is called Stephen reaction.

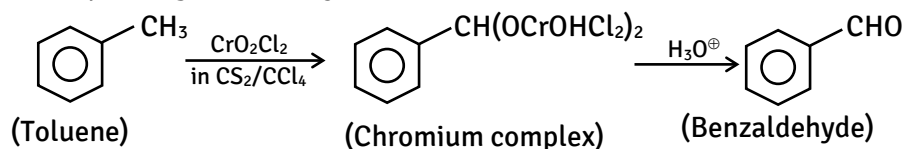
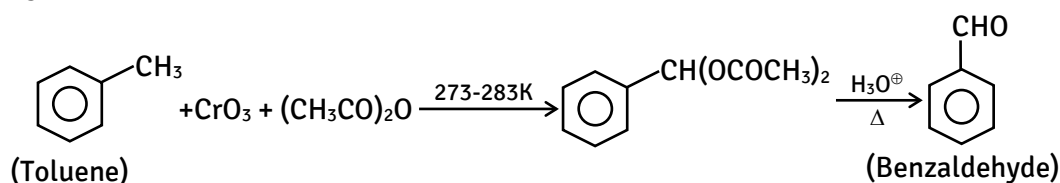
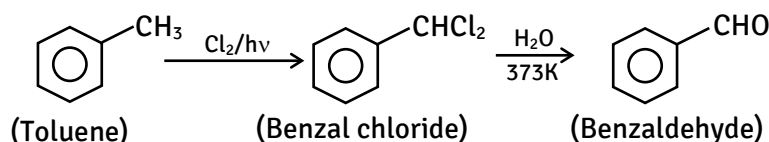
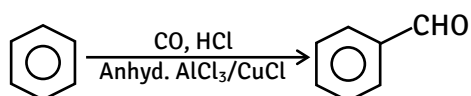
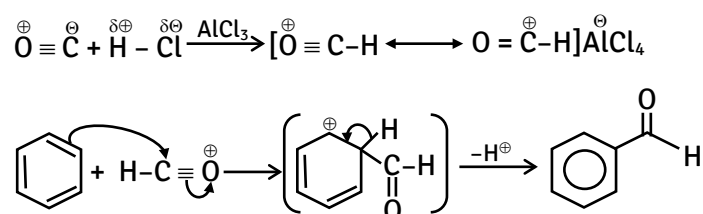
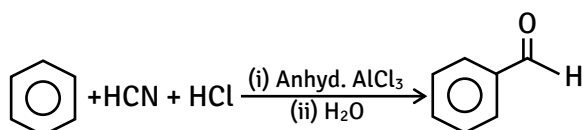
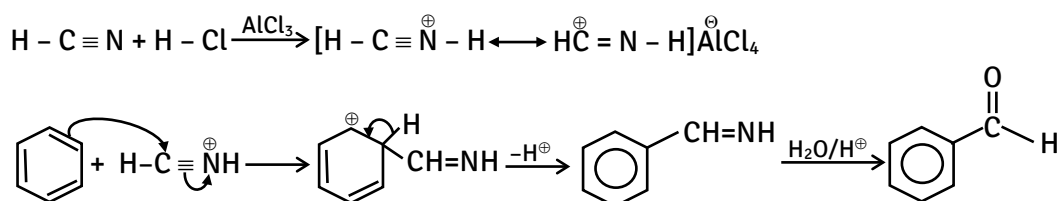


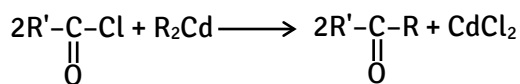
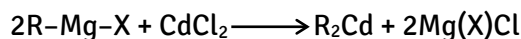
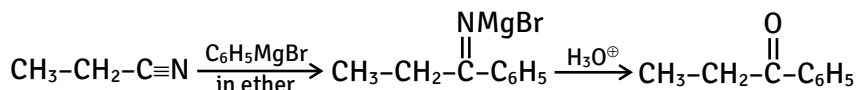
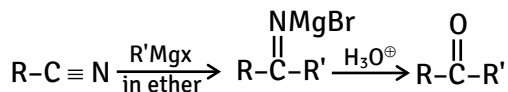
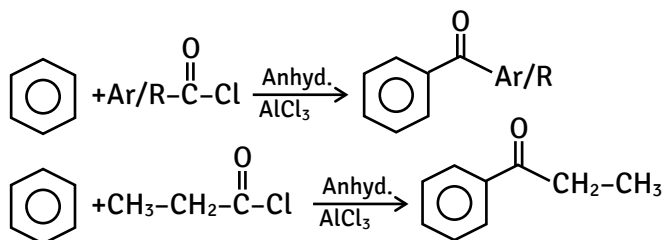
Note. Nitriles are selectively reduced by diisobutyl aluminium hydride (DIBAL-H) to imines followed by hydrolysis to aldehydes

**(3) From esters**

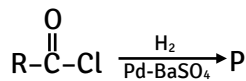
(4) From hydrocarbons**(i) By oxidation of methyl benzene****(a) By chromyl chloride (CrO_2Cl_2):**

Chromyl chloride oxidizes methyl group to a chromium complex which on hydrolysis gives corresponding benzaldehyde. This reaction is called Etard reaction.

**(b) By chromic oxide (CrO_3):****(ii) By side chain chlorination followed by hydrolysis****(iii) By Gattermann – Koch reaction****Mechanism:****(iv) By Gattermann Aldehyde synthesis****Mechanism:**

Specific method of preparation of ketones**(1) From acyl chlorides****(2) From nitriles****(3) From Friedel-Craft acylation reaction****Exercise 1.1**

1. In the following reaction product P is



- (1) RCH_2OH (2) RCOOH
(3) RCHO (4) RCH_3

2. The reagent most suitable for converting a primary alcohol into an aldehyde is:

- (1) $\text{K}_2\text{Cr}_2\text{O}_7$ and H_2SO_4
(2) KMnO_4 and NaOH
(3) $\text{C}_6\text{H}_5\text{NH}^+ \text{ClCrO}_3^-$
(4) HNO_3

3. Stephen reaction is the reaction involving:

- (1) Reduction of alkanoyl chloride with Pd/BaSO_4 .
(2) Reduction of alkyl isocyanide with sodium and alcohol.
(3) Reduction of alkyl cyanide with SnCl_2 and HCl and hydrolysing the intermediate aldimine.
(4) Reduction of carbonyl compound with zinc amalgam and HCl

4. Reduction of benzoyl chloride with Pd and BaSO_4 produces—

- (1) Benzoic acid
(2) Benzaldehyde
(3) Benzoyl cyanide
(4) Benzyl alcohol

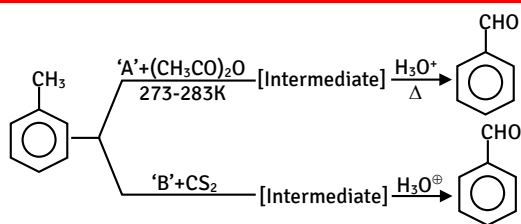
5. $\text{CH}_3-\text{CH}_2-\text{Br} \xrightarrow{\text{KCN}} \text{X} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) SnCl}_2+\text{HCl}} \text{Y}$; Y is

- (1) $\text{CH}_3-\text{CH}_2-\text{CONH}_2$
(2) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$
(3) $\text{CH}_3-\text{CH}_2-\text{CHO}$
(4) $\text{CH}_3-\text{CH}_2-\text{COOH}$

6. Ozonolysis of Isobutylene gives—

- (1) Acetone
(2) Formaldehyde
(3) Acetaldehyde
(4) 1 and 2 both

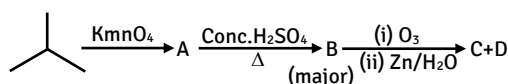
7.



In the above given reaction, identify the reagent 'A' and reagent 'B' respectively

- (1) CrO_3 and CrO_3
- (2) CrO_2Cl_2 and CrO_3
- (3) CrO_3 and CrO_2Cl_2
- (4) CrO_2Cl_2 and CrO_2Cl_2

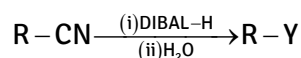
8.



In the above given sequence reaction product C & D are

- (1) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$ and $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{H}$
- (2) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{OH}$ and $\text{CH}_3-\text{C}(=\text{O})-\text{H}$
- (3) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$ and $\text{H}-\text{C}(=\text{O})-\text{H}$
- (4) $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{OH}$ and $\text{H}-\text{C}(=\text{O})-\text{H}$

9.

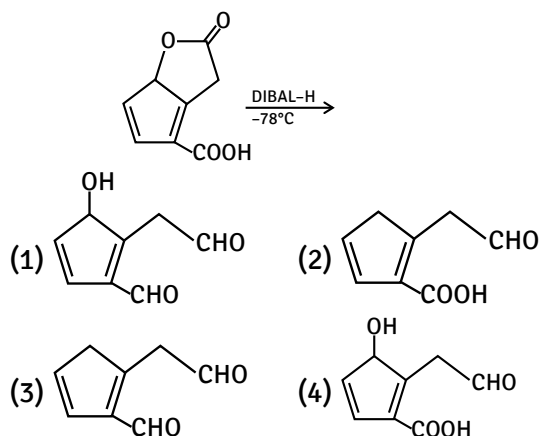


Consider the above reaction and identify "Y"

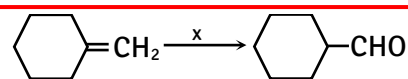
- (1) $-\text{CONH}_2$
- (2) $-\text{CHO}$
- (3) $-\text{COOH}$
- (4) $-\text{CH}_2\text{NH}_2$

10.

The major product obtained in the following reaction is :



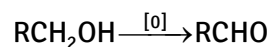
11.



X is :

- (1) (a) BH_3 , THF (b) $\text{H}_2\text{O}_2/\text{OH}^-$
- (2) H_3O^+
- (3) (a) O_3 (b) $\text{Zn}/\text{H}_2\text{O}$
- (4) (a) (i) BH_3/THF (ii) $\text{H}_2\text{O}_2/\text{OH}^-$
(b) Anhydrous CrO_3

12.



Which of the following oxidising agent is most suitable for above reaction?

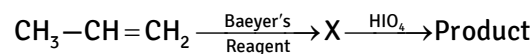
- (1) PCC
- (2) $\text{K}_2\text{Cr}_2\text{O}_7$
- (3) KMnO_4
- (4) All

13.

In which reaction ethyl phenyl ketone is obtained

- (1) $\text{C}_6\text{H}_5-\text{C}\equiv\text{N} \xrightarrow[\text{(ii) H}_2\text{O}/\text{H}^+]{\text{(i) EtMgBr}}$
- (2) $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{Cl} + \text{Ph}_2\text{Cd} \rightarrow$
- (3) + $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{Cl} \xrightarrow[\text{AlCl}_3]{\text{Anhyd.}}$
- (4) All of them

14.



Product is :

- (1) $\text{CH}_3-\text{CH}=\text{O}$
- (2) $\text{H}_2\text{C}=\text{O}$
- (3) $\text{CH}_3-\text{CO}-\text{CH}_3$
- (4) (1) and (2) both

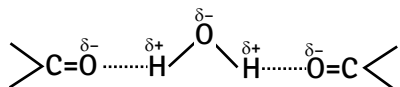
15.

Which of the following on heating with aq. KOH , produce acetaldehyde

- (1) $\text{CH}_2\text{ClCH}_2\text{Cl}$
- (2) CH_3CHCl_2
- (3) $\text{CH}_3-\text{C}(=\text{O})-\text{Cl}$
- (4) $\text{CH}_3-\text{CH}_2-\text{Cl}$

Physical properties of aldehyde and ketones

- Physical state:** Methanal is a pungent smelling gas, ethanal is volatile liquid with b.p.=294 K. Rest of the aldehydes and ketones containing upto eleven carbon atoms are colourless liquids while higher members are solids.
- Solubility:** Aldehydes and ketones upto four carbon atoms are miscible with water due to presence of H-bonding between polar carbonyl group and water molecules.



Solubility of these compounds decreases rapidly on increasing length of alkyl chain and higher members are practically insoluble in water. All these compounds are quite soluble in organic solvents such as CH_3OH , ether, CHCl_3 , benzene etc. Ketones, themselves are good solvents.

- Boiling points:** Boiling points of aldehydes and ketones are higher than those of non-polar compounds i.e., hydrocarbons and weakly polar compounds, such as ether, due to dipole-dipole interactions but lower than that of alcohols of comparable molar masses, due to absence of intermolecular H-bonding.

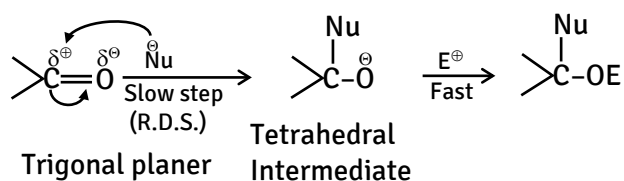
Compound	C_4H_{10} (n-Butane)	$\text{CH}_3\text{OCH}_2\text{CH}_3$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CHO}$	CH_3COCH_3
Molar mass	60	60	60	58	58
B.P. (K)	273	281	370	322	329

- Smell:** Lower aldehydes have exceptionally unpleasant odour whereas other members generally have pleasant odour. As the size of molecule increases smell becomes less pungent and more fragrant. Many naturally occurring aldehydes and ketones are used for blending of many flavoring agents and perfumes.

Chemical reaction of aldehydes and ketones

(1) Nucleophilic addition reactions:

Due to polar nature of carbonyl group, nucleophilic addition reaction are quite common to these compounds in which nucleophile attacks first followed by attack of electrophile



- Stereochemistry of the reaction:**

If chiral carbon is generated on carbonyl carbon during nucleophilic attack then equimolar mixture of the compound having 'R' and 'S' configuration w.r.t. that chiral carbon is formed as product

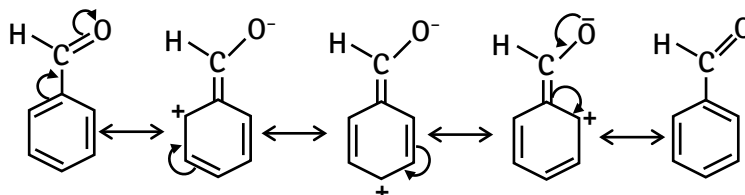
- Reactivity:**

There are two important factors which help in comparing relative reactivity of aldehydes and ketones in nucleophilic addition reactions.

- Steric factor:** Attack of nucleophile changes the trigonal planar carbonyl group to tetrahedral state. Increase in size of alkyl or aryl groups around carbonyl 'C' atom produce steric hindrance to the attacking nucleophile. It is due to the crowding of space around carbonyl carbon atom. Hence greater the number of alkyl/aryl groups or bigger the size of groups, more will be the steric hindrance and lesser will be the tendency of nucleophile to attack the carbonyl carbon.

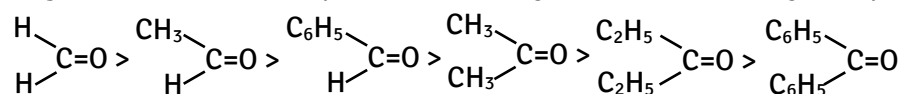
(ii) **Electronic factor:** Alkyl groups present on $-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-$ group show +I effect and tend to increase e^- density on it, due to which attacking tendency of nucleophile decreases. It means, more the +I effect lesser will be reactivity of carbonyl compounds.

In aromatic aldehydes, due to $-M$ effect of carbonyl group their reactivity towards nucleophilic addition decreases.



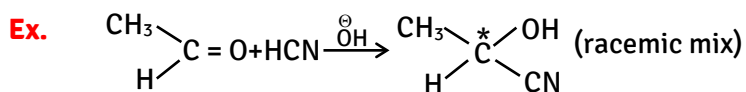
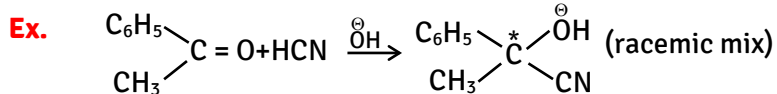
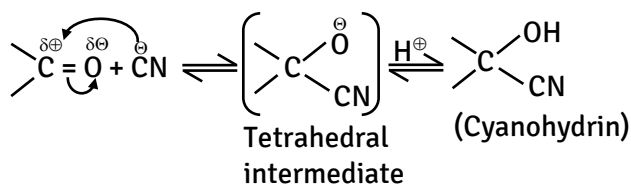
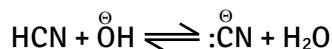
Electron density on the carbon atom of carbonyl group increases, hence the attacking tendency of nucleophile decrease.

By considering both factors the comparative reactivity of different carbonyl compound is

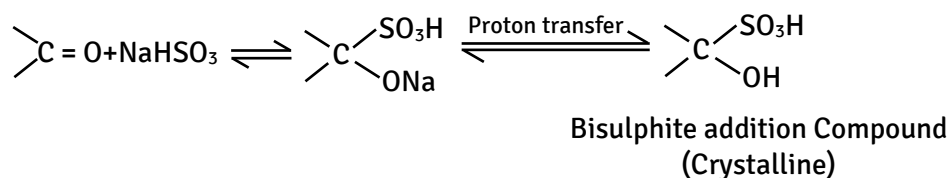


(I) Addition of HCN (hydrogen cyanide):

- Aldehydes & ketones react with HCN to yield cyanohydrin
- reaction occurs very slowly with pure HCN (weak acid) therefore reaction is carried out in basic medium so that it can generate CN^- ion for nucleophilic attack.



(II) Addition of NaHSO_3 (Sodium hydrogen sulphite)



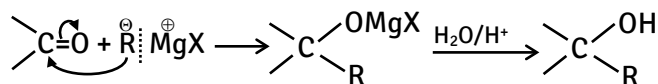
- In most aldehydes equilibrium lies to the right hand side whereas for the most ketones, it lies to the left hand side
- Hydrogen sulphite addition compound is water soluble and can be converted back to the original carbonyl compound by treating it with dilute mineral acid or alkali. Therefore these are useful for separation and purification of aldehydes & ketones.

• **Limitation of the reaction:**

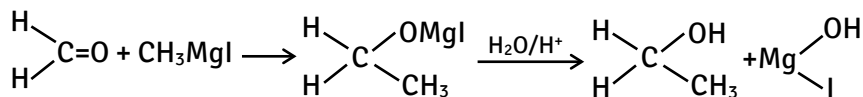
- (i) All aldehyde (Aliphatic & aromatic) show this reaction
- (ii) Methyl ketones $\text{[CH}_3\text{-C(=O)-R]}$ show this reaction

Where R group should not be in resonance with -C(=O)- (i.e. $\text{R} \neq \text{Ar}$)

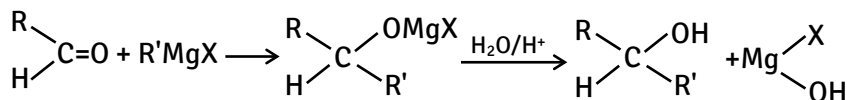
(III) Addition of Grignard reagent: Aldehydes and ketones give addition products with Grignard reagent which produce alcohol on being hydrolysed.



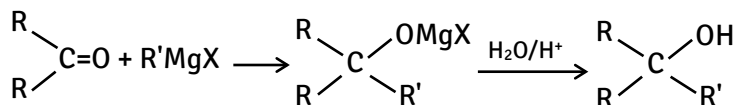
- (a) HCHO gives 1° or primary alcohol



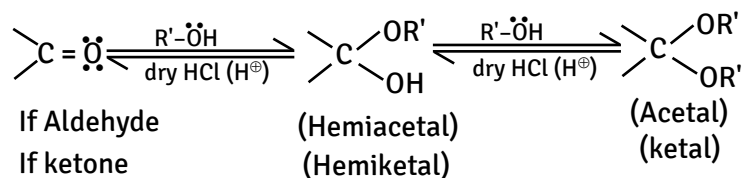
- (b) Higher aldehydes form 2° or secondary alcohol



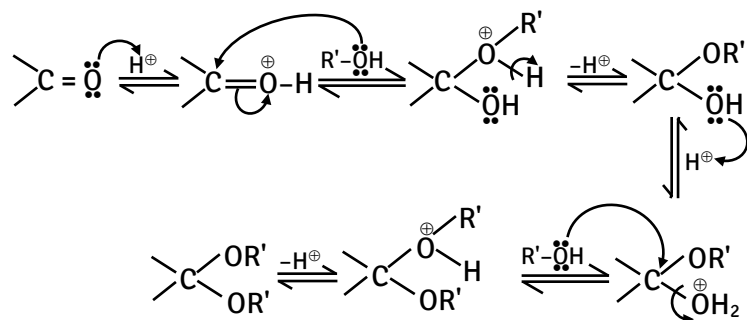
- (c) Ketones form 3° or tertiary alcohols



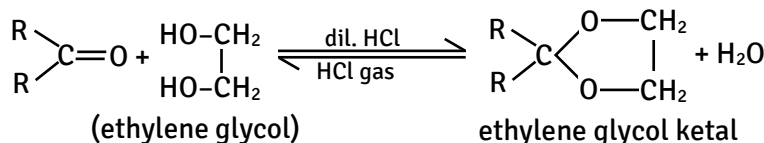
(IV) Addition of alcohols:



Mechanism:



Ex.

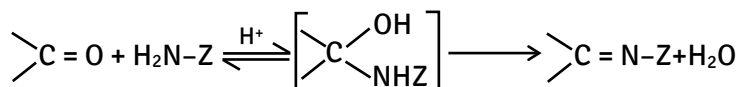


- Dry HCl Protonate the oxygen of carbonyl compound and increases electrophilicity of carbonyl carbon towards the attack of nucleophile
- Acetals and ketals are hydrolysed with aqueous mineral acids to yield corresponding aldehydes and ketones respectively

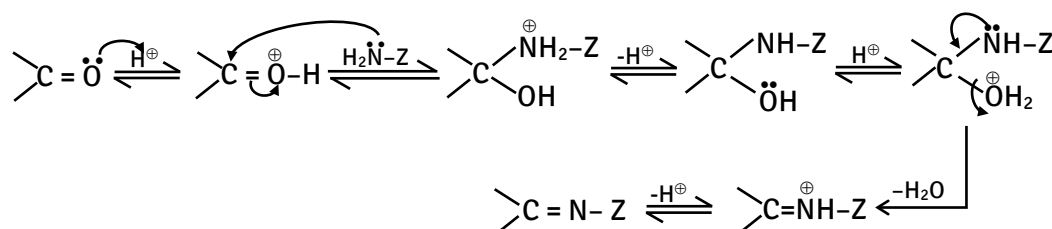
(V) Addition of NH_3 and its derivatives:

Ammonia and its derivatives such as $\text{H}_2\text{N}-\text{Z}$ add to the carbonyl group of aldehydes and ketones. reaction is reversible and catalysed by acid. The equilibrium favours the product formation due to rapid dehydration of intermediate to form $\text{>C}=\text{N}-\text{Z}$.

Here Z can be alkyl, aryl, $-\text{OH}$, $-\text{NH}_2$, $-\text{NHC}_6\text{H}_5$, $-\text{NHCONH}_2$ etc.



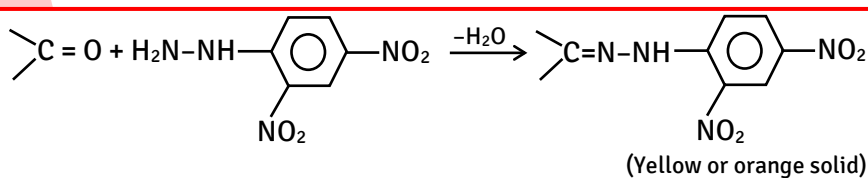
mechanism:



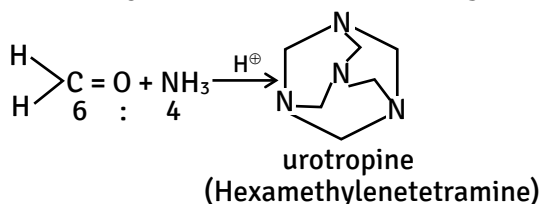
- Reaction follows nucleophilic addition elimination mechanism
- Reaction is carried out in acidic medium (PH value = 4-6)

	Group Z	Reagent	Carbonyl derivative	Product Name
1.	$-\text{H}$	NH_3 Ammonia	$\text{>C}=\text{NH}$	Imine
2.	$-\text{R}$	H_2NR Amine	$\text{>C}=\text{NR}$	Substituted imine or Schiff's base
3.	$-\text{OH}$	H_2NOH (Hydroxyl amine)	$\text{>C}=\text{NOH}$	Oxime
4.	$-\text{NH}_2$	$\text{H}_2\text{N}-\text{NH}_2$ (Hydrazine)	$\text{>C}=\text{N}-\text{NH}_2$	Hydrazone
5.	$-\text{NH}-\text{C}_6\text{H}_5$	$\text{H}_2\text{N}-\text{NH}-\text{C}_6\text{H}_5$ Phenyl hydrazine	$\text{>C}=\text{N}-\text{NH}-\text{C}_6\text{H}_5$	Phenylhydrazone
6.	$-\text{NHCONH}_2$	$\text{H}_2\text{N}-\text{NHCONH}_2$ Semi carbazide	$\text{>C}=\text{N}-\text{NHCONH}_2$	Semi carbazone
7.	$-\text{NH}-\text{C}_6\text{H}_3(\text{NO}_2)_2$	$\text{H}_2\text{N}-\text{NH}-\text{C}_6\text{H}_3(\text{NO}_2)_2$ 2,4-Dinitrophenyl hydrazine	$\text{>C}=\text{N}-\text{NH}-\text{C}_6\text{H}_3(\text{NO}_2)_2$	2,4-Dinitrophenyl hydrazone

Note. 2,4-DNP derivatives (2,4-Dinitrophenyl derivatives) are yellow, orange or red crystalline compounds and it can be used for the identification of aldehyde and ketones.

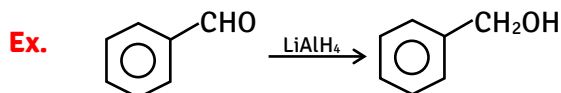
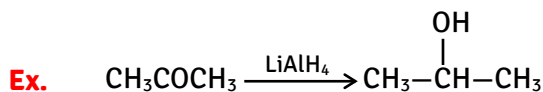
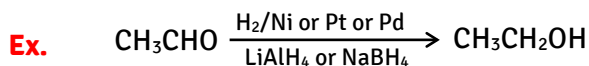
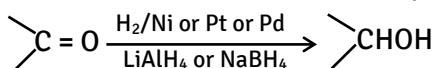


- Carbonyl compound react with primary amine to give Schiff's base (substituted imine)
- except formaldehyde, all other aldehydes react with ammonia to give imine as the final product but formaldehyde react with ammonia to give urotropine $[(\text{CH}_2)_6\text{N}_4]$ as the final product.



(2) Reduction Reactions:

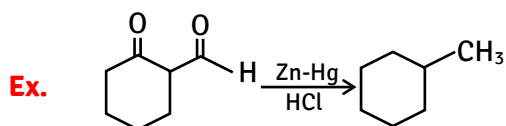
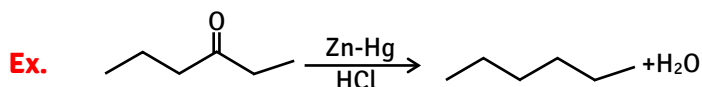
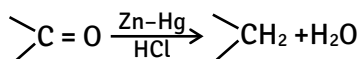
- (I) **Reduction to alcohols:** By catalytic hydrogenation or LiAlH_4 or NaBH_4 , aldehydes and ketones are reduced to 1° and 2° alcohols respectively



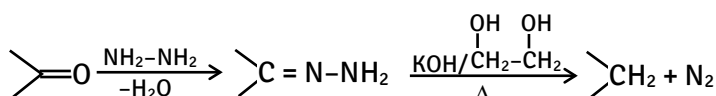
(II) Reduction to hydrocarbons:

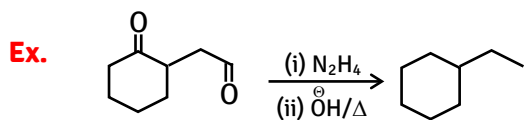
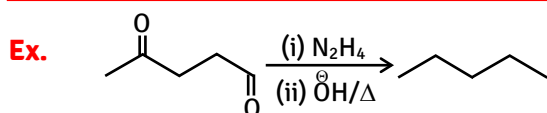
>C=O group of aldehyde and ketones can be reduced to $-\text{CH}_2$ group by use of following reagent

- (i) **Clemmensen reduction:** reduction with Zn-Hg (amalgum) and concentrated HCl

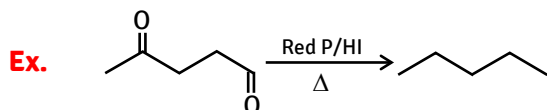
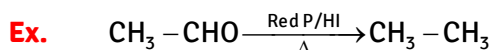


- (ii) **Wolf-kishner reduction:** reduction with hydrazine ($\text{NH}_2\text{-NH}_2$) followed by heating with base (KOH or NaOH) in high boiling solvent such as ethylene glycol.



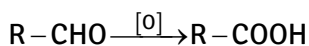


(iii) Reduction with Red P and HI



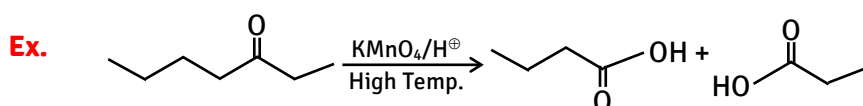
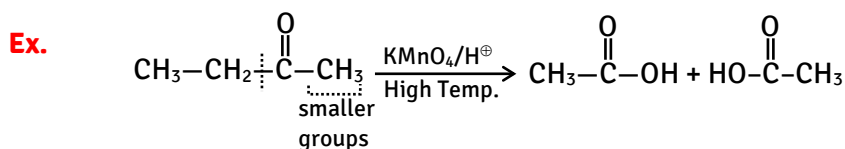
3. Oxidation reaction:

- Aldehyde can easily be oxidised to carboxylic acids on treatment with common oxidising agent like HNO_3 , KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$



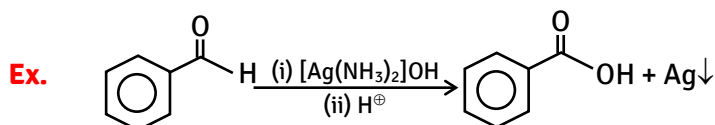
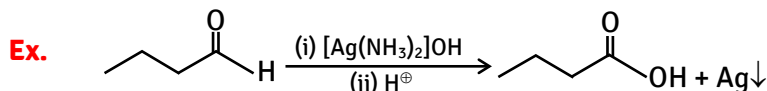
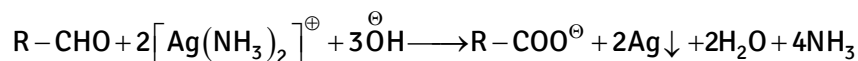
- Ketones are oxidised only under vigorous condition, i.e. by strong oxidising agents and at elevated temperature.

Popoff's rule: during the oxidation of unsymmetrical ketones, the carbon-carbon bond cleavage will always occur such that keto group ($\text{C} = \text{O}$) remains attached with smaller alkyl groups.



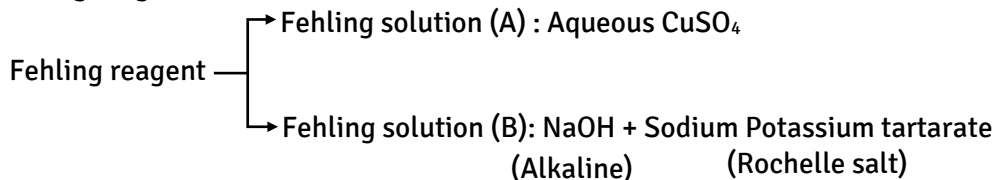
(a) oxidation by Tollen's reagent → Tollen's Test

On warming an aldehyde with freshly prepared ammoniacal silver nitrate solution (Tollen's reagent), a bright silver mirror is produced due to formation of silver metal

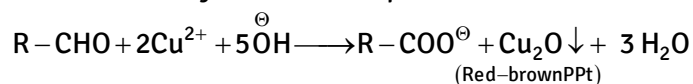


(b) Oxidation by Fehling reagent → Fehling's Test

Fehling reagent is stored in two different bottles as two different solution

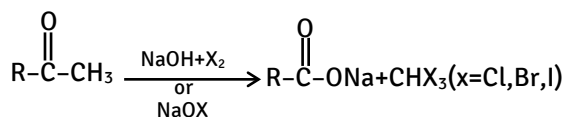
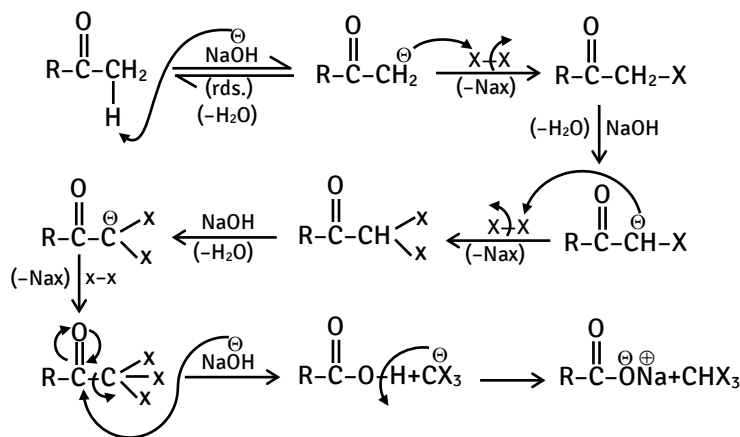
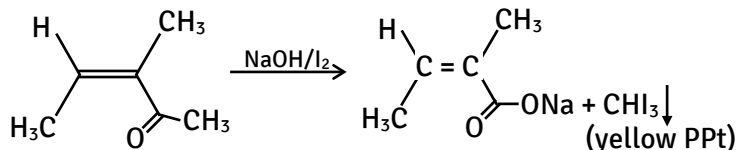
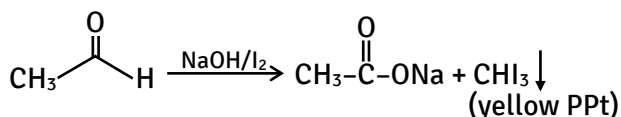


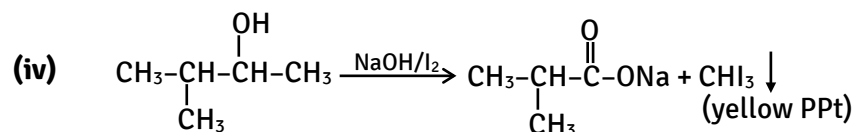
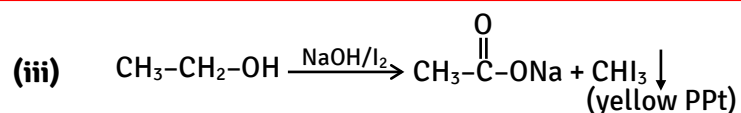
- When Fehling solution (A) and Fehling solution (B) are mixed, dark blue color cu-complex obtained. This cu-Complex (Cu^{2+}) oxidises aldehyde into acid & gives reddish brown ppt of Cu_2O
- Aromatic aldehydes do not respond to this test

**(c) Oxidation of methyl ketones by Haloform reaction:**

- Aldehyde and ketones having at least one methyl group linked to carbonyl carbon atom (i.e having $\begin{array}{c} \text{O} \\ \parallel \\ \text{-C-CH}_3 \end{array}$ group) are oxidised by sodium hypohalite solution (X_2 in presence of NaOH) to give sodium salts of carboxylic acid having one carbon atom less than that of carbonyl compound.

- If reagent is $\text{I}_2 + \text{NaOH}$ or NaOI , then yellow precipitate of CHI_3 is obtained. It is used for detection of $\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3\text{-C-} \end{array}$ group or $\begin{array}{c} \text{OH} \\ | \\ \text{CH}_3\text{-CH-} \end{array}$ group in compound

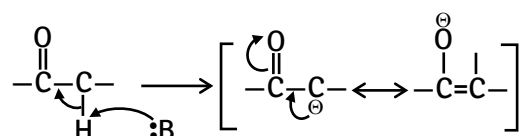
**Mechanism:****Ex. (i)****(ii)**



4. Reactions due to α -hydrogen :

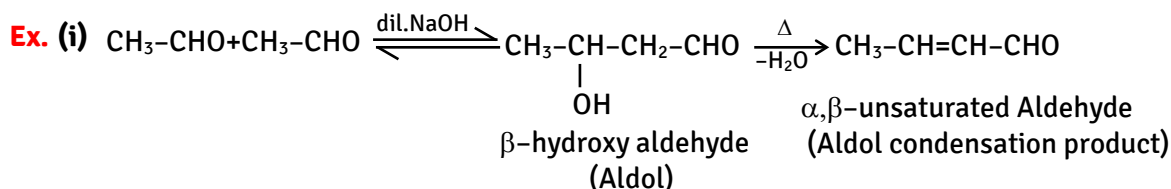
Acidity of α -hydrogen of aldehydes & ketones :

The acidity of α -H atoms of carbonyl compounds is due to strong electron withdrawing effect of the carbonyl group and resonance stabilisation of the conjugate base

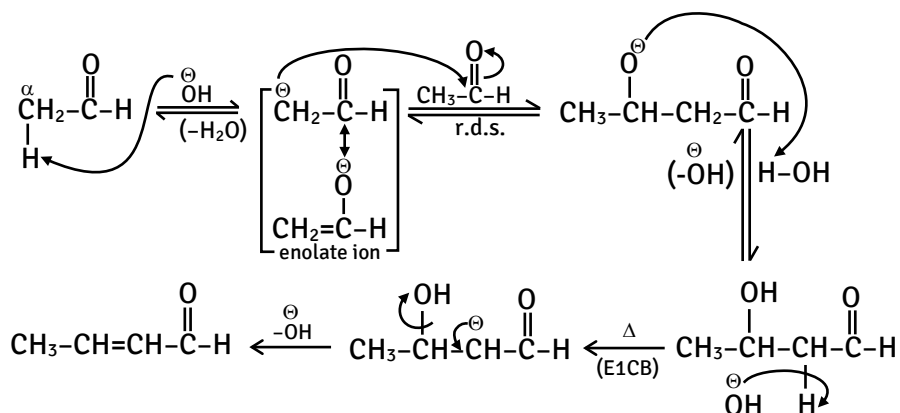


(a) Aldol condensation :

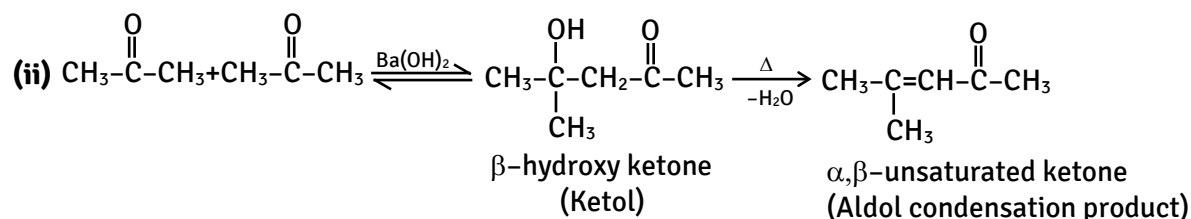
Aldehydes & Ketones having at least one α -hydrogen undergo a reaction in the presence of dilute alkali (eg NaOH/KOH/Na₂CO₃/Ba(OH)₂) as catalyst to form β -hydroxy aldehydes (aldol) or β -hydroxy ketones (Ketol), respectively. This is known as Aldol reaction



Mechanism :

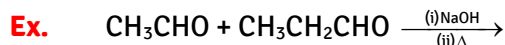


Similarly

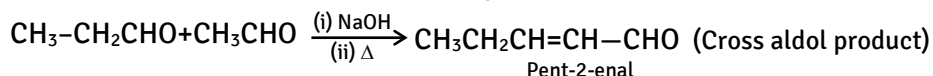
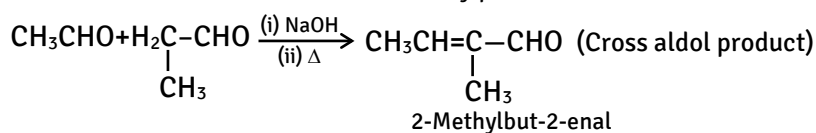
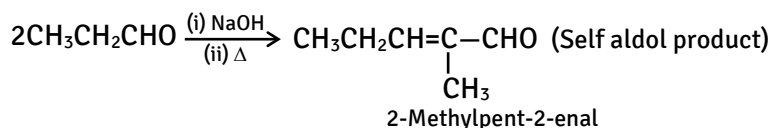
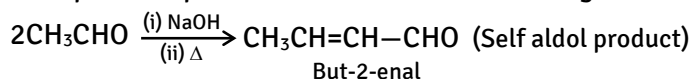


- Carbanion (enolate) is intermediate
- Attack of carbanion is r.d.s.
- The Aldol and ketol readily lose water in the presence of heat to give α,β -unsaturated carbonyl compounds, hence this reaction is called Aldol condensation
- Dehydration takes place by E1-CB mechanism

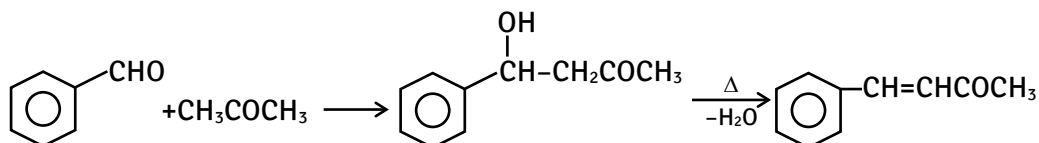
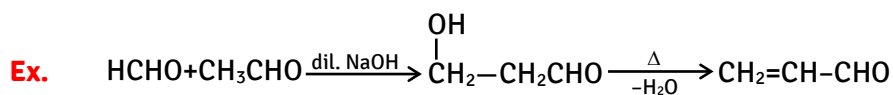
(b) **Cross aldol condensation:** When Aldol condensation is carried out between two different aldehydes or ketones, it is called cross Aldol condensation. If both the molecules have α -H atom, then they can give a mixture of four-condensation products.



Four possible products are formed as following :

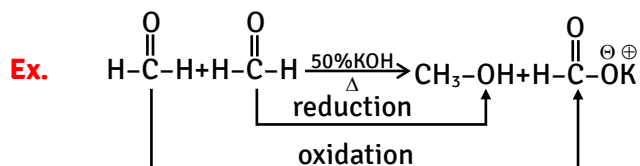


Hence cross aldol condensation becomes useful when one of the carbonyl compound does not contain α -H atom.

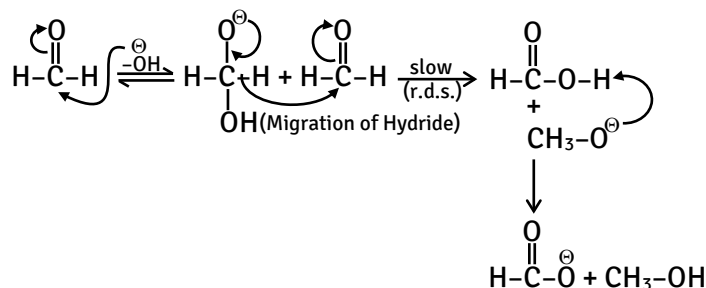


5. **Cannizzaro reaction:** Aldehyde which do not have an α -hydrogen atom, undergo self oxidation and reduction (disproportionation) reaction on heating with concentrated alkali.

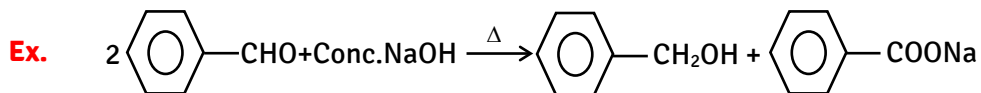
In this reaction, one molecule of the aldehyde is reduced to alcohol while other is oxidised to carboxylic acid salt.



Mechanism:

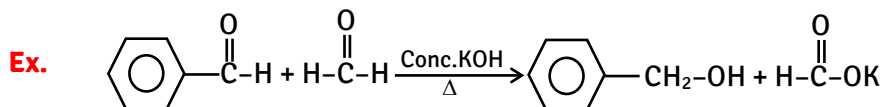


Note. Migration of hydride ion is r.d.s. step

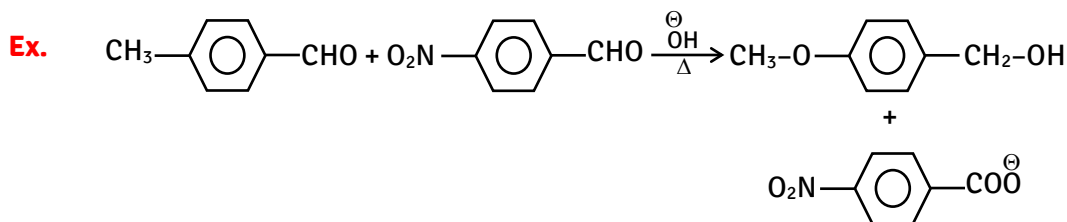
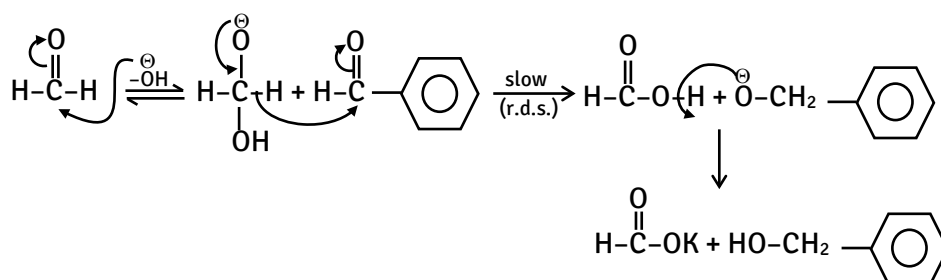


Cross Cannizzaro reaction :

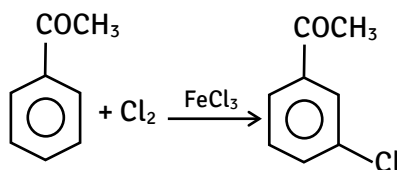
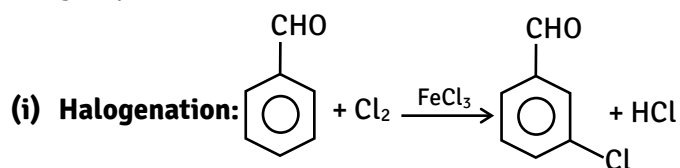
- This reaction takes place between two different aldehydes (both not having any α -H atom) in which aldehyde having more electrophilic carbonyl carbon get oxidised and other get reduced



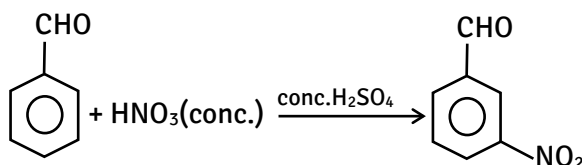
Mechanism:



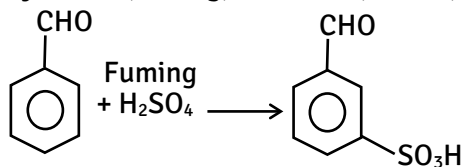
- 6. Electrophilic substitution reaction:** Aromatic aldehydes and ketones undergo electrophilic substitution reaction in the ring in which carbonyl groups act as a deactivating and meta directing group.



- (ii) Nitration:** By conc. HNO_3 and conc. H_2SO_4



Sulphonation: By H_2SO_4 (fuming) or oleum ($\text{H}_2\text{S}_2\text{O}_7$)



Uses of Aldehyde and ketones

Aldehydes and ketones are used in chemical industries as solvents, starting material and reagents for synthesizing the other products, e.g.

- (1) 40% aq. HCHO known as formalin and is used in preserving biological specimens.
- (2) Formaldehyde is also used as disinfectant.
- (3) HCHO is used in manufacturing synthetic polymers like Bakelite.
- (4) Acetaldehyde is used for silvering of mirrors.
- (5) Benzaldehyde as a flavoring agent in perfume industry.
- (6) Acetone is used as a solvent in industries and laboratories.
- (7) Acetone is a common constituent of nail polishes. Many aldehydes and ketones such as butyraldehyde, vanillin, acetophenone, camphor etc. are known for their odours and flavours.

Exercise 1.2

1. Which one of the following on treatment with 50% aq. NaOH yields the corresponding alcohol and acid

- (1) $\text{C}_6\text{H}_5\text{CHO}$ (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
(3) CH_3COCH_3 (4) CH_3CHO

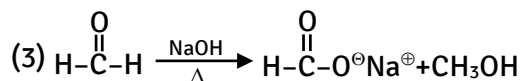
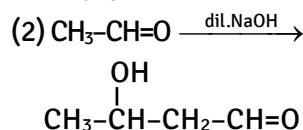
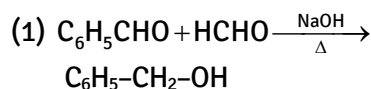
2. Which factors will increase the reactivity of >C=O group?

- (i) Presence of a group with positive inductive effect
(ii) Presence of a group with (-ve) inductive effect
(iii) Acidic medium
(1) Only (i) (2) Only (ii)
(3) Both (i) & (iii) (4) Both (ii) & (iii)

3. Which of the following compound has the largest equilibrium constant for the addition of water?

- (1)
(2)
(3)
(4)

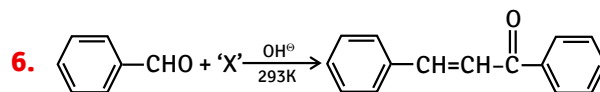
4. Which of the following is an example of aldol reaction?



- (4) All of these

5. Cannizzaro reaction does not take place with:

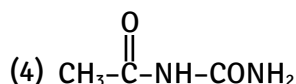
- (1) $(\text{CH}_3)_3\text{CCHO}$ (2)
(3) (4) CH_3CHO



- (1)
(2)
(3)
(4)

7. Acetaldehyde reacts with semi carbazide, product will be:

- (1) $\text{CH}_3\text{CH} = \text{NNH} - \text{CO} - \text{NH}_2$
- (2) $\text{CH}_3\text{CH} = \text{NCONHNH}_2$
- (3) $\text{CH}_3\text{CH} = \text{NHNH}_2$



8. Acetone gives test with:

- (1) 2,4 Dinitro phenyl hydrazine
- (2) Fehling solution
- (3) Schiff's reagent
- (4) All

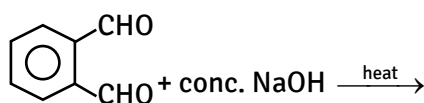
9. Which is most difficult to oxidise:

- (1) HCHO
- (2) CH_3CHO
- (3) CH_3COCH_3
- (4) $\text{CH}_3\text{CH}_2\text{CHO}$

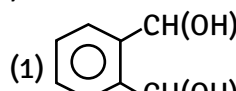
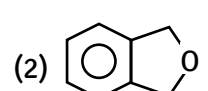
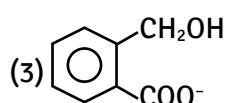
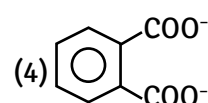
10. Which of the following can be used to differentiate between ethanal and propanal-

- (1) Ammoniacal AgNO_3
- (2) NaHSO_3
- (3) I_2 in presence of base
- (4) 2,4-DNP

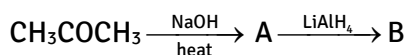
11. The reaction



produces

- (1) 
- (2) 
- (3) 
- (4) 

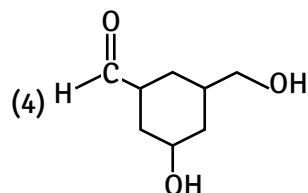
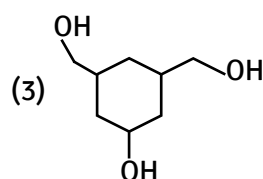
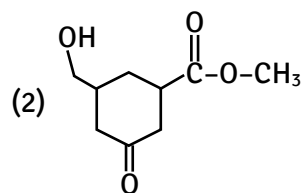
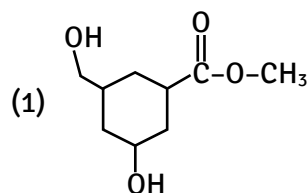
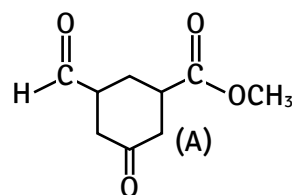
12. Consider the following sequence of reactions.



The final product (B) is :

- (1) $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{COCH}_3$
- (2) $(\text{CH}_3)_2\text{C}=\text{CHCOCH}_3$
- (3) $(\text{CH}_3)_2\text{CHCH}_2\text{CHOHCH}_3$
- (4) $(\text{CH}_3)_2\text{C}=\text{CHCHOHCH}_3$

13. Predict the product when given compound reacts with NaBH_4 .



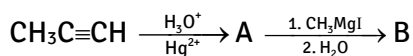
14. Which of the following pairs of carbonyl compounds and Grignard reagent should be chosen as reactants to prepare 2, 4-dimethyl-3-pentanol ?

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$
- (2) $(\text{CH}_3)_2\text{CHCHO}$ and $(\text{CH}_3)_2\text{CHMgBr}$
- (3) $\text{CH}_3\text{CH}_2\text{COCH}_3$ and $(\text{CH}_3)_2\text{CHMgBr}$
- (4) $(\text{CH}_3)_2\text{CHCHO}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$

15. Which of the following forces is correctly described about boiling point of Aldehydes & ketones:

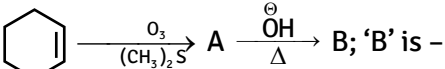
- (1) Hydrogen bond
- (2) Vander wall force
- (3) Dipole-dipole attraction
- (4) None of these

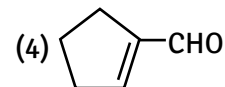
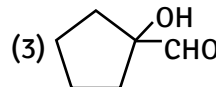
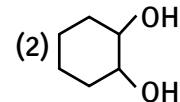
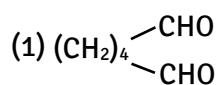
16. Consider the following sequence of reactions



The final product (B) is :

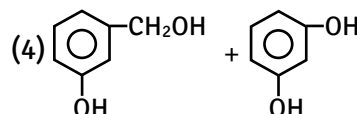
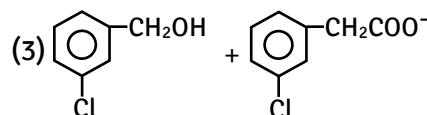
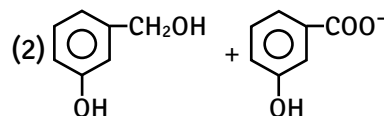
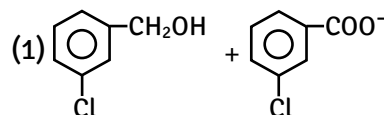
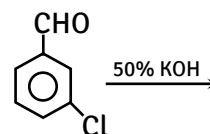
- (1) $\text{CH}_3\text{C}\equiv\text{CCH}_3$
 (2) $\text{CH}_3\text{COCH}_2\text{CH}_3$
 (3) $\text{CH}_3\text{CH}_2\text{CHOHCH}_3$
 (4) $(\text{CH}_3)_3\text{C}-\text{OH}$
17. Carbonyl compounds react with which of the following reagent to form a colorless crystalline solid-
- (1) PCl_5 (2) HCN
 (3) NH_2OH (4) NaHSO_3
18. Which of the following statement is true for Cannizzaro reaction-
- (1) The aldehyde is oxidised as well as reduced
 (2) The aldehydes do not containing α -Hydrogen atoms give the reaction
 (3) The reaction is not given by aldehydes containing α -Hydrogen atoms
 (4) All of these
19. Acetaldehyde and acetone can be distinguished by all the following except-
- (1) Iodine + alkali
 (2) Tollen's reagent
 (3) Fehling solution
 (4) Schiff reagent
20. Which of the following statements is wrong
- (1) All methyl ketones give a positive iodoform test.
 (2) Acetaldehyde is the only aldehyde that gives iodoform test.
 (3) All secondary alcohols give positive iodoform test.
 (4) Any alcohol that can be oxidised to an acetyl group gives a positive iodoform test.

21.  B; 'B' is -



22. The suitable reagent to convert $\text{CH}_3\text{CH}=\text{CH}-\text{CHO}$ into $\text{CH}_3-\text{CH}_2\text{CH}_2-\text{CH}_2-\text{OH}$ is :
- (1) NaBH_4 (2) LiAlH_4
 (3) $\text{Zn}-\text{Hg}/\text{HCl}$ (4) H_2/Pd

23. Predict the products in the given reaction.



24. CH_3CHO and $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$ can be distinguished chemically by :
- (1) Tollen's reagent test
 (2) Fehling solution test
 (3) Benedict test
 (4) Iodoform test

25. Which of the following is useful for characterisation of aldehyde and ketones.
- (1) Derivatives of Semi carbazide
 (2) Derivatives of Phenylhydrazine
 (3) 2, 4-DNP Derivatives
 (4) Derivatives of Hydrazine

26. Which of the following arrangements with respect to their reactivity in nucleophilic addition reaction is correct?

- (1) benzaldehyde < acetophenone
 < p-nitrobenzaldehyde < p-tolualdehyde
- (2) acetophenone < benzaldehyde
 < p-tolualdehyde < p-nitrobenzaldehyde
- (3) acetophenone < p-tolualdehyde
 < benzaldehyde < p-nitrobenzaldehyde
- (4) p-nitrobenzaldehyde < benzaldehyde
 < p-tolualdehyde < acetophenone

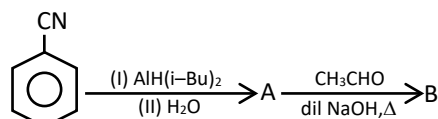
27. $\text{CH}_3\text{--CH}_2\text{--}\underset{\text{CH}_3}{\text{CH}}\text{--CH=O} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) NaBH}_4} \text{P}$; P is

- (1)
- (2)
- (3)
- (4)

28. Which cannot show simple aldol condensation

- (1) $\text{CH}_3\text{--CH=O}$
- (2) $\text{CH}_3\text{--}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{--CH}_3$
- (3) $\text{C}_6\text{H}_5\text{--CH=O}$
- (4) $\text{Ph--}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{--CH}_3$

29. Consider the following reaction sequence:



The Major Product 'B' is

- (1)
- (2)
- (3)
- (4)

30. $\xrightarrow{\text{Br}_2/\text{AlBr}_3}$ 'A' (Major Product)

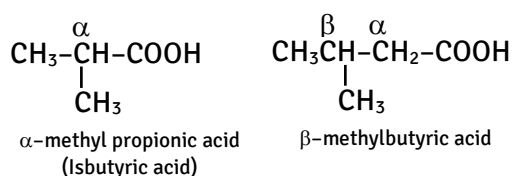
What is the product 'A'?

- (1)
- (2)
- (3)
- (4)

Carboxylic acids:- Introduction

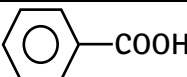
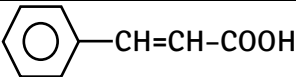
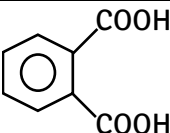
Nomenclature: Common names of carboxylic acids are based on their source of origin e.g., Formic acid (HCOOH) was first obtained from red ants (Latin Formica means ants) similarly acetic acid (CH_3COOH) is so named because it was obtained from vinegar (Latin acetum means vinegar), butyric acid ($\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$) from rancid butter (Latin Butyrum means butter) etc.

In common system position of substituents is indicated by Greek letters α , β , γ , δ etc.



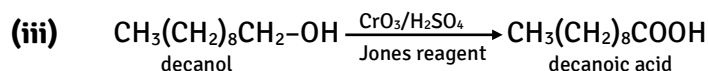
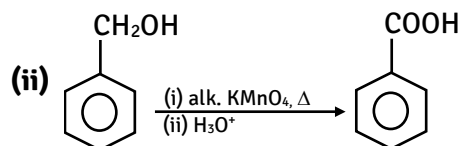
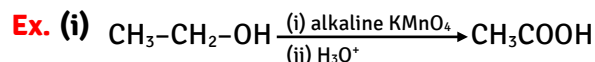
Common Name of some important carboxylic acids

Structure	Common name
HCOOH	Formic acid
$\text{CH}_3\text{--COOH}$	Acetic acid
$\text{CH}_3\text{--CH}_2\text{--COOH}$	Propionic acid

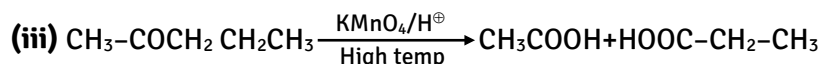
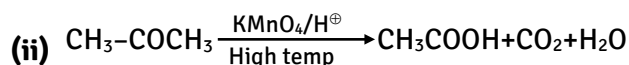
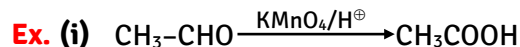
$\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--COOH}$	Butyric acid
$(\text{CH}_3)_2\text{CHCOOH}$	Isobutyric acid
HOOC--COOH	Oxalic acid
$\text{HOOC--CH}_2\text{--COOH}$	Malonic acid
$\text{HOOC--(CH}_2\text{)--COOH}$	Succinic acid
$\text{HOOC--(CH}_2\text{)}_3\text{--COOH}$	Glutaric acid
$\text{HOOC--(CH}_2\text{)}_4\text{--COOH}$	Adipic acid
$\text{HOOC--(CH}_2\text{)}_5\text{--COOH}$	Pimelic acid
	Benzoic acid
	Cinnamic acid
	Phthalic acid

Methods of Preparation of carboxylic acids

(1) From primary alcohol: Primary alcohols are easily oxidised to carboxylic acids with oxidants such as KMnO_4 , in neutral, acidic or alkaline media or by CrO_3 in acidic media or $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic media.

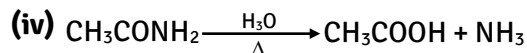
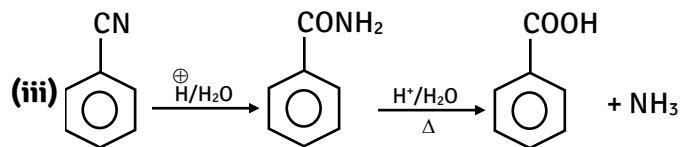


(2) From oxidation of aldehydes and ketones: Aldehydes can be oxidised to carboxylic acids even with mild oxidising agent while ketones are oxidised under vigorous conditions i.e. strong oxidising agents & at elevated temperatures and mixture of acids having lesser number of carbon atoms than parent ketone is obtained.

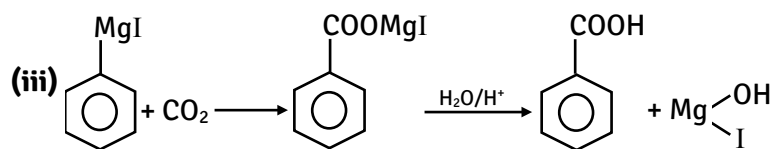
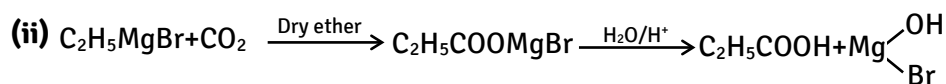
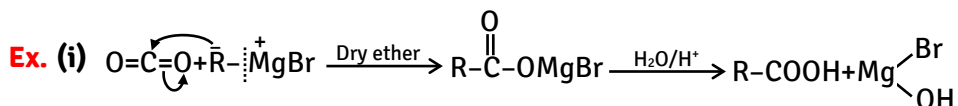


(3) By hydrolysis of nitriles (cyanides) and amides : Nitriles are hydrolysed to amides and then to carboxylic acids in the presence of H^+ or OH^- as catalyst. If conditions used are mild, then reaction can be stopped even at amide stage

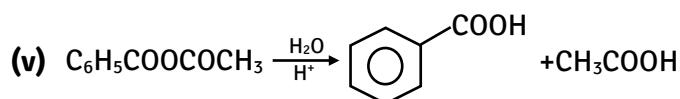
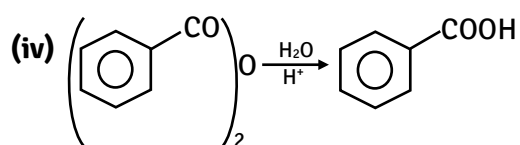
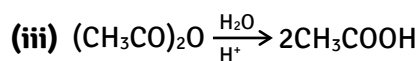
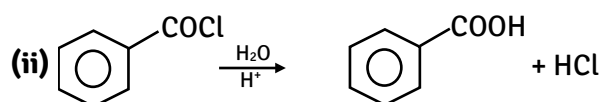
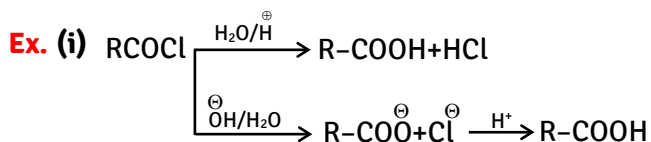




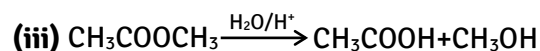
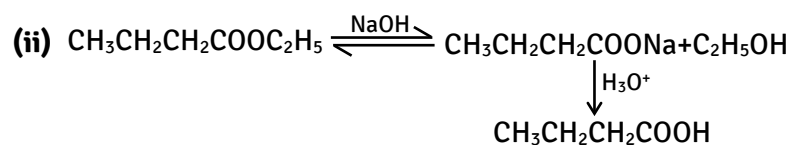
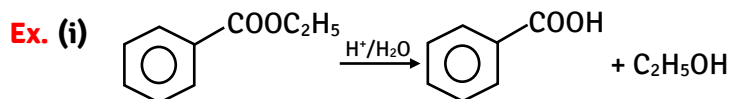
- (4) **From Grignard reagent:** Reaction of Grignard reagent with dry ice (i.e. solid CO_2) in dry ether followed by acidic hydrolysis produces carboxylic acid containing one more carbon atom than Grignard reagent.



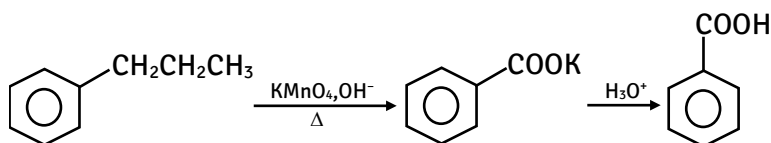
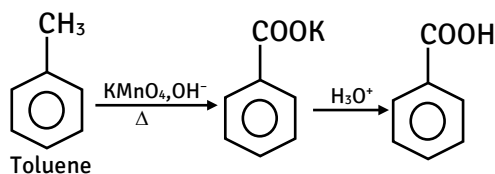
- (5) **By hydrolysis of acid halides and anhydrides:** Acidic hydrolysis of acid halide and anhydride directly gives carboxylic acids while their basic hydrolysis first forms carboxylates, which on acidification form corresponding carboxylic acids.



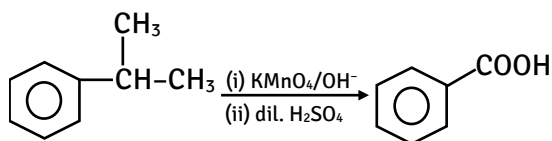
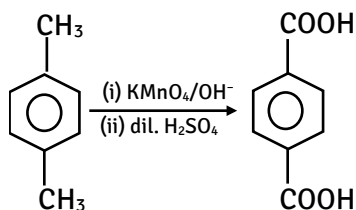
- (6) **By hydrolysis of esters:** Acidic hydrolysis of esters directly gives carboxylic acids while their basic hydrolysis first forms carboxylates, which on acidification form corresponding carboxylic acids.



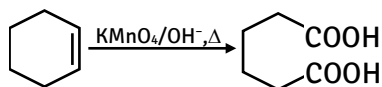
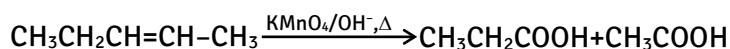
- (7) From alkyl benzenes:** Alkyl benzene containing benzylic H-atom on vigorous oxidation by H_2CrO_4 or H^+/KMnO_4 or $\text{OH}^-/\text{KMnO}_4$ gives Aromatic carboxylic acid.



Note. Entire side chain containing primary and secondary alkyl group of benzene ring is oxidised to $-\text{COOH}$ group, irrespective to the length of side chain while tertiary alkyl group is not affected.



- (8) From oxidation of alkenes:** Alkenes undergo oxidative cleavage by hot acidic KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$ to give carboxylic acid



Exercise 1.3

1. $(\text{CH}_3)_2\text{CO} \xrightarrow[\text{(HCN)}]{\text{NaCN}} \text{A} \xrightarrow{\text{H}_3\text{O}^+} \text{B}$
In the above sequence of reactions A and B are:

- (1) $(\text{CH}_3)_2\text{C}(\text{OH})\text{CN}$, $(\text{CH}_3)_2\text{C}(\text{OH})\text{COOH}$
- (2) $(\text{CH}_3)_2\text{C}(\text{OH})\text{CNH}_2$, $(\text{CH}_3)_2\text{C}(\text{OH})_2$
- (3) $(\text{CH}_3)_2\text{C}(\text{OH})\text{CN}$, $(\text{CH}_3)_2\text{CHCOOH}$
- (4) $(\text{CH}_3)_2\text{C}(\text{OH})\text{CNH}_2$, $(\text{CH}_3)_2\text{C} = \text{O}$

2. By which of the following sequence of steps the alcohol $\text{RCH}_2\text{CH}_2\text{OH}(\text{X})$ can be converted into $\text{RCH}_2\text{CH}_2\text{COOH}(\text{Y})$?

- (1) $\text{X} \xrightarrow{\text{PBr}_3} \xrightarrow{\text{KCN}} \xrightarrow{\text{H}_3\text{O}^+} \text{Y}$
- (2) $\text{X} \xrightarrow{\text{PBr}_3} \xrightarrow{\text{KCN}} \xrightarrow{\text{H}_2/\text{Pt}} \text{Y}$
- (3) $\text{X} \xrightarrow{\text{KCN}} \xrightarrow{\text{H}_3\text{O}^+} \text{Y}$
- (4) $\text{X} \xrightarrow{\text{HCN}} \xrightarrow{\text{PBr}_3} \xrightarrow{\text{H}_3\text{O}^+} \text{Y}$

3. $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{PBr}_3} (\text{A}) \xrightarrow{\text{KCN}} (\text{B}) \xrightarrow{\text{H}_3\text{O}^+} (\text{C})$

In the above sequence of reactions the end product (C) is:

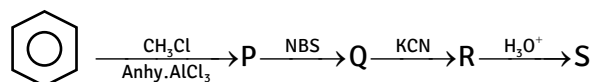
- (1) $\begin{array}{c} \text{CH}_2\text{CH}_2\text{CN} \\ | \\ \text{CH}_2\text{CH}_2\text{CN} \end{array}$ (2) $\begin{array}{c} \text{CH}_2\text{CN} \\ | \\ \text{CH}_2\text{CN} \end{array}$
- (3) $\begin{array}{c} \text{CH}_2\text{COOH} \\ | \\ \text{CH}_2\text{COOH} \end{array}$ (4) $\begin{array}{c} \text{CH}_2\text{CH}_2\text{-NH}_2 \\ | \\ \text{CH}_2\text{CH}_2\text{-NH}_2 \end{array}$

4. $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow{\text{Hg}^{2+}, \text{H}_2\text{SO}_4} \text{A} \xrightarrow[\text{H}_3\text{O}^+]{\text{HCN}} \text{B}$

The end product 'B' will be:

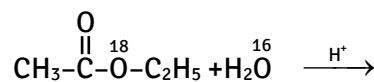
- (1) $\begin{array}{c} \text{OH} \\ | \\ \text{C} \\ | \\ \text{CN} \end{array}$ (2) $\begin{array}{c} \text{O} \\ || \\ \text{HO}-\text{C}-\text{CH}_3 \\ | \\ \text{OH} \end{array}$
- (3) $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{OH} \\ | \\ \text{OH} \end{array}$ (4) $\begin{array}{c} \text{CN} \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{OH} \end{array}$

5. The final product S in the following reaction sequence is



- (1)
- (2)
- (3)
- (4)

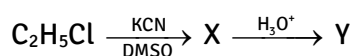
6. Consider the following reaction.



The products formed in the reaction are

- (1) $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{16}{\text{OH}}$ and $\text{C}_2\text{H}_5\overset{18}{\text{OH}}$
- (2) $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{16}{\text{OH}}$ and $\text{C}_2\text{H}_5\overset{16}{\text{OH}}$
- (3) $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{18}{\text{OH}}$ and $\text{C}_2\text{H}_5\overset{16}{\text{OH}}$
- (4) $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{18}{\text{OH}}$ and $\text{C}_2\text{H}_5\overset{18}{\text{OH}}$

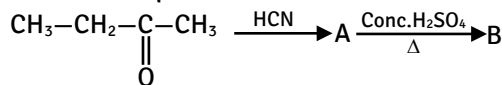
7. Consider the following sequence of reaction:



The products (X) and (Y) are, respectively:

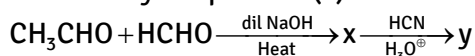
- (1) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$
- (2) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{CONH}_2$
- (3) $\text{C}_2\text{H}_5\text{NC}$ and $\text{C}_2\text{H}_5\text{NHCH}_3$
- (4) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{COOH}$

8. The Final Product 'B' in the following reaction sequence



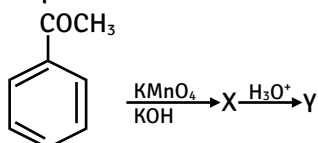
- (1) $\text{CH}_3-\text{CH}=\overset{\text{CH}_3}{\text{C}}-\text{COOH}$
 (2) $\text{CH}_3-\text{CH}=\overset{\text{CH}_3}{\text{C}}-\text{CN}$
 (3) $\text{CH}_3-\text{CH}=\overset{\text{CH}_3}{\text{C}}-\text{CONH}_2$
 (4) $\text{CH}_3-\text{CH}_2-\overset{\text{OH}}{\underset{\text{CH}_3}{\text{C}}}-\text{COOH}$

9. Study the following sequence of reactions and identify the product (Y).



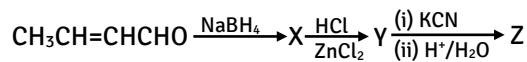
- (1) $\text{CH}_2=\text{CH}-\overset{\text{OH}}{\text{CH}}-\text{COOH}$ (2) $\text{CH}_3-\overset{\text{CN}}{\underset{\text{OH}}{\text{C}}}-\text{COOH}$
 (3) $\text{CH}_3\text{CH}_2-\overset{\text{OH}}{\text{CH}}-\text{COOH}$ (4) $\text{CH}_2=\text{CH}-\overset{\text{CN}}{\text{CH}}-\text{COOH}$

10. Identify (X) and (Y) in the given reaction sequence.



- (1) X = $\text{C}_6\text{H}_5\text{COOH}$, Y = $\text{C}_6\text{H}_5\text{CHO}$
 (2) X = $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$, Y = $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$
 (3) X = $\text{C}_6\text{H}_5\text{COOK}$, Y = $\text{C}_6\text{H}_5\text{COOH}$
 (4) X = $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$, Y = $\text{C}_6\text{H}_5\text{CHO}$

11. The end product (Z) in the given sequence of reactions is



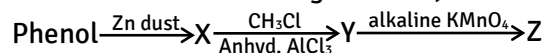
- (1) $\text{CH}_3\text{CH}=\text{CHCH}_2\text{COOH}$
 (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (3) $\text{CH}_3\text{CH}=\text{CHCOOH}$
 (4) $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{COOH}$

12. $\text{C}_6\text{H}_4(\text{C})(\text{B}) \xrightarrow[\text{(iii) H}^+]{\text{(i) Mg(1 Mole)/THF, (ii) CO}_2}$ Product (major)

Product is:-

- (1) $\text{C}_6\text{H}_4(\text{COOH})_2$ (2) $\text{C}_6\text{H}_4(\text{Cl})(\text{COOH})$
 (3) $\text{C}_6\text{H}_3(\text{Br})(\text{COOH})$ (4) $\text{C}_6\text{H}_5\text{Cl}$

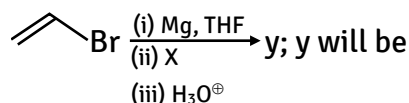
13. Consider the following reaction,



The product Z is

- (1) benzaldehyde (2) benzoic acid
 (3) benzene (4) toluene.

14. $\text{BaCO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{X}(\text{gas})$



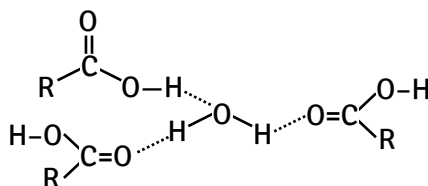
- (1) $\text{CH}_3\text{CH}=\text{CHCOOH}$ (2) $\text{CH}_3\text{CH}=\text{CHCOH}$
 (3) $\text{CH}_3\text{CH}=\text{CHCOBr}$ (4) $\text{CH}_3\text{CH}=\text{CHCHO}$

15. $\text{CH}_3-\text{CH}_2-\text{C}\equiv\text{CH} \xrightarrow[\text{(ii) H}_2\text{O}_2/\text{OH}^-]{\text{(i) BH}_3/\text{THF}} \text{(A)} \xrightarrow{\text{CrO}_3/\text{H}_2\text{SO}_4} \text{(B)}; \text{final product 'B' is-}$

- (1) $\text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3$ (2) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{COOH}$
 (3) $\text{CH}_3\text{CH}_2\overset{\text{OH}}{\text{CH}}-\text{CH}_3$ (4) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{CHO}$

Physical properties of carboxylic acids

- Physical state:** First 3 members are colorless liquids with pungent smell, next 6 members are only liquids with unpleasant odour. Higher acids are colorless waxy solids. Benzoic acid and its homologues are colorless solids.
- Solubility:** Among aliphatic acids first four members are very soluble in water due to the formation of H-bond with water. The solubility decreases with increasing number of carbon atoms. Higher carboxylic acids are practically insoluble in water due to the increased hydrophobic interaction of hydrocarbon part. Benzoic acid, the simplest aromatic carboxylic acid is nearly insoluble in cold water.



- Boiling Points:** boiling points of carboxylic acids are higher than corresponding aldehydes, ketones, ethers or even alcohols of comparable molecular masses.

Compound	CH ₃ COOH	CH ₃ CH ₂ CH ₂ OH	CH ₃ CH ₂ CHO	CH ₃ COCH ₃	CH ₃ CH ₂ CH ₂ CH ₃
Boiling Point (K)	390	370	322	329	309

This is due to more extensive association of carboxylic acid molecules through intermolecular H-bonding. The H-bonds are not broken completely even in the vapour phase. In fact, most carboxylic acids exist as dimer in the vapour phase or in the aprotic solvent



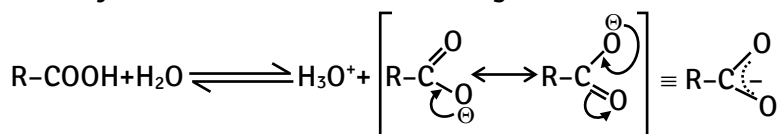
Dimer In vapour phase or in aprotic solvent

Chemical reactions of carboxylic acids

Chemical properties of carboxylic acids can be discussed under the following heads:

- Reactions involving cleavage of O-H bond i.e. acidic nature**

Carboxylic acids dissociate in water to give resonance stabilized carboxylate ion and hydronium ion.



For the above reaction

$$K_{\text{eq}} = \frac{[\text{H}_3\text{O}^+][\text{RCOO}^-]}{[\text{H}_2\text{O}][\text{RCOOH}]}$$

$$K_a = K_{\text{eq}} [\text{H}_2\text{O}] = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}]}$$

Where K_a = acid dissociation constant and

K_{eq} = Equilibrium constant and

$\text{p}K_a = -\log K_a$

Generally comparative acidic strength is studied by comparing their $\text{p}K_a$ values, higher the $\text{p}K_a$, weaker is the carboxylic acid and vice versa.

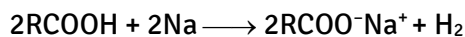
Strong acids have $pK_a < 1$

Moderately strong acids have pK_a values between 1 to 5.

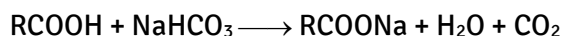
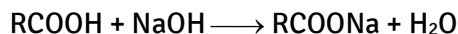
Weak acids have pK_a value between 5 to 15.

Extremely weak acids have $pK_a > 15$

Carboxylic acid as like alcohols react with electropositive metals, evolve hydrogen gas and form salts with alkalis as like phenols. Carboxylic acids can also react with weaker bases such as carbonates and bicarbonates to evolve CO_2 which phenols cannot. This reaction is used to detect presence of carboxyl group in an organic compound.



Sodium carboxylate



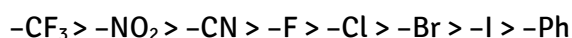
Compound	HCl	CF_3COOH	C_6H_5COOH	CH_3COOH	C_6H_5OH	C_2H_5OH
pK_a values	-7	0.23	4.19	4.76	10	16

Carboxylic acids are more acidic than phenols as carboxylate ion is stabilized by two equivalent resonating structures in which -ve charge is at the more electronegative oxygen atom. In comparison although phenoxide ion is also resonance stabilised but resonating structures are non equivalent in which -ve charge is at less electronegative carbon atom, hence resonating structures of phenoxide are less important than that of carboxylate ions.

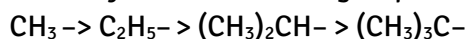
Effect of substituents on acidity of carboxylic acid:

Any substituent which can increase stability of conjugate base of carboxylic acid i.e., carboxylate ion will increase its acidity and a substituent decreasing the stability of carboxylate ion will decrease the acidity of carboxylic acid. It means electron withdrawing group i.e., EWG will increase and electron donating group i.e., EDG will decrease the acidity of carboxylic acid.

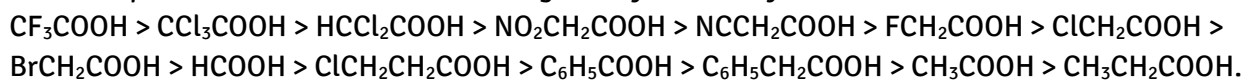
Therefore the effect of the various substituents in increasing acidity order must be



Similarly effect of various groups in decreasing acidity order is



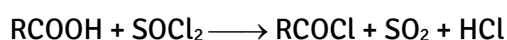
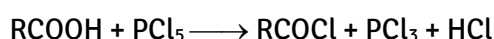
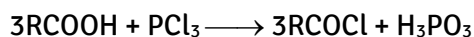
Based on pK_a values the order of decreasing acidity of carboxylic acid are as follows



2. Reactions involving cleavage of C–OH group

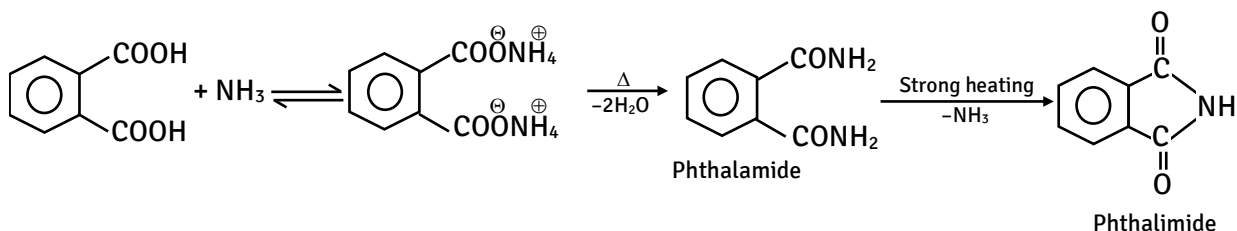
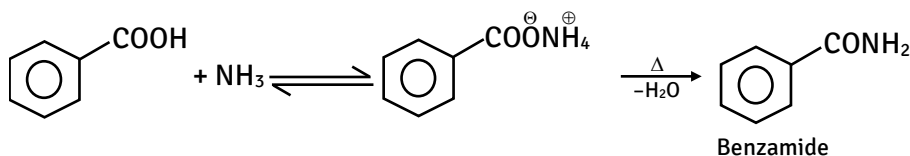
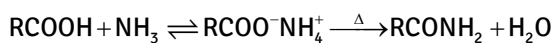
–OH group of carboxylic acids can be replaced by many group such as –Cl, –NH₂; –OR and $-O-\overset{\overset{O}{\parallel}}{C}-R'$ to form acid derivatives i.e., acid halides, acid amides, ester and acid anhydrides.

(a) Reaction with PCl_3 , PCl_5 or $SOCl_2$ forming acid chlorides:

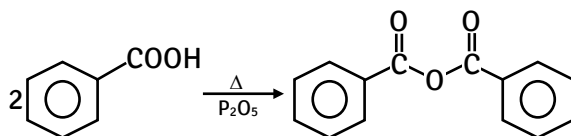
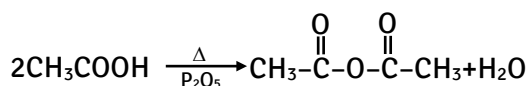
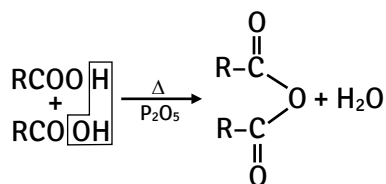


Uses of $SOCl_2$ is preferable over the use of PCl_3 and PCl_5 , because in this reaction the other two products obtained are gaseous i.e., SO_2 and HCl , which escape itself from the container, leaving behind the acid halide.

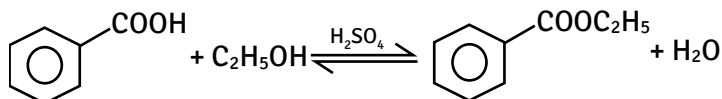
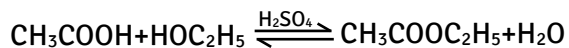
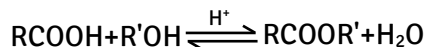
(b) Reaction with NH_3 : Reaction of carboxylic acid with NH_3 forms ammonium salts first which form amide on heating.



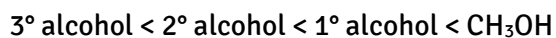
(c) Formation of anhydride: Carboxylic acid on heating in presence of strong dehydrating agent i.e., P_2O_5 or conc. H_2SO_4 give corresponding acid anhydride



(d) Esterification: Reaction of carboxylic acid with alcohols or phenols in the presence of mineral acid i.e., conc. H_2SO_4 or HCl forms esters.



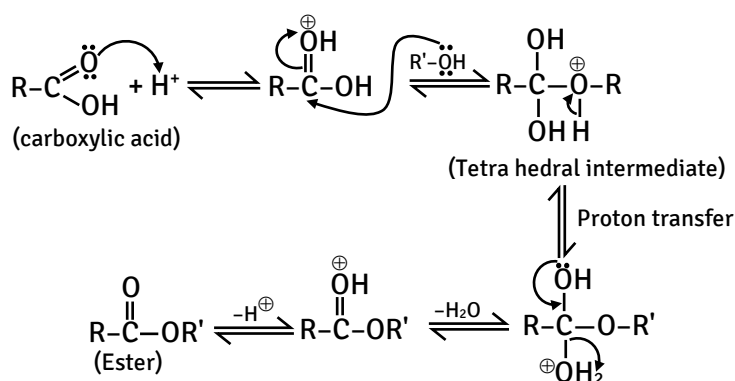
Reactivity of alcohols toward esterification is



and reactivity of carboxylic acids towards esterification is

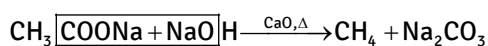
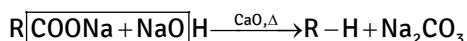


Mechanism of esterification: It is a nucleophilic acyl substitution which is acid catalysed reaction, which can be studied in following steps.

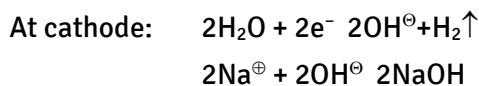
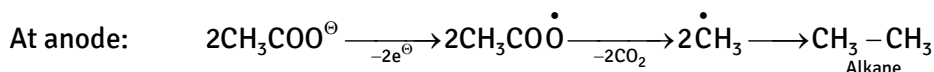
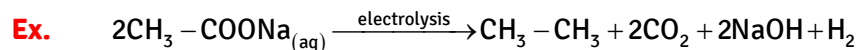


3. Reactions involving $-COOH$ group

(a) **Sodalime decarboxylation:** Carboxylic acids lose CO_2 to form hydrocarbon on heating sodium or potassium salts of carboxylic acid with sodalime ($NaOH + CaO$).

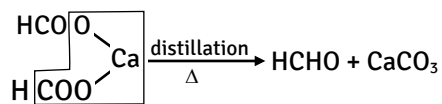


(b) **Kolbe's electrolysis:** Electrolysis of aq. Solution of sodium or potassium salts of carboxylic acids leads to decarboxylation to give alkanes.

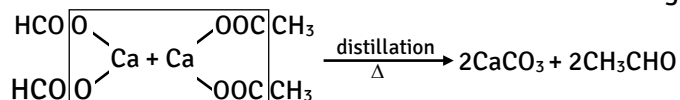


(c) **distillation:** Heating of calcium salts of carboxylic acids forms aldehydes and ketones by distillation.

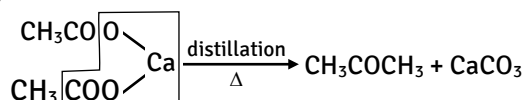
(i) Distillation of calcium formate gives formaldehyde



(ii) Distillation of calcium formate with calcium acetate gives acetaldehyde.

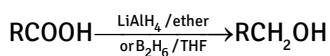


(iii) Distillation of calcium acetate forms acetone.



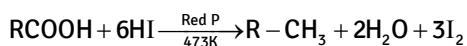
(d) Reduction:

(i) Carboxylic acids are reduced to primary alcohols by LiAlH_4 or B_2H_6 in T.H.F.

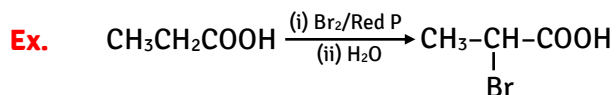
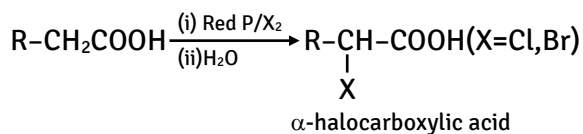


- Diborane does not reduce functional groups such as ester, nitro, halo etc.
- NaBH_4 does not reduce carboxyl group.

(ii) Reduction of carboxylic acids with Red P and HI gives alkane.

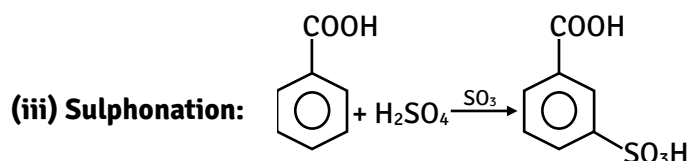
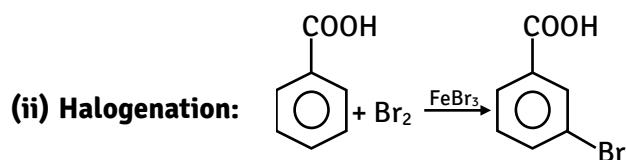
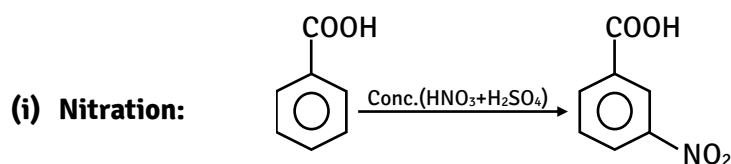
**4. Substitution reaction in the hydrocarbon part:**

(a) Halogenation: Carboxylic acids having atleast one or more α -H atoms react with Cl_2 or Br_2 in the presence of a small amount of Red P to give α -halocarboxylic acids. The reaction is known as Hell-Volhard-Zelinsky reaction

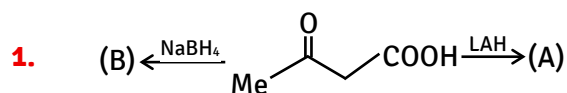


(b) Ring substitution in aromatic acids: $-\text{COOH}$ group in aromatic carboxylic acids is a ring deactivating & meta directing group. They do not undergo Friedel-Craft reaction, because carboxyl group is deactivating and catalyst AlCl_3 (Lewis acid) gets bonded to carboxyl group.

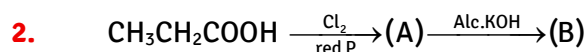
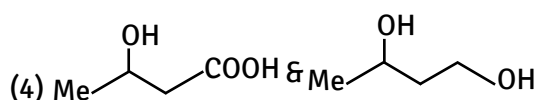
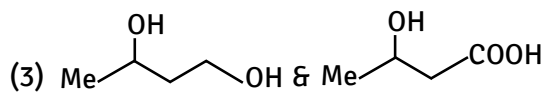
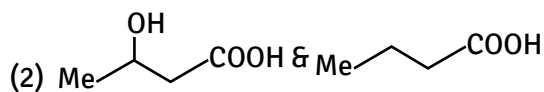
Some common electrophilic substitution reactions of benzoic acid are as following:



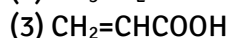
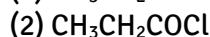
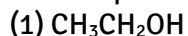
Exercise 1.4



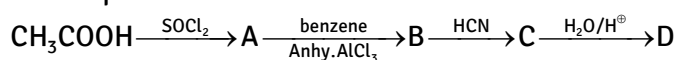
The products (A) and (B) are :



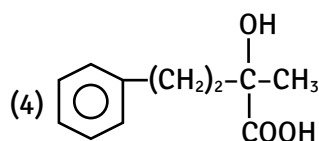
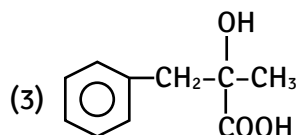
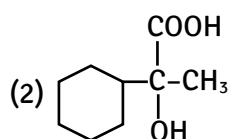
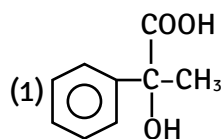
The compound (B) is :



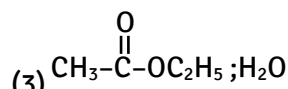
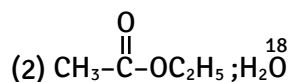
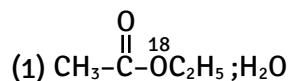
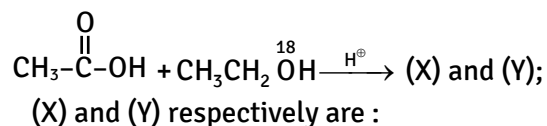
3. In a set of reactions acetic acid yielded a product D.



The structure of D would be

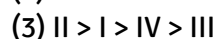
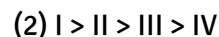


4. Let us consider an esterification of isotopically labelled carboxylic acid :



(4) both (1) and (2)

5. The rate of esterification of HCOOH (I), CH_3COOH (II), $(\text{CH}_3)_2\text{CHCOOH}$ (III) and $(\text{CH}_3)_3\text{CCOOH}$ (IV) with ethanol follows in the order :



6. Ethanol on heating with acetic acid in the presence of a few drops of sulphuric acid gives:

(1) Oil of wintergreen

(2) Oil of mustard

(3) Fruity smell

(4) Oil of bitter almonds

7. $X \xleftarrow[\text{HI}]{\text{red P}} \text{CH}_3\text{COOH} \xrightarrow{\text{LiAlH}_4} Y$. What does not true for X and Y -

(1) X is hydrocarbon of general formula $\text{C}_n\text{H}_{2n+2}$ while Y belong to alkanol

(2) X can be obtained by reducing $\text{CH}_3\text{CH}_2\text{Cl}$ while Y by its hydrolysis

(3) X gives positive litmus test but Y does not

(4) X and Y both belong to different homologous series

8. Acetic acid exists as dimer in C_6H_6 due to:

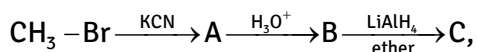
(1) Condensation reaction

(2) Hydrogen bonding

(3) Presence of a carboxyl group

(4) Presence of hydrogen at α -carbon

9. In the following sequence of reactions



the end product (C) is :

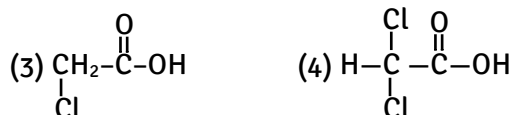
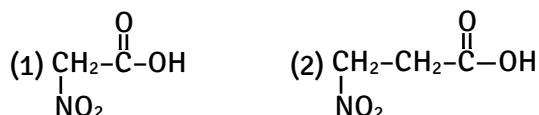
- (1) Acetaldehyde (2) Ethyl alcohol
(3) Acetone (4) Methane

10. $\text{C}_6\text{H}_5\text{N}_2\text{Cl} \xrightarrow{\text{CuCN}} \text{A} \xrightarrow{\text{H}_2\text{O} / \text{H}^+} \text{B} \xrightarrow[\Delta]{\text{NH}_3} \text{C};$

final product is

- (1) $\text{C}_6\text{H}_5 - \text{NH}_2$ (2) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
(3) $\text{C}_6\text{H}_5 - \text{CONH}_2$ (4) $\text{C}_6\text{H}_5 - \text{COOH}$

11. Which of the following acid has Lowest Pka value?



12. $\text{C}_6\text{H}_5\text{CH}_2 - \text{OH} \xrightarrow{\text{CrO}_3 - \text{H}_2\text{SO}_4} (\text{A}) \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) B}_2\text{H}_6} (\text{B})$

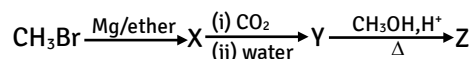
In the above given sequence reaction, final product 'B' is

- (1) $\text{C}_6\text{H}_5 - \text{CHO}$ (2) $\text{C}_6\text{H}_5 - \text{COOH}$
(3) $\text{C}_6\text{H}_5 - \text{CH}_3$ (4) $\text{C}_6\text{H}_5 - \text{CH}_2 - \text{OH}$

13. Hell Volhard Zelinsky reaction involves

- (1) Halogenation of carbonyl compound at $\alpha - \text{C}$
(2) Halogenation of carboxylic ester at $\alpha - \text{C}$
(3) Halogenation of alcohol at $\alpha - \text{C}$
(4) Halogenation of carboxylic acid at $\alpha - \text{C}$

14. Identify the compounds (X), (Y) and (Z) in the following reaction:



- (1) (X) = CH_3MgBr , (Y) = CH_3COOH ,
(Z) = $\text{CH}_3\text{COOCH}_3$
(2) (X) = $\text{CH}_3\text{CH}_2\text{Br}$, (Y) = $\text{CH}_3\text{CH}_2\text{OH}$,
(Z) = $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
(3) (X) = $\text{CH}_3\text{CH}_2\text{MgBr}$, (Y) = $\text{CH}_3\text{CH}_2\text{COOH}$,
(Z) = $\text{CH}_3\text{CH}_2\text{COCH}_3$
(4) (X) = CH_3MgBr , (Y) = $\text{CH}_3\text{CH}_2\text{COOH}$
(Z) = $\text{CH}_3\text{CH}_2\text{COOCH}_3$

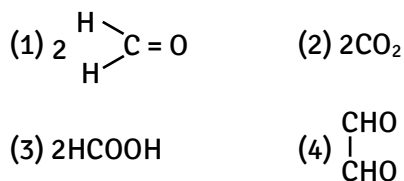
15. $\text{CH}_3 - \text{COOH} \xrightarrow[\Delta]{\text{Reagent}} \text{CH}_3 - \text{CH}_2 - \text{OH}$

The reagent of the given reaction will be :-

- (1) $\text{LiAlH}_4 / \text{H}_3\text{O}^+$
(2) $\text{B}_2\text{H}_6 / \text{H}_3\text{O}^+$
(3) Both (1) and (2)
(4) $\text{NaBH}_4 / \text{H}_3\text{O}^+$

Exercise 2

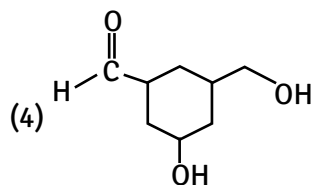
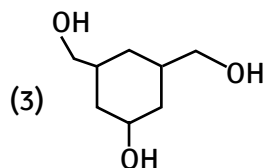
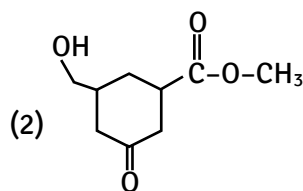
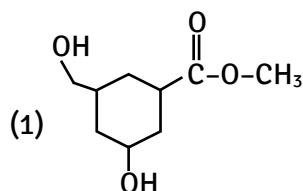
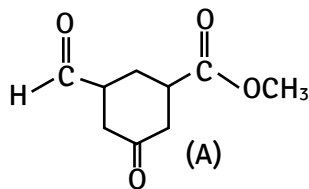
1. $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$ on heating with periodic acid gives :-



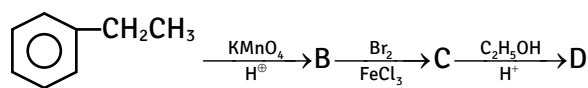
2. Which of the following compounds would undergo the Cannizzaro's reaction-

- (1) Acetaldehyde
(2) Benzaldehyde
(3) Propionaldehyde
(4) Anisole

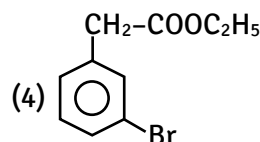
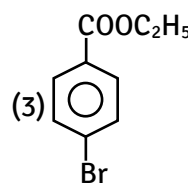
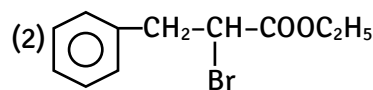
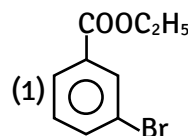
3. Predict the product when given compound reacts with LiAlH_4 :



4. In a set of reactions, ethyl benzene yielded a product D



'D' would be:



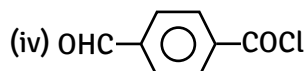
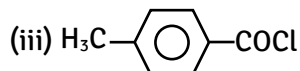
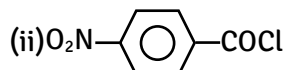
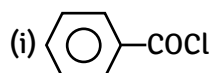
5. In a reaction, (A) $\xrightarrow{\text{NH}_3}$ (B) $\xrightarrow{\text{Heat}}$ (C) $\xrightarrow{\text{P}_2\text{O}_5}$ $\text{C}_2\text{H}_5\text{CN}$; (A), (B) and (C) are :-

- (1) CH_3COOH , $\text{CH}_3\text{COONH}_4$ and CH_3CONH_2
(2) CH_3COCl , CH_3CONH_2 and $\text{CH}_3\text{COONH}_4$
(3) $\text{C}_2\text{H}_5\text{COOH}$, $\text{C}_2\text{H}_5\text{COONH}_4$ and $\text{C}_2\text{H}_5\text{CONH}_2$
(4) $\text{C}_2\text{H}_5\text{CONH}_4$, $\text{C}_2\text{H}_5\text{CONH}_2$ and $\text{C}_2\text{H}_5\text{COOH}$

6. Formic acid and formaldehyde can be distinguished by treating with -

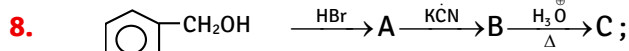
- (1) Benedict's solution
(2) Tollen's reagent
(3) Fehling's solution
(4) NaHCO_3

7. Consider the following compounds :



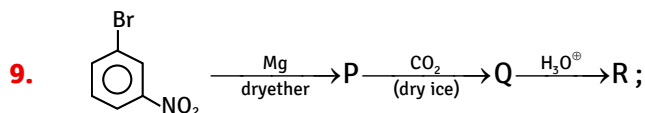
The correct order of reactivity towards hydrolysis is:

- (1) $i > ii > iii > iv$
- (2) $iv > ii > i > iii$
- (3) $ii > iv > i > iii$
- (4) $ii > iv > iii > i$

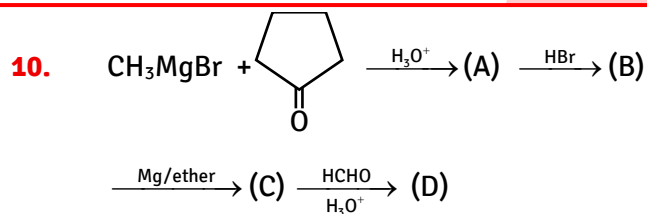
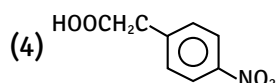
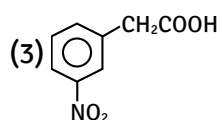
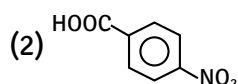
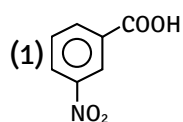


C is—

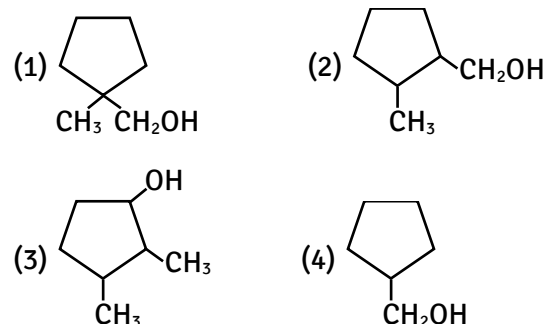
- (1) Phenyl ethanoic acid
- (2) Benzoic acid
- (3) Benzyl amine
- (4) Aniline



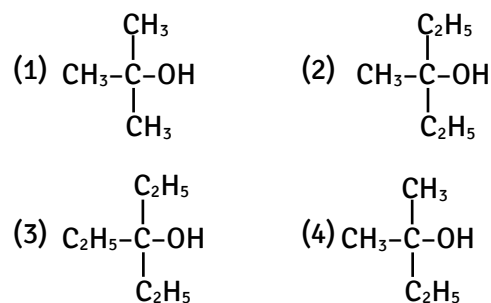
R is—



The end product (D) is :



11. Methyl acetate on treating with excess of $\text{C}_2\text{H}_5\text{MgBr}$ produces :

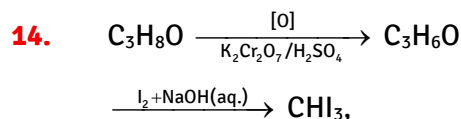


12. Which gives a ketone on treating with a Grignard's reagent—

- (1) Formaldehyde
- (2) Ethyl alcohol
- (3) Methyl cyanide
- (4) Methyl iodide

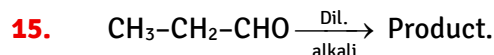
13. Benzaldehyde and formaldehyde differ from acetaldehyde in their reaction with—

- (1) NaOH
- (2) HCN
- (3) 2,4-Dinitrophenyl hydrazine
- (4) Semi carbazide



In this reaction the first compound is –

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (2) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$
 (3) $\text{CH}_3\text{OCH}_2\text{CH}_3$ (4) $\text{CH}_3\text{CH}_2\text{CHO}$

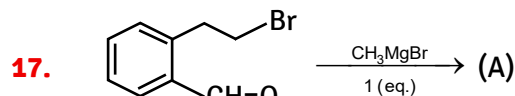


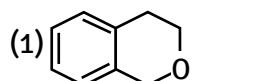
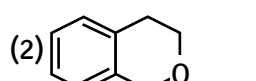
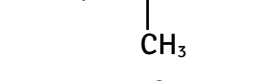
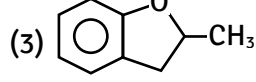
The product in the above reaction is–

- (1) $\text{CH}_3-\text{CH}_2\text{COOH}$
 (2) $\text{CH}_3-\text{CH}_2-\text{CH}_2\text{OH}$
 (3) $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CHO}$
 (4) $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\overset{\text{CH}_3}{\text{CH}}-\text{CHO}$

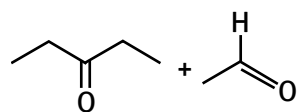
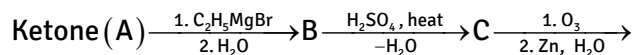
16. An organic compound $\text{C}_3\text{H}_6\text{O}$ does not give a precipitate with 2,4-dinitrophenyl hydrazine reagent also does not react with metallic sodium. It could be:

- (1) $\text{CH}_3-\text{CH}_2-\text{CHO}$ (2) $\text{CH}_2=\text{CH}-\text{OCH}_3$
 (3) $\text{CH}_3-\text{COCH}_3$ (4) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$

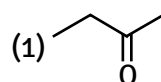
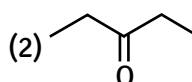
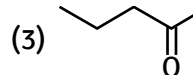
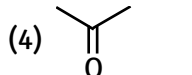


- (1) 
 (2) 
 (3) 
 (4) 

18. Consider the following sequence of reactions:



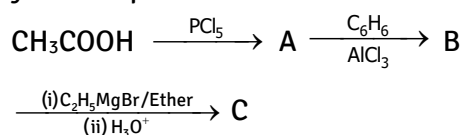
The ketone (A) is :

- (1)  (2) 
 (3)  (4) 

19. Which of the following reactions will not result in the formation of carbon-carbon bonds?

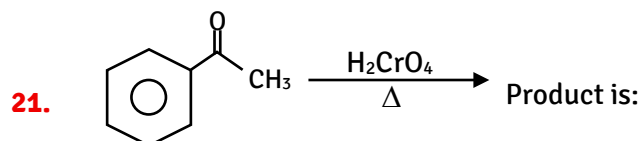
- (1) Friedel-Crafts acylation
 (2) Reimer-Tieman reaction
 (3) Cannizzaro reaction
 (4) Wurtz reaction

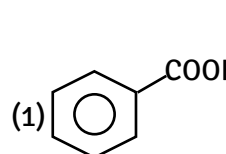
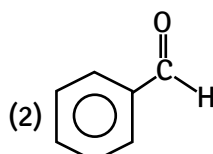
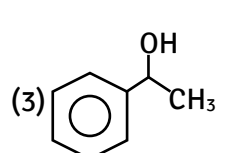
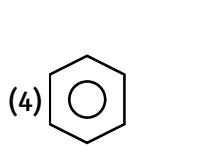
20. In a set of the given reactions, acetic acid yielded a product C.



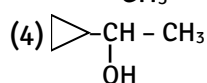
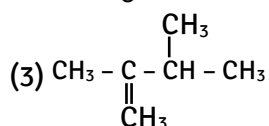
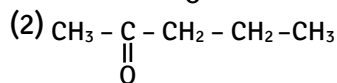
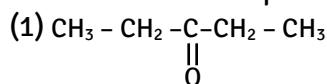
product C would be:

- (1) $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$
 (2) $\text{CH}_3\text{COC}_6\text{H}_5$
 (3) $\text{CH}_3\text{CH}(\text{OH})\text{C}_6\text{H}_5$
 (4) $\text{CH}_3-\underset{\text{C}_2\text{H}_5}{\text{C}}(\text{OH})\text{C}_6\text{H}_5$

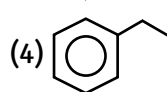
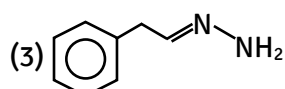
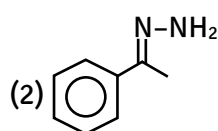
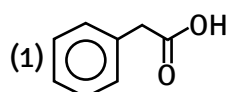
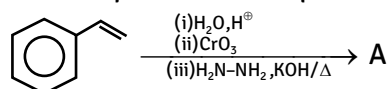


- (1)  (2) 
 (3)  (4) 

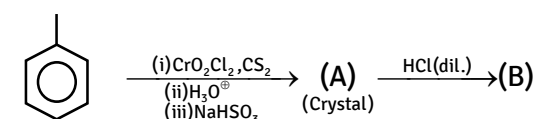
22. An organic compound contains 69.77% carbon, 11.63% hydrogen and rest oxygen. The molecular mass of the compound is 86. It does not reduce Tollen's reagent but forms an addition compound with sodium hydrogensulphite and give positive iodoform test. On vigorous oxidation it gives ethanoic and propanoic acid. Possible structure of the compound is.



23. The final product A formed in the following multistep reaction sequence is:



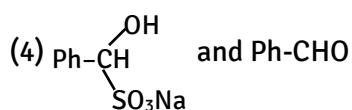
24.



Toluene
(Excess)

A and B formed in the above sequence reaction respectively are :

- (1) Ph-CHO and Ph-COOH
(2) Ph-CH(OCrOHCl₂) and Ph-CHO
(3) Ph-CHO and Ph-COONa

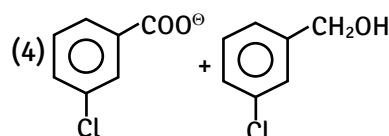
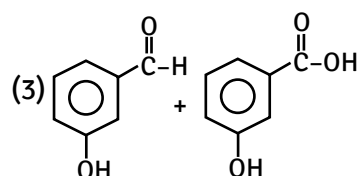
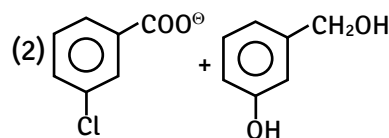
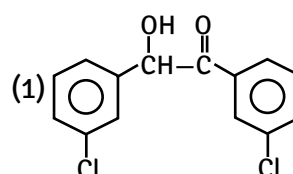


25. From the given below, number of compounds which give positive Fehling's test are.

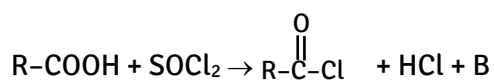
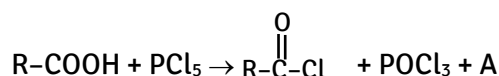
Benzaldehyde, Acetaldehyde, Acetone, acetophenone, methanal.

- (1) 2 (2) 3
(3) 4 (4) 5

26. 3-chloro benzaldehyde on treatment with 50% KOH solutions gives product

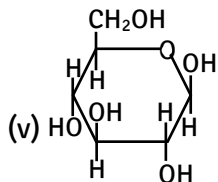
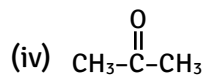
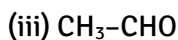
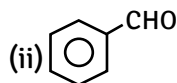
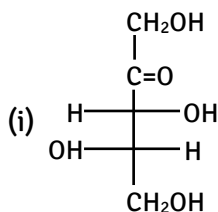


27. The product A and B obtained in the following reaction respectively are :



- (1) H₃PO₃ and SO₂
(2) HCl and SO₃
(3) H₃PO₄ and SO₂
(4) HCl and SO₂

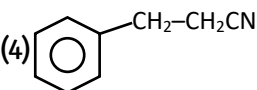
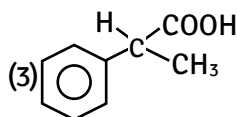
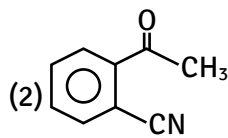
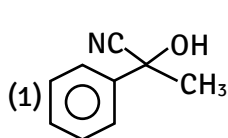
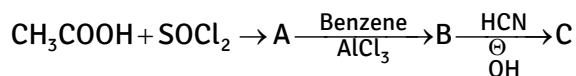
28. Positive Tollen's test is observed in which of the following compound



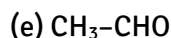
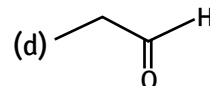
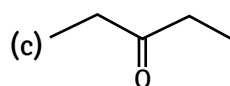
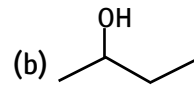
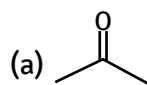
- (1) Only (ii) and (iii)
 (2) Only (i), (ii), (iii) and (v)
 (3) Only (ii), (iii) and (v)
 (4) Only (ii), (iii) and (iv)

29. An organic compound (X) with molecular formula $\text{C}_8\text{H}_8\text{O}$ give positive 2, 4-DNP test. It give a yellow precipitate on reaction with iodine and sodium hydroxide solution. (X) does not reduce Fehling's reagent but on drastic oxidation with chromic acid it gives a carboxylic acid (Y) with molecular formula $\text{C}_7\text{H}_6\text{O}_2$. (X) and (Y) respectively are
 (1) $\text{C}_6\text{H}_5\text{COCH}_3$, $\text{C}_6\text{H}_5\text{COOH}$
 (2) $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$, $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$
 (3) $\text{C}_6\text{H}_5\text{COCH}_3$, CH_3COOH
 (4) $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$, $\text{C}_6\text{H}_5\text{COOH}$

30. The structure of product 'C' formed by the following sequence of reactions is:

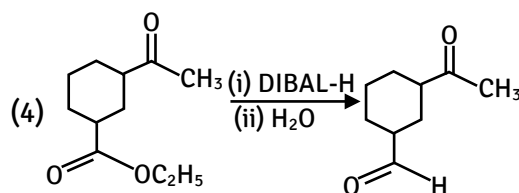
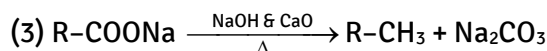
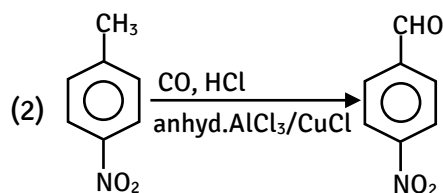
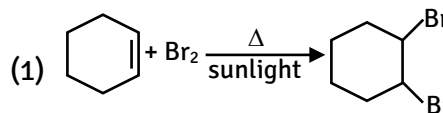


31. Which of the following compounds give yellow precipitate with I_2/NaOH ?

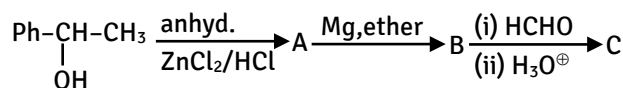


- (1) (a) and (e) only
 (2) (a) and (b) only
 (3) (a), (b) and (e) only
 (4) (a) only

32. Which of the following reaction is correct?

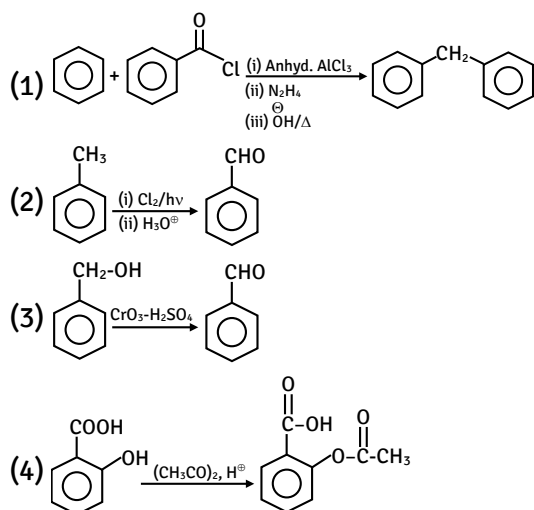


33. In the given reaction, final product is

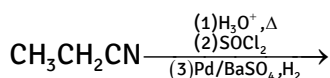


- (1) $\text{Ph-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$
 (2) $\text{Ph-CH(CH}_3\text{)-CHO}$
 (3) $\text{Ph-CH(CH}_3\text{)-CH}_3$
 (4) $\text{Ph-CH(CH}_3\text{)-CH}_2\text{OH}$

34. Which of the following reaction represent incorrect product?

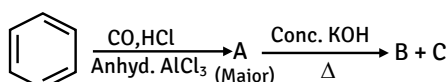


35. The major product of the following chemical reaction is:

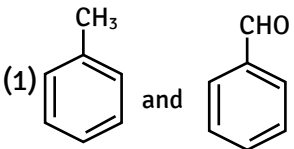
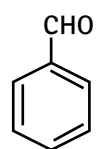
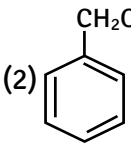
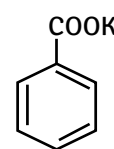
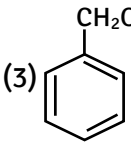
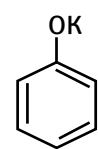
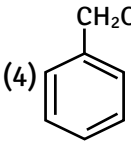
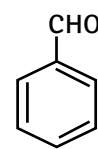


- (1) $\text{CH}_3\text{CH}_2\text{CH}_3$
 (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 (3) $(\text{CH}_3\text{CH}_2\text{CO})_2\text{O}$
 (4) $\text{CH}_3\text{CH}_2\text{CHO}$

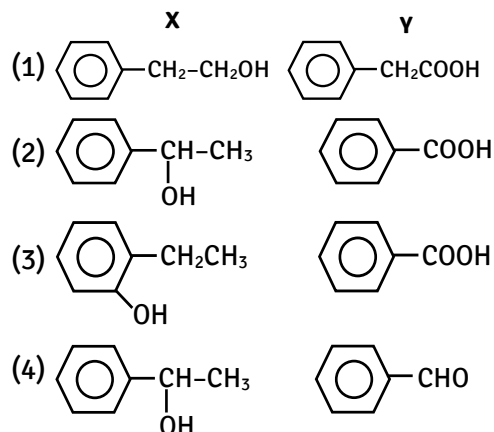
36. Consider the following reaction sequence.



Major products B and C are

- (1)  and 
- (2)  and 
- (3)  and 
- (4)  and 

37. A compound X ($\text{C}_8\text{H}_{10}\text{O}$) upon treatment with alkaline solution of iodine gives a yellow precipitate. The filtrate on acidification gives a white solid Y ($\text{C}_7\text{H}_6\text{O}_2$). Write the structures of X and Y.



38. $\text{C}_6\text{H}_5\text{CHO}$ HCHO CH_3COCH_3
 I II III

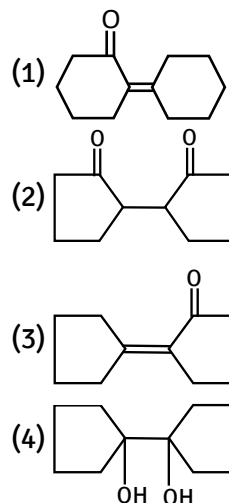
In context with above compounds, consider the following statements.

- a. Compound I reduces Tollens' reagent and Fehling reagent both.
 b. Compound II reduces Tollens' reagent but not Fehling reagent.
 c. Compound III reduces neither Tollens' reagent nor Fehling reagent.

Select the correct statement(s).

- (1) a and b only (2) a and c only
 (3) b and c only (4) c only

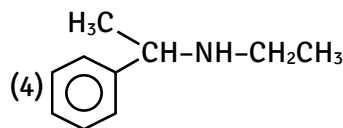
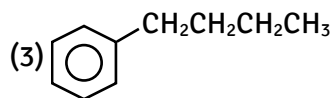
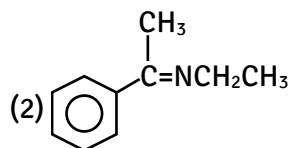
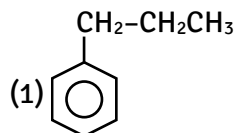
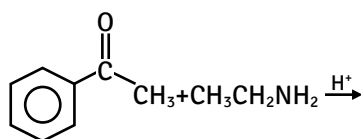
39. Which of the following product is formed when cyclopentanone undergoes aldol condensation reaction on heating with dilute alkali?



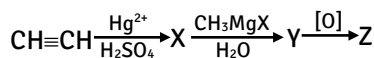
40. Alkene (X) (C_5H_{10}) on ozonolysis gives a mixture of two compounds (Y) and (Z). Compound (Y) gives positive Fehling's test and iodoform test. Compound (Z) does not give Fehling's test but give iodoform test. Compounds (X), (Y) and (Z) are

	X	Y	Z
(1)	$C_6H_5COCH_3$	CH_3CHO	CH_3COCH_3
(2)	$CH_3-CH=C(CH_3)-CH_3$	CH_3CHO	CH_3COCH_3
(3)	$CH_3CH_2CH=CH_2$	CH_3CH_2CHO	$HCHO$
(4)	$CH_3-CH=CH-CH_3$	CH_3CHO	CH_3CHO

41. Find the product of the given reaction:

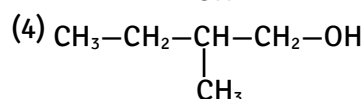
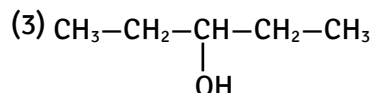
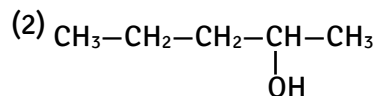
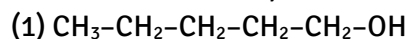


42. In the following sequence of reaction, the final product (Z) is

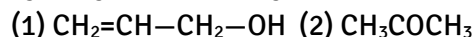


- (1) ethanal
(2) propan-2-ol
(3) propanone
(4) propan-1-ol

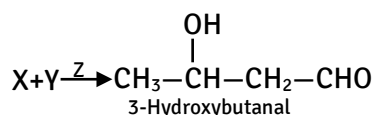
43. A compound 'A' having the molecular formula $C_5H_{12}O$, on oxidation gives a compound 'B' with molecular formula $C_5H_{10}O$. Compound 'B' gave a 2,4-dinitrophenylhydrazine derivative but did not give haloform test or silver mirror test. The structure of compound 'A' is



44. An organic compound of molecular formula C_3H_6O did not give a silver mirror with Tollens' reagent but give an oxime with hydroxylamine. It may be



45. Identify (X), (Y) and (Z) in the given reaction.



	X	Y	Z
(1)	HCHO	CH_3CHO	KOH
(2)	CH_3CHO	CH_3CHO	NaOH
(3)	CH_3CH_2OH	HCHO	H_2SO_4
(4)	CH_3CH_2CHO	HCHO	Dry ether

46.
$$\begin{array}{c} CHO \\ | \\ CHO \end{array} \xrightarrow{NaOH} X$$

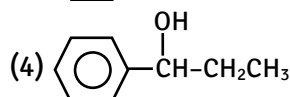
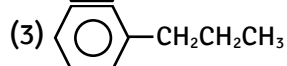
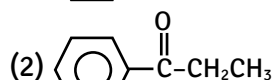
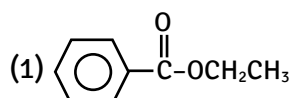
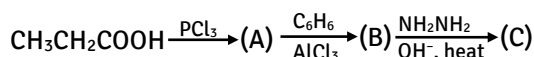
The product (X) will be

- (1) $\begin{array}{c} CH_2ONa \\ | \\ COONa \end{array}$ (2) $\begin{array}{c} COOH \\ | \\ COOH \end{array}$
(3) $\begin{array}{c} CH_2OH \\ | \\ COONa \end{array}$ (4) $\begin{array}{c} CH_2OH \\ | \\ CH_2OH \end{array}$

47. An organic compound (X) with molecular formula $C_9H_{10}O$ gives positive 2,4-DNP and Tollens' tests. It undergoes Cannizzaro reaction and on vigorous oxidation it gives 1,4-benzenedicarboxylic acid. Compound (X) is

(1) benzaldehyde
(2) o-methylbenzaldehyde
(3) p-ethylbenzaldehyde
(4) 2, 2-dimethylhexanal.

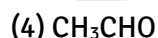
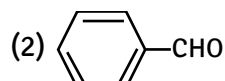
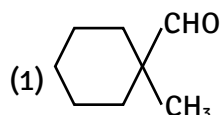
48. The final product (C) obtained in the reaction sequence is



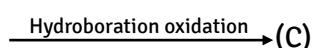
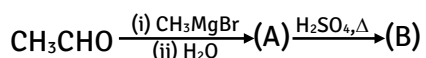
49. The reagent which does not react with both acetone and benzaldehyde is _____.

(1) sodium hydrogensulphite
(2) phenyl hydrazine
(3) Fehling's solution
(4) Grignard reagent

50. Cannizzaro's reaction is not given by _____.

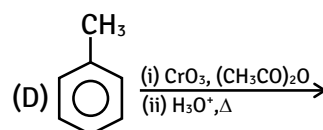
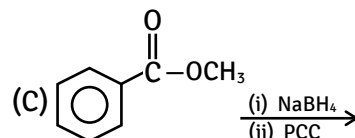
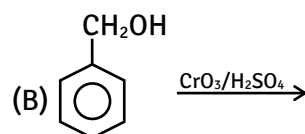
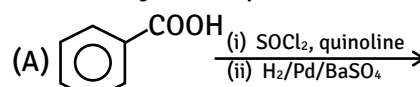


51. Compounds (A) and (C) in the following reactions are



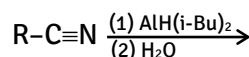
(1) identical
(2) positional isomers
(3) functional isomers
(4) optical isomers.

52. Which of the following reactions will yield benzaldehyde as a product?

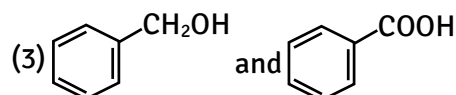
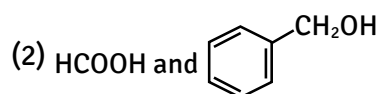
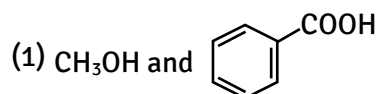
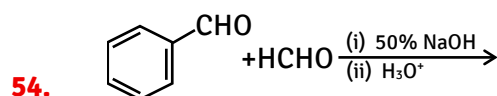


(1) (B) and (C) (2) (C) and (D)
(3) (A) and (D) (4) (A) and (C)

53. The major product of the following reaction is



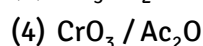
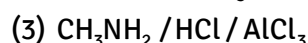
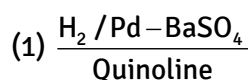
(1) RCOOH (2) RCHO
(3) RCONH₂ (4) RCH₂NH₂



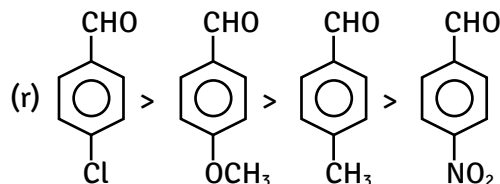
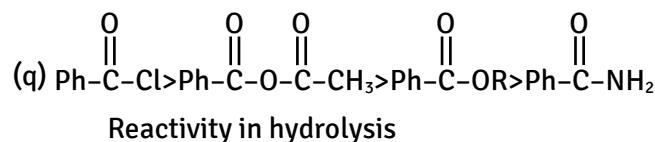
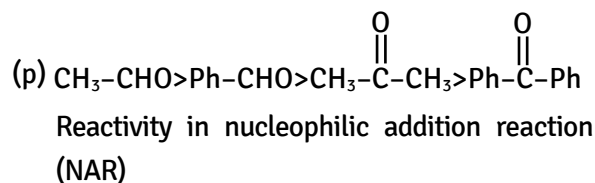
55. Cumene $\xrightarrow[\text{Boil}]{KMnO_4} A \xrightarrow{PCl_5} B \xrightarrow{C}$

Benzaldehyde

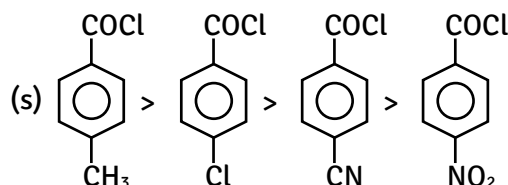
Find "C" :



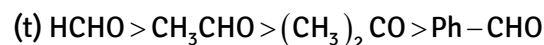
56. Which of the following is correct order:



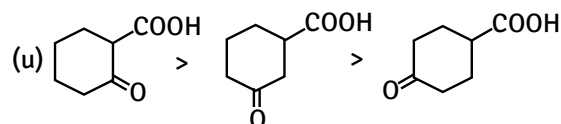
Reactivity in nucleophilic addition reaction (NAR)



Reactivity in hydrolysis



Reactivity towards Grignard reagent

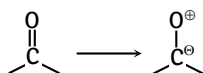


Rate of decarboxylation

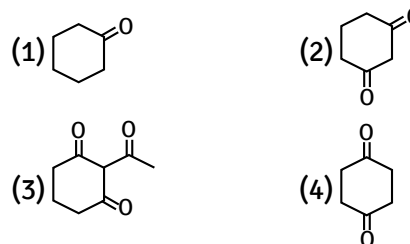
- (1) p, q (2) r, s, t
 (3) p, q, r, s (4) p, q, u

57. Which of the following statement is not true about carbonyl group :

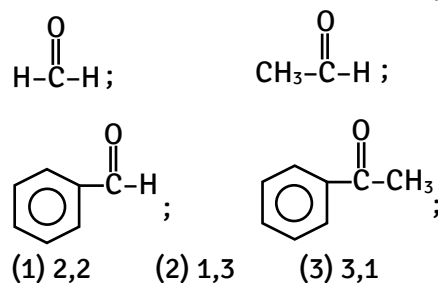
- (1) Carbon-oxygen bond is polarised due to higher electronegativity of oxygen
- (2) Carbonyl carbon is electrophilic while oxygen is nucleophilic center
- (3) Carbonyl group have more dipole moment than ether
- (4) Polarity of carbonyl group as given



58. Which of the compound have most acidic α_{H} :



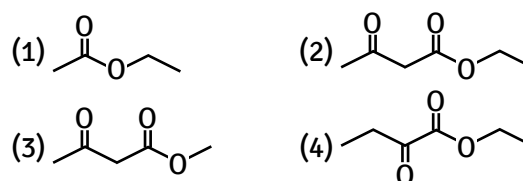
59. How many compounds gives Cannizzaro reaction and aldol condensation respectively:



60. Which reaction shows decrease in number of carbon atoms in product :

- (1) Decarboxylation reaction
- (2) HVZ reaction
- (3) Hoffmann's hypobromamide reaction
- (4) 1 & 3 both

61. $\text{A} \xrightarrow{\text{H}_3\text{O}^+} \text{B} + \text{EtOH}$; $\text{B} \xrightarrow{\Delta} \text{Acetone} + \text{CO}_2$
 A is :



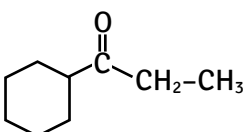
62. Correct statement for nucleophilic addition of sodium bisulphite on carbonyl compounds is:

- (1) This addition is highly sensitive to crowding, steric crowding increases, then addition decreases, so aldehydes are more reactive than ketones in this reaction
- (2) This reaction used for separation of aldehyde and ketone from a mixture containing some other compounds because addition product is a crystalline salt
- (3) This reaction can be reversed in presence of acid or base, so bisulphite addition product can be converted into corresponding carbonyl compound.
- (4) All of these

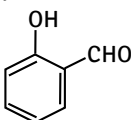
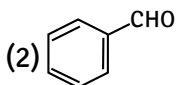
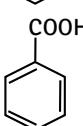
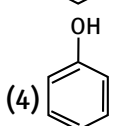
63. Which of the following compound does not form iodoform with $I_2 + NaOH$ or $NaOI$:

- (1) CH_3-CHO (2) CH_3-CH_2-CHO
 (3) CH_3-CH_2-OH (4) $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3$

64. Which of the following compound can form hydrazone and will give iodoform test but will not give Tollens' test?

- (1) 
 (2) $H_3C-\overset{\overset{O}{\parallel}}{C}-CH_2-CH_3$
 (3) $H_3C-\overset{\overset{O}{\parallel}}{C}-CH_2-CHO$
 (4) CH_3-CH_2-OH

65. What will the product, when salicylic acid obtained by Kolbe's reaction is oxidized with Zn powder?

- (1)  (2) 
 (3)  (4) 

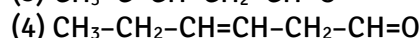
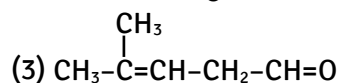
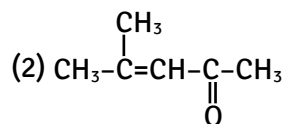
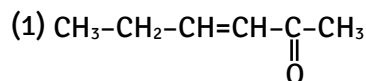
66. Which product of given reaction gives instant turbidity with Lucas reagent :-

- (1) $HCHO \xrightarrow[2. H_3O^+]{1. CH_3MgBr} \text{Product}$
 (2) $CH_3-CHO \xrightarrow[2. H_3O^+]{1. CH_3MgBr} \text{Product}$
 (3) $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3 \xrightarrow[2. H_3O^+]{1. CH_3MgBr} \text{Product}$
 (4) $CH_3-CHO \xrightarrow[2. H_3O^+]{1. (CH_3)_2CH-Br} \text{Product}$

67. Acetaldehyde and benzaldehyde can not be distinguished by -

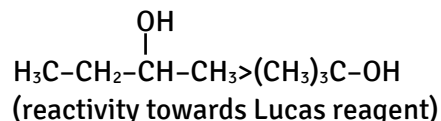
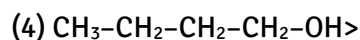
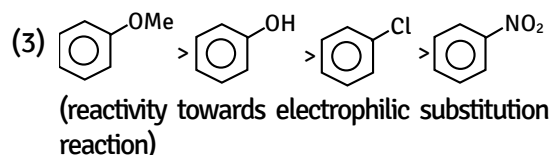
- (1) Tollens' reagent
 (2) Fehling solution
 (3) Benedict solution
 (4) Iodoform test

68. $CH_3-C \equiv CH \xrightarrow[Hg^{2+}]{dil. H_2SO_4} X \xrightarrow[\Delta]{Ba(OH)_2} Y$; Y is

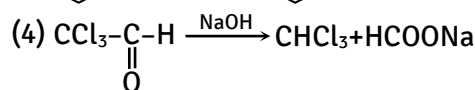
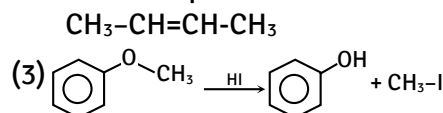
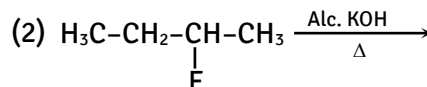
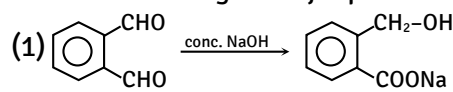


69. Which of the following is correct -

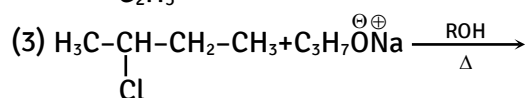
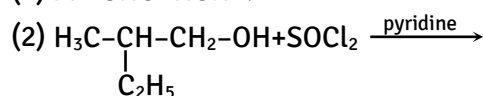
- (1) $CH_3-COCl > CH_3-COOCO-CH_3 > CH_3-COOC_2H_5 > CH_3-CONH_2$
 (reactivity towards hydrolysis)
 (2) $C_6H_5-CH=O > CH_3-CH=O > CH_3-CO-CH_3$
 (reactivity towards cyanohydrin formation)



70. Which of the following reaction is not correct according to major product



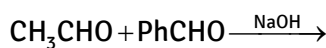
71. In which reaction racemic mixture is obtained as a product?



72. Which of the following do not give aldol reaction with dil. NaOH ?

- (1) CH_3CHO (2) CD_3CHO
(3) CH_3COCH_3 (4) PhCHO

73. How many aldol product will be obtained in the following reaction (Structural only)

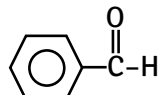


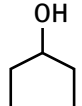
- (1) 2 (2) 3
(3) 4 (4) 1

74. $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{CN} \xrightarrow[2. \text{H}_2\text{O}]{1. \text{DIBAL-H}} ?$

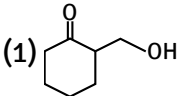
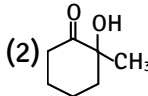
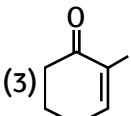
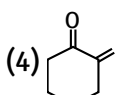
- (1) $\text{CH}_3(\text{CH}_2)_4\text{CN}$
(2) $\text{CH}_3(\text{CH}_2)_5\text{NH}_2$
(3) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{CHO}$
(4) $\text{CH}_3(\text{CH}_2)_3 - \text{CH}_2 - \text{CHO}$

75. Which of the following compound when reacts with CH_3MgBr form 3° alcohol ?

- (1) $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (2) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
(3) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (4) 

76.  $\xrightarrow{\text{PCC}}$ (A) $\xrightarrow[\text{(ii) dil. KOH}/\Delta]{\text{(i) HCHO}}$ (B)

(B) is :

- (1)  (2) 
(3)  (4) 

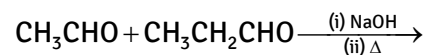
77. Alkene (X) (C_5H_{10}) on ozonolysis gives a mixture of two compounds (Y) and (Z). Compound (Y) and (Z) gives positive Fehling's test, (Z) does not give iodoform test. Compounds (X), (Y) and (Z) are:-

	X	Y	Z
(1)	$\text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{CH}_2\text{CH}_3$	HCHO	$\text{CH}_3-\underset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2\text{CH}_3$
(2)	$\text{CH}_3-\text{CH}=\underset{\text{CH}_3}{\text{C}}-\text{CH}_3$	CH_3CHO	CH_3COCH_3
(3)	$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$	$\text{CH}_3\text{CH}_2\text{CHO}$	HCHO
(4)	$\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_3$	CH_3CHO	$\text{CH}_3\text{CH}_2\text{CHO}$

78. Buta-1,3-diene was subjected to ozonolysis to prepare aldehydes. Which of the following aldehydes will be obtained during the reaction?

- (1) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array} + 2\text{HCHO}$
(2) $\text{CH}_3\text{CHO} + 2\text{HCHO}$
(3) $\text{CH}_3\text{CH}_2\text{CHO} + \text{CH}_3\text{CHO}$
(2) $2\text{CH}_3\text{CH}_2\text{CHO}$

79. How many Aldol condensation products would you expect for the given reaction ? (Exclude stereo isomers)

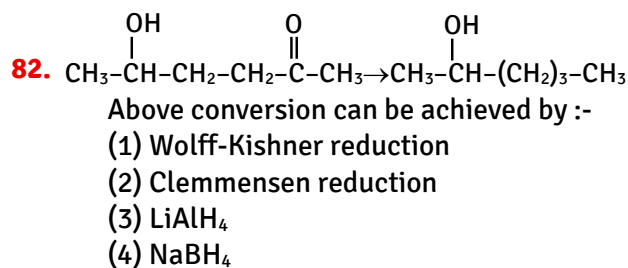
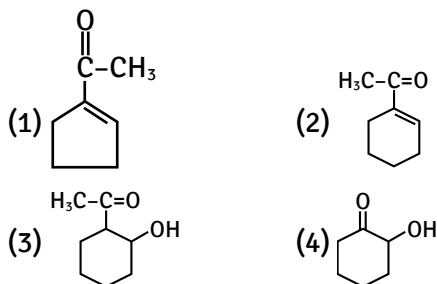
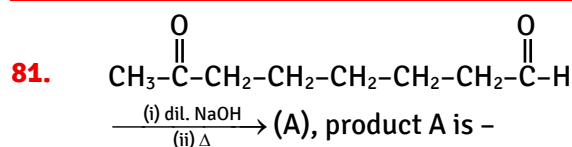


- (1) 4 (2) 3
(3) 1 (4) 2

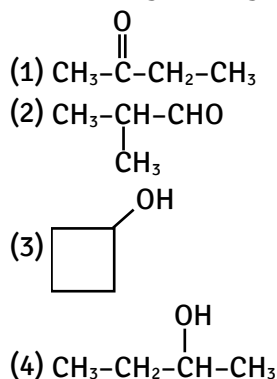
80. $\text{CH}_2-\text{Br} \xrightarrow{\text{alc. KOH}} \text{P} \xrightarrow{\text{NaNH}_2} \text{Q}$
 $\text{CH}_2-\text{Br} \xleftarrow[\text{Anhydrous AlCl}_3]{\text{CH}_3\text{Cl}} \text{R} \xleftarrow[\text{Red hot iron tube (873K)}]{\text{Red hot iron tube (873K)}} \text{Q}$

S is :-

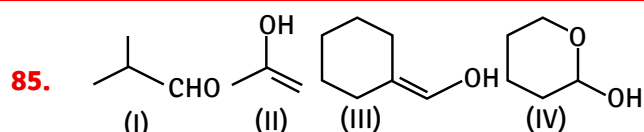
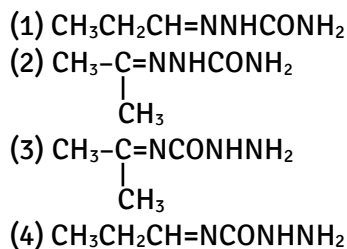
- (1) Chlorobenzene
(2) p-chloro toluene
(3) Toluene
(4) Benzochloride



83. Compound (X) $\text{C}_4\text{H}_8\text{O}$, which reacts with 2, 4-DNP but gives negative haloform test is :-



84. Compound A (molecular formula $\text{C}_3\text{H}_6\text{O}$) is treated with anhydrous CrO_3 to form a product B (molecular formula $\text{C}_3\text{H}_6\text{O}$). B forms a shining silver mirror on warming with ammoniacal silver nitrate. B when treated with an aqueous solution of $\text{H}_2\text{NCONHNH}_2$, gives a product C. Identify the structure of C :-

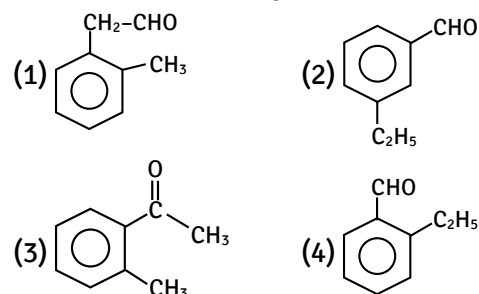


Which among the above compound/s does/do not form Silver mirror when treated with Tollen's reagent

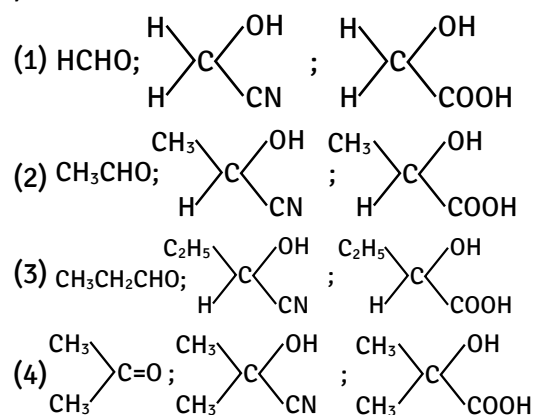
- (1) (II), (III) and (IV) only
 (2) Only (IV)
 (3) Only (II)
 (4) (II) and (III) only

86. An aromatic compound 'X' with molecular formula $\text{C}_9\text{H}_{10}\text{O}$ gives the following chemical tests

- (i) forms 2,4-DNP derivative
 (ii) Reduces Tollens reagent
 (iii) Undergoes Cannizzaro reaction and
 (iv) On vigorous oxidation -1,2-benzenedicarboxylic acid is obtained



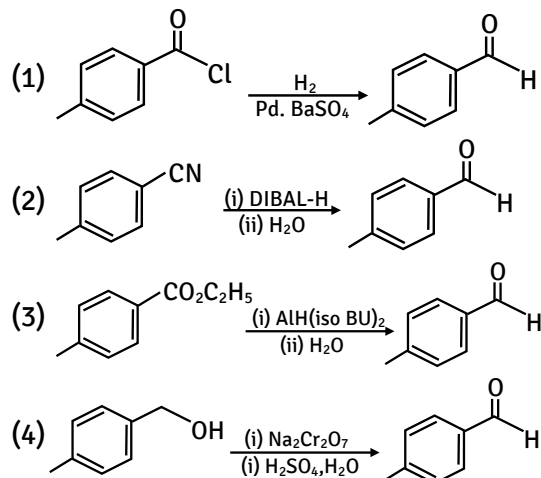
87. A carbonyl compound 'A' reacts with hydrogen cyanide to form a cyanohydrin 'B' which on hydrolysis gives an optically active alpha hydroxy acid 'C'. 'A' gives a positive iodoform test, identify A, B, C product.



88. $\text{CH}_3-\text{CH}_2-\text{CHO} \xrightarrow{\text{OH}^-} \text{A} \xrightarrow{\Delta} \text{B}$
 Product 'A' & 'B' will be respectively :-
 (1) $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CHO}$
 and $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CHO}$
 (2) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CHO}$
 and $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CHO}$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{OH}}{\text{C}}}-\text{CH}_2-\text{CH}_2-\text{CHO}$
 and $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}-\text{CHO}$
 (4) $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\underset{\text{CH}_3}{\text{CH}}-\text{CHO}$
 and $\text{CH}_3-\text{CH}_2-\text{CH}=\underset{\text{CH}_3}{\text{C}}-\text{CHO}$

89. $\text{C}_6\text{H}_5\text{CHO} + \text{HCHO} \xrightarrow[\Delta]{\text{OH}^-} \text{Product (s)}$
 (Major) will be.
 (A) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (B) $\text{C}_6\text{H}_5\text{COO}^-$
 (C) HCOO^- (D) CH_3OH
 (1) A, B and C are correct
 (2) A & B are correct
 (3) C & D are correct
 (4) A & C are correct

90. Which one of the following reactions does not represent correct combination of substrate and product under the given conditions ?



Exercise 3

- 1. Assertion:** Acetaldehyde on treatment with alkaline gives aldol.
Reason: Acetaldehyde molecules contains α hydrogen atom.
- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - (3) If assertion is true but reason is false.
 - (4) If the assertion and reason both are false.
- 2. Assertion:** Acetophenone and benzophenone can be distinguished by iodoform test.
Reason: Acetophenone and benzophenone both are carbonyl compounds.
- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - (3) If assertion is true but reason is false.
 - (4) If the assertion and reason both are false.
- 3. Assertion:** Benzaldehyde is more reactive than ethanal towards nucleophilic attack.
Reason: The overall effect of $-I$ and $+R$ effect of phenyl group decreases the electron density on the carbon atom of $>C=O$ group in benzaldehyde.
- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - (3) If assertion is true but reason is false.
 - (4) If the assertion and reason both are false.
- 4. Assertion:** α -Hydrogen atoms in aldehydes and ketones are acidic.
Reason: The anion left after the removal of α -hydrogen is stabilized by resonance.
- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - (3) If assertion is true but reason is false.
 - (4) If the assertion and reason both are false.
- 5. Assertion:** 2, 2-Dimethyl propanal undergoes Cannizzaro reaction with concentrated NaOH.
Reason: Cannizzaro is a disproportionation reaction.
- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - (3) If assertion is true but reason is false.
 - (4) If the assertion and reason both are false.
- 6. Assertion:** Carboxylic acid exist as dimer.
Reason: Carboxylic acid shows hydrogen bonding.
- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - (3) If assertion is true but reason is false.
 - (4) If the assertion and reason both are false.

7. **Assertion:** Carboxylic acids have higher boiling points than alkanes.

Reason: Carboxylic acids have hydrogen bonding.

- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- (3) If assertion is true but reason is false.
- (4) If the assertion and reason both are false.

8. **Assertion:** Acetals and ketals are unstable in aqueous acidic medium.

Reason: Acetals and ketals are hydrolyzed in acidic medium to yield corresponding aldehydes and ketones.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation for the Assertion
- (2) Both Assertion and Reason are correct but Reason is not correct explanation for Assertion
- (3) Assertion is correct but Reason is incorrect
- (4) Assertion and Reason both are incorrect.

9. **Assertion:** Isopropyl alcohol shows iodoform test

Reason: Sodium hypoiodite oxidises isopropyl alcohol into acetone

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

10. **Assertion:** Aromatic aldehydes and formaldehyde undergo Cannizzaro reaction.

Reason: Aromatic aldehydes and formaldehyde have zero α -H.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

11. **Assertion:** Acetaldehyde is more reactive than acetone in nucleophilic addition reactions.

Reason: Two alkyl groups in acetone reduce the electrophilicity of the carbonyl carbon.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

12. **Assertion:** Benzaldehyde and methanal undergo Cannizzaro reaction.

Reason: Those aldehydes which have α -H atom undergo Cannizzaro reaction.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

- 13.** Given below are two statements
Statement-I: Acidity of α -Hydrogen of aldehydes and ketones is responsible for Aldol reaction
Statement-II: Reaction between benzaldehyde and ethanal will not give cross-Aldol product
Choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are true
(2) Statement I is false but Statement II true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false
- 14.** Given below are two statements:
Statement I: The nucleophilic addition of sodium hydrogen sulphite to an aldehyde or a ketone involves proton transfer to form a stable ion.
Statement II: The nucleophilic addition of hydrogen cyanide to an aldehyde or a ketone yields cyanohydrin as final product.
(1) Both statement I and statement II are true.
(2) Statement I is true but statement II is false.
(3) Statement I is false but statement II is true.
(4) Both statement I and statement II are false.
- 15.** **Statement I:** Addition of ammonia and its derivatives to carbonyl group of aldehyde is a nucleophilic addition-elimination reaction.
Statement II: Addition of hydrazine to acetaldehyde is catalysed by base.
In the light of the above statements, choose the correct answer from the options given below.
(1) Statement I is incorrect but statement II is correct.
(2) Both statement I and statement II are correct.
(3) Both statement I and statement II are incorrect.
(4) Statement I is correct but statement II is incorrect.
- 16.** Given below are two statements:
Statement-I: In Aldol reaction, name aldol is derived from the names of the two functional groups, aldehyde & alcohol, present in the products.
Statement-II: The aldol & ketol do not readily lose water to give aldol condensation product.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and II are correct.
(2) Both statement I and II are incorrect.
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct.
- 17.** Given below are two statements:
Statement-I: The lower members of aldehydes and ketones such as HCHO , CH_3CHO and CH_3COCH_3 are miscible with water in all proportions because they form H-bond with water.
Statement-II: All aldehydes and ketones are fairly soluble in organic solvents like benzene, ether, methanol, chloroform etc.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Both statement I and II are correct.
(4) Statement I is incorrect but statement II is correct.

18. Given below are two statements:
Statement-I: Nitriles are reduced to corresponding imine with stannous chloride in the presence of HCl, which on hydrolysis give corresponding aldehyde.

Statement-II: Nitriles are selectively reduced by DIBAL-H to imines followed by hydrolysis to aldehydes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but statement II is incorrect
- (2) Both statement I and II are incorrect.
- (3) Both statement I and II are correct.
- (4) Statement I is incorrect but statement II is correct.

19. Given below are two statements:
Statement-I: Electronically, Aldehydes are less reactive than ketones towards Nucleophilic Addition Reaction.

Statement-II: Two Alkyl groups increase the electrophilicity of the carbonyl more effectively.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is incorrect but statement II is correct.
- (2) Both statement I and II are incorrect.
- (3) Statement I is correct but statement II is incorrect.
- (4) Both statement I and II are correct.

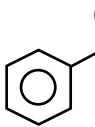
20. Given below are two statements
Statement-I: Ketones react with ethylene glycol to form cyclic products known as ethylene glycol ketals in presence of dry HCl.

Statement-II: Dry hydrogen chloride protonates the oxygen of the carbonyl compounds and therefore, increases the electrophilicity of the carbonyl carbon facilitating the nucleophilic attack of glycol.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and II are correct.
- (2) Both statement I and II are incorrect.
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is

21. Match list-I with list-II

	List-I		List-II
(A)	 $\xrightarrow[\text{(iii) H}_2\text{O/H}^+]{\text{(i) NaOH, (ii) CO}_2}$	(i)	Etard reaction
(B)	 $\xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2}$	(ii)	Gatterman-Koch reaction
(C)	 $\xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) CrO}_2\text{Cl}_2, \text{CS}_2}$	(iii)	Rosenmund reduction
(D)	 $\xrightarrow[\text{anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$	(iv)	Kolbe's reaction

Choose the correct answer from the options give below:

- (1) (A-iv), (B-iii), (C-i), (D-ii)
- (2) (A-iii), (B-iv), (C-ii), (D-i)
- (3) (A-i), (B-iii), (C-ii), (D-iv)
- (4) (A-iii), (B-iv), (C-i), (D-ii)

22. Match the following

	Column-I		Column-II
(p)	$\text{CH}_3 - \text{CH} = \text{O} \xrightarrow[\text{aq.}]{\text{NaOH}}$	(a)	Cannizzaro's reaction
(q)	$\text{CH}_3 - \text{CH}_2 - \text{OH} \xrightarrow[413\text{K}]{\text{Conc. H}_2\text{SO}_4}$	(b)	Aldol condensation
(r)	$\text{C}_6\text{H}_5 - \text{CH} = \text{O} \xrightarrow[\text{KOH, } \Delta]{50\%}$	(c)	Fittig reaction
(s)	 $\xrightarrow[\text{dry ether}]{\text{Na, } \Delta}$	(d)	Williamson's Continuous ether Synthesis

- (1) (p-b), (q-d), (r-a), (s-c)
- (2) (p-d), (q-c), (r-b), (s-a)
- (3) (p-a), (q-d), (r-b), (s-c)
- (4) (p-c), (q-d), (r-b), (s-a)

23. Match the reactions given in list-I with their characteristic reactions/reagents given in list-II and select the correct option.

List-I	List-II
a. $\text{>C=O} \xrightarrow[\text{HCl}]{\text{Zn-Hg}} \text{>CH}_2$	(i) Reduction by DIBAL-H
b. $\text{>C=O} \xrightarrow[\text{(ii) KOH, } \Delta]{\text{(i) NH}_2\text{NH}_2} \text{>CH}_2$	(ii) Reduction by LiAlH_4
c. $\text{-CN} \rightarrow \text{-CHO}$	(iii) Clemmensen reduction
d. $\text{-COOH} \rightarrow \text{-CH}_2\text{OH}$	(iv) Wolff-kishner reduction

- (1) a(iv), b(iii), c(ii), d(i)
 (2) a(ii), b(i), c(iv), d(iii)
 (3) a(iii), b(iv), c(ii), d(i)
 (4) a(iii), b(iv), c(i), d(ii)

24. Match the reagents in column I with products formed by reactions with acetone in column II and mark the appropriate choice.

Column I	
(A)	Hydrazine
(B)	Semicarbazide
(C)	Phenylhydrazine
(D)	Hydroxylamine
Column II	
(i)	$(\text{CH}_3)_2\text{C=NNHCONH}_2$
(ii)	$(\text{CH}_3)_2\text{C=NOH}$
(iii)	$(\text{CH}_3)_2\text{C=NNH}_2$
(iv)	$(\text{CH}_3)_2\text{C=NNHC}_6\text{H}_5$

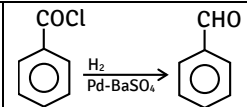
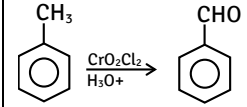
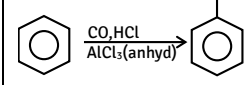
- (1) (A) (i), (B) (ii), (C) (iii), (D) (iv)
 (2) (A) (iv), (B) (iii), (C) (ii), (D) (i)
 (3) (A) (iii), (B) (i), (C) (iv), (D) (ii)
 (4) (A) (ii), (B) (iv), (C) (i), (D) (iii)

25. Match the column I with column II and mark the appropriate choice.

Column I	Column II
(A) Clemmensen reduction	(i) Conc. KOH
(B) Rosenmund reduction	(ii) Zn/Hg+conc. HCl
(C) Iodoform reaction	(iii) $\text{H}_2/\text{Pd-BaSO}_4$
(D) Cannizzaro reaction	(iv) $\text{NaOH}+\text{I}_2$

- (1) (A) $\rightarrow$ (i), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (iv)
 (2) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (iv), (C) $\rightarrow$ (i), (D) $\rightarrow$ (ii)
 (3) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (i)
 (4) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (i), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (iii)

26. Match List-I with List-II.

List-I	
(A)	
(B)	$\text{CH}_3\text{-CN} \xrightarrow[\text{H}_3\text{O}^+]{\text{SnCl}_2/\text{HCl}} \text{CH}_3\text{-CHO}$
(C)	
(D)	
List-II	
(I)	Gatterman-Koch reaction
(II)	Etard reaction
(III)	Stephen reaction
(IV)	Rosenmund reaction

Choose the correct answer from the options given below:

- (1) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
 (2) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
 (3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

27. Match the following column.

	Column-I (Pair of compound)		Column-II (Differentiate method)
(A)	Acetone and acetaldehyde	(P)	$\text{Br}_2 + \text{H}_2\text{O}$ (Bromine water)
(B)	Ethanol and methanol	(Q)	Dil. alkaline KMnO_4 (Bayer's reagent)
(C)	Phenol and Cyclohexanol	(R)	$\text{NH}_4\text{Cl} + \text{AgNO}_3$ (Tollen's reagent)
(D)	n-Pentane and 2-pentene	(S)	NaOI (Iodoform test)

(1) A→S, B→R, C→P, D→Q

(2) A→R, B→S, C→P, D→Q

(3) A→R, B→S, C→Q, D→P

(4) A→R, B→P, C→S, D→Q

28. Match the column

Column-I (Transformations)	Column-II (Reagents)
(a) $\text{R}-\text{COOH} \rightarrow \text{R}-\text{CH}_2\text{OH}$	(p) NH_3/Δ
(b) $\text{Ph}-\text{COOH} \rightarrow \text{Ph}-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{NH}_2$	(q) $\text{Cl}_2/\text{Red P, H}_2\text{O}$
(c) $\text{R}-\text{CH}_2\text{COOH} \rightarrow \text{R}-\underset{\text{Cl}}{\underset{ }{\text{CH}}}-\text{COOH}$	(r) $\text{P}_2\text{O}_5, \Delta$
(d) $\text{CH}_3\text{COOH} \rightarrow (\text{CH}_3\text{CO})_2\text{O}$	(s) $\text{B}_2\text{H}_6/\text{H}_3\text{O}^+$

(1) a-p, b-s, c-r, d-q

(2) a-r, b-p, c-s, d-q

(3) a-s, b-q, c-r, d-p

(4) a-s, b-p, c-q, d-r

29. Match the column

Column-I (Compound)	Column-II (Boiling point)
(a) Propanal	(p) 370
(b) Acetone	(q) 322
(c) 1-Propanol	(r) 273
(d) n-Butane	(s) 329

(1) a-q, b-s, c-p, d-r

(2) a-q, b-p, c-s, d-r

(3) a-p, b-q, c-r, d-s

(4) a-p, b-q, c-s, d-r

30. Match the column:

Column I	Column I
$\text{PhMgBr} + (\text{P}) \xrightarrow{\text{H}^+} 1^\circ \text{ alcohol}$	A $\text{H}-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{H}$
$\text{PhMgBr} + (\text{Q}) \xrightarrow{\text{H}^+} 2^\circ \text{ alcohol}$	B $\text{CH}_3-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{H}$
$\text{PhMgBr} + (\text{R}) \xrightarrow{\text{H}^+} 3^\circ \text{ alcohol}$	C $\text{CH}_3-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{CH}_3$
$\text{PhMgBr} + (\text{S}) \xrightarrow{\text{H}^+}$ 	D EtOH

P Q R S

(1) D A C B

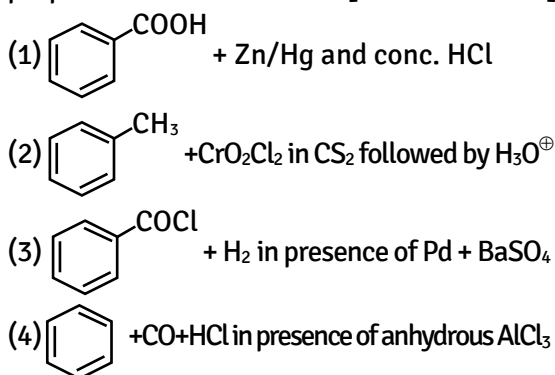
(2) B D A C

(3) B C A D

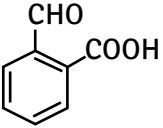
(4) A B C D

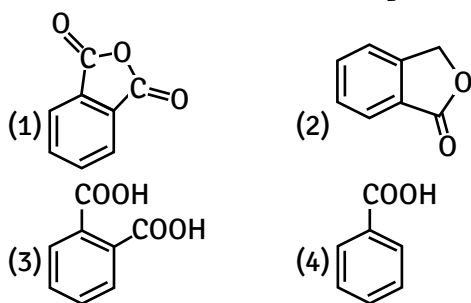
Exercise 4 (Previous Year's Questions)

1. Reaction by which Benzaldehyde cannot be prepared :- **[NEET UG-2013]**

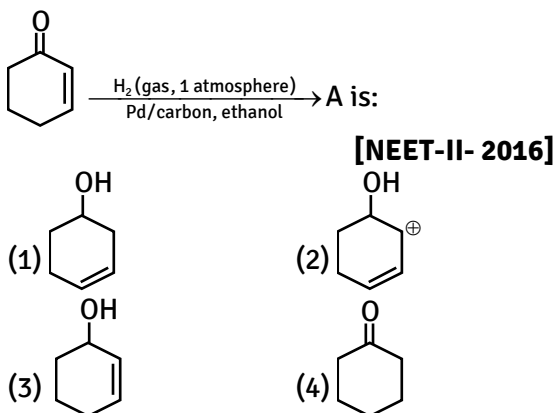


2. An organic compound 'X' having molecular formula C₅H₁₀O yields phenyl hydrazone and gives negative response to the iodoform test and Tollen's test. It produces n-pentane on reduction. 'X' could be :- **[AIPMT-2015]**
- (1) 2-pentanone (2) 3-pentanone
(3) n-amyl alcohol (4) pentanal

3.  $\xrightarrow[\text{(ii) H}^+/\Delta]{\text{(i) [Ag(NH}_3)_2]^+ \text{ OH}^-}$ product
- Product of above reaction is **[AIIMS-2015]**

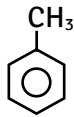
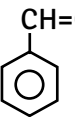


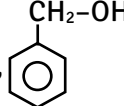
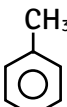
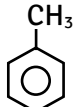
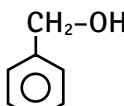
4. The correct structure of the product A formed in the reaction

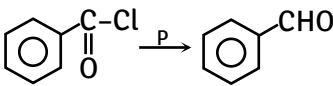


5. The **correct statement** regarding a carbonyl compound with a hydrogen atom on its alpha carbon, is : **[NEET-I-2016]**

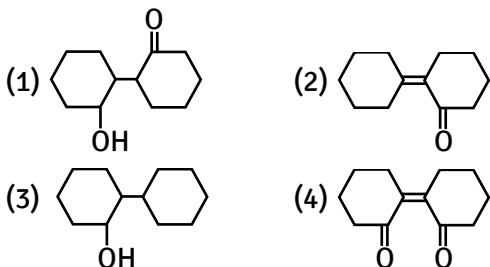
- (1) a carbonyl compound with a hydrogen atom on its alpha-carbon never equilibrates with its corresponding enol.
- (2) a carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as aldehyde-ketone equilibration.
- (3) a carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as carbonylation.
- (4) a carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as keto-enol tautomerism.

6.  $\xrightarrow{\text{P}}$  $\xrightarrow{\text{Zn-Hg/HCl}}$ Q
- P & Q respectively are : **[AIIMS-2016]**

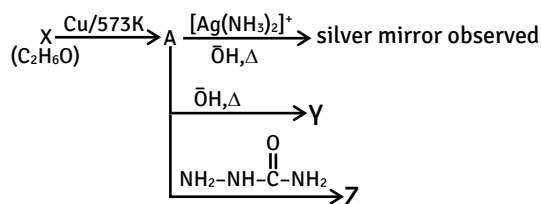
- (1) CrO₃/H₃O⁺, 
- (2) CrO₂Cl₂/H₃O⁺, 
- (3) NaOCl/Δ, 
- (4) KMnO₄/H⁺, 

7.  P will be :- **[NEET-2017]**
 (1) Pd/BaSO₄ + H₂ (2) LiAlH₄
 (3) NaBH₄ (4) Pd + H₂

8. Which of the following product is formed when cyclohexanone undergoes aldol condensation followed by heating? : **[NEET-2017]**



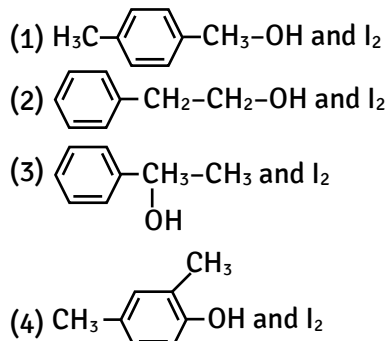
9. Consider the reactions : **[NEET-2017]**



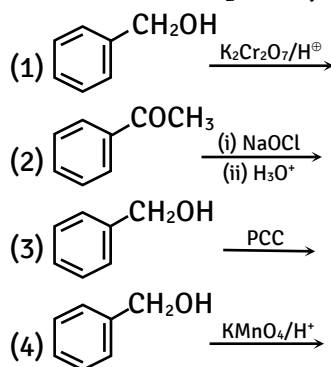
Identify A, X, Y and Z

- (1) A-methoxymethane, X-Ethanoic acid, Y-Acetate ion, Z-hydrazine
 (2) A-methoxymethane, X-Ethanol, Y-Ethanoic acid, Z-Semicarbazide.
 (3) A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone.
 (4) A-Ethanol, X-Acetaldehyde, Y-Butanone, Z-Hydrazone.
10. Carboxylic acid have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their **[NEET(UG) 2018]**
 (1) formation of intramolecular H-bonding
 (2) formation of carboxylate ion
 (3) more extensive association of carboxylic acid via van der Waals force of attraction
 (4) formation of intermolecular H-bonding

11. Compound A, C₈H₁₀O, is found to react with NaOI (Produced by reacting Y with NaOH) and yields a yellow precipitate with characteristic smell. A and Y are respectively: **[NEET(UG) 2018]**



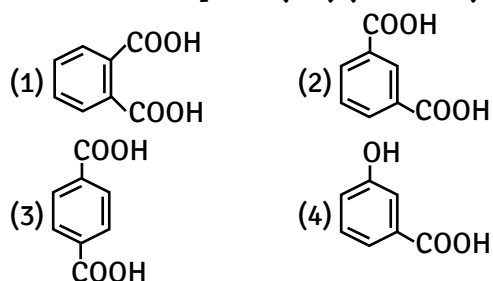
12. The reaction that does not give benzoic acid as the major product is: **[NEET(UG) (Odisha) 2019]**



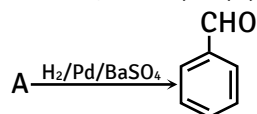
13. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as: **[NEET(UG) 2020]**

- (1) Cross Aldol condensation
 (2) Aldol condensation
 (3) Cannizzaro's reaction
 (4) Cross Cannizzaro's reaction

14. Which of the following acid will form an (a) Anhydride on heating and (b) Acid imide on strong heating with ammonia? **[NEET(UG) (Covid-19) 2020]**

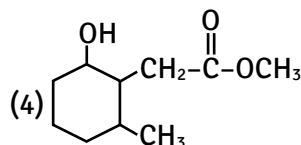
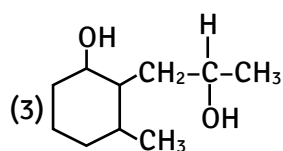
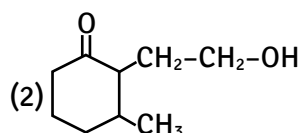
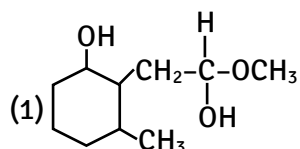
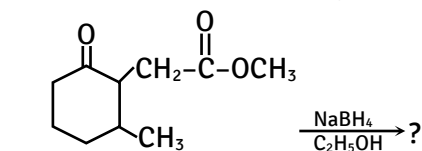


15. Identify compound (A) in the following reaction: **[NEET(UG) (Covid-19) 2020]**



- (1) Benzoyl chloride (2) Toluene
(3) Acetophenone (4) Benzoic acid

16. The product formed in the following chemical reaction is: **[NEET(UG) 2021]**



17. Match List - I with List - II. **[NEET(UG) 2021]**

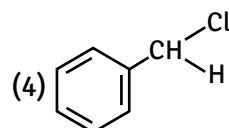
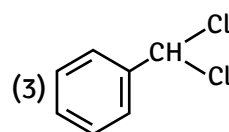
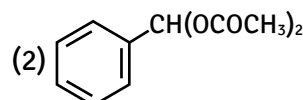
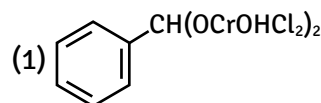
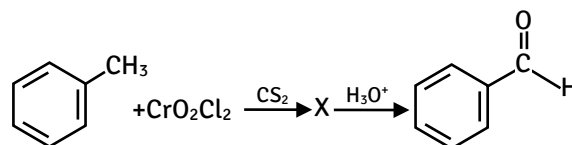
List-I	List-II
a $\xrightarrow[\text{AlCl}_3/\text{CuCl}]{\text{CO, HCl, Anhyd}}$	i Hell-Volhard Zelinsky reaction
b $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$	ii Gattermann-Koch reaction
c $\text{R}-\text{CH}_2-\text{OH} + \text{RCOOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$	iii Haloform reaction
d $\text{R}-\text{CH}_2-\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$	iv Esterification

Choose the correct answer from the options given below.

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

18. The intermediate compound 'X' in the following chemical reaction is:

[NEET(UG) 2021]



19. Match List-I with List-II.

[NEET(UG) 2022]

List-I	List-II
A Cyanohydrin	i NH ₂ OH
B Acetal	ii RNH ₂
C Schiff's base	iii alcohol
d Oxime	iv HCN

Choose the correct answer from the options given below:

- (1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(3) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

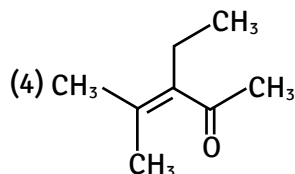
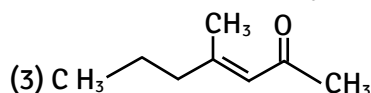
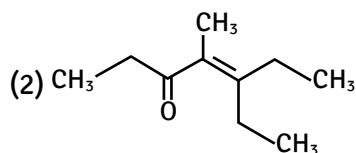
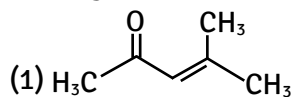
20. $\text{RMgX} + \text{CO}_2 \xrightarrow[\text{ether}]{\text{dry}} \text{Y} \xrightarrow{\text{H}_3\text{O}^+} \text{RCOOH}$

What is Y in the above reaction?

[NEET(UG) 2022]

- (1) $\text{RCOO}^-\text{Mg}^+\text{X}$ (2) $\text{R}_3\text{CO}^-\text{Mg}^+\text{X}$
(3) RCOO^-X^+ (4) $(\text{RCOO})_2\text{Mg}$

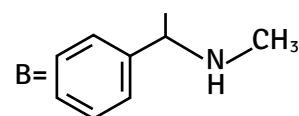
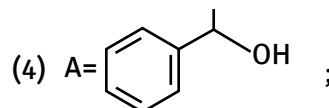
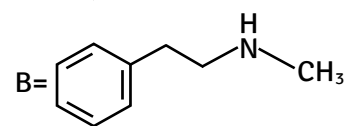
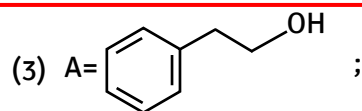
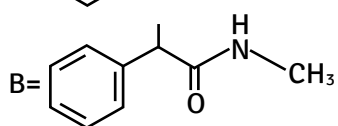
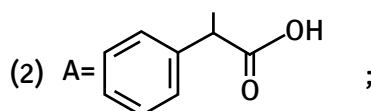
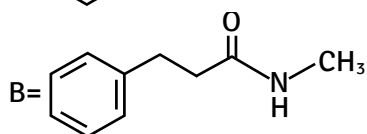
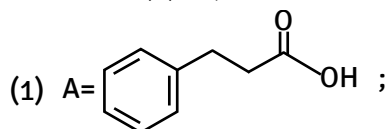
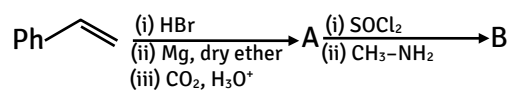
21. Which one of the following is **not** formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating? [NEET(UG) 2022]



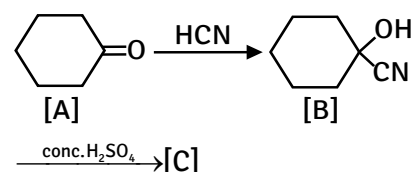
22. Compound X on reaction with O_3 followed by Zn/H_2O gives formaldehyde and 2-methyl propanal as products. The compound X is: [NEET(UG) 2022]

- (1) 3-Methylbut-1-ene
(2) 2-Methylbut-1-ene
(3) 2-Methylbut-2-ene
(4) Pent-2-ene

23. The products A and B in the following reaction sequence are: [Re-NEET (UG) 2022]

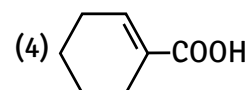
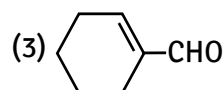
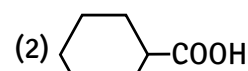
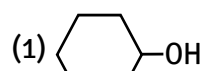


24. Complete the following reaction :

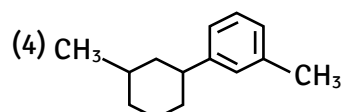
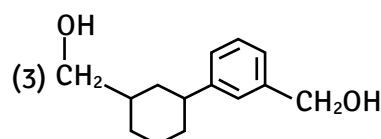
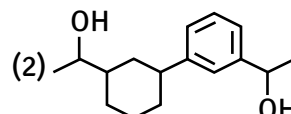
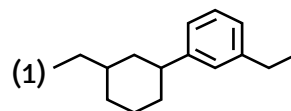
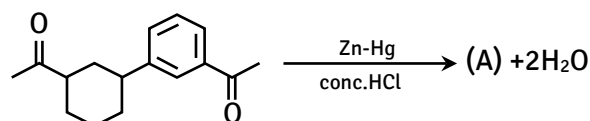


[C] is.

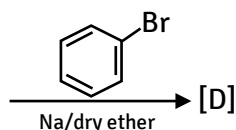
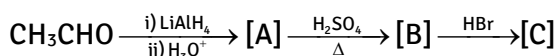
[NEET (UG) 2023]



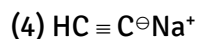
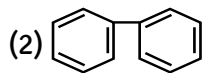
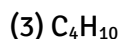
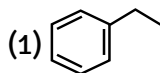
25. Identify product (A) in the following reaction. [NEET (UG) 2023]



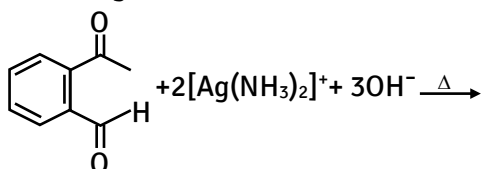
26. Identify the final product [D] obtained in the following sequence of reactions.



[NEET (UG) 2023]

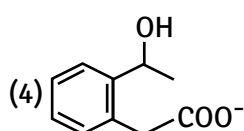
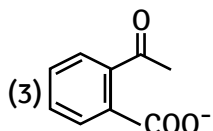
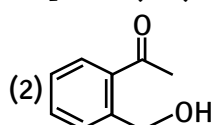
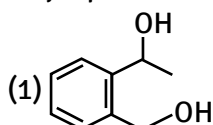


27. Identify the major product obtained in the following reaction:



Major product

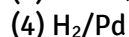
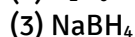
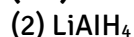
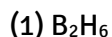
[NEET (UG) 2023]



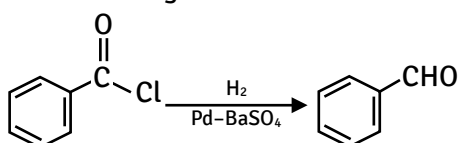
28. $\text{R}-\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O/HCl}]{\text{(i) "X"}} \text{R}-\text{CH}_2\text{OH}$
 $\text{R}-\text{CH}=\text{CH}_2 \xrightarrow[\text{(ii) H}_2\text{O, NaOH, H}_2\text{O}_2]{\text{(i) "X"}} \text{R}-\text{CH}_2-\text{CH}_2-\text{OH}$

Identify 'X' in above reactions

[NEET (UG)-Manipur-2023]



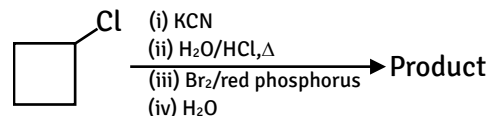
29. The following conversion is known as:



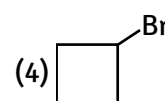
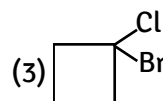
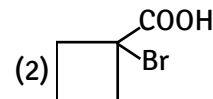
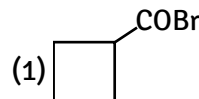
[NEET (UG)-Manipur-2023]

- (1) Stephen reaction
 (2) Gattermann-Koch reaction
 (3) Etard reaction
 (4) Rosenmund reaction

30. Identify the product in the following reaction

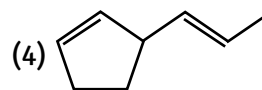
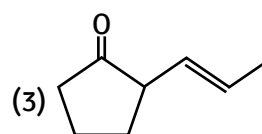
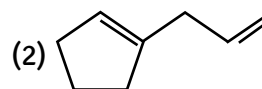
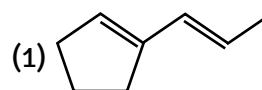
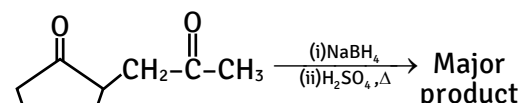


[NEET (UG)-Manipur-2023]



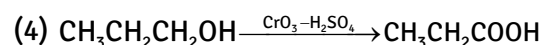
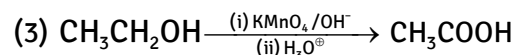
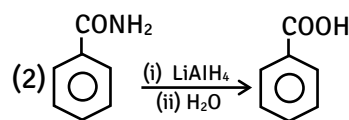
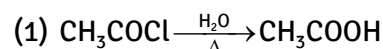
31. The major product formed in the following conversion is _____.

[NEET (UG)-Manipur-2023]



32. Select the incorrect reaction among the following :

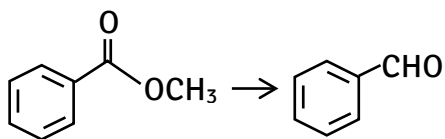
[RE-NEET-2024]



33. Fehling's solution 'A' is [NEET-2024]

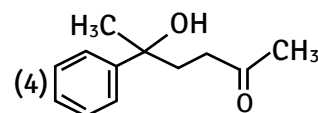
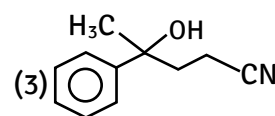
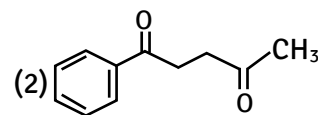
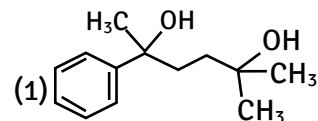
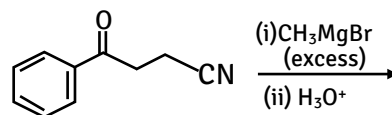
- (1) alkaline solution of sodium potassium tartrate (Rochelle's salt)
- (2) aqueous sodium's citrate
- (3) aqueous copper sulphate
- (4) alkaline copper sulphate

34. Identify the suitable reagent for the following conversion. [NEET - 2025]



- (1) (i) NaBH_4 (ii) $\text{H}^+/\text{H}_2\text{O}$
- (2) $\text{H}_2/\text{Pd}-\text{BaSO}_4$
- (3) (i) LiAlH_4 (ii) $\text{H}^+/\text{H}_2\text{O}$
- (4) (i) $\text{AlH}(\text{iBu})_2$ (ii) H_2O

35. The major product of the following reaction is: [NEET - 2025]



ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	3	3	3	2	3	4	3	3	2	4	4	1	4	4	2	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	2	2	4	3	1	1	3	3	3	4	1	2	3	4	4	4	1	3
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	4	4	1	4	3	3	4	3	2	1										

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	1	1	3	2	3	1	4	1	1	3	1	2	2	1	2	

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	3	3	1	1	2	3	3	2	2	3	4	4	4	1	3	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	3	1	3	4	3	1	1	1	2	3	1	2	4	2	2	2	3	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	2	4	4	1	4	4	2	1	1	3	4	4	3	4	2	2	4	3	2
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	3	2	2	3	3	3	3	4	2	3	2	2	1	4	4	3	1	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Ans.	2	4	2	2	3	3	1	2	1	2	1	4	1	3	3	4	4	1	1	3
Que.	81	82	83	84	85	86	87	88	89	90										
Ans.	2	1	2	1	3	4	2	4	4	4										

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	4	1	2	1	1	1	1	1	1	3	3	1	4	3	3	3	2	1
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	1	1	4	3	3	1	2	4	1	4										

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	1	4	4	2	1	2	3	4	3	3	1	1	1	4	4	1	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
Ans.	2	1	2	4	1	1	3	1	4	2	1	2	3	4	4					

Chapter 5

AMINES

Chapter Summary

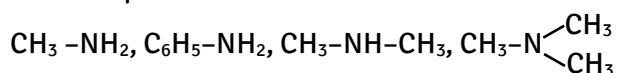
- Introduction
- Preparation of amines
- Physical properties
- Chemical Reactions
- Diazonium Salts

Introduction

Definition:

Amines can be considered as derivatives of ammonia, obtained by replacement of one, two or all the three hydrogen atoms by alkyl and/or aryl groups.

For example:



Structures of Amines-

Like ammonia, nitrogen atom of amines is trivalent and carries an unshared pair of electrons. Nitrogen orbitals in amines are therefore, sp^3 hybridised and the geometry of amines is pyramidal. Each of the three sp^3 hybridised orbitals of nitrogen overlap with orbitals of hydrogen or carbon depending upon the composition of the amines. The fourth orbital of nitrogen in all amines contains an unshared pair of electrons. Due to the presence of unshared pair of electrons, the angle C-N-E, (where E is C or H) is less than 109.5° ; for instance, it is 108° in case of trimethylamine as shown in Fig.

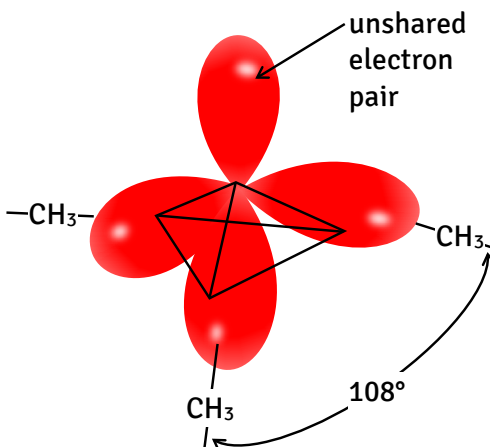
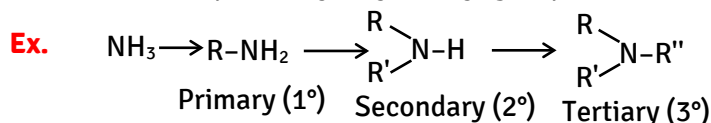


Fig. Pyramidal shape of trimethylamine

Classification

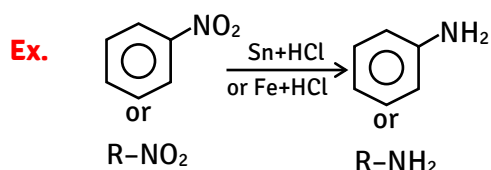
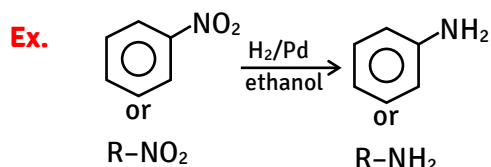
Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of 'H' atoms replaced by alkyl or aryl groups in ammonia (NH_3) molecule.

**Preparation of amines**

Amines are prepared by following methods

(1) Reduction of nitro compounds:

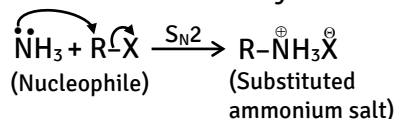
Nitro compounds are reduced to amines by passing H_2 gas in the presence of finely divided Ni, Pd or Pt and also by reduction with metal in acidic medium



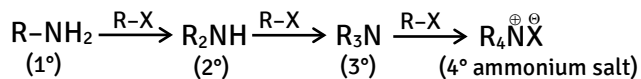
Note. Reduction with Fe scrap and HCl is Preferred because FeCl_2 formed gets hydrolysed to release hydrochloric acid during the reaction thus only a small amount of HCl is required to initiate the reaction

(2) Ammonolysis of alkyl halides:

The process of cleavage of the C-X bond by NH_3 molecule is known as ammonolysis. The reaction is carried out in a sealed tube at 373 K. The 1° amine thus obtained behaves as a nucleophile and can further react with alkyl halide to form 2° and 3° amines and finally 4° ammonium salt



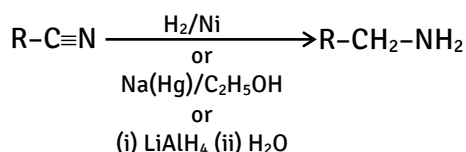
The nucleophilic R-NH_2 reacts with R-X to give 2° , 3° amines and some 4° salt resulting in a complex mixture.



Note. (i) Ammonolysis has the disadvantage of yielding a mixture of primary, secondary, tertiary amines and also a quaternary ammonium salt
(ii) Primary amine is obtained as a major product by taking large excess of ammonia

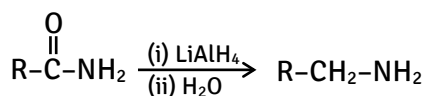
3. Reduction of nitriles:

Nitriles on reduction with LiAlH_4 or catalytic hydrogenation produce 1° amines. This reaction is used for ascent of amine series

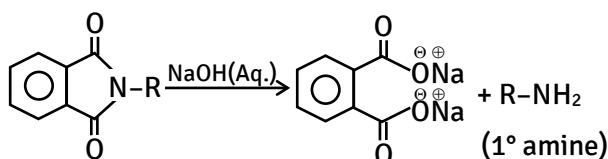
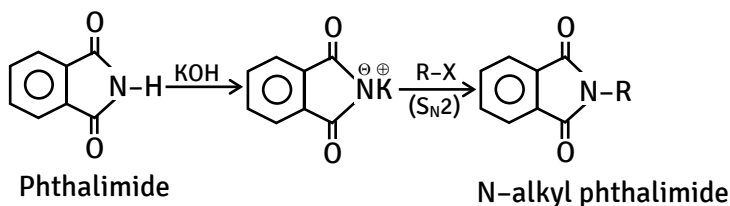


4. Reduction of amides:

The amides on reduction with LiAlH_4 yield amine

**5. Gabriel phthalimide synthesis:**

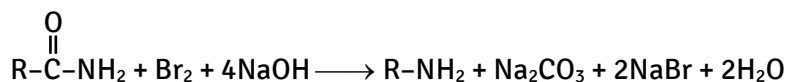
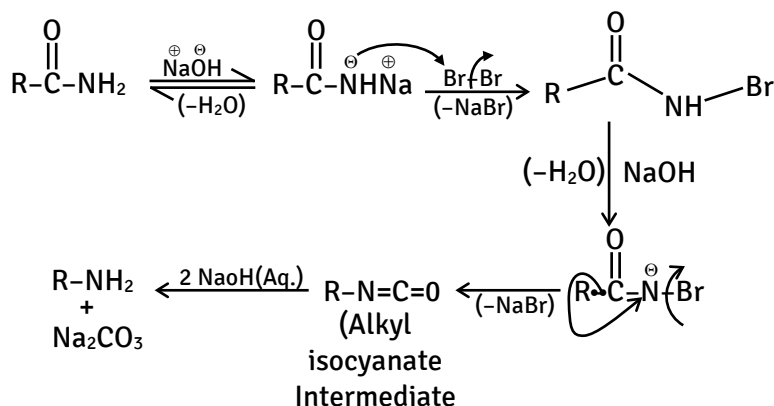
Phthalimide on treatment with ethanolic KOH forms potassium salt of phthalimide which on heating with alkyl halide followed by alkaline hydrolysis produces the corresponding primary amines



- Note.** (i) Gabriel synthesis is used for the preparation of primary amines
 (ii) Aromatic primary amines can not be prepared by this method because aryl halide do not undergo nucleophilic substitution with the anion formed by phthalimide

6. Hoffmann bromamide degradation reaction:

Amides on reaction with Bromine in an aqueous or ethanolic solution of sodium hydroxide give primary amine with one 'C' atom less than that present in the amide

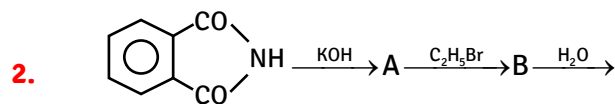
**Mechanism:**

- Note.** In this reaction migration of an alkyl or aryl group takes place from carbonyl carbon of the amide to the nitrogen atom.

Exercise 1.1

1. $C_2H_5NH_2$ cannot be prepared by the reduction of:

- (1) $C_2H_5NO_2$ (2) $CH_3CH=NOH$
(3) C_2H_5NC (4) CH_3CN



C + D, C and D in the sequence are:

- (1) Benzoic acid + aniline
(2) phthalic acid + ethylamine
(3) Phthalic acid + aniline
(4) Benzoic acid + ethylamine
3. Which of the following is obtained by reducing methyl cyanide with $Na + C_2H_5OH$:
(1) Methyl alcohol (2) Acetic acid
(3) Ethyl amine (4) Methane
4. Which of the following would undergo Hoffmann bromide reaction to form primary amine:
(1) $RCONHCH_3$ (2) $RCOONH_4$
(3) $RCONH_2$ (4) $RCONHOH$
5. Gabriel reaction for the synthesis of amines, involves the use of
(1) 1° amide (2) 2° amide
(3) Imides (4) Aliphatic amide
6. In the Hoffmann bromamide degradation reaction, the number of moles of $NaOH$ and Br_2 used per mole of amine produced are -
(1) Four moles of $NaOH$ and two moles of Br_2
(2) Two moles of $NaOH$ and two moles of Br_2
(3) Four moles of $NaOH$ and one mole of Br_2
(4) One mole of $NaOH$ and one mole of Br_2
7. Which of the following is a 3° amine?
(1) 1-methylcyclohexanamine
(2) Triethylamine
(3) Tert-butylamine
(4) N-methylaniline

8. The best reagent for converting, 2-phenylpropanamide into 1-phenylethanamine is ____.

- (1) excess H_2/Pt
(2) $NaOH/Br_2$
(3) $NaBH_4$ /methanol
(4) $LiAlH_4$ /ether

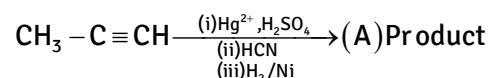
9. Which of the following will not give a primary amine?

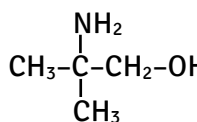
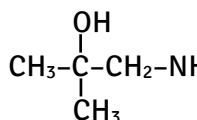
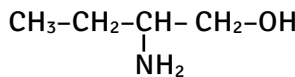
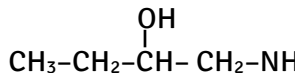
- (1) $CH_3CONH_2 \xrightarrow{Br_2, KOH}$
(2) $CH_3CN \xrightarrow{LiAlH_4}$
(3) $CH_3NC \xrightarrow{LiAlH_4}$
(4) $CH_3CONH_2 \xrightarrow{LiAlH_4}$

10. Amine produced by Hoffmann degradation of benzamide is

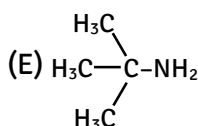
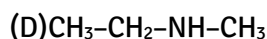
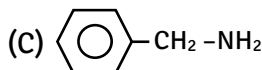
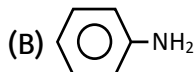
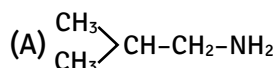
- (1) Benzyl amine
(2) Aniline
(3) Aromatic primary amine
(4) 2 & 3 Both

11. The product (A) formed in the following reaction sequence is



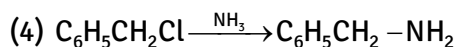
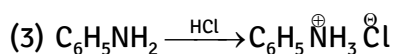
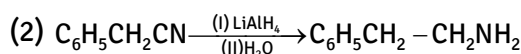
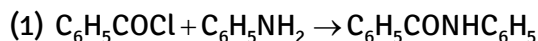
- (1) 
(2) 
(3) 
(4) 

12. Which of following amines cannot synthesized by Gabriel Phthalimide synthesis method

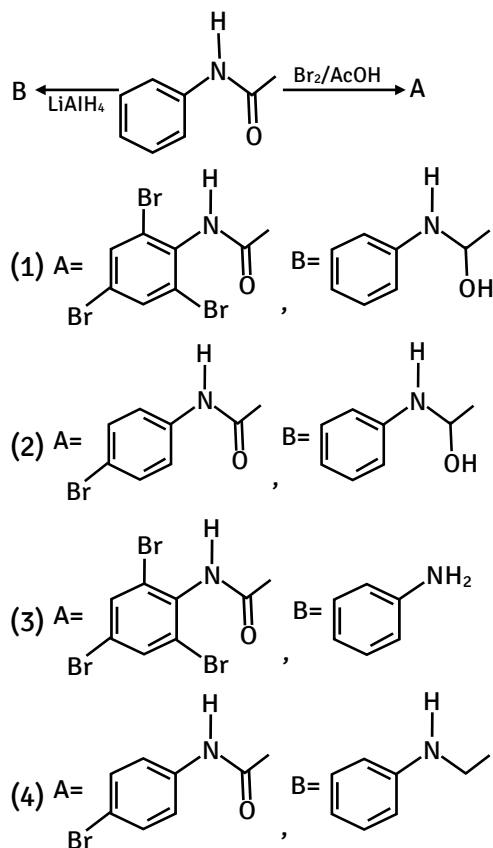


- (1) only B (2) Only B and D
(3) Only A and C (4) Only B, D and E

13. Which of the following reaction is an example of ammonolysis?



14. The major products A and B from the following reactions are:



15. Reduction of $\text{CH}_3\text{CH}_2\text{NC}$ with hydrogen in presence of Ni or Pt as catalyst gives

- (1) $\text{CH}_3\text{CH}_2\text{NH}_2$ (2) $\text{CH}_3\text{CH}_2\text{NHCH}_3$
(3) $\text{CH}_3\text{CH}_2\text{NHCH}_2\text{CH}_3$ (4) $(\text{CH}_3)_3\text{N}$

Physical properties

(a) **Physical state:** The lower aliphatic amines are gasses with fishy odour. Primary amines with 3 or more 'c' atoms are liquid and higher ones are solid. Aniline and other arylamines are usually colourless but get colored on storage due to atmospheric oxidation.

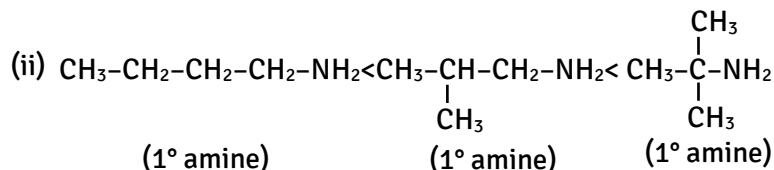
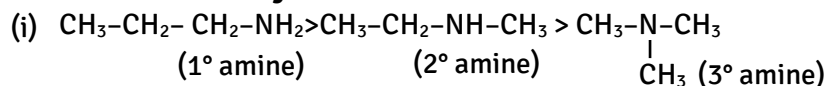
(b) **Solubility:** Lower aliphatic amines are soluble in H_2O because they form H-bonding with H_2O . But solubility decreases with increase in molecular mass due to increase in size of the hydrophobic alkyl part. Higher amines are insoluble in H_2O .

Amines are soluble in organic solvents such as alcohol, benzene, and ether.

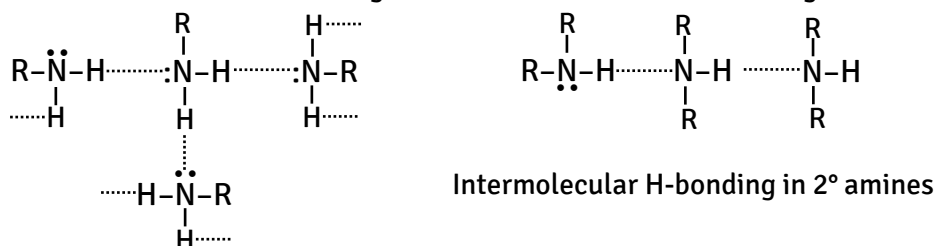
Butan-1-ol is more soluble in H_2O than butan-1-amine due to greater EN of O atom than that of N atom (EN of O atom in alcohol is 3.5, whereas EN of N atom in amine is 3.0). therefore alcohols are more polar than amines and form stronger intermolecular H-bonds than amines.

$$\text{Solubility} \propto \frac{1}{\text{Molecular weight (Hydrophobic part)}}$$

Solubility $\propto$ Branching (When MW & types of amines is same)

Ex. order of solubility:-

- (c) **Boiling Point:** 1° amines form stronger intermolecular H-bonding between N atom of one and H atom of another molecule, than 2° amines, since 1° amines have two H atoms for H-bonding; that is why boiling points of 1° amines are greater than those of 2° amines of the same molar masses. Similarly, 2° amines form stronger intermolecular H-bonding than 3° amines, since 2° amines have one H atom for H-bonding, whereas 3° amines do not form H-bonding due to the absence of H atom available for H-bond. Thus, the boiling points of 2° amines are higher than those of 3° amines of the same molar masses. The order of boiling points of isomeric amines is: 1° > 2° > 3° amines. The intermolecular H-bonding in 1° and 2° amines is shown in Fig.

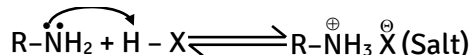


Intermolecular H-bonding in 1° amines

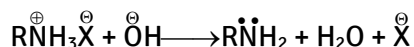
Table: Comparison of boiling points of amines, alcohols, and alkanes of comparable molar masses.

S.NO.	Compound	Molar mass	Boiling point (K)
1.	n-C ₄ H ₉ NH ₂ (1° amine)	73	350.8
2.	Et ₂ NH (2° amine)	73	329.3
3.	EtNMe ₂ (3° amine)	73	310.5
4.	(Alkane)	72	300.8
5.	n-C ₄ H ₉ OH (Alcohol)	74	390.3

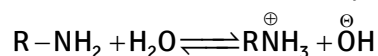
- (d) **Basic Nature of Amines:** Due to basic nature, they react with acid to form salts.



Amine salts on treatment with bases such as NaOH regenerate the parent amine.



Amines have an unshared pair of electrons on N atom due to which they behave as Lewis base. Basic characters of amines are expressed in terms of their K_b and pK_b values.



$$K_{eq} = \frac{[\text{RNH}_3^+][\text{OH}^-]}{[\text{RNH}_2][\text{H}_2\text{O}]}, \quad K_b = K_{eq} [\text{H}_2\text{O}] = \frac{[\text{RNH}_3^+][\text{OH}^-]}{[\text{RNH}_2]} \quad pK_b = -\log K_b$$

The larger the value of K_b or smaller the value of pK_b , the stronger is the base.

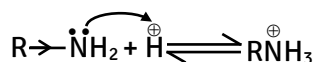
- i. Aliphatic amines are stronger bases than NH_3 ($\text{pK}_b = 4.75$) due to +I effect of the alkyl groups leading to high e^- density on the N atom. Their pK_b values lie between 3 and 4.22.
- ii. Aromatic amines are weaker bases than NH_3 due to $\ominus$ withdrawing nature of the aryl group.
- iii. Besides the inductive effect, there are other effects such as solvation effect, steric hindrance, etc., which affect the basic strength of amines.

iv. **Comparison of basic strength of amines.**

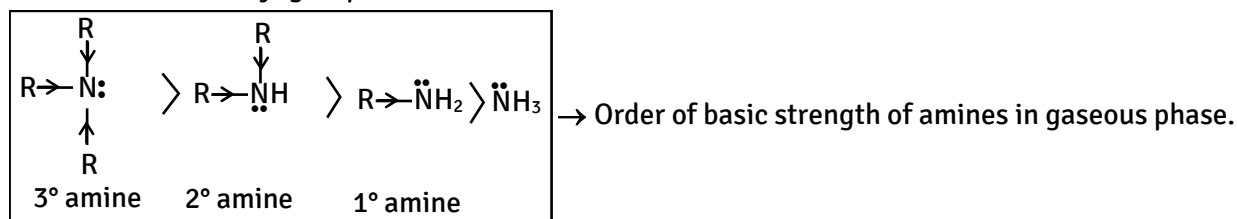
(a) **Decreasing order of basicity of amines in gaseous phase:**

$[\text{3}^\circ \text{ amine } (\text{R}_3\text{N}) > \text{2}^\circ \text{ amine } (\text{R}_2\text{NH}) > \text{1}^\circ \text{ amine } (\text{RNH}_2) > \text{NH}_3]$

This is due to +I effect of alkyl groups.



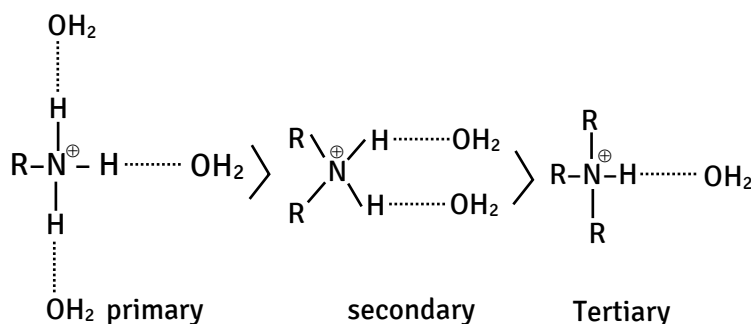
The alkyl group (R) pushes e^- density towards N atom and thus makes the LP more available for sharing with the H^+ of the acid. Moreover, RNH_3^+ is stabilised due to dispersal of the positive charge by the +I effect of the alkyl group. Thus, the basic character of the aliphatic amines increases with the increase in the number of alkyl groups.



(b) **Basic character of aliphatic amine in polar solvent:**

In the aqueous phase, the substituted ammonium cations get stabilised not only by +I effect of alkyl group but also by solvation with H_2O molecule.

The greater the size of the ion, the lesser will be the solvation and less stabilised is the ion.



The decreasing order of extent of H-bonding in water and order of stability of ions by solvation

The greater the stability of substituted ammonium cation, the stronger is the base. So, the order of basicity should be $1^\circ > 2^\circ > 3^\circ$ amine but according to +I effect of R group, the basicity order of amine is $3^\circ > 2^\circ > 1^\circ$.

Both the results do not match with the observed basicity of amine. So, some other factor, steric hindrance, is also responsible in determining the basic character of amines.

When the alkyl group is small [such as ($-\text{CH}_3$) group], there is no steric hindrance to H bonding. If the alkyl group is bigger than ($-\text{CH}_3$) group, there would be steric hindrance to H bonding. Hence, inductive effect, solvation effect, and steric hindrance of alkyl group decide the basic strength of alkyl amines in the aqueous phase.

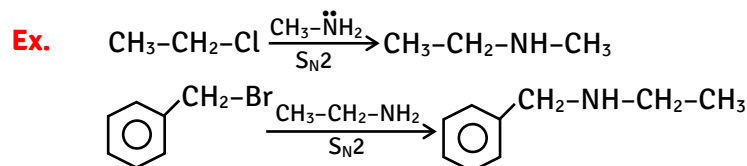
Result :

Decreasing order of basic character of alkyl substituted amines in aqueous solution:

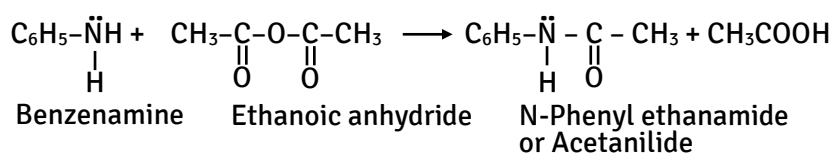
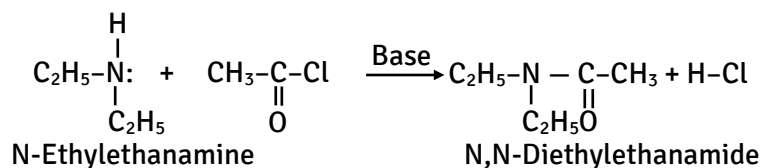
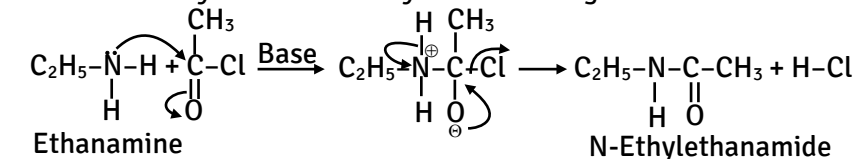
(R)-	Relative basic strength
CH_3-	$2^\circ > 1^\circ > 3^\circ > \text{NH}_3$
C_2H_5-	$2^\circ > 3^\circ > 1^\circ > \text{NH}_3$

Chemical Reactions

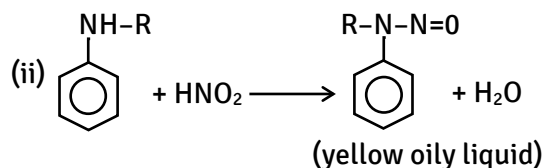
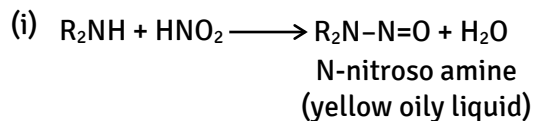
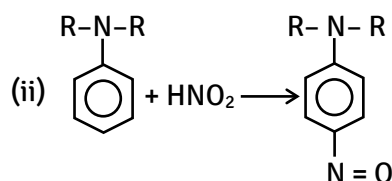
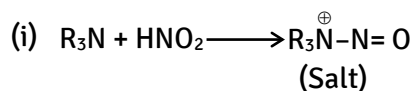
- (1) **Alkylation:** Reaction of an amine with alkyl halide, result in alkylation of the nitrogen. The reaction takes place by $\text{S}_{\text{N}}2$ mechanism with the amine acting as a nucleophile.



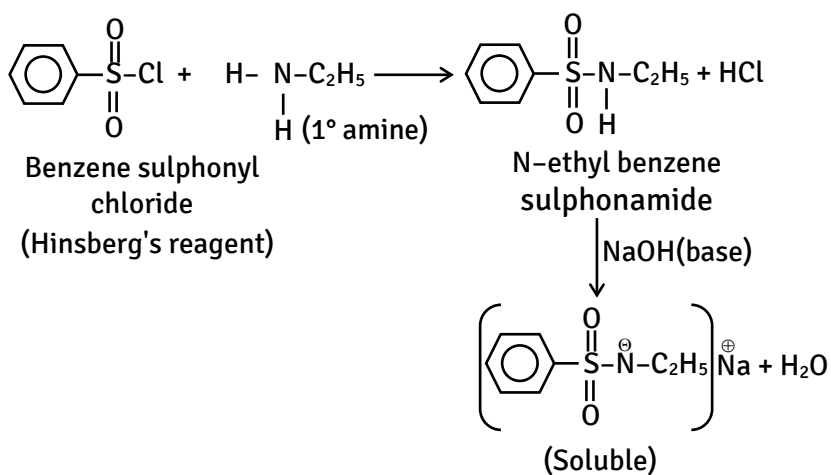
- (2) **Acylation:** Primary and secondary amines undergo nucleophilic acyl substitution reaction when treated with acyl halides or anhydrides forming N-substituted amides

**Points to ponder:**

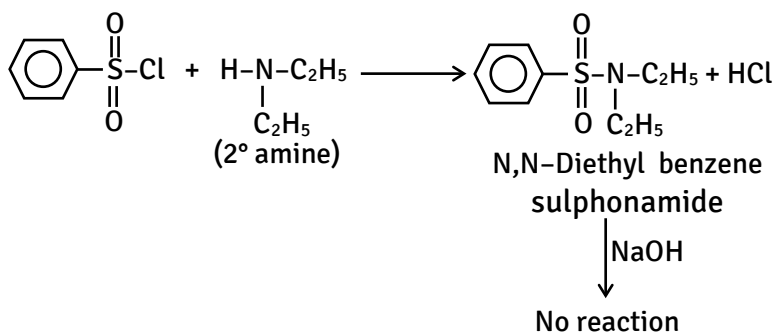
- Aliphatic and aromatic primary and secondary amines react with acid chlorides, anhydrides and esters by nucleophilic acyl substitution reaction. This reaction is known as acylation.
 - You can consider this reaction as the replacement of hydrogen atom of $-\text{NH}_2$ or $>\text{N}-\text{H}$ group by the acyl group.
 - The products obtained by acylation reaction are known as amides.
 - The reaction is carried out in the presence of a base stronger than the amide, like pyridine, which removes HCl and shifts the equilibrium to the right hand side.
 - Amines also react with benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$). This reaction is known as benzoylation.
- $$\text{CH}_3\text{NH}_2 + \text{C}_6\text{H}_5\text{COCl} \longrightarrow \text{CH}_3\text{NHCOC}_6\text{H}_5 + \text{HCl}$$
- Methanamine Benzoyl chloride N-Methylbenzamide
- Benzoylation is also known as **Schotten-Baumann reaction**.

(c) Secondary amines (Aromatic and Aliphatic):**(d) Tertiary amines (Aromatic and Aliphatic):****(5) Reaction with aryl sulphonyl chloride: Hinsberg's test**

- This reaction is used to distinguish 1°, 2° and 3° amine

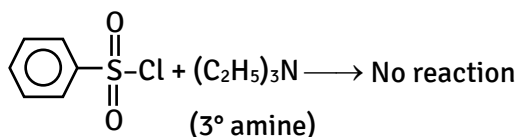
(a) Reaction with primary amines:

- The hydrogen attached to nitrogen in sulphonamide is strongly acidic due to the presence of strong electron withdrawing sulphonyl group. Hence it is soluble in alkali

(b) Reaction with secondary amines:

- Since N,N-diethyl benzene sulphonamide does not contain any hydrogen atom attached to nitrogen atom therefore it is not acidic and insoluble in alkali

(c) Reaction with Tertiary amines:

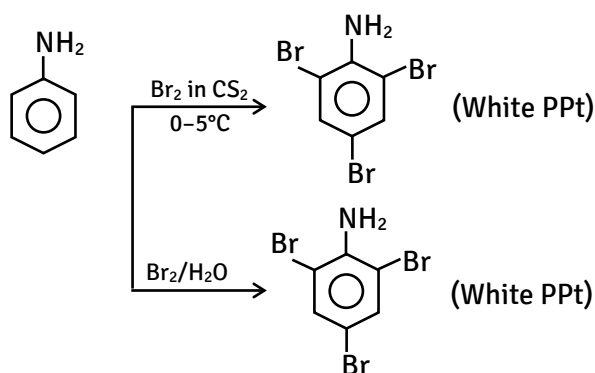


- Tertiary amines do not react with benzene sulphonyl chloride (Hinsberg reagent)

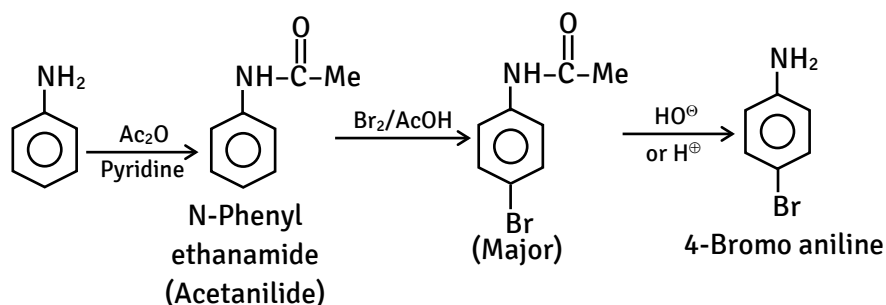
Note. However these days benzene sulphonyl chloride is replaced by P-toluene sulphonyl chloride

6. Electrophilic substitution reactions:

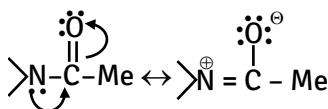
- (i) Bromination:** Aniline reacts with Br_2 water at room temperature or with brominating reagent to give a white precipitate of 2,4,6-tribromo aniline.



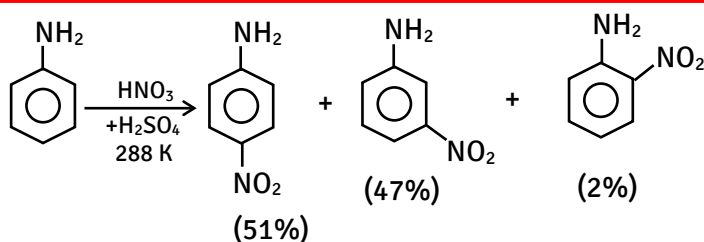
Since aromatic amines have high reactivity, so to prepare mono-substituted aniline derivative, ($-\text{NH}_2$) group is protected by acetylation with acetic anhydride (Ac_2O), then the desired substitution is carried out followed by hydrolysis of the substituted amide to obtain the substituted amine.



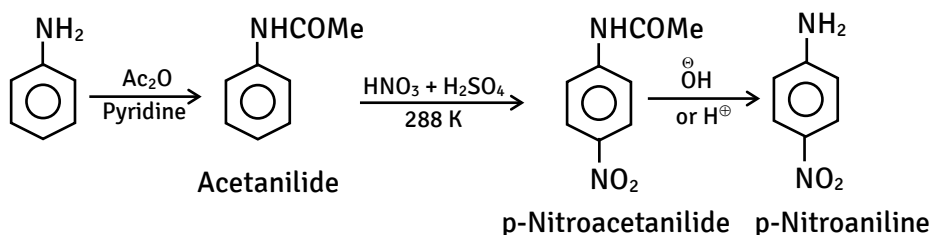
The lone Pair on N atom of acetanilide interact with O atom due to resonance (i.e., cross conjugation). Hence, lone pair on N atom are less available for donation to benzene ring by resonance, thereby decreasing the activating effect of ($-\text{NH}_2$) group.



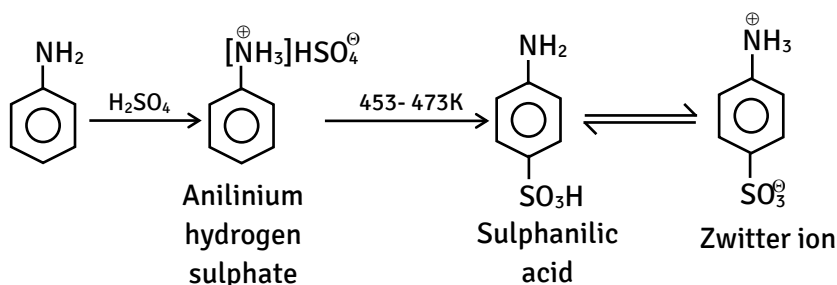
- (ii) Nitration:** The direct nitration of aniline gives tarry oxidation products along with nitro derivatives, In the acidic medium, aniline is protonated to give anilinium ion which is meta directing that's why m-derivative along with o- and p-products are obtained.



If ($-\text{NH}_2$) group is protected by acetylation, p-nitro derivative is obtained as the major product.



- (iii) **Sulphonation:** Aniline on sulphonation with concentrated H_2SO_4 first gives anilinium hydrogen sulphate, which on heating with H_2SO_4 at 453-473 K yields p-aminobenzene sulphonic acid (sulphanilic acid) as the major product.



- (iv) **Friedel-Crafts Reaction:** Aniline does not undergo F.C. reaction (both alkylation and acetylation) due to salt formation with AlCl_3 (Lewis acid) which acts as a catalyst. Due to this, N atom of aniline acquires positive charge and hence acts as a strong deactivating group for further reaction.

Exercise 1.2

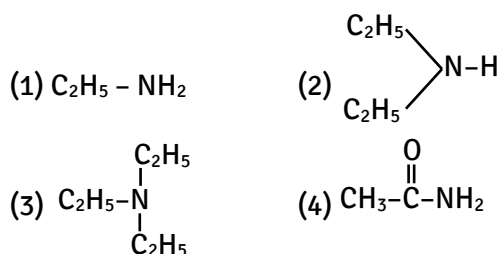
- Among the following, the strong base is –
 - $\text{C}_6\text{H}_5\text{NH}_2$
 - p- $\text{NO}_2\text{C}_6\text{H}_4\text{NH}_2$
 - m- $\text{NO}_2\text{C}_6\text{H}_4\text{NH}_2$
 - $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
- Which of the following should be most volatile?

(I) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$	(II) $(\text{CH}_3)_3\text{N}$
(III) $\begin{array}{c} \text{CH}_3-\text{CH}_2 \\ \quad \quad \quad \diagup \\ \quad \quad \quad \text{NH} \\ \quad \quad \quad \diagdown \\ \quad \quad \quad \text{CH}_3 \end{array}$	(IV) $\text{CH}_3\text{CH}_2\text{CH}_3$

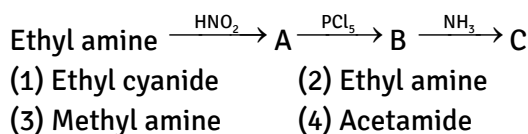
 - II
 - IV
 - I
 - III
- The presence of primary amines can be confirmed by:
 - Reaction with HNO_2
 - Reaction with CHCl_3 and alc. KOH
 - Reaction with Hinsberg reagent
 - All
- A compound (X) having the molecular formula $\text{C}_3\text{H}_9\text{N}$ reacts with benzenesulphonyl chloride to form a solid that is insoluble in alkalis. The compound (X) is:

(1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$	(2) $(\text{CH}_3)_2\text{CHNH}_2$
(3) $\text{CH}_3\text{CH}_2\text{NHCH}_3$	(4) $(\text{CH}_3)_3\text{N}$

5. Which one of the following will not react with the Grignard reagent ($\text{C}_2\text{H}_5\text{MgBr}$):



6. The end-product in the reaction sequence would be:

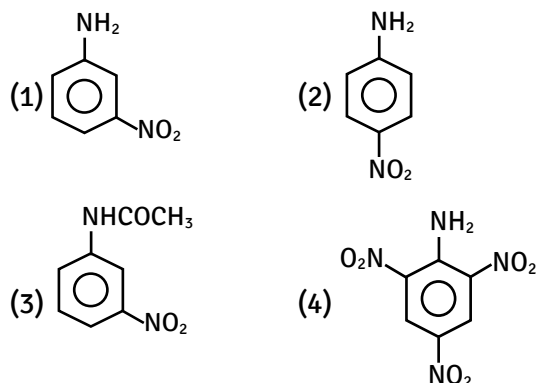
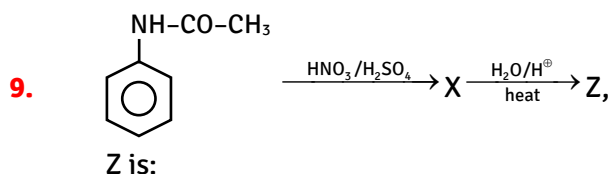


7. Which one of the following tests can be used to identify primary amino group in a given organic compound:-

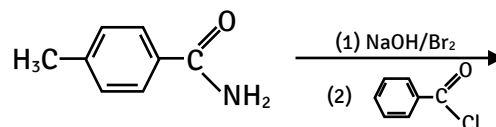
- (1) Iodoform test
(2) Victor Meyer's test
(3) Carbylamine reaction
(4) Schiff's Test

8. Amine whose product is insoluble in NaOH (in Hinsberg test)-

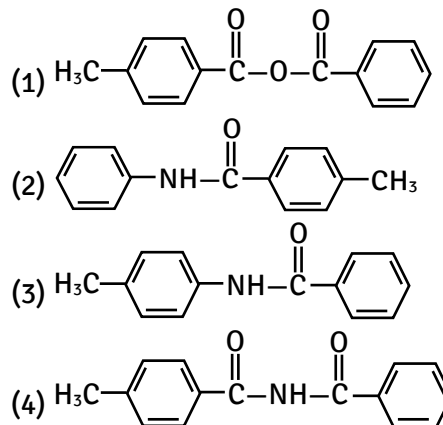
- (1) $\text{CH}_3\text{CH}_2\text{NH}_2$ (2) $(\text{CH}_3)_2\text{CHNH}_2$
(3) $(\text{CH}_3)_2\text{NH}$ (4) $(\text{CH}_3)_3\text{N}$



10. In the reaction the structure of the product [X] is:



[X]



11. Aniline on treatment with bromine water yields white precipitate of:-

- (1) o-Bromoaniline
(2) p-Bromoaniline
(3) 2, 4, 6-Tribromoaniline
(4) m-Bromoaniline

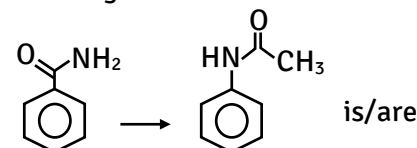
12. Acetanilide when treated with bromine in acetic acid mainly gives:-

- (1) o-Bromoacetanilide
(2) N-Bromoacetanilide
(3) p-Bromoacetanilide
(4) m-Bromoacetanilide

13. Reaction of nitrous acid with aliphatic primary amine in the cold gives

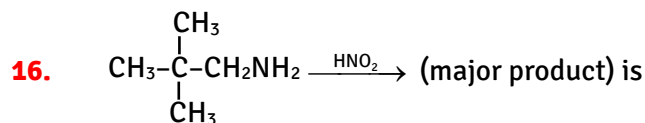
- (1) A diazonium salt (2) An alcohol
(3) A nitrite (4) A dye

14. The reagents needed to convert:



- (1) $\text{KOH} + \text{Br}_2; \text{LiAlH}_4$
(2) $\text{KOH} + \text{Br}_2; \text{CH}_3\text{COCl} / \text{Py}$
(3) $\text{HONO}, \text{Cu}_2\text{Cl}_2, (\text{CH}_3\text{CO})_2\text{O}$
(4) $\text{KOH} + \text{CHCl}_3, \text{CH}_3\text{COCl} / \text{Py}$

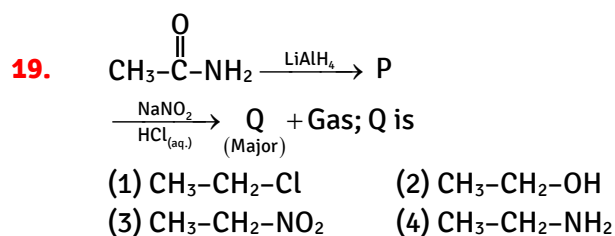
15. Gabriel phthalimide synthesis is used for the preparation of
- (1) Primary aromatic amines
 - (2) Primary aliphatic amines
 - (3) Both (1) and (2)
 - (4) Secondary amines



- (1) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_2\text{OH}$
- (2) $\text{CH}_3-\underset{\text{OH}}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_2\text{CH}_3$
- (3) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_2\text{N}_2^+$
- (4) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}=\text{CH}_2$

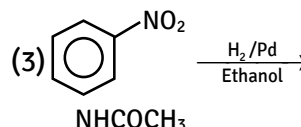
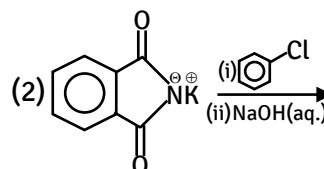
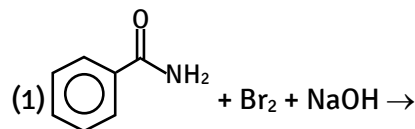
17. The compound 'A' when treated with HNO_3 (in presence of H_2SO_4) gives compound 'B' which is then reduced with Sn and HCl to aniline. The compound 'A' is
- (1) Toluene
 - (2) Benzene
 - (3) Ethane
 - (4) Acetamide

18. Which of the following amine will not react with benzoyl chloride -
- (1) isopropyl amine
 - (2) p-toluidine
 - (3) N, N-Diethyl aniline
 - (4) N-methyl aniline

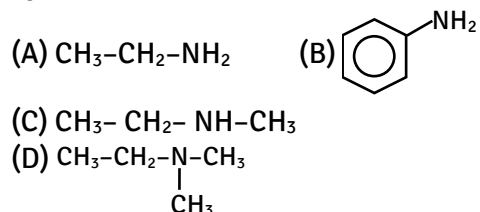


20. Which Statement is true
- (1) Aniline undergoes Friedel craft alkylation at faster rate than benzene
 - (2) Aniline reacts with bromine water at room temperature to give 2,4,6 - tribromo aniline with yellow ppt.
 - (3) Anilinium ion is meta directing
 - (4) All

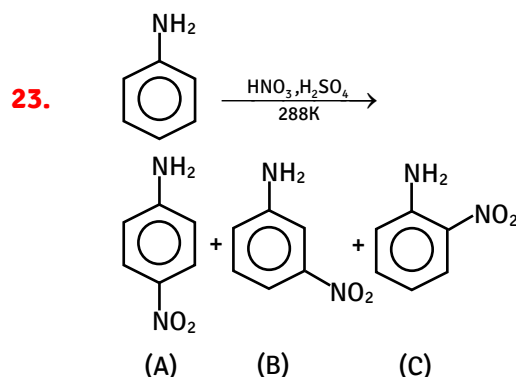
21. Which of the transformation given below would not result in aromatic amines?



22. Which among the following compounds gives foul smelling products on heating with Chloroform and ethanolic potassium hydroxide?



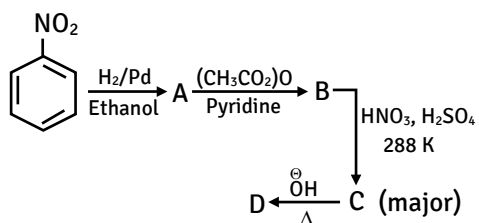
- (1) Only A
- (2) Only A & B
- (3) Only A, C and D
- (4) Only A & C



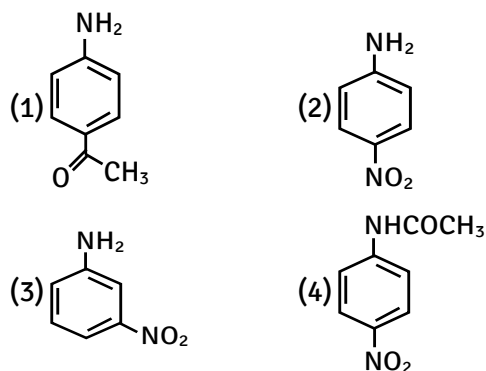
Incorrect statement about the given chemical reaction is :

- (1) In strongly acidic medium, aniline is protonated to form the anilinium ion which is meta directing.
- (2) % yield of 'B' is greater than 'C'
- (3) % yield of 'B' is greater than 'A'
- (4) Reaction is possible and compound A will be major product.

24. Consider the following reaction sequence



Final Product 'D' is-

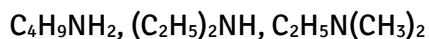


25. $\text{R-NH}_2 + \text{CHCl}_3 + \text{KOH} \xrightarrow{\text{Heat}} (\text{Y}) + \text{KCl} + \text{H}_2\text{O}$

Product (Y) has foul smell. The reaction is known as

- (1) Wurtz reaction
- (2) Kolbe's reaction
- (3) Hell-Volhard-Zelinsky reaction
- (4) Carbylamine reaction

26. The correct order of boiling points of the following isomeric amines is

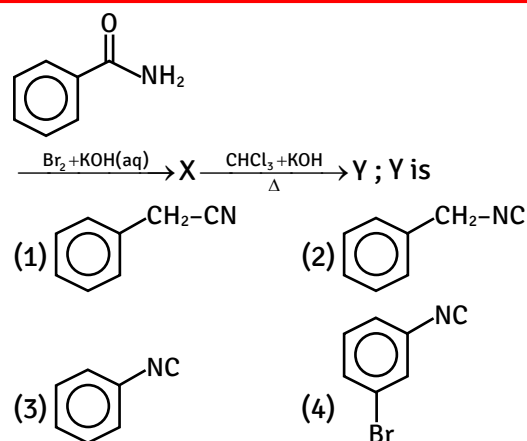


- (1) $\text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2 > (\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_4\text{H}_9\text{NH}_2$
- (2) $(\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2 > \text{C}_4\text{H}_9\text{NH}_2$
- (3) $\text{C}_4\text{H}_9\text{NH}_2 > (\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2$
- (4) $(\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_4\text{H}_9\text{NH}_2 > \text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2$

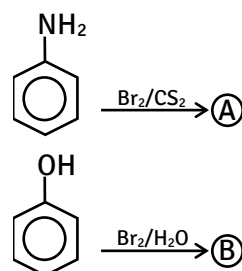
27. Ethyl amine and aniline can be distinguished by :-

- (1) Isocyanide test
- (2) Azo dye test
- (3) Hinsberg's reagent
- (4) Grignard reagent

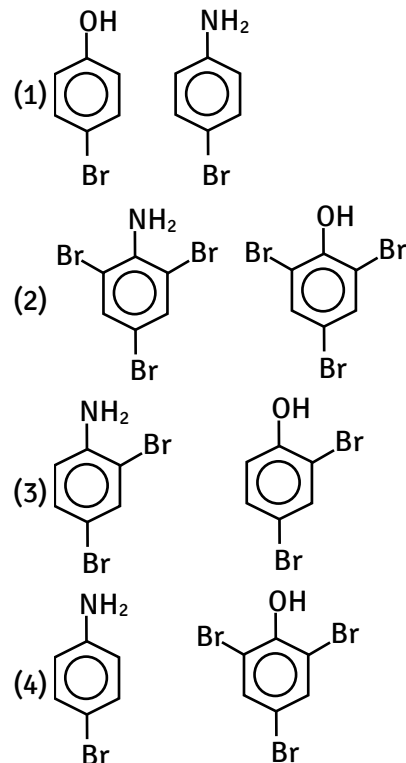
28.



29.



A & B respectively are -

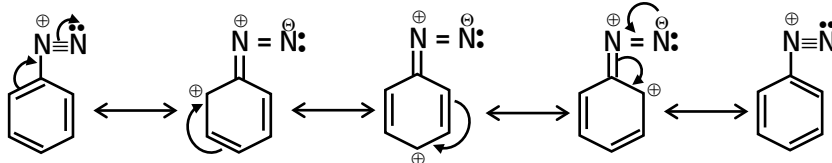


30. Primary, secondary and tertiary amines can be separated using :-

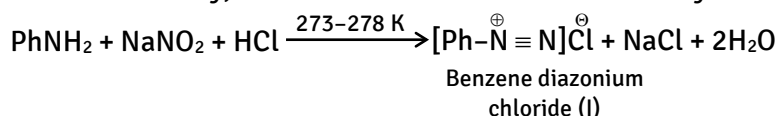
- (1) Para-Toluene sulphonyl chloride
- (2) Chloroform and KOH
- (3) Benzene sulphonic acid
- (4) Acetyl amide

Diazonium Salts

Introduction: 1° aromatic amines form arene diazonium salts which are stable for a short time in solution at low temperature (273-278 K) unlike 1° aliphatic amines which form highly unstable alkyl diazonium salts. The stability of arenediazonium ion is due to resonance stabilisation.



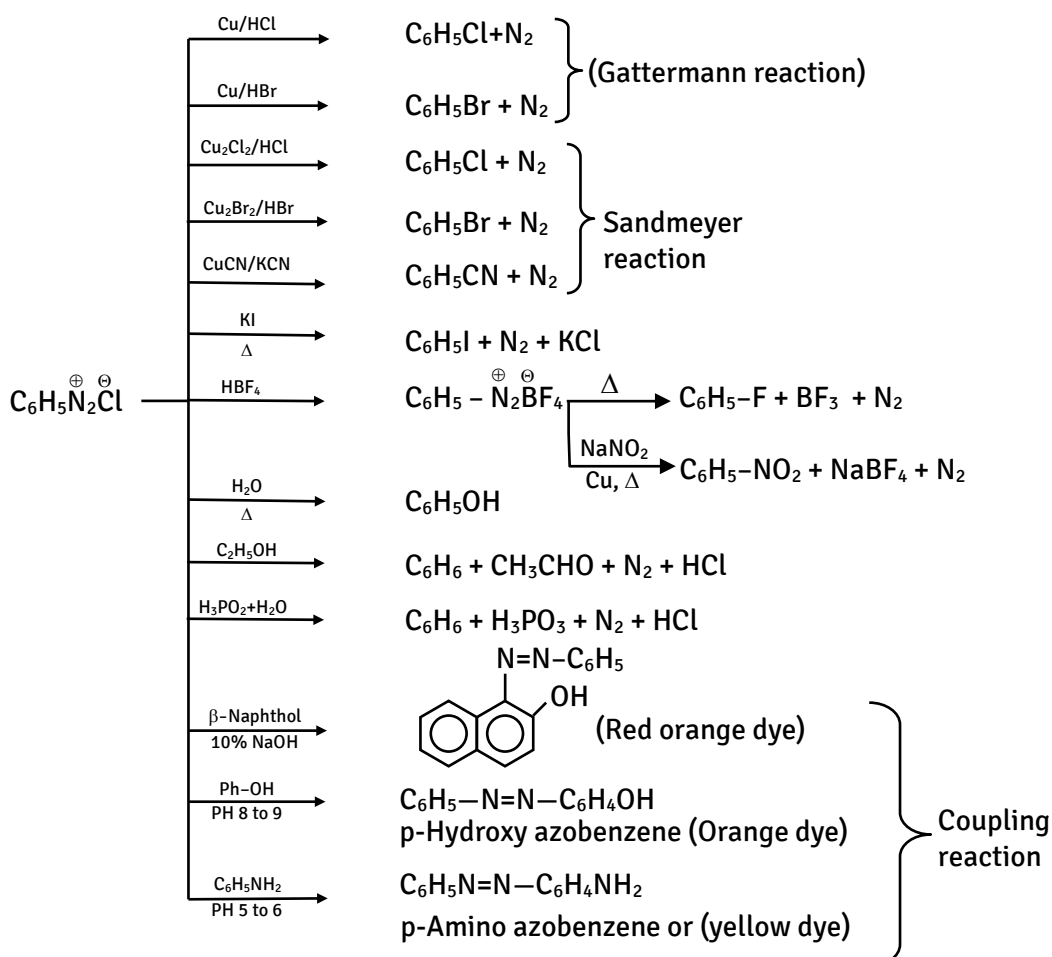
Diazotisation: The conversion of 1° aromatic amine into diazonium salt with HNO_2 (nitrous acid) is known as diazotisation. HNO_2 is produced in the reaction mixture by the reaction of NaNO_2 and HCl . Due to instability, diazonium salts are used immediately after their preparation.



Physical properties of benzene diazonium salt :

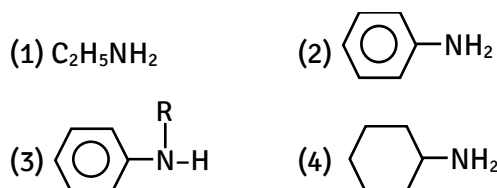
Benzenediazonium chloride is a colourless crystalline solid. It is readily soluble in water and is stable in cold but reacts with water when warmed. It decomposes easily in the dry state. Benzenediazonium fluoroborate is water insoluble and stable at room temperature.

Chemical properties of benzene diazonium salt :



Exercise 1.3

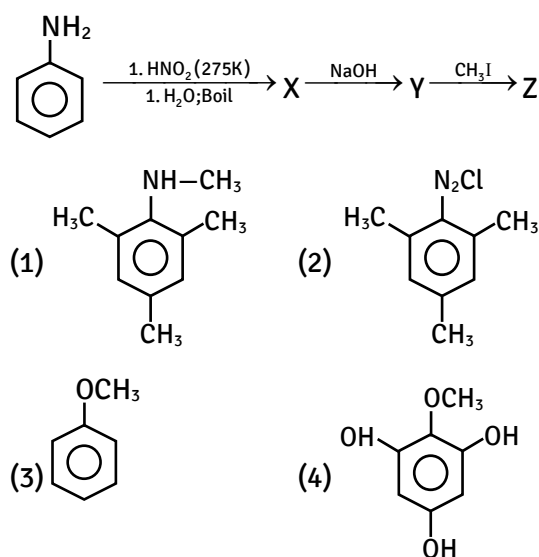
1. Which of the following amines give N-nitroso derivative with NaNO_2 and HCl :-



2. Which statement is incorrect -

- (1) $-\text{NH}_2$ group is present in both ethylamine and aniline
 (2) Both ethylamine and aniline give isocyanide test
 (3) Ethylamine and aniline give unpleasant smell with CHCl_3 and KOH
 (4) Both ethylamine and aniline give hydroxy compound with HNO_2 at 0°C

3. Identify 'Z' in the reaction given below



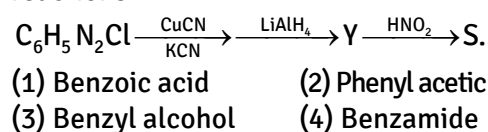
4. $\text{ArN}_2^+\text{Cl}^- \xrightarrow{\text{Cu/HCl}} \text{ArCl} + \text{N}_2 + \text{CuCl}$;
 the reaction is named as _____.

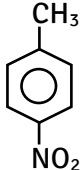
- (1) Sandmeyer reaction
 (2) Gattermann reaction
 (3) Balz Schiemann reaction
 (4) Carbylamine reaction

5. Compound A has a molecular formula $\text{C}_7\text{H}_7\text{NO}$. On treatment with Br_2 and KOH , A gives an amine B which gives carbylamine test. B upon diazotisation and coupling with phenol gives an azo dye. A can be -

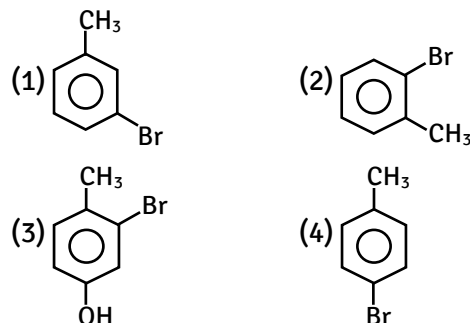
- (1) $\text{C}_6\text{H}_5\text{CONHCOCH}_3$
 (2) $\text{C}_6\text{H}_5\text{CONH}_2$
 (3) $\text{C}_6\text{H}_5\text{NO}_2$
 (4) o, m or p- $\text{C}_6\text{H}_4(\text{NH}_2)\text{CHO}$

6. Identify S in the following sequence of reactions

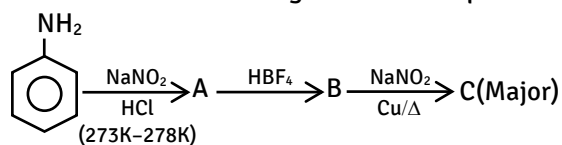


7.  $\xrightarrow[\text{(iv) H}_3\text{PO}_2, \text{H}_2\text{O}]{\text{(i) Br}_2, \text{Fe} \rightarrow \text{(ii) Sn, HCl} \rightarrow \text{(iii) NaNO}_2, \text{HCl, 273K}}$ (A)

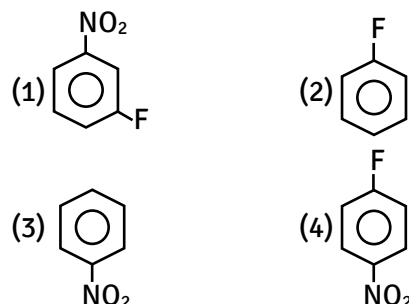
Final product (A) in the above sequence reaction is



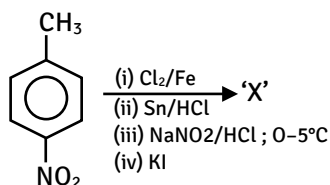
8. Consider the following reaction sequence



Product 'C' is -

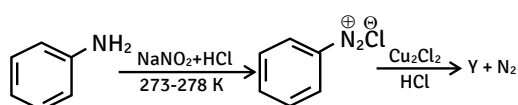


9. The Major Product 'X' formed in the following reaction sequence is



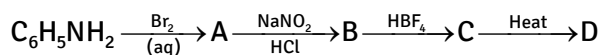
- (1) (2)
- (3) (4)

10. Identify the compound Y in the following reaction.



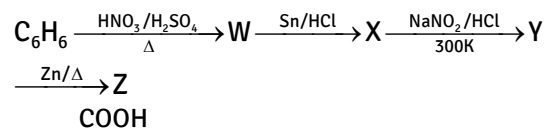
- (1) (2)
- (3) (4)

11. The product 'D' in the following sequence of reactions is



- (1) 2, 4, 6-tribromofluorobenzene
 (2) fluorobenzene
 (3) p-bromofluorobenzene
 (4) tribromobenzene.

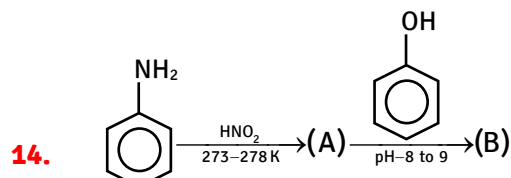
12. 'Z' in the following sequence of reactions is



- (1) (2)
- (3) (4)

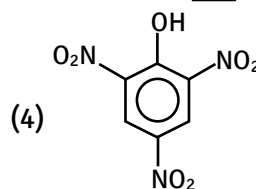
13. Which is Sandmeyer reaction among the following :

- (1) $\text{C}_6\text{H}_5\text{N}_2\text{Cl} \xrightarrow[\Delta]{\text{HBF}_4} \text{C}_6\text{H}_5\text{F}$
 (2) $\text{C}_6\text{H}_5\text{N}_2\text{Cl} \xrightarrow{\text{Cu}/\text{HCl}} \text{C}_6\text{H}_5\text{Cl}$
 (3) $\text{C}_6\text{H}_5\text{N}_2\text{Cl} \xrightarrow{\text{Cu}_2\text{Cl}_2/\text{HCl}} \text{C}_6\text{H}_5\text{Cl}$
 (4) All of these

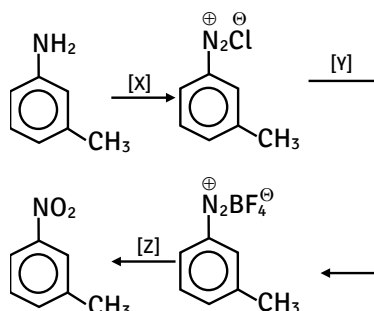


Final product ?

- (1) $\text{Ph}-\text{N}=\text{N}-\text{C}_6\text{H}_4-\text{NH}_2$
 (2) $\text{Ph}-\text{N}=\text{N}-\text{Ph}$
 (3) $\text{Ph}-\text{N}=\text{N}-\text{C}_6\text{H}_4-\text{OH}$



- 15.



X, Y, Z are-

- (1) NaNO_2/Cu , Δ , HBF_4 , $\text{NaNO}_2+\text{HCl}/0-5^\circ\text{C}$
 (2) $\text{NaNO}_2+\text{HCl}/0-5^\circ\text{C}$, $\text{B}_2\text{H}_6/\text{THF}$, NaNO_2/Cu , Δ
 (3) $\text{NaNO}_2+\text{HCl}/0-5^\circ\text{C}$, AgF , NaNO_2/Cu , Δ
 (4) $\text{NaNO}_2+\text{HCl}/0-5^\circ\text{C}$, HBF_4 , NaNO_2/Cu , Δ

Exercise 2

1. Which one of the following compound gives a secondary amine on reduction

(1) Nitromethane
(2) Nitrobenzene
(3) Methyl isocyanide
(4) Methyl cyanide

2. $\text{C}_6\text{H}_5\text{COCl} \xrightarrow{\text{NH}_3} \text{X} \xrightarrow{\text{P}_2\text{O}_5} \text{Y} \xrightarrow[\text{H}_2]{\text{Ni}} \text{Z}$, Z is

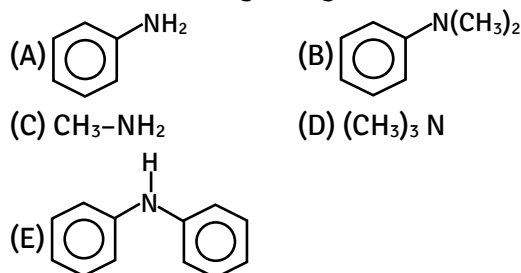
The end product in the above sequence of reaction is -

(1) Benzoic acid (2) Aniline
(3) Benzyl amine (4) Benzonitrile

3. Amongst the given set of reactants, the most appropriate for preparing 2° amine is _____.

(1) 2° R-Br + NH₃
(2) 2° R-Br + NaCN followed by H₂/Pt
(3) 1° R-NH₂ + RCHO followed by H₂/Pt
(4) 1° R-Br (2 mol) + potassium phthalimide followed by H₃O⁺/heat.

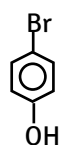
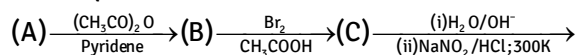
4. Which among the following compounds react with Hinsberg's reagent?



Choose the correct answer from the options given below:

(1) B and D only (2) C and D only
(3) A, B and E only (4) A, C and E only

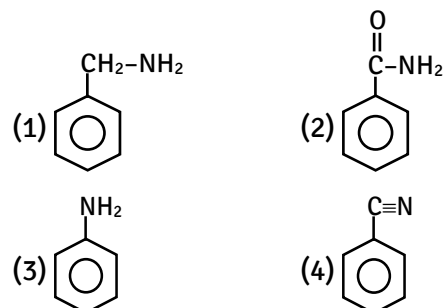
5. Compound A from the following reaction sequence is:



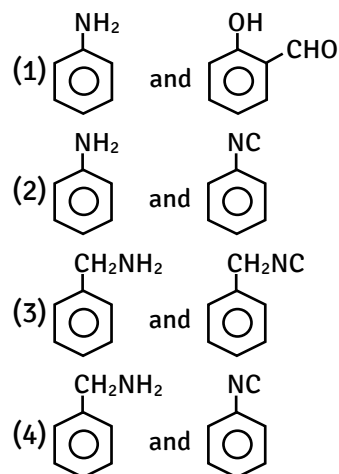
(1) Benzoic acid (2) Aniline
(3) Salicylic acid (4) Phenol

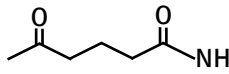
6. $\text{PhMg Br} \xrightarrow[\text{(iii) NH}_3, \Delta]{\text{(i) CO}_2, \text{(ii) H}_3\text{O}^+} \text{(A)} \xrightarrow{\text{Br}_2, \text{NaOH}} \text{(B)}$

What would be the final product of above sequence reaction?

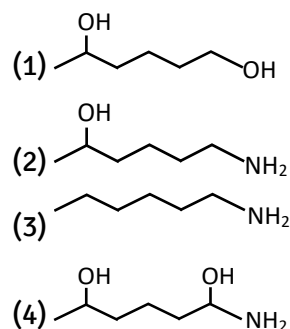


7. Hoffmann bromamide degradation of benzamide gives product 'A' which upon heating with CHCl₃ and NaOH gives product 'B'. The structure of 'A' and 'B' respectively are :

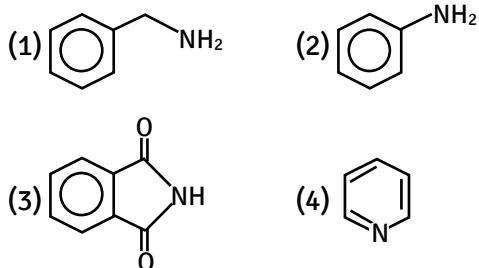


8.  $\xrightarrow[\text{(III) H}_3\text{O}^+]{\text{(I) Zn-Hg/HCl, (II) LiAlH}_4}$ Product

major product is-

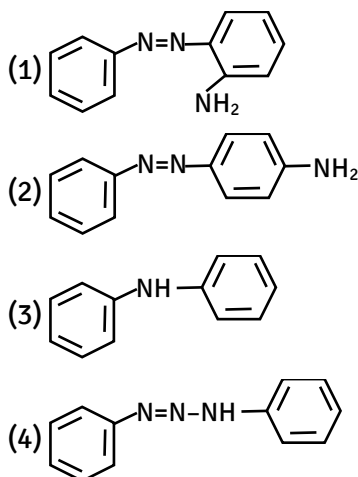


9. Which of the following is least basic

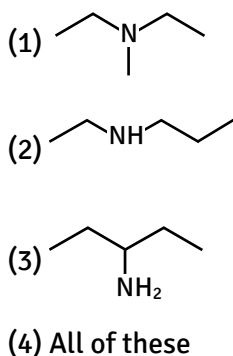


10. $\text{C}_6\text{H}_5\text{NO}_2 \xrightarrow{\text{Sn}+\text{HCl}} \text{'A'} \xrightarrow[\text{H}^+]{\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-} \text{'P'}$ (Yellow coloured)

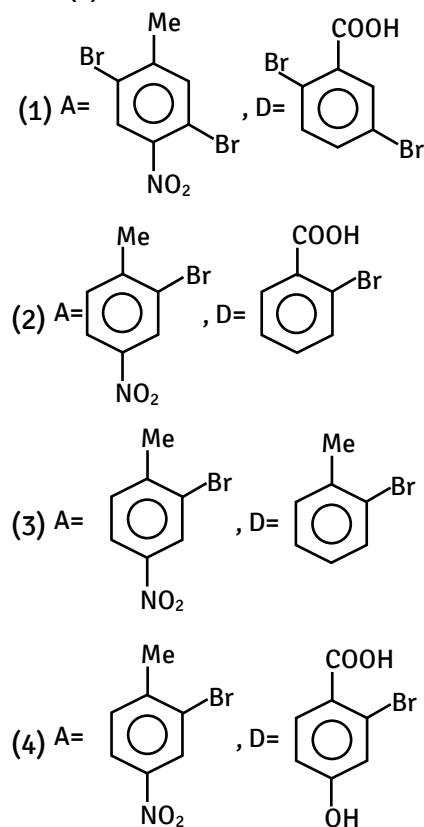
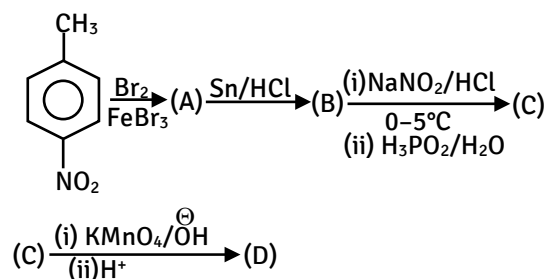
Consider the above reaction, the product 'P' is :



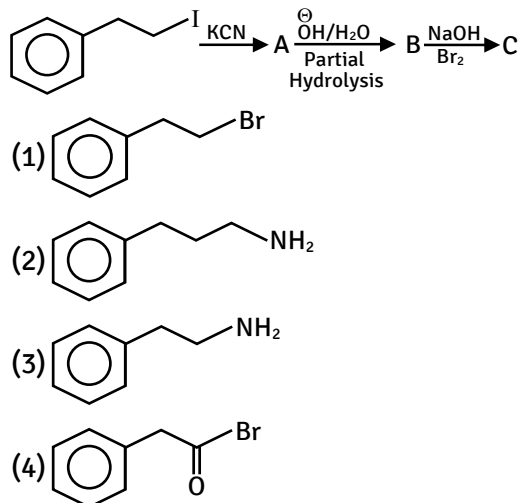
11. A compound 'X' having molecular formula $\text{C}_5\text{H}_{13}\text{N}$ on reaction with Hinsberg's reagent gives compound 'Y' which does not dissolve in aqueous alkali. The structure of the compound 'X' could be.



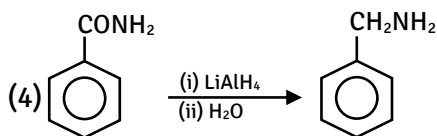
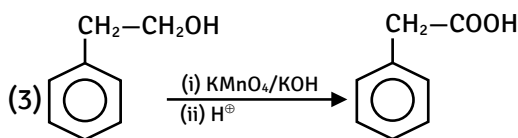
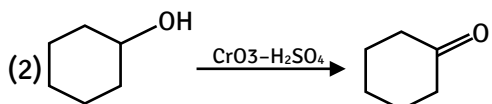
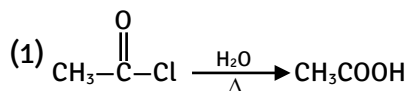
12. Identify the product A and D formed in the following reaction



13. Identify the Major Product 'C' formed in the following reaction sequence :



14. Select the incorrect product among the following reactions:



15. Hoffmann bromamide degradation reaction is shown by which of the following?

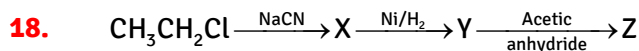
- (1) ArNH_2 (2) ArCONH_2
(3) ArNO_2 (4) ArCH_2NH_2

16. Amine that cannot be prepared by Gabriel phthalimide synthesis is

- (1) aniline (2) benzyl amine
(3) methyl amine (4) iso-butylamine.

17. The decreasing order of boiling points of isomeric amines is p-amine > s-amine > t-amine. This trend of boiling point can be explained as

- (1) boiling point increases with increase in molecular mass
(2) tertiary amines have highest boiling point due to highest basicity
(3) intermolecular hydrogen bonding is maximum in primary amines and absent in tertiary amines
(4) intramolecular hydrogen bonding is present in tertiary amines.



Z in the above reaction is

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NHCOCH}_3$
(2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
(3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONHCH}_3$
(4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONHCOCH}_3$

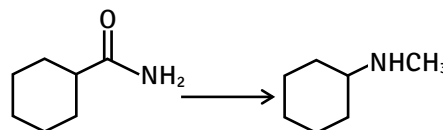
19. Which of the following amines does not react with Hinsberg's reagent?

- (1) $\text{CH}_3\text{CH}_2\text{NH}_2$ (2) CH_3NHCH_3
(3) $(\text{CH}_3\text{CH}_2)_3\text{N}$ (4) All of these

20. Which of the following compounds cannot be identified by carbylamine test?

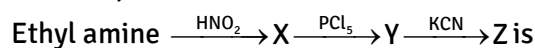
- (1) $\text{CH}_3\text{CH}_2\text{NH}_2$
(2) $(\text{CH}_3)_2\text{CHNH}_2$
(3) $\text{C}_6\text{H}_5\text{NH}_2$
(4) $\text{C}_6\text{H}_5\text{NHC}_6\text{H}_5$

21. Which of the following methods can be used to carry out the following conversions?



	(i)	(ii)	(iii)
(1)	Br_2/KOH	CHCl_3/KOH	H_2/Pd
(2)	KCN	H_2/Pd	Sn/HCl
(3)	CuCN	$\text{H}_2\text{O}/\text{H}^+$	H_2/Pd
(4)	$\text{HNO}_3/\text{H}_2\text{SO}_4$	$(\text{CH}_3\text{CO})_2\text{O}$	Fe/HCl

22. The end product Z of the reaction :

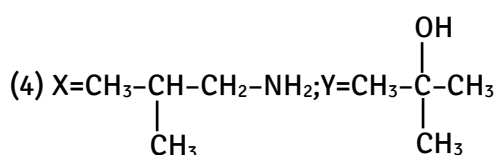
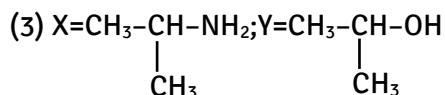
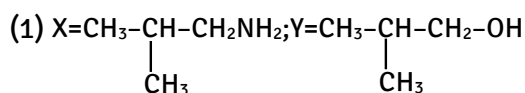
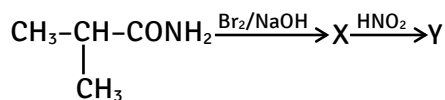


- (1) propanenitrile (2) triethylamine
(3) diethylamine (4) propylamine.

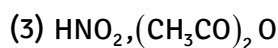
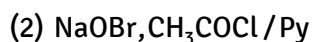
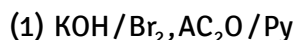
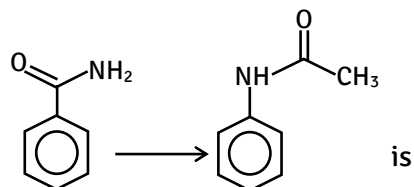
23. Which of the following tests is suitable to differentiate between aniline and benzylamine?

- (1) Aniline gives dye test on diazotisation and reaction with β -naphthol while benzylamine gives alcohol.
(2) Benzylamine gives green dye with β -naphthol after diazotisation while aniline gives orange dye.
(3) Aniline gives carbylamine reaction while benzylamine does not.
(4) Benzylamine gives carbylamine reaction while aniline does not.

24. Identify X and Y in the reaction.



25. The reagent required to convert



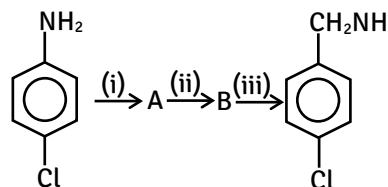
26. An organic compound (X) was treated with sodium nitrite and HCl in ice cold conditions. Bubbles of nitrogen gas were seen coming out. The compound (X) may be

(1) a secondary aliphatic amine
 (2) a primary aromatic amine
 (3) a primary aliphatic amine
 (4) a tertiary amine.

27. Which of the following reactions is not correctly matched?

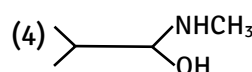
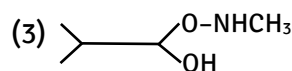
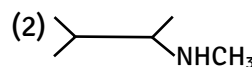
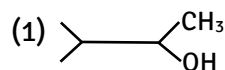
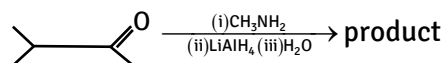
(1)	Reaction used to convert amide into primary amine with one carbon atom less	Hoffmann Bromamide reaction
(2)	Reaction used to convert primary amines into isocyanides	Carbylamine reaction
(3)	Reaction used to distinguish primary, secondary and tertiary amines	Hinsberg's reaction
(4)	Preparation of primary amines using phthalimide	Victor Meyer's synthesis

28. Mark the correct route of the conversion of p-chloroaniline to p-chlorobenzylamine.



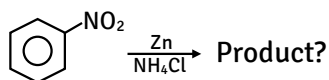
	(i)	(ii)	(iii)
(1)	Alkylation	KCN	H_2/PT
(2)	Diazotisation	CuCN	H_2/PT
(3)	Oxidation	H_2/Pt	Hydrolysis
(4)	Diazotisation	$\text{H}_2\text{O}/\text{H}^+$	Sn/HCl

29. The major organic product formed from the following reaction



30. The best reagent for converting 2-phenylpropanamide into 2-phenylpropanamine is _____.
 (1) excess H_2
 (2) Br_2 in aqueous NaOH
 (3) iodine in the presence of red phosphorus
 (4) $LiAlH_4$ in ether.

31. What is the product obtained in the following reaction:



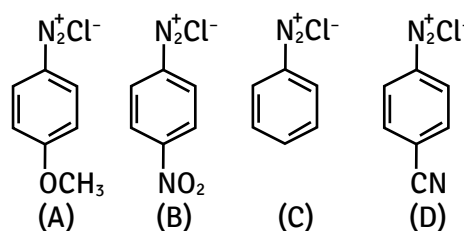
- (1) (2)
 (3) (4)

32. Best method for preparing primary amines from alkyl halides without changing the number of carbon atoms in the chain is
 (1) Hoffmann bromamide reaction
 (2) Gabriel phthalimide synthesis
 (3) Carbyl amine reaction
 (4) Ammonolysis of alkyl halide

33. Which of the following methods of preparation of amines will not give same number of carbon atoms in the chain of amines as in the reactant?

- (1) Reaction of nitrite with $LiAlH_4$.
 (2) Reaction of amide with $LiAlH_4$ followed by treatment with water.
 (3) Heating alkyl halide with potassium salt of phthalimide followed by hydrolysis.
 (4) Treatment of amide with bromine in aqueous solution of sodium hydroxide.

34. The correct stability order of the following diazonium salt is

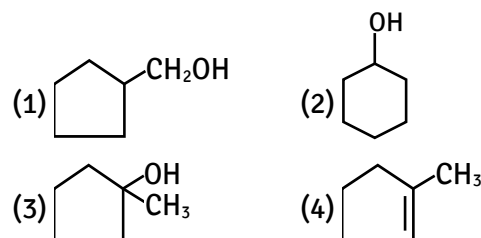


- (1) (A) > (B) > (C) > (D)
 (2) (A) > (C) > (D) > (B)
 (3) (C) > (A) > (D) > (B)
 (4) (C) > (D) > (B) > (A)

35. The correct sequential addition of reagents in the preparation of 3-nitrobenzoic acid from benzene is

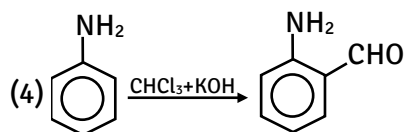
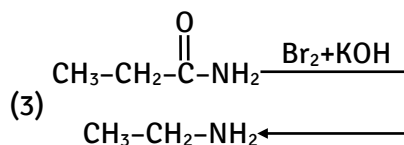
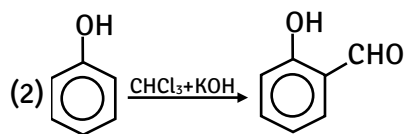
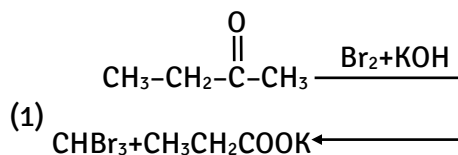
- (1) $Br_2 / AlBr_3, HNO_3 / H_2SO_4, Mg / ether, CO_2, H_3O^+$
 (2) $Br_2 / AlBr_3, NaCN, H_3O^+, HNO_3 / H_2SO_4$
 (3) $Br_2 / AlBr_3, HNO_3 / H_2SO_4, NaCN, H_3O^+$
 (4) $HNO_3 / H_2SO_4, Br_2 / AlBr_3, Mg / ether, CO_2, H_3O^+$

36. Find out major product :

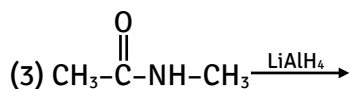
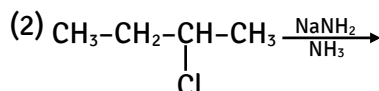
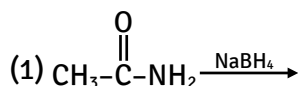


37. $CH_3CH_2COOH \xrightarrow{SOCl_2} A \xrightarrow[\Delta]{NH_3} B \xrightarrow[KOH]{Br_2} C$
 Structure of compound C is -
 (1) $CH_3CH_2CH_2NH_2$
 (2) $CH_3CH_2NH_2$
 (3) $CH_3CH_2NHCH_3$
 (4) $CH_3CH_2CONH_2$

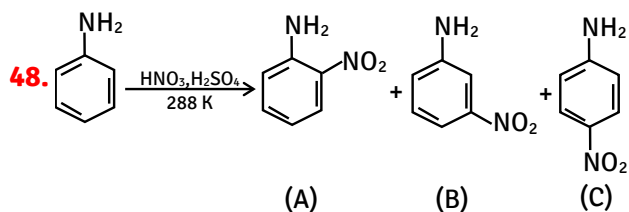
46. Incorrect reaction is :-



47. Amine is obtained as major product in



(4) All of these



Consider the given reaction,
percentage yield of products:

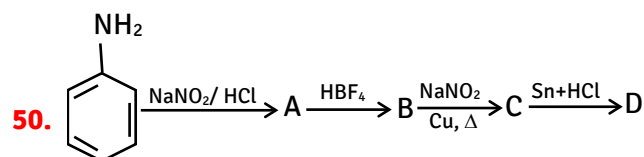
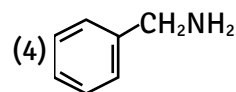
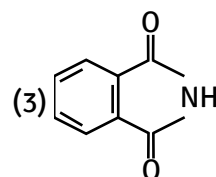
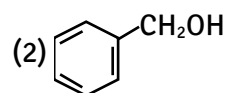
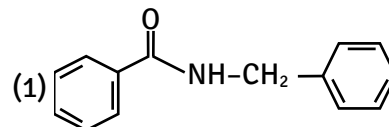
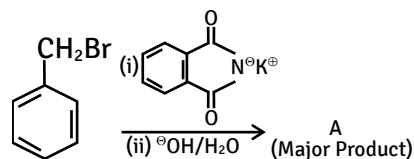
(1) $\text{C} > \text{A} > \text{B}$

(2) $\text{B} > \text{C} > \text{A}$

(3) $\text{A} > \text{C} > \text{B}$

(4) $\text{C} > \text{B} > \text{A}$

49. What is A in the following reaction



Which statement is correct

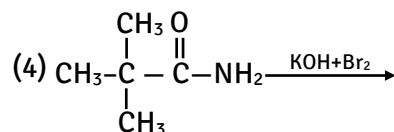
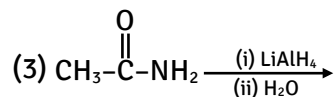
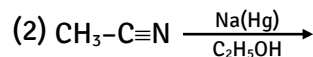
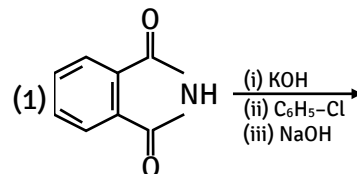
(1) A gives orange color with phenol

(2) B is fluorobenzene

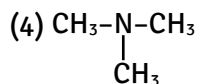
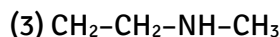
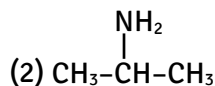
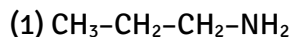
(3) D is benzylamine

(4) D is more basic than pyridine

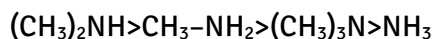
51. 1° Amine is not formed by which reaction.



52. Which of the following amine among C_3H_9N will not react with Hinsberg reagent



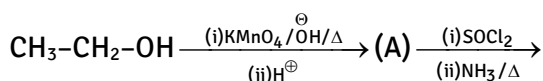
53. Some statements are given about following order of basic strength in aqueous medium.



- (a) More substituted ammonium ion gets stabilized due to +I effect of alkyl group.
 (b) The greater the size of ammonium ion, more will be solvation, so more stable ion so more basicity.
 (c) More the alkyl groups, more steric hindrance to H-bonding in ammonium ion.
 (d) Basic strength in aqueous medium is explained by more than one type of effects. Correct statements is/are

- (1) a, b, c, d
 (2) a, b, d only
 (3) a, c, d only
 (4) a and d only

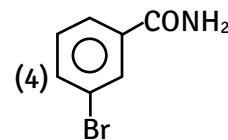
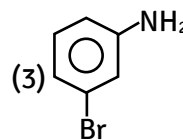
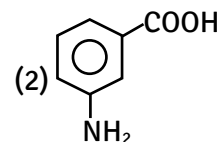
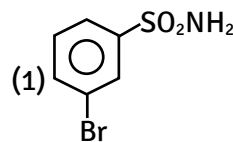
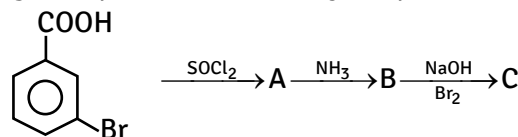
54. In the given reaction sequence:



(B) $\xrightarrow{Br_2 / KOH}$ (C) will be -

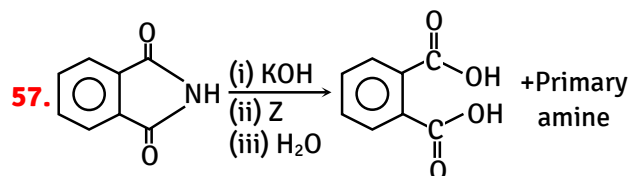
- (1) Methylamine (2) Ethylamine
 (3) Propylamine (4) Acetamide

55. In a set of reactions m-bromobenzoic acid gave a product 'C'. Identify the product 'C'



56. An organic compound 'A' on treatment with NH_3 gives 'B' which on heating gives 'C'. 'C' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is:

- (1) CH_3CH_2COOH
 (2) CH_3COOH
 (3) $CH_3CH_2CH_2COOH$
 (4) $CH_3-\underset{\substack{| \\ CH_3}}{CH}COOH$



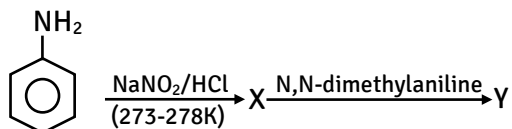
Z may not be :

- (1) CH_3-Cl (2) $CH_3-\underset{\substack{| \\ CH_3}}{CH}-Cl$
 (3) (4)

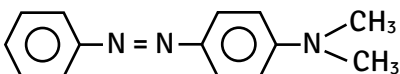
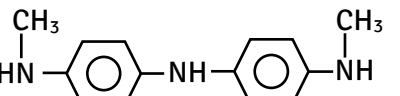
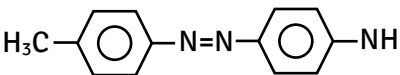
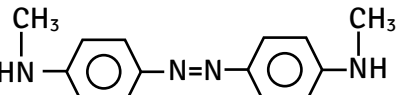
58. The end product (B) formed in the reaction
- $$CH_2=CH_2 \xrightarrow[2. KCN (excess)]{1. Br_2 / CCl_4} A \xrightarrow[2. H_3O^+]{1. LiAlH_4} B$$

- (1) $CH_2=CHCH_2NH_2$
 (2) $H_2N(CH_2)_4NH_2$
 (3) $CH_3NH(CH_2)_2NHCH_3$
 (4) $NC(CH_2)_2CN$

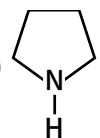
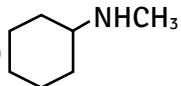
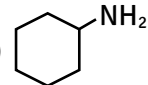
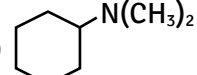
59. Aniline in a set of the following reactions yielded a coloured product 'Y'



The structure of 'Y' would be

- (1) 
- (2) 
- (3) 
- (4) 

60. Among the following compounds which one will produce a Schiff base on reaction with cyclohexanone:

- (1) 
- (2) 
- (3) 
- (4) 

Exercise 3

1. **Assertion:** Hoffmann's bromamide reaction is given by primary amines.
Reason: Primary amines are more basic than secondary amines.
- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
 - (2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion
 - (3) Both Assertion and Reason are incorrect.
 - (4) Assertion is not correct but Reason is correct.
2. **Assertion:** N-Ethylbenzene sulphonamide is soluble in alkali.
Reason: Hydrogen attached to nitrogen in sulphonamide is strongly acidic.
- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
 - (2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion
 - (3) Both Assertion and Reason are incorrect.
 - (4) Assertion is not correct but Reason is correct.
3. **Assertion:** Only a small amount of HCl is required in the reduction of nitro compounds with iron scrap and HCl in the presence of steam.
Reason: FeCl_2 formed gets hydrolysed to release HCl during the reaction.
- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
 - (2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion
 - (3) Both Assertion and Reason are incorrect.
 - (4) Assertion is not correct but Reason is correct.
4. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).
Assertion: The correct boiling point order of isomeric amines is -
Primary > Secondary > Tertiary
Reason: The intermolecular H-bonding in primary amine is maximum due to two hydrogen available for more association but in tertiary amine there is no hydrogen available for H-bond.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 - (3) (A) is correct but (R) is not correct.
 - (4) (A) is not correct but (R) is correct.
5. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).
Assertion: The alkyl diazonium salt are unstable but the arenediazonium salt are stable for a short time at low temperature ($0^\circ - 5^\circ \text{C}$)
Reason: Due to resonance in arenediazonium salt carbon-nitrogen bond becomes stronger than alkyl diazonium salt.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 - (3) (A) is correct but (R) is not correct.
 - (4) (A) is not correct but (R) is correct.

6. **Assertion :** tert-Butyl amine can be formed by Gabriel phthalimide synthesis.

Reason: It follows S_N1 mechanism.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

7. **Assertion:** Gabriel phthalimide synthesis cannot be used to prepare aromatic primary amines.

Reason: Aryl halides do not undergo nucleophilic substitution reaction with the anion formed by phthalimide.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

8. **Assertion:** Experimental reaction of CH_3Cl with aniline and anhydrous AlCl_3 does not give o and p-methylaniline.

Reason: The -NH_2 group of aniline becomes deactivating because of salt formation with anhydrous AlCl_3 and hence yields m-methyl aniline as the product.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

9. **Assertion :** Carbylamine reaction is used in identification of primary amines but not secondary and tertiary amines.

Reason: Primary amines have boiling points higher than isomeric secondary and tertiary amines.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

10. **Assertion:** N,N-diethyl benzene sulphonamide is insoluble in alkali.

Reason: sulphonyl group attached to nitrogen atom is strong electron withdrawing group.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- (3) If the Assertion is correct but Reason is incorrect.
- (4) If the Assertion is incorrect and Reason is correct.

11. Given below are two statements:

Statement-I: Aryl fluoride should not be prepared by direct halogenation of benzene.

Statement-II: This is easily obtained by reaction of Benzene diazonium salt with fluoroboric acid/ Δ .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I and Statement-II are correct.
- (2) Both Statement-I and II are incorrect.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement-II is correct.

12. Statement-I: Butan-1-ol has boiling point higher than Butan-1-amine

Statement-II: Solubility of ethanol is greater than ethanal in polar solvent.

- (1) Both statement I and statement II are true
- (2) Both statement I and statement II are false
- (3) Statement I is true, but statement II is false
- (4) Statement I is false but statement II is true

13. Given below are two statements :

Statement I : -NHCOCH_3 group is also o/p-directing like -NH_2 group but it is less activating than -NH_2 group.

Statement II: Lone pair of nitrogen in -NHCOCH_3 group is in resonance with carbonyl group.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but statement II is incorrect.
- (2) Both statement I and statement II are correct.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both statement I and statement II are incorrect.

14. Given below are two statements:

Statement-I: Mixing the solution of freshly prepared diazonium salt with cuprous chloride or cuprous bromide results in the replacement of the diazonium group by -Cl or -Br .

Statement-II: Replacement of the diazonium group by iodine does not require the presence of cuprous halide and is done simply shaking the diazonium salt with potassium iodide.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Both Statement-I and II are incorrect
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both statements-I and II are correct.

15. Given below two statement:

Statement-I: Aniline react with conc. H_2SO_4 followed by heating at 453-473 K gives p-amino benzene sulphonic acid which gives blood red color in the "Lassaigne's test".

Statement-II : In Friedel-craft alkylation and acylation reactions, aniline forms salt with AlCl_3 catalyst due to this nitrogen of aniline acquires a positive charge and act as deactivating group.

Choose the correct answer from the option given below:

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false.

16. Match the following

Column-I		Column-II	
a)	$\text{R-NH}_2 + \text{CHCl}_3 + \text{KOH} \rightarrow$	p)	Cannizzaro reaction
b)	$\text{ArN}_2\text{X} + \text{Cu}_2\text{Cl}_2 / \text{HCl} \rightarrow$	q)	Stephen's reaction
c)	$2\text{Ar-CHO} + \text{Conc. NaOH} \xrightarrow{\Delta}$	r)	Carbylamine reaction
d)	$\text{R-CN} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) SnCl}_2 + \text{HCl}}$	s)	Sandmeyer's reaction

- (1) (a-r), (b-s), (c-p), (d-q)
- (2) (a-s), (b-r), (c-q), (d-p)
- (3) (a-q), (b-s), (c-p), (d-r)
- (4) (a-p), (b-q), (c-r), (d-s)

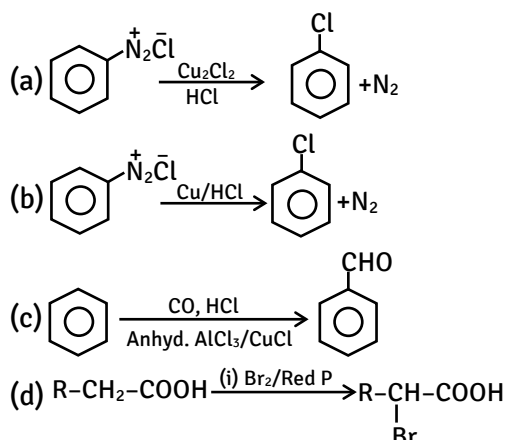
17. Match the column

Column-I (Compound)		Column-II ($\text{P}K_b$ value)	
(i)	Methan amine	(p)	4.75
(ii)	N-Methyl methanamine	(q)	3.38
(iii)	N, N-Dimethyl methanamine	(r)	3.27
(iv)	Ammonia	(s)	4.22

- (1) (i)-(p), (ii)-(q), (iii)-(r), (iv)-(s)
- (2) (i)-(p), (ii)-(r), (iii)-(s), (iv)-(q)
- (3) (i)-(s), (ii)-(r), (iii)-(q), (iv)-(p)
- (4) (i)-(q), (ii)-(r), (iii)-(s), (iv)-(p)

18. Match the list-I with list-II

list-I



list-II

- (p) Gatterman-koch reaction
 (q) Sandmeyer reaction
 (r) Hell-volhard-zelinsky reaction
 (s) Gatterman reaction
 (1) (a-p), (b-q), (c-s), (d-r)
 (2) (a-q), (b-p), (c-s), (d-r)
 (3) (a-s), (b-q), (c-p), (d-r)
 (4) (a-q), (b-s), (c-p), (d-r)

19. Match the list-I with list-II

List-I		List-II	
Name of reaction		Reagent used	
A.	Decarboxylation	(i)	CHCl ₃ /KOH
B.	Finkelstein reaction	(ii)	AgF
C.	Swarts reaction	(iii)	NaI/dry acetone
D.	Carbyl amine reaction	(iv)	NaOH & CaO, Δ

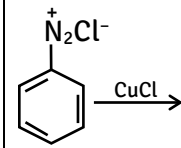
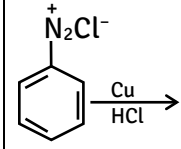
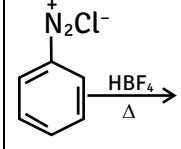
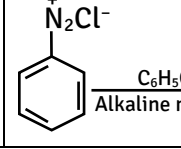
- (1) (A-iv), (B-ii), (C-iii), (D-i)
 (2) (A-iv), (B-iii), (C-ii), (D-i)
 (3) (A-i), (B-iii), (C-ii), (D-iv)
 (4) (A-iii), (B-ii), (C-i), (D-iv)

20. Watch the compounds given in column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	Benzenesulphonyl chloride	(i)	Zwitter ion
(B)	Sulphanilic acid	(ii)	Hinsberg's reagent
(C)	Alkyldiazonium salts	(iii)	Dyes
(D)	Aryldiazonium salts	(iv)	Conversion to alcohols

- (1) (A) → (iv), (B) → (iii), (C) → (i), (D) → (ii)
 (2) (A) → (ii), (B) → (iv), (C) → (iii), (D) → (i)
 (3) (A) → (ii), (B) → (i), (C) → (iv), (D) → (iii)
 (4) (A) → (ii), (B) → (iii), (C) → (iv), (D) → (i)

21. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)		(i)	Coupling reaction
(B)		(ii)	Balz-Schiemann reaction
(C)		(iii)	Gatterman reaction
(D)		(iv)	Sandmeyer reaction

- (1) (A) → (iv), (B) → (iii), (C) → (ii), (D) → (i)
 (2) (A) → (iii), (B) → (iv), (C) → (i), (D) → (ii)
 (3) (A) → (ii), (B) → (iii), (C) → (iv), (D) → (i)
 (4) (A) → (i), (B) → (ii), (C) → (iii), (D) → (iv)

22. Match the column :-

Column-I		Column-II	
Compound		pK _b value	
(A)	Benzenamine	(p)	9.38
(B)	Phenyl methanamine	(q)	4.70
(C)	Ethanamine	(r)	3.29
(D)	N, N-dimethyl aniline	(s)	8.92

(1) A-p, B-q, C-r, D-s

(2) A-q, B-p, C-r, D-s

(3) A-p, B-q, C-s, D-r

(4) A-r, B-s, C-q, D-p

23. Match List -I and List-II

List-I		List-II	
(a)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \rightarrow \text{R}-\text{CHO}$	(i)	Br_2/NaOH
(b)	$\text{R}-\text{CH}_2-\text{COOH} \rightarrow \text{R}-\underset{\text{Cl}}{\underset{ }{\text{CH}}}-\text{COOH}$	(ii)	$\text{H}_2/\text{Pd}-\text{BaSO}_4$
(c)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(iii)	$\text{Zn}(\text{Hg})/\text{Conc. HCl}$
(d)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \rightarrow \text{R}-\text{CH}_2-\text{CH}_3$	(iv)	$\text{Cl}_2/\text{Red P, H}_2\text{O}$

Choose the correct answer from the options given below :

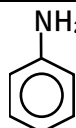
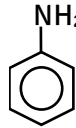
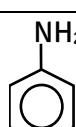
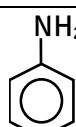
(1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

24. Match the column-I with Column-II

Column-I (Reaction)		Column-II (Process)	
(a)	 + $\text{CH}_3-\text{CH}_2-\text{Cl}$	(p)	Acetylation
(b)	 + $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$	(q)	Alkylation
(c)	 + $\text{CHCl}_3 + \text{Alc. KOH}$	(r)	Diazotization
(d)	 + $\text{HNO}_2 \xrightarrow{0^\circ-5^\circ\text{C}}$	(s)	Carbylamine reaction
		(t)	Halogenation

Correct match is

(1) (a) $\rightarrow$ (q), (b) $\rightarrow$ (p), (c) $\rightarrow$ (s), (d) $\rightarrow$ (r)

(2) (a) $\rightarrow$ (t), (b) $\rightarrow$ (p), (c) $\rightarrow$ (s), (d) $\rightarrow$ (r)

(3) (a) $\rightarrow$ (q), (b) $\rightarrow$ (p), (c) $\rightarrow$ (t), (d) $\rightarrow$ (s)

(4) (a) $\rightarrow$ (t), (b) $\rightarrow$ (q), (c) $\rightarrow$ (s), (d) $\rightarrow$ (r)

25. Match the column

Column-I Reaction		Column-II Name/Process	
(a)	$\text{ArN}_2^+\text{X}^- \xrightarrow[\text{HCl}]{\text{Cu}_2\text{Cl}_2}$	(P)	Gattermann reaction
(b)	$\text{ArN}_2^+\text{X}^- \xrightarrow{\text{Cu/HCl}}$	(Q)	Coupling reaction
(c)	$\text{ArN}_2^+\text{X}^- \xrightarrow[\text{H}_2\text{O}]{\text{H}_3\text{PO}_2}$	(R)	Sandmeyer reaction
(d)	$\text{ArN}_2^+\text{X}^- \xrightarrow[0^\circ-5^\circ\text{C}]{\text{Ph-OH}}$	(S)	Reduction

Correct match is -

(1) (a) $\rightarrow$ (R), (b) $\rightarrow$ (P), (c) $\rightarrow$ (S), (d) $\rightarrow$ (Q)

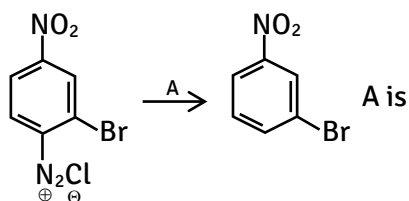
(2) (a) $\rightarrow$ (R), (b) $\rightarrow$ (P), (c) $\rightarrow$ (Q), (d) $\rightarrow$ (S)

(3) (a) $\rightarrow$ (P), (b) $\rightarrow$ (R), (c) $\rightarrow$ (S), (d) $\rightarrow$ (Q)

(4) (a) $\rightarrow$ (Q), (b) $\rightarrow$ (R), (c) $\rightarrow$ (P), (d) $\rightarrow$ (S)

Exercise 4 (Previous Year's Questions)

1. In the reaction



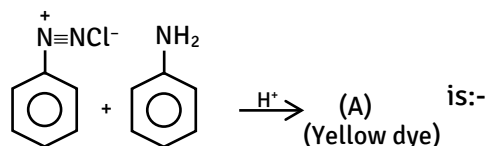
[NEET UT - 2013]

- (1) $\text{H}^+/\text{H}_2\text{O}$ (2) $\text{HgSO}_4/\text{H}_2\text{SO}_4$
(3) Cu_2Cl_2 (4) H_3PO_2 and H_2O

2. Nitrobenzene on reaction with conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ at $80-100^\circ\text{C}$ forms which one of the following products? [NEET UT - 2013]

- (1) 1, 2, 4-Trinitrobenzene
(2) 1,2-Dinitrobenzene
(3) 1,3-Dinitrobenzene
(4) 1,4-dinitrobenzene

3. In the following reaction, the product (A) [AIPMT - 2014]

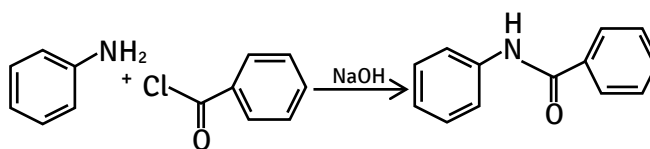


- (1)
- (2)
- (3)
- (4)

4. Which of the following will be most stable diazonium salt? [AIPMT - 2014]

- (1) $\text{CH}_3\text{N}_2^+\text{X}^-$
(2) $\text{C}_6\text{H}_5\text{N}_2^+\text{X}^-$
(3) $\text{CH}_3\text{CH}_2\text{N}_2^+\text{X}^-$
(4) $\text{C}_6\text{H}_5\text{CH}_2\text{N}_2^+\text{X}^-$

5. The following reaction

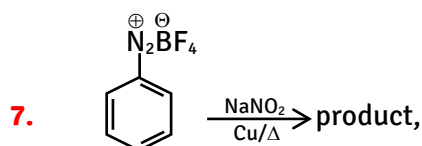


is known by the name: [RE-AIPMT - 2015]

- (1) Acetylation reaction
(2) Schotten-Baumann reaction
(3) Friedel-Craft's reaction
(4) Perkin's reaction

6. Method by which Aniline cannot be prepared is: [RE-AIPMT - 2015]

- (1) reduction of nitrobenzene with H_2/Pd in ethanol
(2) potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
(3) hydrolysis of phenylisocyanide with acidic solution
(4) degradation of benzamide with bromine in alkaline solution



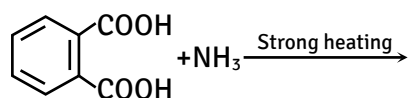
product of given reaction is [AIIMS - 2015]

- (1)
- (2)
- (3)
- (4)

8. Which of the following reactions is appropriate for converting acetamide to methanamine? [NEET - 2017]

- (1) Carbylamine reaction
- (2) Hoffmann hypobromite reaction
- (3) Stephens reaction
- (4) Gabriel phthalimide synthesis

9. The major product of the following reaction is: [NEET(UG)-2019]



- (1)
- (2)
- (3)
- (4)

10. The amine that reacts with Hinsberg's reagent to give an alkali insoluble product is: [NEET(UG)-(Odisha)2019]

- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{NH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
- (2) $\text{CH}_3-\text{CH}_2-\underset{\text{CH}_2\text{CH}_3}{\text{N}}-\text{CH}_2\text{CH}_3$
- (3) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{NH}_2}{\text{C}}}-\text{CH}_2\text{CH}_2\text{CH}_3$
- (4) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\underset{\text{CH}_3}{\text{CH}}-\text{NH}_2$

11. Which of the following amine will give the carbylamine test? [NEET(UG)-2020]

- (1)
- (2)
- (3)
- (4)

12. Reaction of propanamide with sodium hydroxide and bromine will give [NEET(UG)-2020(Covid-2019)]

- (1) Ethylamine
- (2) Methylamine
- (3) Propylamine
- (4) Aniline

13. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali. [NEET(UG)-2021]

- (1) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{O}_2$
- (2) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{H}-\text{CH}_3$
- (3) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{H}_2$
- (4) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$

14. Given below are two statements :

[NEET-2022]

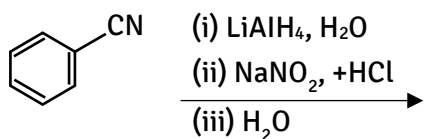
Statements I : Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

Statements II: Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

In the light of the above statements, choose the most appropriate answer from the options given below -

- (1) Statement I is incorrect but Statement II is correct
- (2) Both Statement I and Statement II are correct
- (3) Both Statement I and Statement II are incorrect
- (4) Statement I is correct but Statement II is incorrect

15. The product formed from the following reaction sequence is **[NEET(UG)-2022]**



- (1)
 (2)
- (3)
 (4)

16. Which of the following reaction will not give primary amine as the product –

[NEET(UG)-2023]

- (1) $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{Br}_2/\text{KOH}}$ Product
 (2) $\text{CH}_3\text{CN} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4}$ Product
 (3) $\text{CH}_3\text{NC} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4}$ Product
 (4) $\text{CH}_3\text{CONH}_2 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4}$ Product

17. Given below are two statements:

Statement I: Aniline does not undergo Friedel-Crafts alkylation reaction.

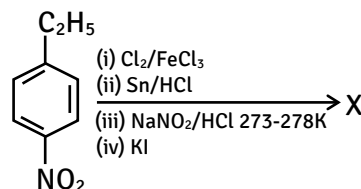
Statement II: Aniline cannot be prepared through Gabriel synthesis. In the light of the above statements, choose the correct answer from the options given below:

[NEET-2024]

- (1) Statement I is correct, but Statement II is false.
 (2) Statement I is incorrect, but Statement II is true.
 (3) Both Statement I and Statement II are true.
 (4) Both Statement I and Statement II are false.

18. The major product X formed in the following reaction sequence is :

[RE-NEET-2024]



- (1)
 (2)
- (3)
 (4)

19. The compound that does not undergo Friedel-Crafts alkylation reaction but gives a positive carbylamines test is :

[RE-NEET-2024]

- (1) aniline
 (2) pyridine
 (3) N-methylaniline
 (4) triethylamine

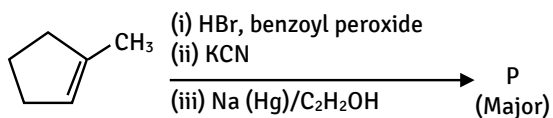
20. Which one of the following compounds does not decolourize bromine water?

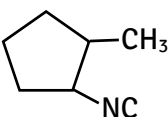
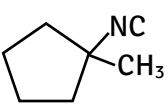
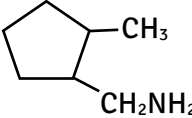
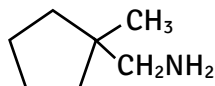
[NEET-2025]

- (1)
 (2)
- (3)
 (4)

21. Predict the major product 'P' in the following sequence of reactions –

[NEET-2025]



- (1) 
- (2) 
- (3) 
- (4) 

22. Given below are two statements:

Statement I: Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at 273-278 K. It decomposes easily in the dry state.

Statement II: Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

In the light of the above statements, choose the most appropriate answer from the options given below : [NEET-2025]

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	3	2	3	3	3	3	2	2	3	4	2	4	4	4	2	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	4	3	3	2	3	3	2	3	3	3	2	2	2	2	2	3	2	3
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	2	2	3	2	4	3	2	3	2	1										

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	3	4	3	2	2	3	2	3	1	1	1	2	3	3	4	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	3	4	2	3	2	3	3	2	2	2	3	3	2	1	3	1	3	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	1	1	3	4	3	4	2	2	4	2	2	4	2	4	2	2	1	1	2
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	4	2	4	4	3	4	4	1	1	4	3	1	3	1	4	2	1	3

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	1	1	1	4	1	3	2	2	1	1	2	4	3	1	4	4	2	3
Que.	21	22	23	24	25															
Ans.	1	1	3	1	1															

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	4	2	2	2	2	2	2	1	2	1	3	4	4	3	3	3	1	3
Que.	21	22																		
Ans.	3	3																		

Chapter 6

BIOMOLECULES

Chapter Summary

- Introduction
- Classification of Carbohydrates
- Monosaccharides
- Chemical reactions of Glucose
- Cyclic structure of Glucose
- Fructose (Ketohehexose)
- Disaccharides
- Polysaccharides
- Proteins
- Structure of Proteins
- Enzymes
- Vitamins
- Nucleic Acids
- Hormones

Introduction

Complex organic compound which governs the common activities of the living organism are called biomolecules. Living systems are made up of various complex biomolecules like carbohydrates, proteins, nucleic acids, lipids, etc. In addition, some simple molecules like vitamins and mineral salts also play an important role in the functions of organisms.

Carbohydrates:

Carbohydrates are primarily produced by plants and form a very large group of naturally occurring organic compounds.

Most of the carbohydrates have general formula $C_x(H_2O)_y$ and considered as hydrates of carbon.

Ex. $C_6(H_2O)_6$ (Glucose, Fructose)
 $C_{12}(H_2O)_{11}$ (Maltose, Lactose, Sucrose)

But some compounds which have formula according to $C_x(H_2O)_y$ are not known as carbohydrate

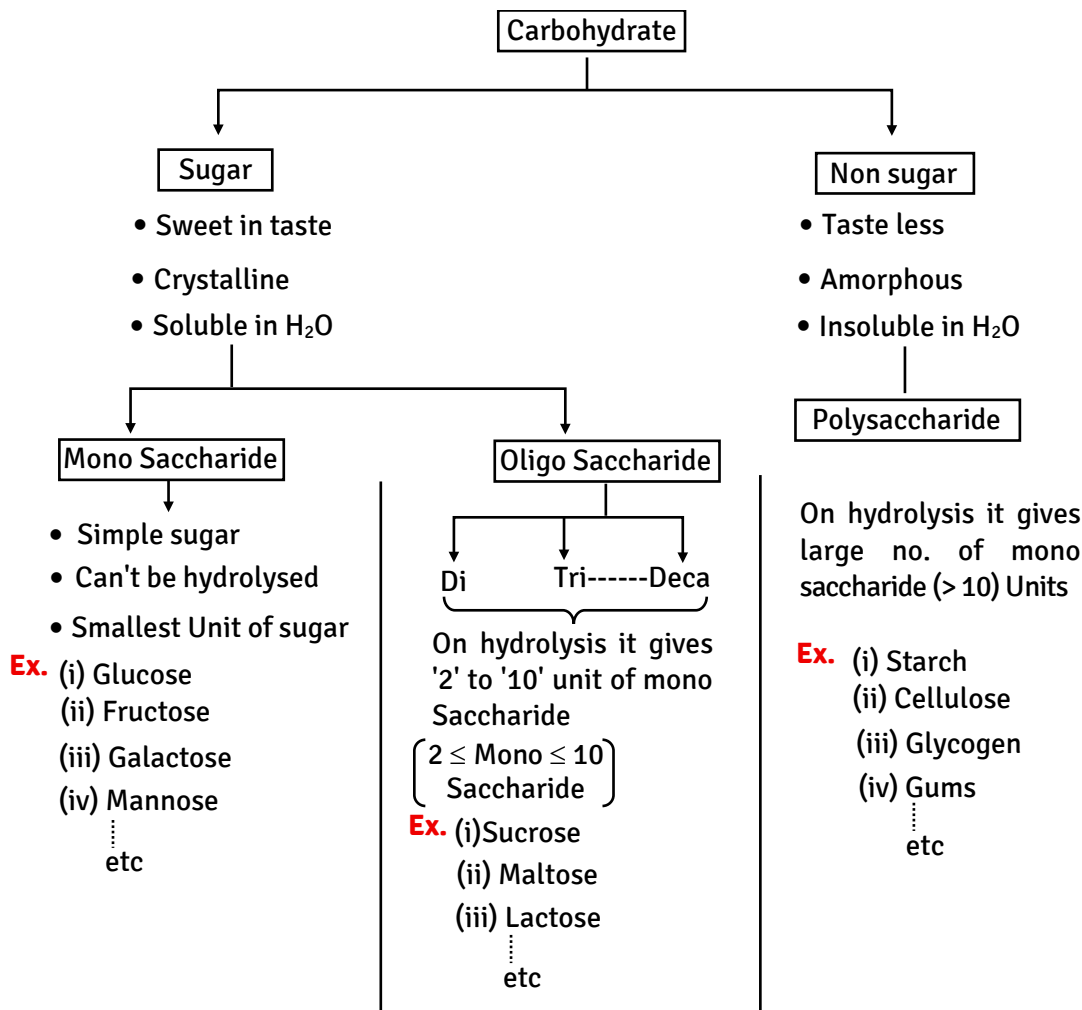
Ex. CH_2O (Formaldehyde)
 $C_2(H_2O)_2$ (Acetic acid)
 $C_3(H_2O)_3$ (Lactic acid)

Some carbohydrates are also known which do not have formula according to $C_x(H_2O)_y$

Ex. $C_6H_{12}O_5$ (Rhamnose)

Modern definition of carbohydrates/saccharides: poly hydroxy aldehyde/ketones are known as carbohydrate

Classification of Carbohydrates



Monosaccharides

A carbohydrate that cannot be hydrolysed further to give simplest unit of polyhydroxy aldehyde or ketone is called monosaccharide.

Ex. Glucose, fructose, ribose, etc.

Monosaccharides are further classified on the basis of number of carbon atoms and the functional group present in them.

If Aldehyde group is present in monosaccharide, then it is known as **aldose**.

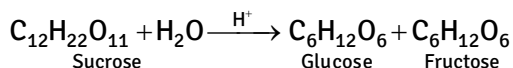
If keto group is present in monosaccharide, then it is known as **ketose**.

Different types of Monosaccharides

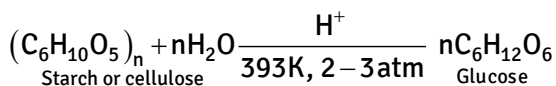
Carbon Atom	General term	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose	Ketopentose
6	Hexose	Aldohexose	Ketohexose
7	Heptose	Aldoheptose	Ketoheptose

Glucose (Aldohexose): Preparation of Glucose:

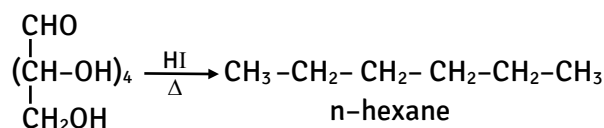
1. **From sucrose (Cane sugar):** If sucrose is boiled with dilute HCl or H_2SO_4 in alcoholic solution, glucose and fructose are obtained in equal amounts.



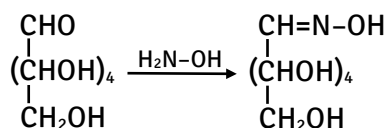
2. **From starch:** Commercially glucose is obtained by hydrolysis of starch by boiling it with dilute H_2SO_4 at 393 K under pressure.

**Chemical reactions of Glucose**

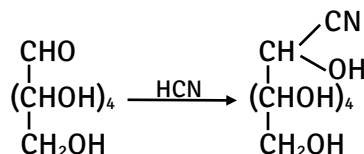
1. On prolonged heating with HI or Red/HI, it form n-hexane which confirm linear carbon chain in glucose



2. Glucose react with $\text{NH}_2\text{-OH}$ (Hydroxylamine) to form an oxime

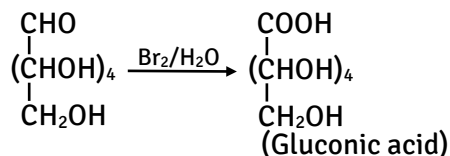


3. It react with HCN (Hydrogen cyanide) to give cyanohydrin



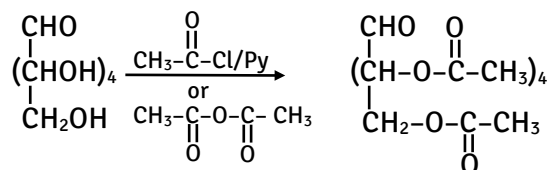
These reactions confirm the presence of a carbonyl group ($>\text{O}$) in glucose

4. On reaction with $\text{Br}_2/\text{H}_2\text{O}$, gluconic acid is obtained since $\text{Br}_2/\text{H}_2\text{O}$ is mild oxidizing agent can oxidize only Aldehyde (Aldose)

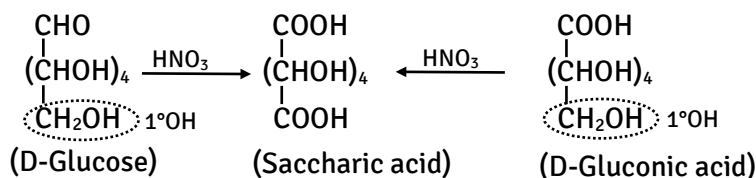


It means fructose doesn't react with $\text{Br}_2/\text{H}_2\text{O}$ therefore glucose & fructose can't be differentiated by bromine water

5. On Acylation of glucose, glucose Penta acetate is obtained which confirm the presence of Five-OH groups

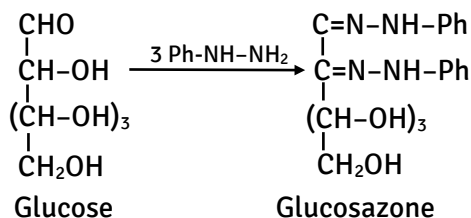


6. On reaction of glucose and gluconic acid with Nitric acid (HNO_3), Both gives saccharic acid.

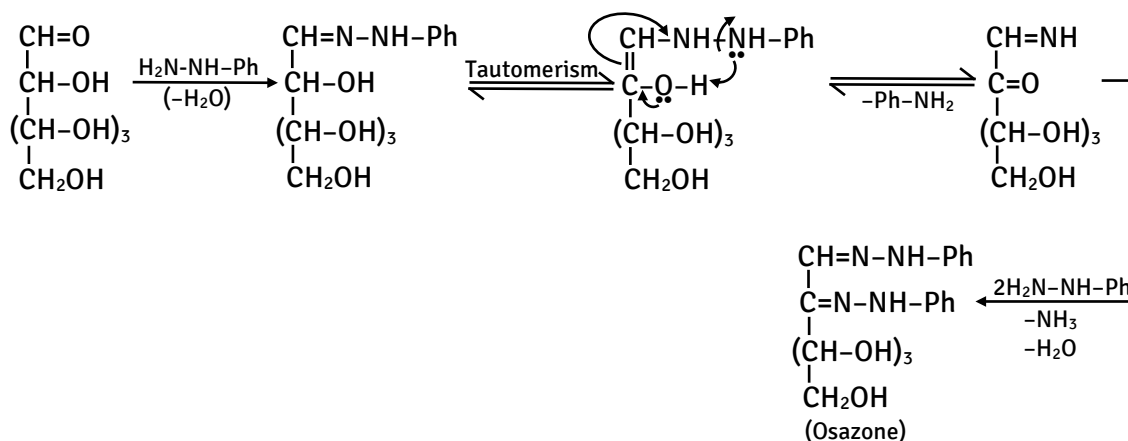


It confirm the presence of 1°-OH

7. **Reaction with phenyl hydrazine : (formation of osazone):**

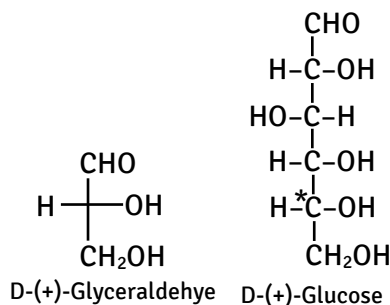


Mechanism:



Configuration in monosaccharides:

For assigning the configuration of monosaccharides, it is the last asymmetric carbon atom (as shown below) which is compared. As in (+) glucose, -OH on the last asymmetric carbon is on the right hand side which is comparable to (+) glyceraldehyde, so it is assigned D-configuration.



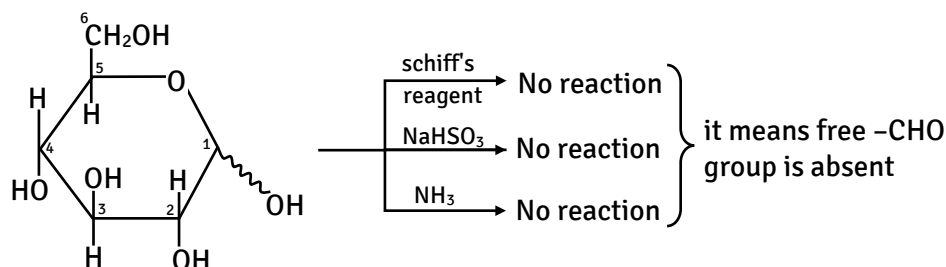
- 'D' and 'L' are relative configuration hence they don't have relation with the optical activity of the compound.
- 'D' and 'L' are also not related with 'd' (+) and 'l' (-)

$$\left. \begin{array}{l} \text{d} \rightarrow \text{dextrorotatory } / (+) \\ \text{l} \rightarrow \text{Laevorotatory } / (-) \end{array} \right\} \begin{array}{l} \text{Direction of optical} \\ \text{rotation of the compound} \end{array}$$
- Glucose is a dextrorotatory

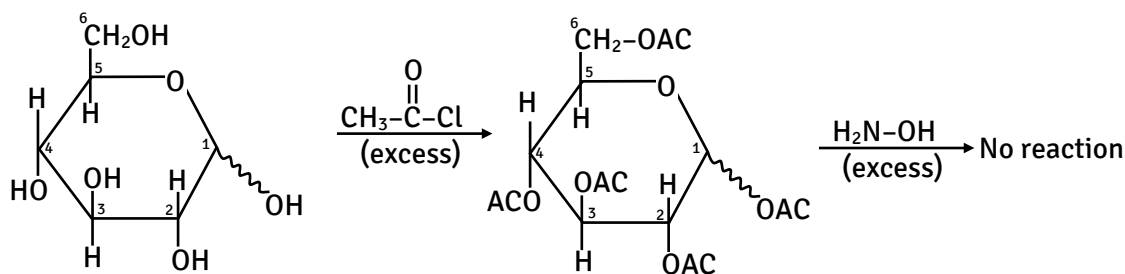
Cyclic structure of Glucose

The following reactions and facts can't be explained by open chain structure it can only explained by cyclic structure of glucose

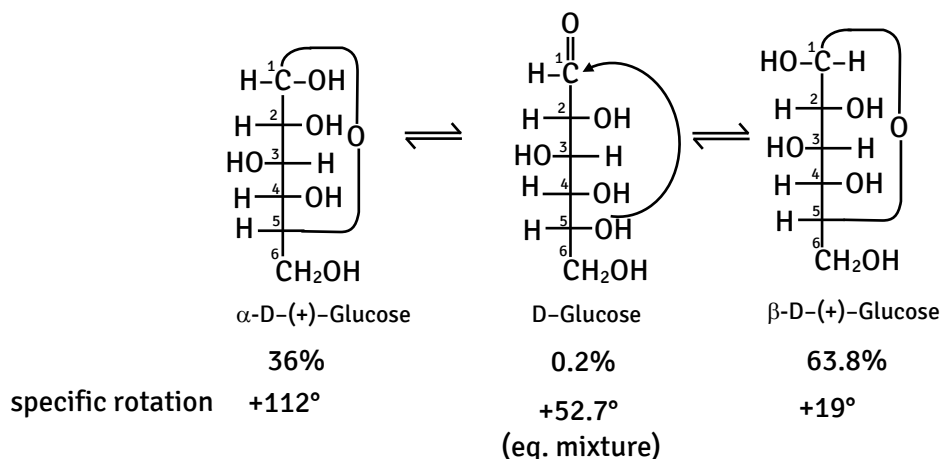
- (i) Glucose doesn't give schiff's test
- (ii) It doesn't react with NH_3 (ammonia) and NaHSO_3 (sodium hydrogen sulphite)



- (iii) Glucose Penta acetate doesn't react with $\text{H}_2\text{N}-\text{OH}$ (hydroxyl amine) which indicate the absence of free $-\text{CHO}$ group



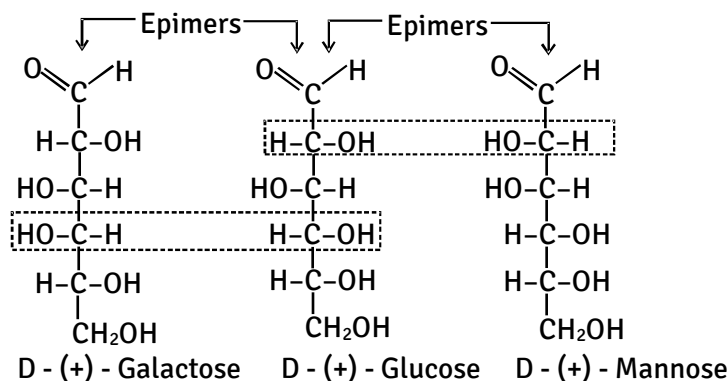
Note: It was found that glucose forms a six-membered ring in which $-\text{OH}$ at C-5 is involved in ring formation. This explains the absence of $-\text{CHO}$ group and also existence of glucose in two forms as shown below. These two cyclic forms exist in equilibrium with open chain structure.



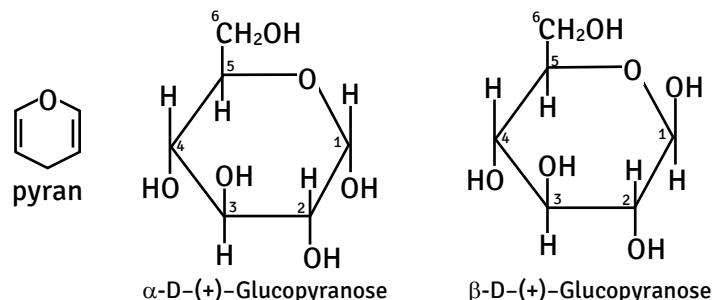
Anomers: The two sugars that differs in configuration only on the carbon that was the carbonyl carbon in the open chain form is called as anomers and that carbon is called anomeric carbon.

α -glucose and β -glucose are anomers of glucose and their equilibrium mixture contains 36% $\alpha\text{-D}$ -glucose, 63.8% $\beta\text{-D}$ -glucose and 0.2% open chain form.

Epimers: A pair of diastereomers that differ only in the configuration about of a single carbon atom are said to be epimers. D(+)-glucose is epimeric with D(+)-Mannose and D(+)-galactose.

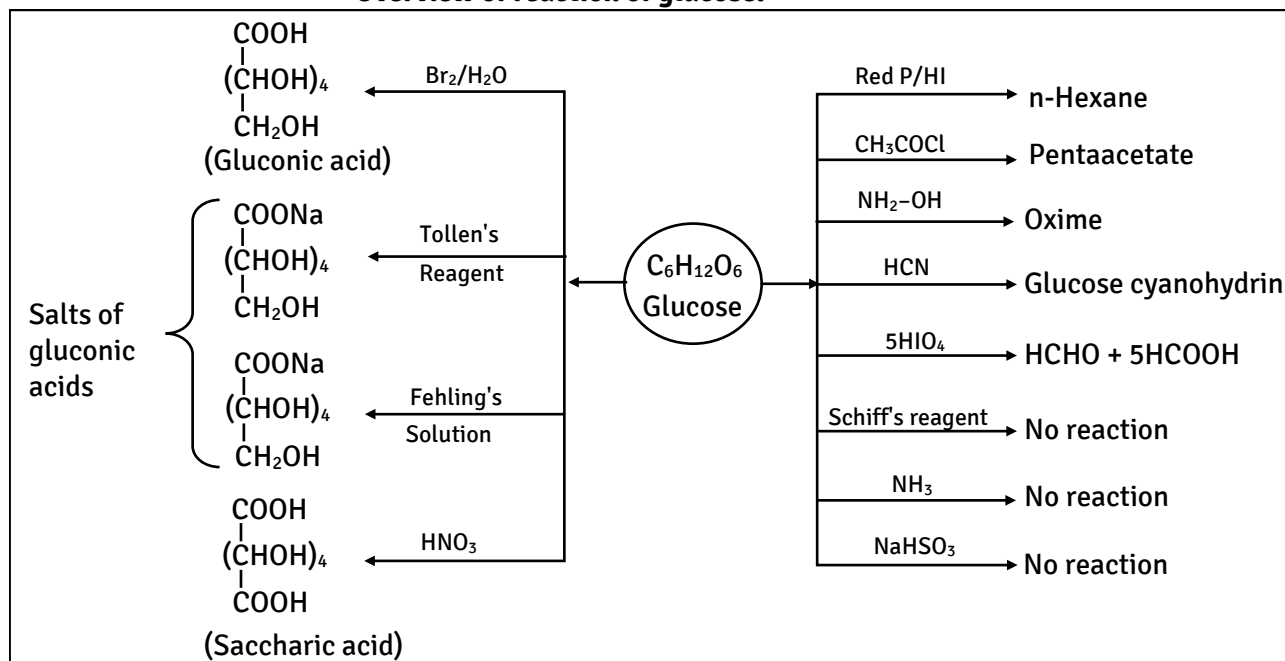


Haworth Projection: The six membered cyclic structure of glucose is called pyranose structure (α - or β -), in analogy with pyran. The cyclic structure of glucose is more correctly represented by Haworth structure as given below.



- Note:** (i) Tollen's reagent and Fehling solution is used in Basic condition in which α & β - Form (Hemi acetal) get decomposed and increase open chain aldehyde and oxidise this form into gluconic acid or we can say that sugar reduce Tollen's and Fehling solution, hence it is known as reducing sugar
- (ii) all monosaccharides are reducing sugar

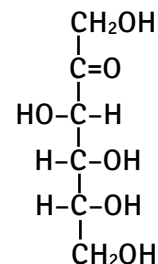
Overview of reaction of glucose:



Fructose (Ketohehexose)

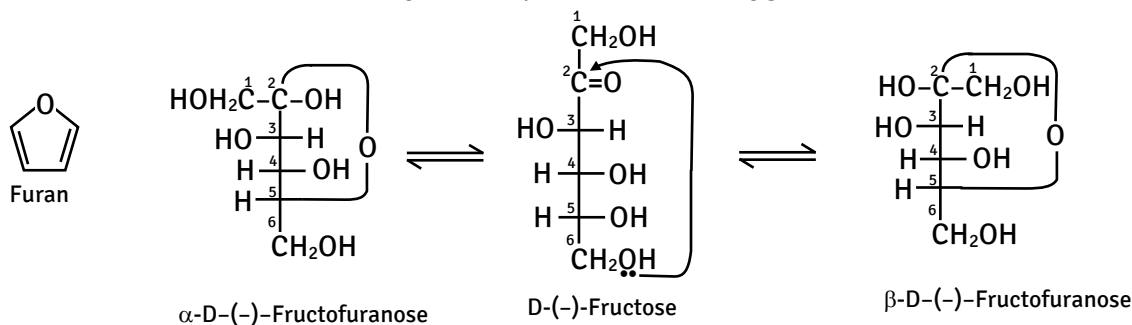
Structure of Fructose:

Fructose also has the molecular formula $C_6H_{12}O_6$ and on the basis of its reactions it was found to contain a ketonic functional group at carbon number 2 and six carbons in straight chain as in the case of glucose. It belongs to D-series and is a laevorotatory compound. It is appropriately written as D-(-)-fructose. Its open chain structure is as shown below.

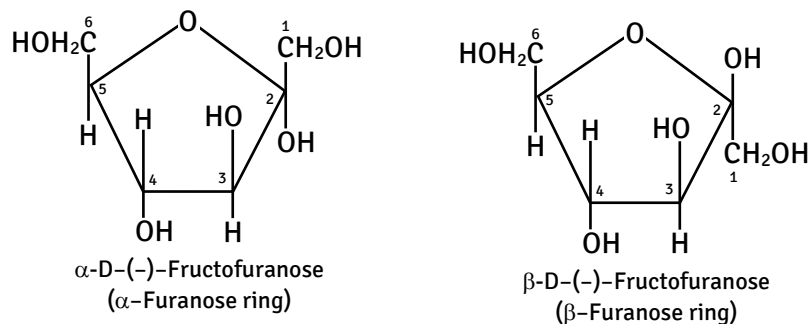


D-(-)-Fructose

It also exists in two cyclic forms which are obtained by the addition of -OH at C-5 to the $(>C=O)$ group. The ring, thus formed is a five membered ring and is named as furanose with analogy to the compound furan. Furan is a five membered cyclic compound with one oxygen and four carbon atoms.



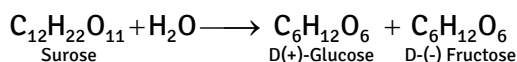
Haworth projections: The cyclic structures of two anomers of fructose are represented by Haworth structures as given.



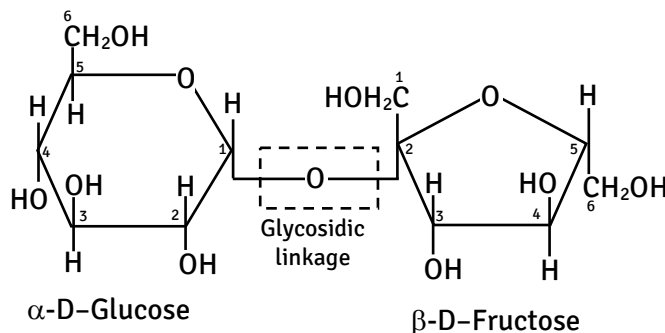
Disaccharides

The two monosaccharides are joined together by an oxide linkage formed by the loss of a water molecule. Such a linkage between two monosaccharide units through oxygen atom is called glycosidic linkage.

- (i) **Sucrose:** One of the common disaccharides is sucrose which on hydrolysis gives equimolar mixture of D- (+)-glucose and D-(-) fructose.



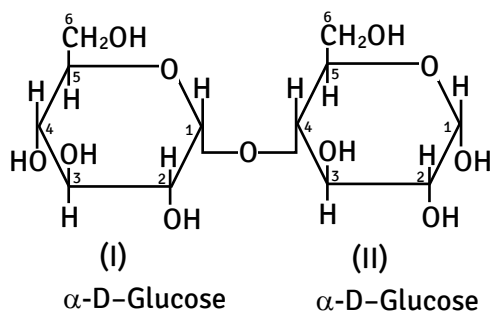
These two monosaccharides are held together by a glycosidic linkage between C_1 of α -glucose and C_2 of β -fructose. Since the reducing groups of glucose and fructose are involved in glycosidic bond formation hence, sucrose is a non reducing sugar.



Sucrose

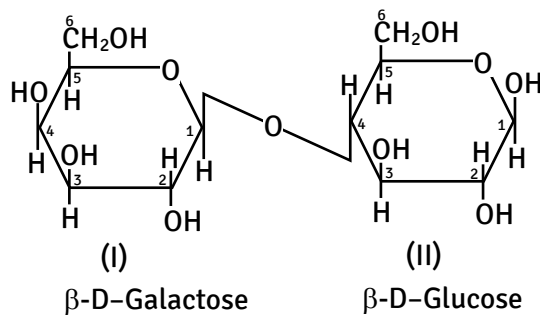
Sucrose is dextrorotatory but after hydrolysis gives dextrorotatory glucose and laevorotatory fructose. Since the laevorotation of fructose (-92.4°) is more than dextrorotation of glucose ($+52.5^\circ$), the mixture is laevorotatory. Thus, Hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to laevo (-) and the product is named as invert sugar.

- (ii) **Maltose:** Another disaccharide, maltose is composed of two α -D-glucose units in which C_1 of one glucose (I) is linked to C_4 of another glucose unit (II). The free aldehyde group can be produced at C_1 of second glucose in solution and it shows reducing properties so it is a reducing sugar.



Maltose

- (iii) **Lactose:** It is more commonly known as milk sugar since this disaccharide is found in milk. It is composed of β -D-galactose and β -D-glucose. The linkage is between C_1 of galactose and C_4 of glucose. Hence it is also a reducing sugar.



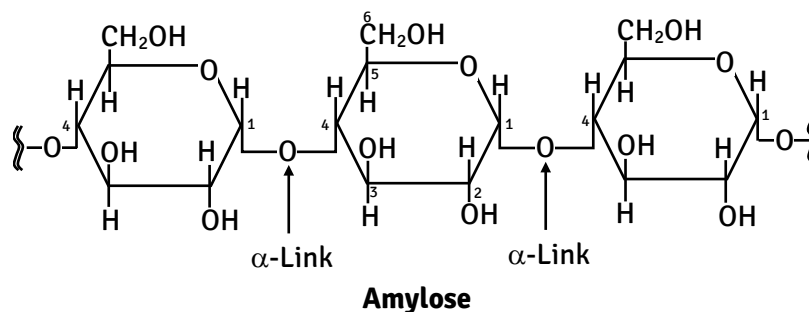
Lactose

Polysaccharides

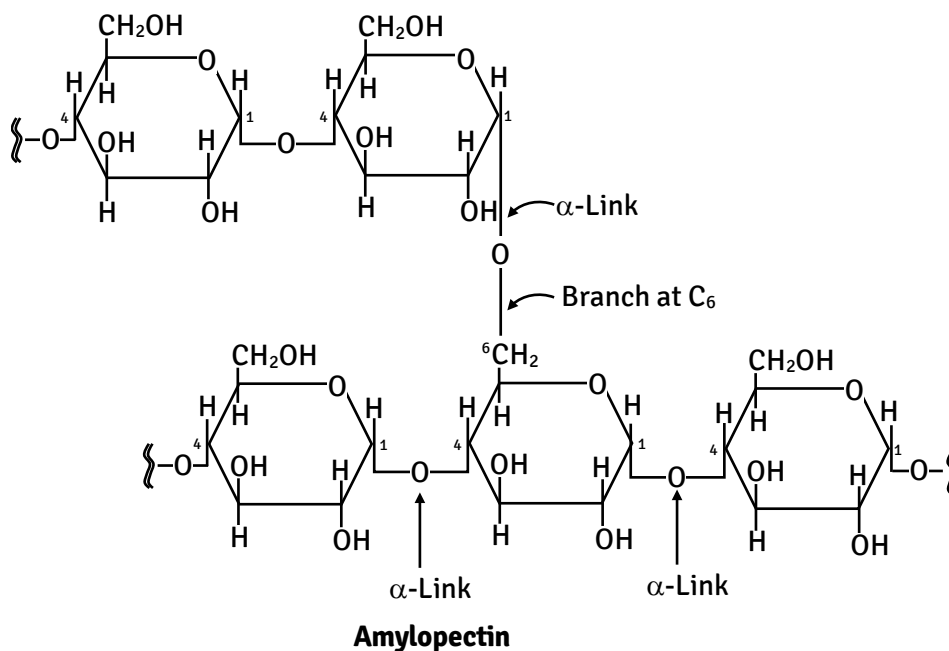
Polysaccharides contain a large number of monosaccharide units joined together by glycosidic linkages. They mainly act as the food storage or structural materials.

- (i) **Starch:** Starch is the main storage polysaccharide of plants. It is the most important dietary source for human beings. High content of starch is found in cereals, roots, tubers and some vegetables. It is a polymer of α -glucose and consists of two components-amylose and amylopectin

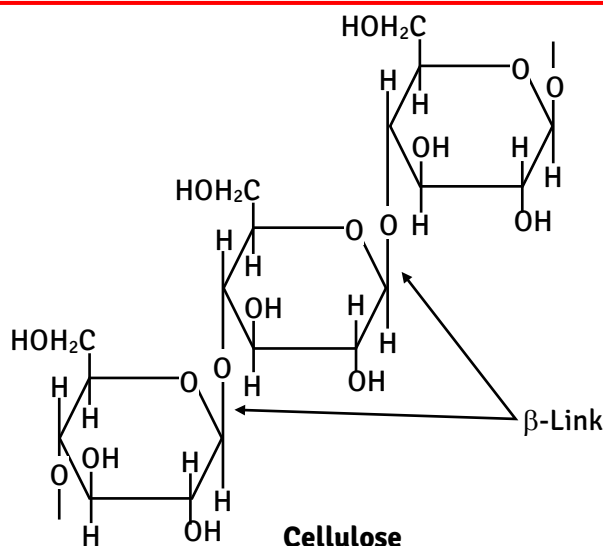
Amylose: Amylose is hot water soluble component which constitutes about 15-20 % of starch. Chemically amylose is a long unbranched chain with 200-1000 α -D-(+)-glucose units held by $C_1 - C_4$ glycosidic linkage.



Amylopectin: Amylopectin is insoluble in hot water and constitutes about 80–85% of starch. It is a branched chain polymer of α -D-glucose units in which chain is formed by $C_1 - C_4$ glycosidic linkage whereas branching occurs by $C_1 - C_6$ glycosidic linkage.



- (ii) **Cellulose:** Cellulose occurs exclusively in plants and it is the most abundant organic substance in plant kingdom. It is a predominant constituent of cell wall of plant cells. Cellulose is a straight chain polysaccharide composed only of β -D-glucose units which are joined by glycosidic linkage between C_1 of one glucose unit and C_4 of the next glucose unit.



(iii) Glycogen: The carbohydrates are stored in animal body as glycogen. It is also known as animal starch because its structure is similar to amylopectin and is rather more highly branched. It is present in liver, muscles and brain. When the body needs glucose, enzymes break the glycogen down to glucose. Glycogen is also found in yeast and fungi.

Exercise 1.1

1. Carbohydrates may be:
 - (1) Sugars
 - (2) Starch
 - (3) Polyhydroxy aldehyde/ ketones
 - (4) All
2. Sucrose on hydrolysis gives:
 - (1) Glucose and galactose
 - (2) Maltose and galactose
 - (3) Glucose and fructose
 - (4) None of these
3. When glucose is heated with nitric acid, the product is :

(1) Gluconic acid	(2) Saccharic acid
(3) Glycolic acid	(4) Oxalic acid
4. Ring structure of glucose is due to formation of hemiacetal and ring formation between:

(1) C ₁ and C ₅	(2) C ₁ and C ₄
(3) C ₁ and C ₃	(4) C ₂ and C ₄
5. Which of the following is a ketohexose :

(1) Glucose	(2) Fructose
(3) Sucrose	(4) Starch
6. A disaccharide which give only glucose on hydrolysis:

(1) Lactose	(2) Fructose
(3) Sucrose	(4) Maltose
7. Which of the following is not a reducing sugar-

(1) Sucrose	(2) Galactose
(3) Glucose	(4) Lactose
8. All the following are composed exclusively of glucose except:

(1) Lactose	(2) Amylose
(3) Cellulose	(4) Maltose
9. Glycogen is a branched chain polymer of α -D-glucose units in which chain is formed by C₁—C₄ glycosidic linkage whereas branching occurs by the formation of C₁-C₆ glycosidic linkage. Structure of glycogen is similar to.

(1) Amylose	(2) Amylopectin
(3) Cellulose	(4) Glucose

- 10.** Which of the following monosaccharides are present as five membered cyclic structure (furanose structure)?
 (i) Ribose (ii) Glucose
 (iii) Fructose (iv) Galactose
 (1) iii, iv only (2) i, ii, iii only
 (3) i, iii only (4) i, iv only
- 11.** Benedict's solution is not reduced by :
 (1) Glucose (2) Fructose
 (3) Maltose (4) Sucrose
- 12.** Which of the following polymer is stored in the liver of animals?
 (1) Amylose (2) Cellulose
 (3) Amylopectin (4) Glycogen
- 13.** Select the **incorrect** statement.
 (1) Starch is a homopolymer of glucose forming an -glycosidic chain.
 (2) Maltose and Lactose are disaccharidase.
 (3) Cellulose has only D-glucose unit which are joined by glycosidic linkage between C₁ of one glucose unit and C₄ of another glucose unit.
 (4) The structure of glycogen is similar to amylose.
- 14.** Which of the following statement is not true for glucose?
 (1) Glucose exists in two crystalline forms α -D-glucose and β -D-glucose
 (2) Glucose gives Schiff's test for aldehyde
 (3) Glucose reacts with hydroxylamine to form oxime
 (4) The pentaacetate of glucose does not react with hydroxylamine to give oxime
- 15.** Which of the following statements is not true about glucose?
 (1) It is an aldohexose.
 (2) On heating with HI it forms n-hexane.
 (3) It is present in furanose form.
 (4) It does not give 2,4-DNP test.

Proteins

- The word protein is derived from Greek word, "proteios" which means primary or of prime importance.
- Proteins are the most abundant biomolecules of the living system. Chief sources of proteins are milk, cheese, pulses, peanuts, fish, meat etc.
- All proteins are polymers of α -amino acids.

Amino acids : Amino acids contain amino ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) functional groups. Depending upon the relative position of amino group with respect to carboxyl group, the amino acids can be classified as α , β , γ , δ and so on. Only α -amino acids are obtained on hydrolysis of proteins. They may contain other functional groups also.

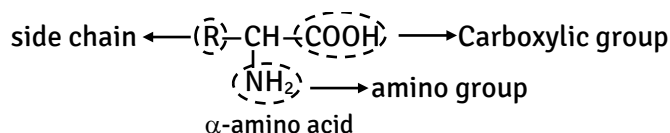


Table : Natural Amino Acids

$$\begin{array}{c} \text{COOH} \\ | \\ \text{H}_2\text{N}-\text{C}-\text{H} \\ | \\ \text{R} \end{array}$$

	Name of the amino acids	Characteristic feature of side chain, R	Three letter Symbol	One letter code
1.	Glycine	H	Gly	G
2.	Alanine	-CH ₃	Ala	A
3.	Valine*	(H ₃ C) ₂ CH-	Val	V
4.	Leucine*	(H ₃ C) ₂ CH-CH ₂ -	Leu	L
5.	Isoleucine*	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{CH}- \\ \\ \text{CH}_3 \end{array}$	Ile	I
6.	Arginine*	$\begin{array}{c} \text{HN}=\text{C}-\text{NH}-(\text{CH}_2)_3- \\ \\ \text{NH}_2 \end{array}$	Arg	R
7.	Lysine*	H ₂ N-(CH ₂) ₄ -	Lys	K
8.	Glutamic acid	HOOC-CH ₂ -CH ₂ -	Glu	E
9.	Aspartic acid	HOOC-CH ₂ -	Asp	D
10.	Glutamine	$\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{N}-\text{C}-\text{CH}_2-\text{CH}_2- \end{array}$	Gln	Q
11.	Asparagine	$\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{N}-\text{C}-\text{CH}_2- \end{array}$	Asn	N
12.	Threonine*	H ₃ C-CHOH-	Thr	T
13.	Serine	HO-CH ₂ -	Ser	S
14.	Cysteine	HS-CH ₂ -	Cys	C
15.	Methionine*	H ₃ C-S-CH ₂ -CH ₂ -	Met	M
16.	Phenylalanine*	C ₆ H ₅ -CH ₂ -	Phe	F
17.	Tyrosine	(p) HO-C ₆ H ₄ -CH ₂ -	Tyr	Y
18.	Tryptophan*	$\begin{array}{c} -\text{CH}_2 \\ \\ \text{Indole ring} \end{array}$	Trp	W
19.	Histidine*	$\begin{array}{c} -\text{CH}_2 \\ \\ \text{Imidazole ring} \end{array}$	His	H
20.	Proline ^(a)	$\begin{array}{c} \text{COOH}^a \\ \\ \text{HN}-\text{C}-\text{H} \\ \\ \text{CH}_2 \end{array}$	Pro	P

* essential amino acid, a= entire structure

Classification of Amino Acids: Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule.

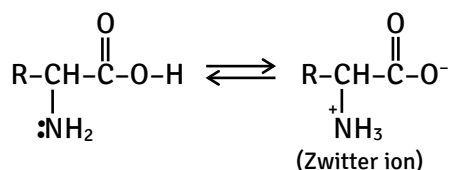
- The amino acids, which can be synthesized in the body, are known as non-essential amino acids.

Ex.	(1) Glycine	(2) Alanine	(3) Tyrosine	(4) Serine
	(5) Asparagine	(6) Proline	(7) Glutamine	(8) Cysteine
	(9) Aspartic acid	(10) Glutamic acid		

- Those amino acids which cannot be synthesized in the body and must be obtained through diet, are known as essential amino acids.

Ex.	(1) Valine	(2) Leucine	(3) Isoleucine	(4) Phenylalanine	(5) Tryptophan
	(6) Threonine	(7) Methionine	(8) Lysine	(9) Arginine	(10) Histidine

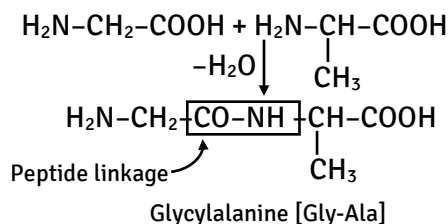
- Amino acids are usually colourless, crystalline solids. These are water-soluble, high melting solids and behave like salts rather than simple amines or carboxylic acids. This behaviour is due to the presence of both acidic (carboxyl group) and basic (amino group) groups in the same molecule.



- In aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton, giving rise to a dipolar ion known as zwitter ion. This is neutral but contains both positive and negative charges.
- In zwitter ionic form, amino acids show amphoteric behaviour as they react both with acids and bases.
- Except glycine, all other naturally occurring α -amino acids are optically active, since the α carbon atom is asymmetric. These exist both in 'D' and 'L' forms. Most naturally occurring amino acids have L-configuration. L-Aminoacids are represented by writing the $-\text{NH}_2$ group on left hand side.

Structure of Proteins

Proteins are the polymers of α -amino acids and they are connected to each other by peptide bond or peptide linkage. Chemically, peptide linkage is an amide formed between $-\text{COOH}$ group and $-\text{NH}_2$ group.



The reaction between two molecules of similar or different amino acids, proceeds through the combination of the amino group of one molecule with the carboxyl group of the other. This results in the elimination of a water molecule and formation of a peptide bond $-\text{CO}-\text{NH}-$.

Proteins can be classified into two types on the basis of their molecular shape.

(a) Fibrous proteins:

When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed. Such proteins are generally insoluble in water. Some common examples are keratin (present in hair, wool, silk) and myosin (present in muscles), etc.

(b) Globular proteins:

- This structure results when the chains of polypeptides coil around to give a spherical shape. These are usually soluble in water. Insulin and albumins are the common examples of globular proteins.
- Structure and shape of proteins can be studied at four different levels, i.e., primary, secondary, tertiary and quaternary, each level being more complex than the previous one.

(i) Primary structure of proteins:

Proteins may have one or more polypeptide chains. Each polypeptide in a protein has amino acids linked with each other in a specific sequence and this sequence of amino acids that is said to be the primary structure of that protein. Any change in this primary structure (i.e., the sequence of amino acids) creates a different protein.

(ii) Secondary structure of proteins:

- The secondary structure of protein refers to the shape in which a long polypeptide chain can exist.
- They are found to exist in two different types of structures viz. α -helix and β -pleated sheet structure.
- These structures arise due to the regular folding of the backbone of the polypeptide chain due to

hydrogen bonding between >C=O and -NH- groups of the peptide bond.

- (A) α -Helix:** α -Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a right handed screw (helix) with the -NH group of each amino acid residue hydrogen bonded to the >C=O of an adjacent turn of the helix as shown in figure.

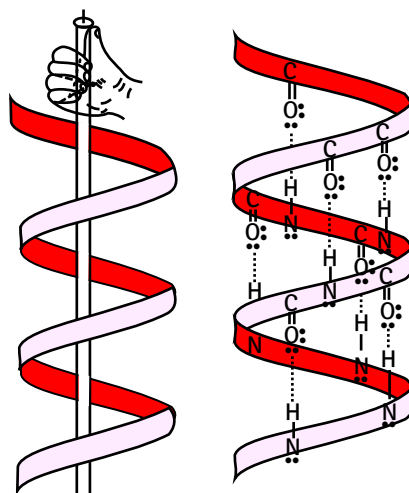
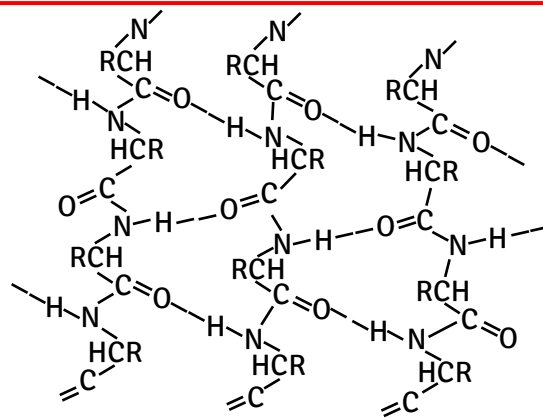


Fig. α -Helix structure of proteins

- (B) β -pleated sheet:** In β -pleated sheet structure, all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together by intermolecular hydrogen bonds. The structure resembles the pleated folds of drapery and therefore is known as β -pleated sheet.

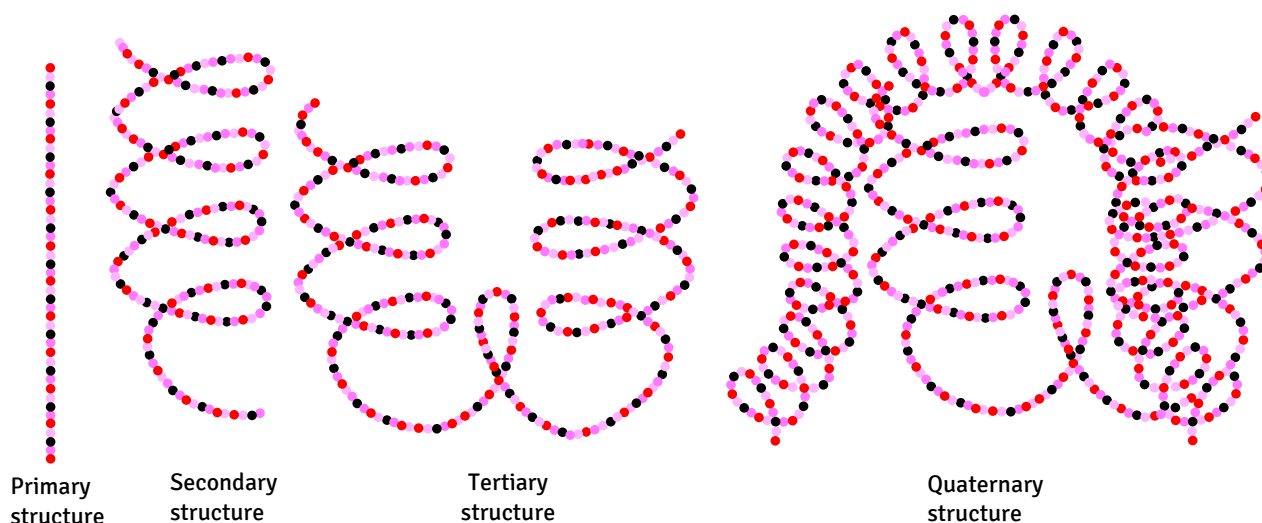


(β -Pleated sheet structure of proteins)

(iii) Tertiary structure of proteins: The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, van der Waals and electrostatic forces of attraction.

(iv) Quaternary structure of proteins: Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. The spatial arrangement of these subunits with respect to each other is known as quaternary structure.

A diagrammatic representation of all these four structures is given in figure.

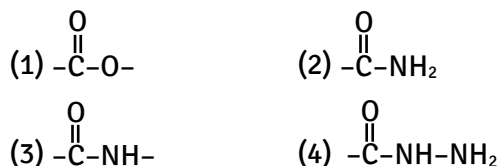


Denaturation of Proteins:

- Physical change like change in temperature or chemical change like change in pH, the hydrogen bonds are disturbed. Due to this, globules unfold and helix get uncoiled and protein loses its biological activity. This is called denaturation of protein.
- The coagulation of egg white on boiling is a common example of denaturation. Another example is curdling of milk which is caused due to the formation of lactic acid by the bacteria present in milk.
- During denaturation secondary and tertiary structures are destroyed but primary structure remains unchanged/intact.

Exercise 1.2

1. Peptide linkage is -



2. Select the correct statement.

- (1) Proteins upon hydrolysis gives α -amino acid.
- (2) Except glycine, all other naturally occurring α -amino acids are optically active.
- (3) In fibrous proteins polypeptide chains are held together by hydrogen and disulphide bonds.
- (4) All

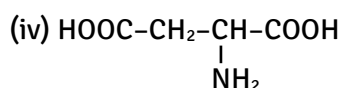
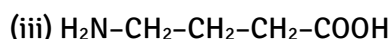
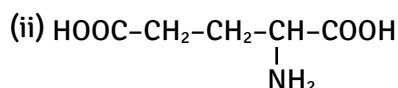
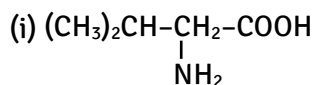
3. Which of the following is an essential amino acid?

- (1) Serine
- (2) Cysteine
- (3) Glycine
- (4) Phenylalanine

4. α -Helical structure refers to the:

- (1) Primary structure of protein
- (2) Secondary structure of protein
- (3) Tertiary structure of protein
- (4) Quaternary structure of proteins

5. Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule. Which of the following are acidic?



- (1) i, ii, iii
- (2) ii, iii
- (3) ii, iv
- (4) ii only

6. Each polypeptide in a protein has aminoacids linked with each other in a specific sequence. This sequence of amino acids is said to be.

- (1) Primary structure of proteins.
- (2) Secondary structure of proteins.
- (3) Tertiary structure of proteins.
- (4) Quaternary structure of proteins.

7. Out of the following which type of interaction is responsible for the stabilisation of α -helix structure of proteins?

- (1) Ionic bonding
- (2) Hydrogen bonding
- (3) Covalent bonding
- (4) van der Waals forces

8. Which of the following statement is/are correct for denaturation of proteins?

- (1) Can be caused by change in temperature.
- (2) Can be caused by change in pH.
- (3) During denaturation secondary and tertiary structure are destroyed but primary structure remains intact.
- (4) All of the above are correct.

9. Proteins can be classified into two types on the basis of their molecular shape i.e., fibrous proteins and globular proteins. Examples of globular proteins are :

- (i) Insulin
- (ii) Keratin
- (iii) Albumin
- (iv) Myosin
- (1) i, iii only
- (2) i, ii, iii only
- (3) iii, iv only
- (4) ii, iv only

10. Lysine, $\text{H}_2\text{N}-(\text{CH}_2)_4-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$

is.

- (i) α -Amino acid
- (ii) Basic amino acid
- (iii) Amino acid synthesised in body
- (iv) β -Amino acid
- (1) i, ii only
- (2) i, ii, iii only
- (3) ii, iv only
- (4) iii, iv only

- 11.** Which of the following is non-essential amino acid?
 (1) Glycine (2) Valine
 (3) Leucine (4) Threonine
- 12.** Identify the essential amino acids from below:
 (A) Valine (B) Proline
 (C) Lysine (D) Threonine
 (E) Tyrosine
 Choose the correct answer from the options given below:
 (1) (A), (C) and (D) only
 (2) (A), (C) and (E) only
 (3) (B), (C) and (E) only
 (4) (C), (D) and (E) only
- 13.** Amino acids generally exist in the form of zwitter ions. This means they contain
 (1) basic —NH_2 group and acidic —COOH group
 (2) the basic —NH_3^+ group and acidic —COO^- group
 (3) basic —NH_2 and acidic —H^+ group
 (4) basic —COO^- group and acidic —NH_3^+ group.
- 14.** Keratin, a structural protein is present in
 (1) hair (2) wool
 (3) silk (4) all of these.
- 15.** Most common types of secondary structures of proteins are
 (1) α -helix and β -helix structures
 (2) α -helix and β -pleated sheet structures
 (3) right and left hand twisted structures
 (4) globular and fibrous structures.

Enzymes

Enzymes are biological catalysts. almost all enzymes are globular proteins. enzyme are very specific for a particular reaction and for a particular substrate. Some important enzymes and their functions are given below:

Enzyme	Reaction catalysed
(i) Invertase or sucrase	Sucrose $\longrightarrow$ Glucose + Fructose
(ii) Maltase	Maltose $\longrightarrow$ Glucose + Glucose
(iii) Lactase	Lactose $\longrightarrow$ Glucose + Galactose
(iv) Amylase	Starch $\longrightarrow$ n Glucose
(v) Pepsin	Proteins $\longrightarrow$ Amino acids
(vi) Trypsin	Proteins $\longrightarrow$ Amino acids
(vii) Nucleases	DNA or RNA $\longrightarrow$ Nucleotides
(viii) DNA polymerase	Deoxynucleotide triphosphates DNA
(ix) RNA polymerase	Ribonucleotide triphosphates RNA

Vitamins

- Certain organic compounds are required in small amounts in our diet but their deficiency causes specific diseases. These compounds are called vitamins.
- Most of the vitamins cannot be synthesised in our body but plants can synthesise almost all of them, so they are considered as essential food factors. However, the bacteria of the gut can produce some of the vitamins required by us.
- All the vitamins are generally available in our diet. Different vitamins belong to various chemical classes and it is difficult to define them on the basis of structure. They are generally regarded as organic

compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth and health of the organism.

- Vitamins are designated by alphabets A, B, C, D, etc. Some of them are further named as sub-groups e.g. B₁, B₂, B₆, B₁₂, etc. Excess of vitamins is also harmful and vitamin pills should not be taken without the advice of doctor.
 - The term “Vitamine” was coined from the word vital + amine since the earlier identified compounds had amino groups. Later work showed that most of them did not contain amino groups, so the letter ‘e’ was dropped and the term vitamin is used these days.
 - Vitamins are classified into two groups depending upon their solubility in water or fat.
- (i) **Fat soluble vitamins:** Vitamins which are soluble in fat and oils but insoluble in water are kept in this group. These are vitamins A, D, E and K. They are stored in liver and adipose (fat storing) tissues.
- (ii) **Water soluble vitamins:** B group vitamins and vitamin C are soluble in water so they are grouped together. Water soluble vitamins must be supplied regularly in diet because they are readily excreted in urine and cannot be stored (except vitamin B₁₂) in our body.

Some important vitamins, their sources and their deficiency diseases

S.No.	Vitamin	Sources	Deficiency diseases
1	Vitamin A	Fish liver oil, carrots, butter and milk	Xerophthalmia (hardening of cornea of eye) Night blindness
2	Vitamin B ₁ (Thiamine)	Yeast, milk, green vegetables and cereals.	Beri-Beri (loss of appetite, retarded growth)
3	Vitamin B ₂ (Riboflavin)	Milk, egg white, liver, kidney.	Cheilosis (fissuring at corners of mouth and lips), digestive disorders and burning sensation of the skin
4	Vitamin B ₆ (Pyridoxine)	Yeast, milk, egg yolk, cereals and grams	Convulsions
5	Vitamin B ₁₂	Meat, fish, egg and curd	Pernicious anaemia (RBC deficient in haemoglobin)
6	Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7	Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteomalacia (soft bones and joint pain in adults)
8	Vitamin E	Vegetable oils like wheat germ oil, sunflower oil, etc.	Increased fragility of RBCs and muscular weakness
9	Vitamin K	Green leafy vegetables	Increased blood clotting time

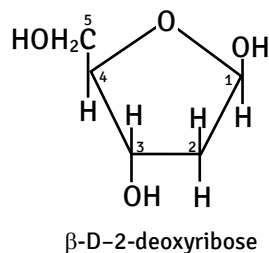
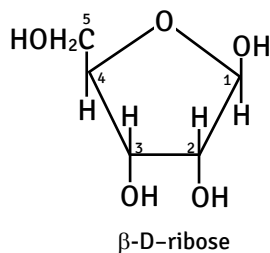
Nucleic Acids

Introduction:

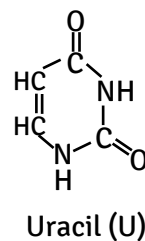
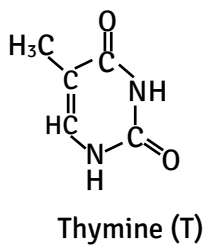
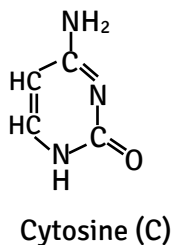
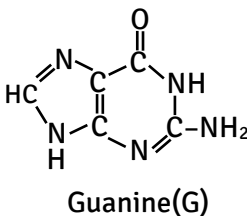
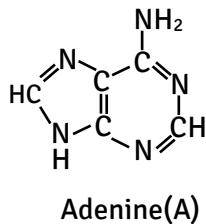
It has been observed that nucleus of a living cell is responsible for the transmission of inherent characters, also called heredity. The particles in nucleus of the cell, responsible for heredity, are called chromosomes which are made up of proteins and another type of biomolecules called nucleic acids. These are mainly of two types, the deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). Since nucleic acids are long chain polymers of nucleotides, so they are also called polynucleotides.

Chemical composition of Nucleic Acids:

Complete hydrolysis of DNA (or RNA) yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds (called bases). In DNA molecules, the sugar moiety is β -D-2-deoxyribose whereas in RNA molecule, it is β -D-ribose.



DNA contains four bases viz. adenine (A), guanine (G), cytosine (C) and thymine (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is uracil (U).

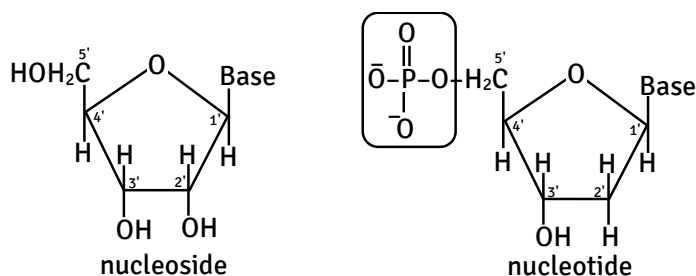


Structure of Nucleic Acids:

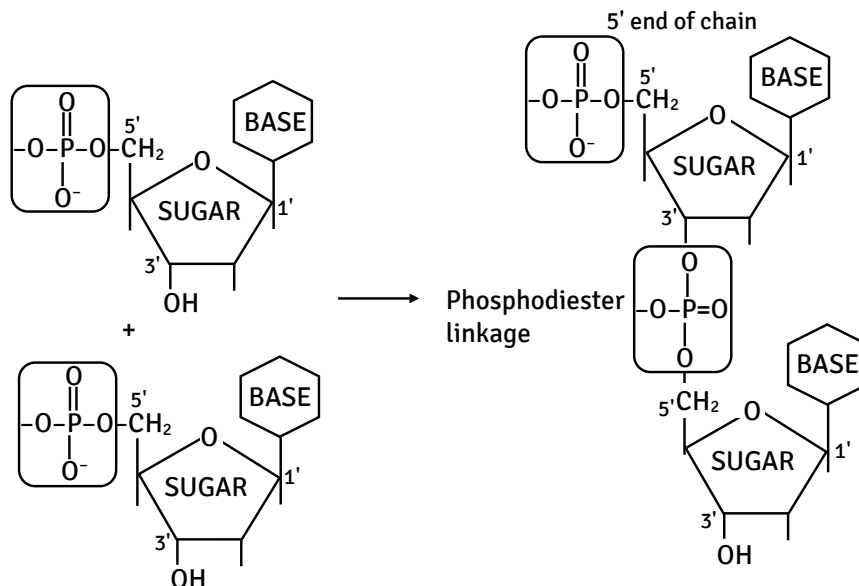
Nucleoside: A unit formed by the attachment of a base to 1' position of sugar is known as nucleoside. It contains only two components: a pentose sugar and a nitrogenous base.

Nucleotide: When nucleoside is linked to phosphoric acid at 5' -position of sugar moiety, we get a nucleotide.

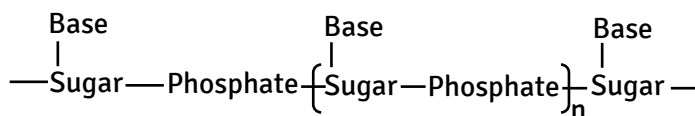
It contains all the three components: a pentose sugar, a nitrogenous base and a phosphoric acid group.



Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar. The formation of a typical dinucleotide is shown in below Fig.

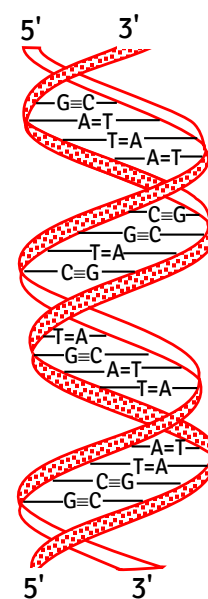


A simplified version of nucleic acid chain is as shown below.



Information regarding the sequence of nucleotides in the chain of a nucleic acid is called its primary structure. Nucleic acids have a secondary structure also. James Watson and Francis Crick gave a double strand helix structure for DNA (Fig.). Two nucleic acid chains are wound about each other and held together by hydrogen bonds between pairs of bases. The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine.

In secondary structure of RNA, helices are present which are only single stranded. Sometimes they fold back on themselves to form a double helix structure. RNA molecules are of three types and they perform different functions. They are named as messenger RNA (m-RNA), ribosomal RNA (r-RNA) and transfer RNA (t-RNA).



Double strand helix structure for DNA

DNA Fingerprinting:

It is known that every individual has unique fingerprints. These occur at the tips of the fingers and have been used for identification for a long time but these can be altered by surgery. A sequence of bases on DNA is also unique for a person and information regarding this is called DNA fingerprinting. It is same for every cell and cannot be altered by any known treatment. DNA fingerprinting is now used

- (i) in forensic laboratories for identification of criminals.
- (ii) to determine paternity of an individual.
- (iii) to identify the dead bodies in any accident by comparing the DNA's of parents or children.
- (iv) to identify racial groups to rewrite biological evolution.

Biological Functions of Nucleic Acids:

DNA is the chemical basis of heredity and may be regarded as the reserve of genetic information. DNA is exclusively responsible for maintaining the identity of different species of organisms over millions of years. A DNA molecule is capable of self duplication during cell division and identical DNA strands are transferred to daughter cells. Another important function of nucleic acids is the protein synthesis in the cell. Actually, the proteins are synthesised by various RNA molecules in the cell but the message for the synthesis of a particular protein is present in DNA.

Hormones

Hormones are molecules that act as intercellular messengers. These are produced by endocrine glands in the body and are poured directly in the blood stream which transports them to the site of action.

Chemical Nature of Hormones:

1. Steroids:- Estrogens, androgens
 2. Polypeptides:- Insulin, endorphins
 3. Amino acid derivatives:- Epinephrine, norepinephrine
- The role of insulin is keeping the blood glucose level within the narrow limit. Insulin is released in response to the rapid rise in blood glucose level. Hormone glucagon tends to increase the glucose level in the blood. The two hormones together regulate the glucose level in the blood.
 - Epinephrine and norepinephrine mediate responses to external stimuli. Growth hormones and sex hormones play role in growth and development.
 - Thyroxine produced in the thyroid gland is an iodinated derivative of amino acid tyrosine. Abnormally low level of thyroxine leads to hypothyroidism which is characterised by lethargy and obesity. Increased level of thyroxine causes hyperthyroidism. Low level of iodine in the diet may lead to hypothyroidism and enlargement of the thyroid gland. This condition is largely being controlled by adding sodium iodide to commercial table salt ("Iodised" salt).
 - Steroid hormones are produced by adrenal cortex and gonads (testes in males and ovaries in females). Glucocorticoids control the carbohydrate metabolism, modulate inflammatory reactions, and are involved in reactions to stress. The mineralocorticoids control the level of excretion of water and salt by the kidney. If adrenal cortex does not function properly then one of the results may be Addison's disease characterised by hypoglycemia, weakness and increased susceptibility.
 - Hormones released by gonads are responsible for development of secondary sex characters. Testosterone is the major sex hormone produced in males. It is responsible for development of secondary male characteristics (deep voice, facial hair, general physical constitution) and estradiol is the main female sex hormone. It is responsible for development of secondary female characteristics and participates in the control of menstrual cycle. Progesterone is responsible for preparing the uterus for implantation of fertilised egg.

Exercise 1.3

1. Which of the following bases is not present in DNA ?
(1) Cytosin (2) Gaunine
(3) Uracil (4) Adenine
2. In DNA, the complimentary bases are :
(1) Adenine and thymine; guanine and uracil
(2) Adenine and guanine; thymine and cytosine
(3) Uracil and adenine; cytosine and guanine
(4) Adenine and thymine; guanine and cytosine
3. Nucleic acids are the polymers of.
(1) Nucleosides (2) Nucleotides
(3) Bases (4) Sugars
4. Which one of the following vitamins is water soluble :
(1) Vitamin A (2) Vitamin B
(3) Vitamin E (4) Vitamin K
5. Deficiency of vitamin B₁ causes the disease:
(1) Cheilosis (2) Sterility
(3) Convulsions (4) Beri-Beri
6. Which one of the following is a peptide hormone:
(1) Glucagon (2) Testosterone
(3) Thyroxin (4) Adrenaline
7. Which of the following vitamin is helpful in delaying the blood clotting:
(1) Vitamin C (2) Vitamin B
(3) Vitamin E (4) Vitamin K
8. Vitamin C must be supplied regularly in diet because
(1) it is water soluble hence excreted in urine and can't be stored in the body
(2) it is fat soluble hence stored in the body and cannot be used on regular basis
(3) it is required in a large amount by the body hence supplied regularly
(4) it is water soluble hence used by the body on daily basic and it to be supplied regularly.
9. Bases common to RNA and DNA are
(1) adenine, guanine, cytosine
(2) adenine, uracil, cytosine
(3) adenine, guanine, thymine
(4) guanine, uracil, thymine.
10. A unit in nucleic acid which contains 'base-sugar-phosphate' unit is called
(1) nucleotide (2) nucleoside
(3) phosphotide (4) polypeptide.
11. Which of the following statements is not true about RNA?
(1) It is present in the nucleus of the cell.
(2) It has always double stranded α -helix structure.
(3) It controls the synthesis of protein.
(4) It usually does not replicate.
12. A metal present in vitamin B-12 is:
(1) Aluminium (2) Zinc
(3) Iron (4) Cobalt
13. Which of the following is/are incorrect?
(1) Nucleoside is linked to phosphoric acid at 5 position of sugar moiety get nucleotide.
(2) DNA contain four bases adenine guanine, cytosine and thymine.
(3) RNA contain four bases adenine guanine, cytosine and uracil.
(4) Glucose form the hydrogen sulphite addition product with NaHSO₃
14. The monomer of nucleic acids are held together by :
(1) Phosphodiester linkage
(2) Amide linkage
(3) Glycosidic linkage
(4) Peptide linkage
15. Insulin is secreted by:
(1) Pancreas (2) Stomach
(3) Thyroid (4) Adrenal medulla

Exercise 2

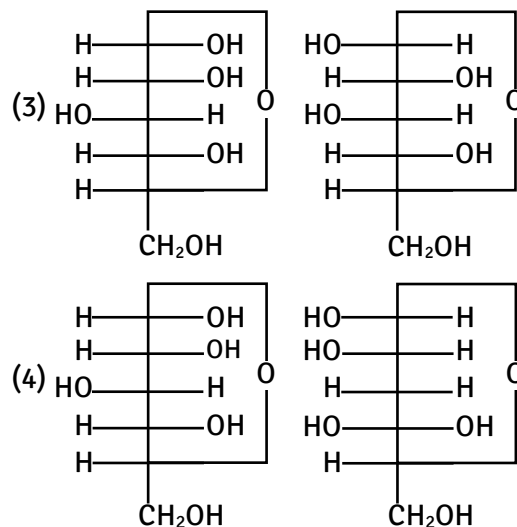
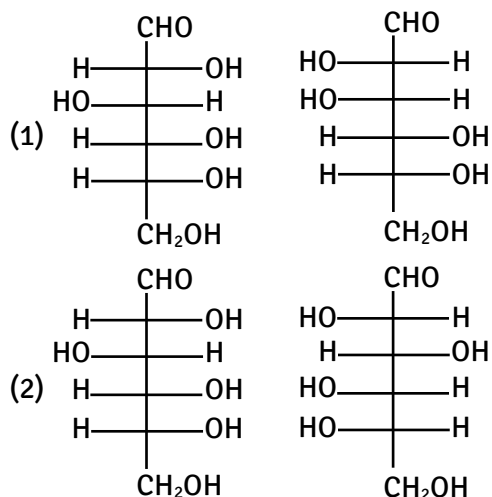
1. Starch molecules are polymer with repeating glucose units. Select the correct statement(s).

- (1) Glucose units are joined through - glycosidic linkage.
- (2) The branches of amylopectin are linked to the chain with -1,6'-glycosidic linkages.
- (3) Amylose has an unbranched skeleton of glucose molecules with -1,4' glycosidic linkages.
- (4) All of the above

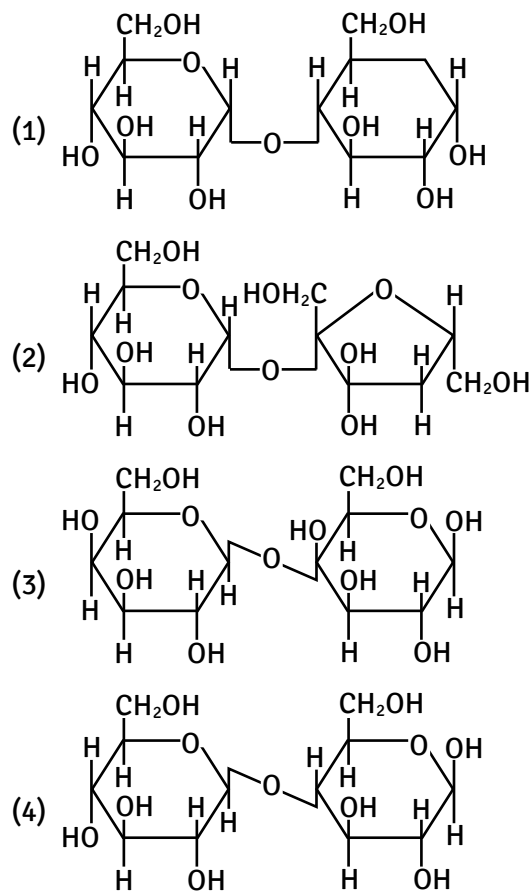
2. Which of the following reactions of glucose can be explained only by its cyclic structure.

- (1) Glucose forms penta acetate
- (2) Glucose reacts with hydroxylamine to form an oxime
- (3) Penta acetate of glucose does not react with hydroxylamine
- (4) Glucose is oxidised by nitric acid to gluconic acid

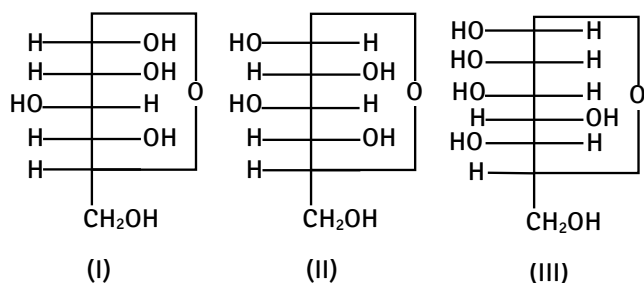
3. Which of the following pairs represents anomers?



4. In disaccharides, if the reducing groups of monosaccharides i.e. aldehydic or ketonic groups are bonded, these are non-reducing sugars. Which of the following disaccharide is a non-reducing sugar?

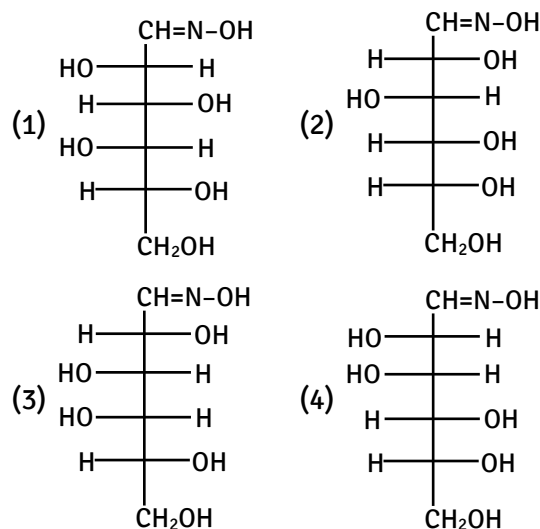


5. Three cyclic structures of monosaccharides are given below which of these are anomers.



- (1) I and II
 (2) II and III
 (3) I and III
 (4) III is anomer of I and II
6. Which of the following carbohydrates are branched polymer of glucose?
- (i) Amylose (ii) Amylopectin
 (iii) Cellulose (iv) Glycogen
- (1) ii, iv only (2) i, ii, iv only
 (3) iii, iv only (4) ii, iii, iv only
7. Amylopectin is composed of :
- (1) α -D-Glucose, C_1-C_4 and C_1-C_6 glycosidic linkage
 (2) α -D-Glucose, C_1-C_4 glycosidic linkage
 (3) β -D-Glucose, C_1-C_4 and C_1-C_6 glycosidic linkage
 (4) β -D-Glucose, C_1-C_4 glycosidic linkage
8. Identify the incorrect statement from the following:
- (1) Glycogen is called as animal starch
 (2) Glycosidic linkage makes cellulose polymer
 (3) Amylose is a branched chain polymer of glucose
 (4) Starch is a polymer of D-glucose
9. When glucose is heated with nitric acid, the product is
- (1) Gluconic acid (2) Saccharic acid
 (3) n-hexane (4) oxalic acid

10. D(+) glucose react with hydroxyl amine and yield an oxime. The structure of the oxime would be

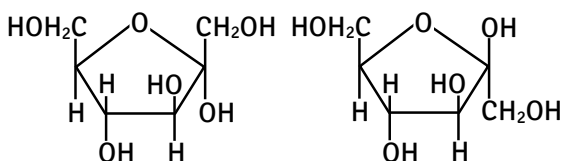


11. Identify the incorrect statement from the following
- (1) Amylose is a branched chain Polymer of glucose
 (2) Starch is a Polymer of α -D glucose
 (3) β -Glycosidic linkage makes cellulose polymer
 (4) Glycogen is called as animal starch
12. Which of the following is non reducing sugar
- (1) Starch (2) Cellulose
 (3) Sucrose (4) All
13. Which of the following statement is not true about glucose?
- (1) It is an aldohexose
 (2) On heating with HI, it forms n-hexane
 (3) It does not give schiff's test
 (4) It is present in furanose form
14. Which of the glycosidic linkage between galactose and glucose is present in lactose?
- (1) C-1 of galactose and C-4 of glucose
 (2) C-1 of glucose and C-6 of galactose
 (3) C-1 of glucose and C-4 of galactose
 (4) C-1 of galactose and C-6 of glucose

15. On oxidation with a mild oxidising agent like $\text{Br}_2/\text{H}_2\text{O}$, the glucose is oxidised to
 (1) saccharic acid (2) glucaric acid
 (3) gluconic acid (4) valeric acid.

16. During acetylation of glucose it needs x moles of acetic anhydride. The value of x would be
 (1) 3 (2) 5 (3) 4 (4) 1

17. Five-membered ring structures of fructose are given below. Mark the incorrect statement.

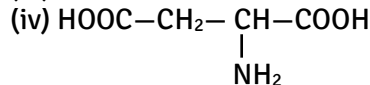
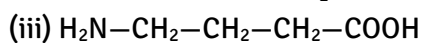
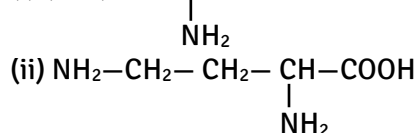
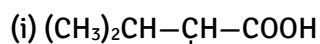


- (1) The five-membered ring structures are named as furanose structures.
 (2) The cyclic structures represent two anomers of fructose.
 (3) Five-membered ring structures are named as pyranose structures.
 (4) These are also called Haworth structures.
18. Which one of the following sets of monosaccharides forms sucrose?
 (1) α -D-Galactopyranose and α -D-glucopyranose
 (2) α -D-Glucopyranose and β -D-fructofuranose
 (3) β -D-Glucopyranose and α -D-fructofuranose
 (4) α -D-Glucopyranose and β -D-fructopyranose
19. Invert sugar is
 (1) a type of cane sugar
 (2) optically inactive form of sugar
 (3) mixture of glucose and galactose
 (4) mixture of glucose and fructose in equimolar quantities.
20. The glycosidic linkage involved in linking the glucose units in amylose part of starch is
 (1) $\text{C}_1\text{-C}_4$ β -linkage (2) $\text{C}_1\text{-C}_6$ β -linkage
 (3) $\text{C}_1\text{-C}_6$ α -linkage (4) $\text{C}_1\text{-C}_4$ α -linkage

21. Starch is composed of two polysaccharides which are
 (1) amylopectin and glycogen
 (2) amylose and glycogen
 (3) amylose and amylopectin
 (4) cellulose and glycogen.

22. Carbohydrates are stored in human body as the polysaccharide
 (1) starch (2) glycogen
 (3) cellulose (4) amylose.

23. Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule. Which of the following are basic?



- (1) (ii) only (2) (iii) and (iv)
 (3) (i) and (ii) (4) (ii) and (iii)
24. Which of the following statements is not correct?
 (1) Only α -amino acids are obtained on hydrolysis of proteins.
 (2) The amino acids which are synthesised in the body are known as non-essential amino acids.
 (3) There are 20 essential amino acids.
 (4) L-amino acids are represented by writing the $-\text{NH}_2$ group on the left side.
25. Which of the following statements is not correct?
 (1) Proteins are polyamides formed from amino acids.
 (2) Except glycine, all other amino acids show optical activity.
 (3) Natural proteins are commonly made up of L-isomer of amino acids.
 (4) In α -amino acids, $-\text{NH}_2$ and $-\text{COOH}$ groups are attached to different carbon atoms.

26. The peptide linkage formed between glycine ($\text{NH}_2\text{CH}_2\text{COOH}$) & alanine ($\text{NH}_2\text{CH}(\text{CH}_3)\text{COOH}$) to

give glycylalanine can be shown as

- (1) $\text{NH}_2\text{—CH}_2\text{—NH—CH}(\text{CH}_3)\text{—COOH}$
 (2) $\text{NH}_2\text{—CH}_2\text{—CONH—CH}(\text{CH}_3)\text{—COOH}$
 (3) $\text{H}_2\text{NCOCH}_2\text{—CH}(\text{CH}_3)\text{—CONH}_2$
 (4) $\text{HOOC—CH}_2\text{—NH—NH—CH}(\text{CH}_3)\text{—COOH}$

27. In fibrous proteins, polypeptide chains are held together by

- (1) van der Waals' forces
 (2) electrostatic forces of attraction
 (3) hydrogen bonds
 (4) covalent bonds.

28. Globular proteins are present in

- (1) blood (2) eggs
 (3) milk (4) all of these.

29. Denaturation of protein leads to loss of its biological activity by

- (1) formation of amino acids
 (2) loss of primary structure
 (3) loss of both primary and secondary structure
 (4) loss of both secondary and tertiary structures.

30. Mark the wrong statement about denaturation of proteins.

- (1) The primary structure of the protein does not change.
 (2) Globular proteins are converted into fibrous proteins.
 (3) Fibrous proteins are converted into globular proteins.
 (4) The biological activity of the protein is lost.

31. Which of the statements about "Denaturation" given below are correct ?

- (a) Denaturation of proteins causes loss of secondary and tertiary structures of the protein
 (b) Denaturation leads to the conversion of double strand of DNA into single strand.
 (c) Denaturation affects primary structure which gets distorted.

- (1) (a), (b) and (c) (2) (b) and (c)
 (3) (a) and (c) (4) (a) and (b)

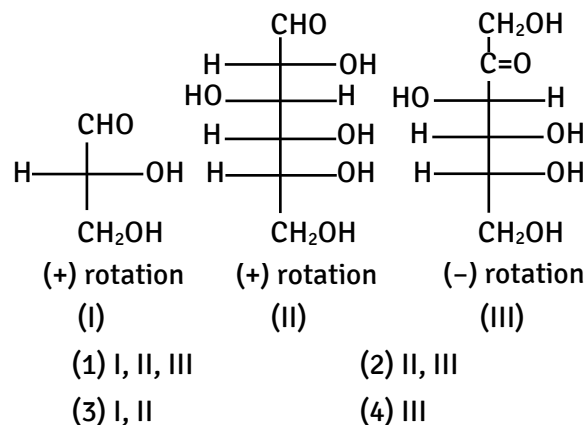
32. Each polypeptide in a protein has amino acids linked with each other in a specific sequence. This sequence of amino acids is said to be ____.

- (1) primary structure of proteins
 (2) secondary structure of proteins
 (3) tertiary structure of proteins
 (4) quaternary structure of proteins

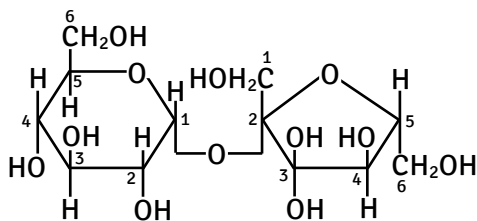
33. DNA and RNA contain four bases each. Which of the following bases is not present in RNA?

- (1) Adenine (2) Uracil
 (3) Thymine (4) Cytosine

34. Optical rotations of some compounds along with their structures are given below. Which of them have D-configuration?



35. Structure of a disaccharide formed by glucose and fructose is given below. Identify anomeric carbon atoms in monosaccharide units.



- (1) '1' carbon of glucose and '1' carbon of fructose.
 (2) '1' carbon of glucose and '5' carbon of fructose.
 (3) '1' carbon of glucose and '2' carbon of fructose.
 (4) '6' carbon of glucose and '6' carbon of fructose.

36. The structure of protein that is unaffected by heating is

- (1) secondary structure
 (2) tertiary structure
 (3) primary structure
 (4) quaternary structure.

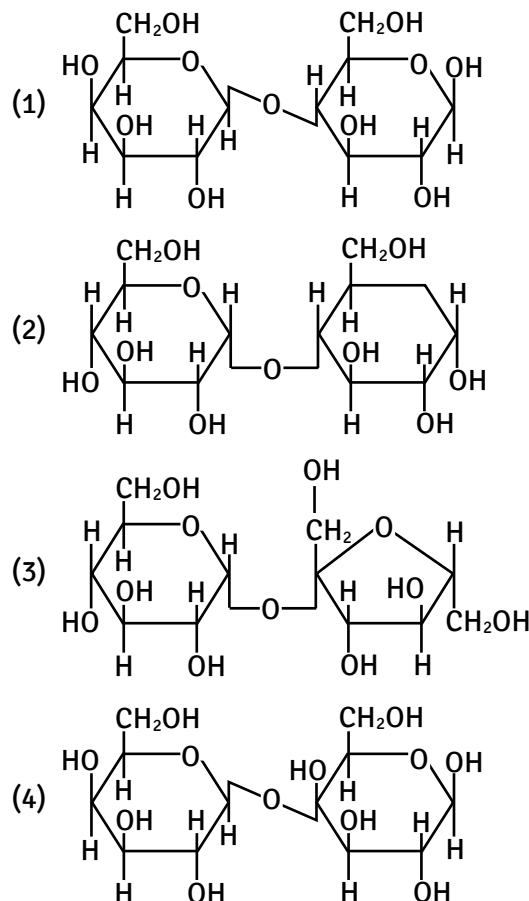
37. The secondary structure of protein is stabilised by

- (1) van der Waals' forces
 (2) peptide bond
 (3) hydrogen bonding
 (4) glycosidic bond.

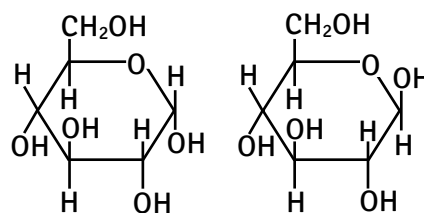
38. Which of the given statements is incorrect about glycogen?

- (1) It is present in animal cells.
 (2) Only α -linkages are present in the molecule.
 (3) It is a straight chain polymer similar to amylose.
 (4) It is present in some yeast and fungi.

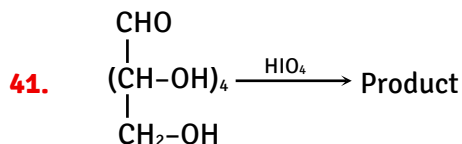
39. Which of the following disaccharide is a nonreducing sugar :



40. Relation between (A) and (B) is :



- (1) Enantiomers
 (2) Anomers
 (3) Positional isomers
 (4) Functional isomers



- (1) 5 $\text{HCHO} + \text{HCOOH}$
 (2) 5 $\text{HCOOH} + \text{HCHO}$
 (3) $\text{CH}_3\text{COOH} + 5\text{HCHO}$
 (4) 5 $\text{CH}_3\text{CHO} + \text{HCHO}$

42. Which of the following is sulphur containing amino acid :

- (a) Methionine
- (b) Cysteine
- (c) Leucine
- (d) Glutamic acid

- (1) a, b, c (2) a, b
- (3) b, c (4) c, d

43. Which of the following are essential aminoacids :

- (1) Tryptophane, Lysine and Histidine
- (2) Serine, Lysine and Tryptophane
- (3) Serine, Histidine and Tryptophane
- (4) Serine, Histidine and Lysine

44. Which of the following are non reducing in nature-

- (A) Cellulose (B) Sucrose
- (C) Lactose (D) Starch

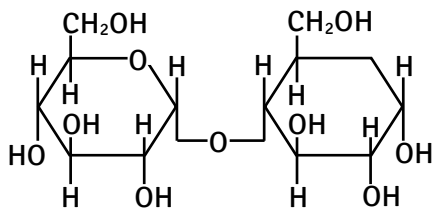
- (1) A and B are correct
- (2) A, B and C are correct
- (3) A, B and D are correct
- (4) A, C and D are correct

45. Which one of the following sets of monosaccharides forms sucrose ?

- (1) D-Glucofuranose and - D-fructofuranose
- (2) D-Glucopyranose and - D-fructopyranose
- (3) D-Galactopyranose and - D-glucopyranose
- (4) D-Glucopyranose and - D-Fructofuranose

Exercise 3

1. **Assertion:** -glycosidic linkage is present in maltose.



Reason: Maltose is composed of two α -D-glucose units in which C-1 of one glucose unit is linked to C-4 of another glucose unit.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - (3) If Assertion is True but the Reason is False.
 - (4) If Assertion & Reason are false.
2. **Assertion:** Oxidation of glucose by Br_2 -water gluconic acid.
- Reason:** Br_2 -water oxidizes $-\text{CHO}$ group into $-\text{COOH}$ group.
- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - (3) If Assertion is True but the Reason is False.
 - (4) If Assertion & Reason are false.
3. **Assertion:** Glucose does not give 2,4 -DNP test.
- Reason:** Glucose exists in cyclic hemiacetal form,
- (1) If both assertion and reason are true and reason is the correct explanation of assertion,
 - (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) If assertion is true but reason is false.
 - (4) If both assertion and reason are false.

4. **Assertion:** Maltose and lactose are examples of reducing sugars.

Reason: Maltose and lactose reduce Fehling's solution and Tollens' reagent.

- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct.
5. **Assertion:** Glucose is correctly named as D-(+)-glucose.
- Reason:** 'D' before the name of glucose represents its dextrorotatory nature.
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct.
6. **Assertion:** Hydrolysis of sucrose brings about a change in sign of rotation from dextro to laevo.
- Reason:** Hydrolysis always changes the optical rotation of a compound.
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct.

7. **Assertion:** Sucrose is a disaccharide and a non-reducing sugar.
Reason: Sucrose involves glycosidic linkage between C₁ of β -glucose and C₂ of α -fructose. Choose the most appropriate answer from the options given below:
- (1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 - (2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 - (3) If the Assertion is correct but Reason is incorrect.
 - (4) If the Assertion is incorrect and Reason is correct.
8. Given below are two statements
Statement-I: D-(+)-Glucose and D-(-)-Fructose are formed on hydrolysis of sucrose
Statement-II: Sucrose is an invert sugar
Choose the most appropriate answer from the options given below:
- (1) Both statement I and statement II are true
 - (2) Statement I is false but Statement II true
 - (3) Statement I is true but Statement II is false
 - (4) Both Statement I and Statement II are false
9. Given below are two statements:
Statement-I: The pentaacetate of glucose does not react with hydroxylamine indicating the absence of free -CHO group.
Statement-II: Despite having the aldehyde group, glucose does not give schiff's test and it does not form the hydrogen-sulphite addition product with NaHSO₃.
In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct and statement II is incorrect
 - (2) Statement-I and statement-II both are correct.
 - (3) Statement I and statement II both are incorrect
 - (4) Statement I is incorrect and statement II is correct.

10. Given below are two statements:
Statement-I: Amylose is water soluble component which constitutes about 15- 20% of starch. Chemically amylose is a long unbranched chain with 200-1000 α -D-(+)-glucose units held together by C₁-C₄ glycosidic linkage.
Statement-II: Amylopectin is insoluble in water and constitutes about 80-85% of starch. It is a branched chain polymer of α -D-glucose units in which chain is formed by C₁-C₄ glycosidic linkage whereas branching occurs by C₁-C₆ glycosidic linkage.
In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement-I and statement-II both are correct.
 - (2) Statement I is correct and statement II is incorrect
 - (3) Statement I is incorrect and statement II is correct.
 - (4) Statement I and statement II both are incorrect
11. Given below are two statements:
Statement-I: Proteins are the polymer of α -amino acids.
Statement-II : The amino acids which can be synthesized in body are known as essential amino acids.
In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Both statement I & II are correct.
 - (2) Both statement I & II are incorrect.
 - (3) Statement I is incorrect but statement II is correct
 - (4) Statement I is correct but statement II is incorrect

12. Statement-I: Enzymes which catalyze the oxidation of one substrate with simultaneous reduction of another substrate are named as oxidoreductase enzymes.

Statement-II: Enzymes are needed only in small quantity for the progress of a reaction.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct.
- (3) Both statements I and statements II are correct.
- (4) Both statements I and statements II are incorrect.

13. Given below are two statements:

Statement-I: All carbohydrates which reduce Fehling's solution and Tollen's reagent are referred to as reducing sugars.

Statement-II: All disaccharides whether aldose or ketose are reducing sugars.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is incorrect and statement II is correct.
- (2) Statement-I and statement-II both are correct.
- (3) Statement I is correct and statement II is incorrect
- (4) Statement I and statement II both are incorrect

14. Given below are two statements:

Statement-I: Sucrose is levorotatory but after hydrolysis gives dextrorotatory glucose and levorotatory fructose.

Statement-II: Hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to laevo (-) and the product is named as invert sugar.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct and statement II is incorrect
- (2) Statement I is incorrect and statement II is correct.
- (3) Statement-I and statement-II both are correct.
- (4) Statement I and statement II both are incorrect

15. Given below are two statements:

Statement-I: Only α -Amino acids are obtained on hydrolysis of proteins.

Statement-II: The amino acids which can be synthesized in body are known as essential amino acids.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I & II are correct.
- (2) Both statement I & II are incorrect.
- (3) Statement I is incorrect but statement II is correct
- (4) Statement I is correct but statement II is incorrect

16. Match the column.

Column-I (Reaction of Glucose)		Column-II (Product)	
a	Glucose $\xrightarrow{\text{HI}, \Delta}$	p	Oxime
b	Glucose $\xrightarrow{\text{Br}_2 \text{ Water}}$	q	n-Hexane
c	Glucose $\xrightarrow{\text{Nitric acid}}$	r	Gluconic acid
d	Glucose $\xrightarrow[\text{H}^+]{\text{NH}_2\text{OH}}$	s	Saccharic acid

- (1) a - q, b - s, c - p, d - r
- (2) a - q, b - p, c - r, d - s
- (3) a - q, b - s, c - r, d - p
- (4) a - q, b - r, c - s, d - p

17. Match List-I with List-II

List-I (Saccharides)		List-II (Glycosidic-linkages)	
(A)	Sucrose	(I)	$\alpha 1-4$
(B)	Maltose	(II)	$\alpha 1-4$ and $\alpha 1-6$
(C)	Lactose	(III)	$\alpha 1-\beta 2$
(D)	Amylopectin	(IV)	$\beta 1-4$

Choose the correct answer from the options given below

- (1) (A-III), (B-I), (C-IV), (D-II)
- (2) (A-IV), (B-II), (C-I), (D-III)
- (3) (A-II), (B-IV), (C-III), (D-I)
- (4) (A-I), (B-II), (C-III), (D-IV)

18. Match the vitamins given in column I with the deficiency diseases caused by it given in column II and mark the appropriate choice.

Column I		Column II	
(A)	Vitamin B ₁	(i)	Convulsions
(B)	Vitamin B ₂	(ii)	Pernicious anaemia
(C)	Vitamin B ₁₂	(iii)	Beri-beri
(D)	Vitamin B ₆	(iv)	Cheilosis

- (1) (A) → (iv), (B) → (iii), (C) → (i), (D) → (ii)
 (2) (A) → (i), (B) → (iv), (C) → (iii), (D) → (ii)
 (3) (A) → (ii), (B) → (i), (C) → (iv), (D) → (iii)
 (4) (A) → (iii), (B) → (iv), (C) → (ii), (D) → (i)

19. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	Nucleoside	(i)	Sugar + base + phosphoric acid group
(B)	Nucleotide	(ii)	Cytosine + uracil
(C)	DNA	(iii)	Sugar + base
(D)	RNA	(iv)	Cytosine + thymine

- (1) (A) → (iii), (B) → (i), (C) → (iv), (D) → (ii)
 (2) (A) → (i), (B) → (iv), (C) → (iii), (D) → (ii)
 (3) (A) → (ii), (B) → (iii), (C) → (i), (D) → (iv)
 (4) (A) → (iv), (B) → (ii), (C) → (i), (D) → (iii)

20. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	Peptide linkage	(i)	Inversion
(B)	Nucleic acid	(ii)	Polysaccharide
(C)	Hydrolysis of cane sugar	(iii)	Proteins
(D)	Starch	(iv)	Nucleotides

- (1) (A) → (ii), (B) → (i), (C) → (iii), (D) → (iv)
 (2) (A) → (iv), (B) → (i), (C) → (ii), (D) → (iii)
 (3) (A) → (iii), (B) → (iv), (C) → (i), (D) → (ii)
 (4) (A) → (i), (B) → (iii), (C) → (iv), (D) → (ii)

21. Match the vitamin in column I with their scientific name in column II and choose the correct option-

Vitamin		Chemical name	
(a)	Vitamin B ₆	(i)	Riboflavin
(b)	Vitamin C	(ii)	Pyridoxine
(c)	Vitamin B ₂	(iii)	Ascorbic acid
(d)	Vitamin B ₁	(iv)	Thiamine

- (1) a(ii), b(iii), c(i), d(iv)
 (2) a(ii), b(i), c(iv), d(iii)
 (3) a(iii), b(i), c(iv), d(ii)
 (4) a(i), b(ii), c(iii), d(iv)

22. Match the following column-I with product given in the column-II and select the correct option from the code given below :-

Column- I		Column- II	
(A)	$\begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow[\text{Red P}]{\text{HI ; } \Delta}$	(p)	$\begin{array}{c} \text{COOH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array}$
(B)	$\begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{NH}_2\text{OH}}$	(q)	$\begin{array}{c} \text{CN} \\ \\ \text{CH} \\ \quad \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array}$
(C)	$\begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow[\text{Alkaline}]{\text{HCN}}$	(r)	$\begin{array}{c} \text{CH=N-OH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array}$
(D)	$\begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{Br}_2 \text{ water}}$	(s)	CH ₃ -(CH ₂) ₄ -CH ₃

Codes :

	A	B	C	D
(1)	(s)	(r)	(q)	(p)
(2)	(s)	(r)	(p)	(q)
(3)	(r)	(s)	(q)	(p)
(4)	(s)	(q)	(p)	(r)

23. Match the column I with II and mark the appropriate choice:

Column I		column II	
(A)	Nucleoside	(i)	Sugar + base + phosphoric acid group
(B)	Nucleotide	(ii)	Cytosine + Uracil
(C)	DNA	(iii)	Sugar + base
(D)	RNA	(iv)	Cytosine + thymine

- (1) A - iii; B - i; C - iv; D - ii
 (2) A - i; B - iv; C - iii; D - ii
 (3) A - ii; B - iii; C - i; D - iv
 (4) A - iv; B - ii; C - i; D - iii

24. Match the column.

Column-I (Carbohydrates)		Column-II (Source or name)	
(a)	Sucrose	(p)	Animal starch
(b)	Lactose	(q)	Cereals, roots, tubers
(c)	Glycogen	(r)	Milk sugar
(d)	Starch	(s)	Cane sugar

- (1) a-r, b-s, c-q, d-p
 (2) a-s, b-r, c-p, d-q
 (3) a-s, b-p, c-r, d-q
 (4) a-p, b-q, c-r, d-s

25. Match the following:

Column-I		Column-II	
(i)	Riboflavin	(A)	Beriberi
(ii)	Thiamine	(B)	Scurvy
(iii)	Pyridoxine	(C)	Cheilosis
(iv)	Ascorbic Acid	(D)	Convulsions

- (1) i C; ii A; iii D; iv B
 (2) i C; ii D; iii A; iv B
 (3) i A; ii D; iii C; iv B
 (4) i D; ii B; iii A; iv C

26. Match the column

Column-I (Carbohydrates)		Column-II (Source or name)	
a	Sucrose	p	Animal starch
b	Lactose	q	Cereals, roots, tubers
c	Glycogen	r	Milk sugar
d	Starch	s	Cane sugar

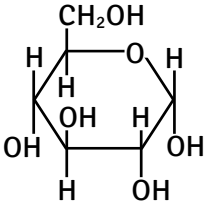
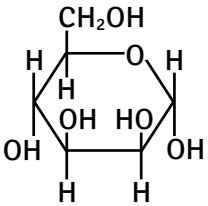
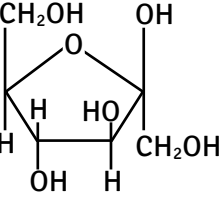
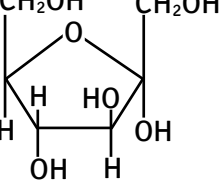
- (1) a - r, b - s, c - q, d - p
 (2) a - s, b - r, c - p, d - q
 (3) a - s, b - p, c - r, d - q
 (4) a - p, b - q, c - r, d - s

27. Match the column.

Column-I (Carbohydrates)		Column-II (Monosaccharide unit)	
a	Maltose	p	β -D-Glucose
b	Lactose	q	Two α -D-Glucose
c	Cellulose	r	β -D-Galactose & β -D-Glucose
d	Sucrose	s	α -D-Glucose & β -D-Fructose

- (1) a - q, b - s, c - r, d - p
 (2) a - q, b - r, c - p, d - s
 (3) a - r, b - s, c - p, d - q
 (4) a - r, b - p, c - q, d - s

28. Match the column.

Column-I (Structure)		Column-II (name)	
a		p	α -D-Mannose
b		q	β -D-Fructose
c		r	α -D-Glucose
d		s	α -D-Fructose

- (1) a - p, b - r, c - q, d - s
 (2) a - r, b - p, c - q, d - s
 (3) a - p, b - r, c - s, d - q
 (4) a - r, b - p, c - s, d - q

29. Match the column I & II:

Column-I (Vitamins)		Column-II (Sources)	
a	Vitamin-A	p	Exposure to sunlight
b	Vitamin-B ₁₂	q	Yeast, milk
c	Vitamin-B ₆	r	Meat, fish, curd
d	Vitamin-D	s	Fish liver oil

- (1) a - s, b - r, c - q, d - p
 (2) a - s, b - q, c - p, d - r
 (3) a - q, b - s, c - r, d - p
 (4) a - r, b - p, c - q, d - s

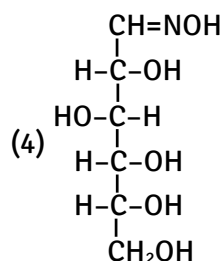
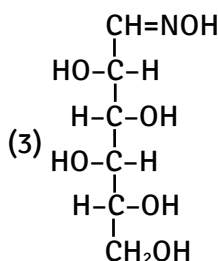
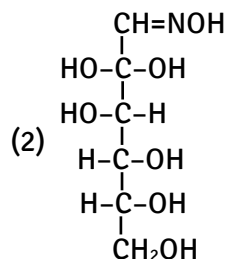
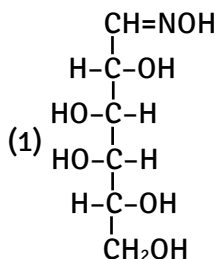
30. Match the column:

Column-I		Column-II	
a	Fat soluble vitamin	p	Fingerprinting
b	Water soluble vitamin	q	Vitamin A and vitamin D
c	DNA	r	Uracil
d	RNA	s	Vitamin B and vitamin C

- (1) a - s, b - s, c - r, d - p
 (2) a - q, b - s, c - p, d - r
 (3) a - q, b - r, c - p, d - s
 (4) a - q, b - s, c - r, d - p

Exercise 4 (Previous Year's Questions)

1. D(+) glucose reacts with hydroxylamine and yields an oxime. The structure of the oxime would be: **[AIPMT - 2014]**



2. In a protein molecule various amino acids are linked together by **[NEET-I- 2016]**

- (1) Glycosidic bond
(2) Glycosidic bond
(3) Peptide bond
(4) Dative bond

3. The correct statement regarding RNA and DNA, respectively is **[NEET-I- 2016]**

- (1) The sugar component in RNA is arabinose and the sugar component in DNA is 2'-deoxyribose.
(2) The sugar component in RNA is ribose and the sugar component in DNA is 2'-deoxyribose.
(3) The sugar component in RNA is arabinose and the sugar component in DNA is ribose.
(4) The sugar component in RNA is 2'-deoxyribose and the sugar component in DNA is arabinose.

4. Which one given below is a non-reducing sugar? **[NEET-I- 2016]**

- (1) Maltose (2) Lactose
(3) Glucose (4) Sucrose

5. The central dogma of molecular genetics states that the genetic information flows from **[NEET-II-2016]**

- (1) DNA Carbohydrates Proteins
(2) DNA RNA Proteins
(3) DNA RNA Carbohydrates
(4) Amino acids Proteins DNA

6. Which of the following statements is not correct? **[NEET- 2017]**

- (1) Insulin maintains sugar level in the blood of a human body
(2) Ovalbumin is a simple food reserve in egg white
(3) Blood proteins thrombin and fibrinogen are involved in blood clotting
(4) Denaturation makes the proteins more active.

7. The difference between amylose and amylopectin is **[NEET(UG) 2018]**

- (1) Amylopectin have 1,4 -linkage and 1,6 -linkage
(2) Amylose have 1,4 -linkage and 1,6 -linkage
(3) Amylopectin have 1,4 -linkage and 1,6 -linkage
(4) Amylose is made up of glucose and galactose

8. Which of the following compound can form a zwitterion? **[NEET(UG) 2018]**

- (1) Aniline (2) Acetanilide
(3) Benzoic acid (4) Glycine

9. The non-essential amino acid among the following is **[NEET(UG) 2019]**

- (1) Valine (2) Leucine
(3) Alanine (4) Lysine

10. Which structure(s) of proteins remains(s) intact during denaturation process ?

[NEET(UG) 2019(Odisha)]

- (1) Both secondary and tertiary structures
- (2) Primary structure only
- (3) Secondary structure only
- (4) Tertiary structure only

11. Sucrose on hydrolysis gives

[NEET(UG) 2020]

- (1) D-Fructose + -D-Fructose
- (2) D-Glucose + -D-Fructose
- (3) D-Glucose + -D-Glucose
- (4) D-Glucose + -D-Fructose

12. Which of the following is a basic amino acid:

[NEET(UG) 2020]

- (1) Lysine
- (2) Serine
- (3) Alanine
- (4) Tyrosine

13. The RBC deficiency is deficiency disease of:

[NEET(UG) 2021]

- (1) Vitamin B₁₂
- (2) Vitamin B₆
- (3) Vitamin B₁
- (4) Vitamin B₂

14. The **incorrect** statement regarding enzymes is:

[NEET(UG) 2022]

- (1) Enzymes are biocatalysts.
- (2) Like chemical catalysts enzymes reduce the activation energy of bio processes.
- (3) Enzymes are polysaccharides.
- (4) Enzymes are very specific for a particular reaction and substrate.

15. Given below are two statements:

Statement I: a unit formed by the attachment of a base of 1' position of sugar is known as nucleoside.

Statement II: When nucleoside is linked to phosphorous acid at 5' -position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the option given below :

[NEET(UG) 2023]

- (1) Both Statement I and Statement II are true.
- (2) Both Statement I and Statement II are false.
- (3) Statement I is true but Statement II is false.
- (4) Statement I is false but Statement II is true.

16. Cheilosis occurs due to deficiency of

[NEET(UG)-Manipur 2023]

- (1) Thiamine
- (2) Nicotinamide
- (2) Pyridoxamine
- (4) Riboflavin

17. The reagents with which glucose does **not** react to give the corresponding tests/products are

[NEET-2024]

- A. Tollen's reagent
- B. Schiff's reagent
- C. HCN
- D. NH₂OH
- E. NaHSO₃

Choose the correct options from the given below:

- (1) B and E
- (2) E and D
- (3) B and C
- (4) A and D

18. Given below are two statements :

[RE-NEET-2024]

Statements I: Glycogen is similar to amylose in its structure.

Statements II: Glycogen is found in yeast and fungi also.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true.
- (3) Both Statement I and Statement II are true.
- (4) Both Statement I and Statement II are false.

19. Sugar 'X' [NEET-2025]

- A. is found in honey
- B. is a keto sugar
- C. exists in a and b anomeric forms
- D. is laevorotatory

'X' is :

- (1) Maltose (2) Sucrose
- (3) D-Glucose (4) D-Fructose

20. Match List I with List II. [NEET-2025]

List I (Name of Vitamin)		List II (Deficiency disease)	
A.	Vitamin B ₁₂	I.	Cheilosis
B.	Vitamin D	II.	Convulsions
C.	Vitamin B ₂	III.	Rickets
D.	Vitamin B ₆	IV.	Pernicious anemia

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-III, C-II, D-IV
- (4) A-IV, B-III, C-I, D-II

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	4	3	2	1	2	4	1	1	2	3	4	4	4	2	3	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	3	4	4	2	3	1	2	4	1	1	1	1	4	4	2	

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Ans.	3	4	2	2	4	1	4	1	1	1	2	4	4	1	1	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	2	1	1	1	3	2	2	1	3	4	1	3	2	3	2	4	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	2	1	3	4	2	3	4	4	3	4	1	3	1	3	3	3	3	3	2
Que.	41	42	43	44	45															
Ans.	2	2	1	3	4															

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	1	1	3	3	3	1	2	1	4	3	3	2	4	4	1	4	1	3
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	1	1	1	2	1	2	2	2	1	2										

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	2	4	2	4	1	4	3	2	4	1	1	3	3	4	1	2	4	4

